ALGEBRA I · COMMON CORE HIGH SCHOOL MATH

Question Collection

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

84 question sets • 2100 questions

CONTENTS

Set 1	**HSN-RN.A.1**	Explain the meaning of rational exponents from the properties of integer exponen	7
Set 2	**HSN-RN.A.2**	Rewrite expressions with radicals and rational exponents	38
Set 3	**HSN-RN.B.3**	Prove that the sum of a rational number and an irrational number is irrational	69
Set 69	**HSN-RN.B.3**	Prove that the product of a nonzero rational number and an irrational number is ir	100
Set 70	**HSN-RN.B.3**	Prove that sums and products of two rational numbers are rational	131
Set 4	**HSN-Q.A.1**	Use units to guide multistep solutions and to choose scales for graphs	162
Set 5	**HSN-Q.A.2**	Define appropriate quantities for descriptive modeling	193
Set 6	**HSN-Q.A.3**	Choose a level of accuracy that fits the limits of measurement	224
Set 7	**HSA-SSE.A.1.a**	Interpret the terms, factors, and coefficients of an expression in context	255
Set 8	**HSA-SSE.A.1.b**	Interpret complicated expressions by viewing one or more of their parts as a singl	286
Set 9	**HSA-APR.A.1**	Add and subtract polynomials and explain why the result is a polynomial	317
Set 10	**HSA-APR.A.1**	Multiply polynomials	348
Set 11	**HSA-SSE.A.2**	Use structure to factor with a greatest common factor and the difference of squar	379
Set 12	**HSA-SSE.A.2**	Factor quadratic trinomials	410
Set 13	**HSA-SSE.A.2**	Use structure to factor perfect squares and expressions in quadratic form	441
Set 14	**HSA-SSE.B.3.a**	Factor a quadratic expression to reveal the zeros of the function it defines	472
Set 15	**HSA-SSE.B.3.b**	Complete the square to reveal the maximum or minimum value of a quadratic fu	503

CONTENTS (continued)

Set 16	**HSA-SSE.B.3.c**	Use the properties of exponents to transform exponential expressions	534
Set 17	**HSA-REI.A.1**	Explain each step in solving an equation and justify the solution method	565
Set 18	**HSA-REI.B.3**	Solve linear equations in one variable, including equations with letter coefficients	596
Set 19	**HSA-REI.B.3**	Solve linear inequalities and compound inequalities in one variable	627
Set 20	**HSA-CED.A.1**	Create linear equations and inequalities in one variable and use them to solve pr	658
Set 21	**HSA-CED.A.4**	Rearrange formulas to highlight a quantity of interest	689
Set 22	**HSA-REI.B.4.a**	Complete the square to solve quadratic equations and derive the quadratic formu	720
Set 71	**HSA-REI.B.4.a**	Derive the quadratic formula by completing the square	751
Set 23	**HSA-REI.B.4.b**	Solve quadratic equations by inspection, square roots, and factoring	782
Set 24	**HSA-REI.B.4.b**	Solve quadratic equations with the quadratic formula and use the discriminant	813
Set 25	**HSA-CED.A.1**	Create quadratic and exponential equations in one variable and use them to solve	844
Set 26	**HSA-REI.C.5**	Show that replacing an equation by its sum with a multiple of the other keeps the	875
Set 72	**HSA-REI.C.5**	Prove that replacing an equation by its sum with a multiple of the other gives an e	906
Set 27	**HSA-REI.C.6**	Solve systems of linear equations exactly by substitution and elimination	937
Set 28	**HSA-REI.C.6**	Solve systems of linear equations in context, including systems with no solution or	968
Set 29	**HSA-REI.C.7**	Solve a system of a linear equation and a quadratic equation algebraically	999
Set 30	**HSA-REI.D.10**	Understand the graph of an equation as the set of all its solutions	1030

CONTENTS (continued)

Set 31	**HSA-REI.D.11**	Explain why intersections of y = f(x) and y = g(x) solve f(x) = g(x), and approxim	1061
Set 73	**HSA-REI.C.7**	Solve linear-quadratic systems by graphing and check the solutions algebraically	1092
Set 84	**HSA-REI.C.6**	Solve systems of linear equations by graphing, including approximate solutions	1123
Set 32	**HSA-REI.D.12**	Graph the solutions of linear inequalities and systems of linear inequalities in two	1154
Set 33	**HSA-CED.A.2**	Create equations in two or more variables and graph them on labeled axes	1185
Set 34	**HSA-CED.A.3**	Represent constraints with equations and inequalities and decide whether solution	1216
Set 35	**HSF-IF.A.1**	Understand a function as a rule that assigns exactly one output to each input, wit	1247
Set 36	**HSF-IF.A.2**	Use function notation, evaluate functions, and interpret statements in function not	1278
Set 37	**HSF-IF.A.3**	Recognize that sequences are functions whose domain is a set of integers	1309
Set 38	**HSF-IF.B.4**	Interpret key features of graphs and tables of functions in context	1340
Set 78	**HSF-IF.B.4**	Sketch graphs that show key features given in a verbal description	1371
Set 39	**HSF-IF.B.5**	Relate the domain of a function to the quantity it describes	1402
Set 40	**HSF-IF.B.6**	Calculate, interpret, and estimate the average rate of change of a function over a	1433
Set 79	**HSF-IF.B.6**	Estimate the average rate of change of a function from its graph	1464
Set 41	**HSF-IF.C.7.a**	Graph linear functions and show their intercepts	1495
Set 42	**HSF-IF.C.7.a**	Graph quadratic functions and show intercepts, maxima, and minima	1526
Set 43	**HSF-IF.C.7.b**	Graph absolute value functions and show their key features	1557

CONTENTS (continued)

Set 74	**HSF-IF.C.7.b**	Graph square root and cube root functions	1588
Set 75	**HSF-IF.C.7.b**	Graph piecewise-defined and step functions	1619
Set 44	**HSF-IF.C.7.e**	Graph exponential functions and show intercepts and end behavior	1650
Set 45	**HSF-IF.C.8.a**	Factor and complete the square to show zeros, extreme values, and symmetry, an	1681
Set 46	**HSF-IF.C.8.b**	Use the properties of exponents to interpret growth and decay rates	1712
Set 47	**HSF-IF.C.9**	Compare properties of two functions represented in different ways	1743
Set 48	**HSF-BF.A.1.a**	Write an explicit expression or a recursive process for a function from a context	1774
Set 80	**HSF-BF.A.1.a**	Build quadratic functions that model situations	1805
Set 49	**HSF-BF.A.1.b**	Combine standard function types using arithmetic operations	1836
Set 50	**HSF-BF.A.2**	Write arithmetic and geometric sequences recursively and with an explicit formula	1867
Set 81	**HSF-BF.A.2**	Model situations with arithmetic and geometric sequences and translate between	1898
Set 51	**HSF-BF.B.3**	Identify the effect of replacing f(x) by f(x) + k, kf(x), f(kx), and f(x + k), and recogn	1929
Set 82	**HSF-BF.B.3**	Experiment with transformations of graphs and recognize even and odd functions f	1960
Set 52	**HSF-BF.B.4.a**	Solve f(x) = c for a simple function f and write an expression for the inverse	1991
Set 53	**HSF-LE.A.1.a**	Prove that linear functions grow by equal differences and exponential functions gr	2022
Set 54	**HSF-LE.A.1.b**	Recognize situations in which a quantity changes at a constant rate per unit interv	2053
Set 55	**HSF-LE.A.1.c**	Recognize situations in which a quantity grows or decays by a constant percent ra	2084

CONTENTS (continued)

Set 56	**HSF-LE.A.2**	Construct linear and exponential functions from a graph, a description, two points	2115
Set 57	**HSF-LE.A.3**	Observe that exponential growth eventually exceeds linear, quadratic, and polyn	2146
Set 58	**HSF-LE.B.5**	Interpret the parameters of a linear or exponential function in context	2177
Set 59	**HSS-ID.A.1**	Represent data with dot plots, histograms, and box plots	2208
Set 76	**HSS-ID.A.1**	Make histograms from data	2239
Set 77	**HSS-ID.A.1**	Make box plots from data	2270
Set 60	**HSS-ID.A.2**	Compare the center and spread of two or more data sets	2301
Set 61	**HSS-ID.A.3**	Interpret differences in shape, center, and spread, accounting for outliers	2332
Set 62	**HSS-ID.B.5**	Summarize categorical data in two-way frequency tables and interpret relative fre	2363
Set 63	**HSS-ID.B.6.a**	Fit a linear, quadratic, or exponential function to data and use it to solve problems	2394
Set 83	**HSS-ID.B.6.a**	Construct scatter plots from paired data and describe how the variables are relat	2425
Set 64	**HSS-ID.B.6.b**	Assess the fit of a function by plotting and analyzing residuals	2456
Set 65	**HSS-ID.B.6.c**	Fit a linear function to a scatter plot that suggests a linear association	2487
Set 66	**HSS-ID.C.7**	Interpret the slope and the intercept of a linear model in the context of the data	2518
Set 67	**HSS-ID.C.8**	Compute (using technology) and interpret the correlation coefficient of a linear fit	2549
Set 68	**HSS-ID.C.9**	Distinguish between correlation and causation	2580

SET 1 • HSN-RN.A.1 • NUMBER & QUANTITY

Explain the meaning of rational exponents from the properties of integer exponents

MAIN	Algebra I main question	**HSN-RN.A.1**
BACKWARD 1	Apply the properties of integer exponents	**8.EE.A.1**
BACKWARD 2	Evaluate square roots and cube roots	**8.EE.A.2**
FORWARD 1	Express the solution of an exponential equation as a logarithm	**HSF-LE.A.4**
FORWARD 2	Use the inverse relationship between exponents and logarithms	**HSF-BF.B.5**

SET 1 • HSN-RN.A.1

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

For the power-of-a-power property to hold, $(7^{1/4})^4$ must equal $7^{(1/4)\cdot 4} = 7$.

What number must $7^{1/4}$ be? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q1/5

Use the properties of exponents to explain why $8^{2/3} = 4$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q2/5

For $x > 0$, which expression is equal to $x^{3/4}$?

A. $x^3 \div 4$

B. the fourth root of x, cubed

C. the cube root of x, to the fourth power

D. $3x \div 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q3/5

True or false?

$16^{1/2} = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q4/5

True or false?

For $x > 0$, $x^{1/2} \cdot x^{1/2} = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSN-RN.A.1

BACKWARD 1

Apply the properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$2^3 \cdot 2^5$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q1/5

Write as a single power.

$(3^2)^4$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q2/5

Write as a single power.

$5^6 \div 5^2$

A. 1^4

B. 5^4

C. 5^3

D. 5^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q3/5

True or false?

$(7^2)^3 = 7^6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q4/5

True or false?

$4^0 = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSN-RN.A.1

BACKWARD 2

Evaluate square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Find the value.

$\sqrt{81}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

Find the value.

$\sqrt[3]{27}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

Find the value.

$\sqrt[3]{64}$

A. 32

B. 4

C. 8

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

True or false?

$\sqrt[3]{(-8)} = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

True or false?

$\sqrt{25} = \pm 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSN-RN.A.1

FORWARD 1

Express the solution of an exponential equation as a logarithm • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $10^x = 500$. Write the solution as a logarithm, then use a calculator to round it to the nearest hundredth.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q1/5

A colony of 400 bacteria grows by a factor of 10 every hour: $N = 400 \cdot 10^t$. Solve $400 \cdot 10^t = 3{,}000$. Give the exact logarithm and round to the nearest hundredth. (log means base 10.)

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q2/5

Which expression is the solution of $3 \cdot 10^x = 60$? (log means base 10.)

A. $x = \log 60$

B. $x = 20$

C. $x = \log 57$

D. $x = \log 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q3/5

True or false?

The solution of $2 \cdot 10^t = 1{,}000$ is $t = \log 500 \approx 2.70$. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q4/5

True or false?

The solution of $10^x = -10$ is $x = -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSN-RN.A.1

FORWARD 2

Use the inverse relationship between exponents and logarithms • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

$f(x) = 10^x$ and $g(x) = \log x$ (log means base 10).

Find $f(g(100))$ and $g(f(2))$, and say what they show.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q1/5

A sound level is $L = 10 \log(I \div I_0)$ decibels, where log means base 10.

Solve for the intensity I in terms of L.

$I =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q2/5

Which function is the inverse of $f(x) = 5^x$?

A. $f^{-1}(x) = x \div 5$

B. $f^{-1}(x) = x^5$

C. $f^{-1}(x) = \log_5 x$

D. $f^{-1}(x) = 5^{-x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q3/5

True or false?

$10^{\log_{10} 7} = 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q4/5

True or false?

$\log(10^x) = 10x$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q5/5

SET 2 • HSN-RN.A.2 • NUMBER & QUANTITY

Rewrite expressions with radicals and rational exponents

MAIN	Algebra I main question	**HSN-RN.A.2**
BACKWARD 1	Solve equations of the form $x^2 = p$ and $x^3 = p$	**8.EE.A.2**
BACKWARD 2	Use the meaning of rational exponents	**HSN-RN.A.1**
FORWARD 1	Solve radical equations and identify extraneous solutions	**HSA-REI.A.2**
FORWARD 2	Add, subtract, and multiply complex numbers	**HSN-CN.A.2**

SET 2 • HSN-RN.A.2

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Assume $x \geq 0$. Write $x^{5/2}$ using a radical.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q1/5

Write $\sqrt[3]{y^2}$ using a rational exponent.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q2/5

Simplify.

$\sqrt{50} \cdot \sqrt{2}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q3/5

Which expression is equal to $(27x^6)^{1/3}$ for every real number x?

A. $27x^2$

B. $9x^2$

C. $3x^3$

D. $3x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q4/5

True or false?

$\sqrt{(x^2)} = x$ for every real number x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q5/5

SET 2 • HSN-RN.A.2

BACKWARD 1

Solve equations of the form $x^2 = p$ and $x^3 = p$ • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve for x.

$x^3 = 125$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q1/5

Solve for x. Give all solutions.

$x^2 = 36$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q2/5

Solve for x.

$x^3 = -8$

A. ± 2

B. -2

C. -4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q3/5

True or false?

$x^2 = 7$ has the solutions $\sqrt{7}$ and $-\sqrt{7}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q4/5

True or false?

$x^3 = 27$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSN-RN.A.2

BACKWARD 2

Use the meaning of rational exponents • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

Find the value.

$25^{1/2}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q1/5

Find the value.

$27^{2/3}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q2/5

Find the value.

$32^{1/5}$

A. 160

B. 2

C. 6.4

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-B2-Q3**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$9^{3/2} = 27$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q4/5

True or false?

$8^{1/3} = \frac{8}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSN-RN.A.2

FORWARD 1

Solve radical equations and identify extraneous solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $\sqrt{(x + 3)} = 5$. Check your solution.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q1/5

Solve $\sqrt{(2x + 1)} = x - 1$. Check for extraneous solutions.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q2/5

What are the solutions of $\sqrt{x} = -3$?

A. $x = 3$

B. $x = 9$

C. $x = -9$

D. There is no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q3/5

True or false?

Squaring both sides of an equation can introduce extraneous solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q4**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q4/5

True or false?

$x = -4$ is a solution of $\sqrt{(x + 8)} = x + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q5**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSN-RN.A.2

FORWARD 2

Add, subtract, and multiply complex numbers • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2

Add. Write the answer in the form a + bi.

$(3 + 2i) + (1 - 5i)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q1/5

Multiply. Write the answer in the form a + bi.

$(2 + i)(3 - i)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q2/5

Simplify.

$i^2 \cdot i$

A. 1

B. $-i$

C. i

D. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q3**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q3/5

True or false?

$(1 + i)^2 = 2i$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q4/5

True or false?

$i \cdot i = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q5/5

SET 3 • HSN-RN.B.3 • NUMBER & QUANTITY

Prove that the sum of a rational number and an irrational number is irrational

MAIN	Algebra I main question	**HSN-RN.B.3**
BACKWARD 1	Classify numbers as rational or irrational	**8.NS.A.1**
BACKWARD 2	Write rational numbers as quotients of integers	**7.NS.A.2.b**
FORWARD 1	Find exact lengths and perimeters on the coordinate plane	**HSG-GPE.B.7**
FORWARD 2	Write numbers in the complex number system as a + bi	**HSN-CN.A.1**

SET 3 • HSN-RN.B.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why $5 + \sqrt{2}$ is irrational. (Suppose $5 + \sqrt{2} = r$ for some rational number r.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q1/5

Explain why $\sqrt{3} - 4$ is irrational. (Suppose $\sqrt{3} - 4 = r$ for some rational number r.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q2/5

Prove that if a is rational and x is irrational, then a + x is irrational.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q3/5

Which sum is irrational?

A. $\sqrt{2} + (-\sqrt{2})$

B. $0.25 + \frac{3}{4}$

C. $\sqrt{16} + 1.5$

D. $\sqrt{5} + 1.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q4/5

True or false?

The sum of two irrational numbers is always irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q5/5

SET 3 • HSN-RN.B.3

BACKWARD 1

Classify numbers as rational or irrational • Grade 8 • The Number System • 8.NS.A.1

Is $\sqrt{49}$ rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q1/5

The number 0.1010010001... continues with one more 0 before each 1. Is it rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q2/5

Which number is rational?

A. $\sqrt{2}$

B. $\sqrt{10}$

C. π

D. 0.2727... (the digits 27 repeat)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

Every integer is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

$\sqrt{20}$ is rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSN-RN.B.3

BACKWARD 2

Write rational numbers as quotients of integers • Grade 7 • The Number System • 7.NS.A.2.b

Write 1.25 as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q1/5

Write $-3\frac{1}{3}$ as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q2/5

Which quotient is equal to 0.6?

A. $\frac{6}{100}$

B. $\frac{5}{3}$

C. $\frac{3}{5}$

D. $\frac{1}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q3/5

True or false?

$$-7 = \frac{-7}{1}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q4/5

True or false?

$\frac{2}{0}$ is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSN-RN.B.3

FORWARD 1

Find exact lengths and perimeters on the coordinate plane • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the exact perimeter of the triangle with vertices (0, 0), (3, 0), and (0, 3).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q1/5

A square has vertices (0, 0), (1, 1), (0, 2), and (−1, 1). Find its exact perimeter, and say whether it is rational or irrational.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q2/5

What is the area of the triangle with vertices (0, 0), (4, 0), and (4, 3)?

A. 6

B. 12

C. 7

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q3/5

True or false?

The triangle with vertices (0, 0), (1, 0), and (0, 1) has an irrational perimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q4/5

True or false?

The triangle with vertices (0, 0), (6, 0), and (0, 8) has an irrational perimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSN-RN.B.3

FORWARD 2

Write numbers in the complex number system as a + bi • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{-49}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q1/5

Name the real part and the imaginary part of $4 - 3i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q2/5

Which of these is a real number?

A. i

B. i^2

C. $2i$

D. $1 + i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q3**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q3/5

True or false?

Every real number is a complex number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q4/5

True or false?

$i^2 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q5/5

SET 69 • HSN-RN.B.3 • NUMBER & QUANTITY

Prove that the product of a nonzero rational number and an irrational number is irrational

MAIN	Algebra I main question	**HSN-RN.B.3**
BACKWARD 1	Classify numbers as rational or irrational	**8.NS.A.1**
BACKWARD 2	Multiply and divide rational numbers	**7.NS.A.2.c**
FORWARD 1	Find exact lengths, perimeters, and areas with irrational side lengths	**HSG-GPE.B.7**
FORWARD 2	Factor sums of squares over the complex numbers	**HSN-CN.C.8**

SET 69 • HSN-RN.B.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why $3\sqrt{7}$ is irrational. (Suppose $3\sqrt{7} = r$ for some rational number r.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q1/5

Prove that if a is a nonzero rational number and x is irrational, then ax is irrational.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q2/5

Why does the statement "a rational number times an irrational number is irrational" need the rational number to be nonzero?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q3/5

Which product is irrational?

A. $\sqrt{5} \cdot \sqrt{5}$

B. $\sqrt{2} \cdot \sqrt{8}$

C. $0 \cdot \sqrt{5}$

D. $\frac{2}{3}\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q4/5

True or false?

The product of two irrational numbers is always irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q5/5

SET 69 • HSN-RN.B.3

BACKWARD 1

Classify numbers as rational or irrational • Grade 8 • The Number System • 8.NS.A.1

Is √64 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q1/5

The decimal 0.1010010001... has one more 0 between the 1s each time. Is it rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q2/5

Which of these numbers is irrational?

A. $\sqrt{10}$

B. $-\frac{4}{7}$

C. $\sqrt{100}$

D. 0.333...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q3/5

True or false?

Every terminating decimal is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q4/5

True or false?

$\pi = \frac{22}{7}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSN-RN.B.3

BACKWARD 2

Multiply and divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find $(-\frac{2}{3}) \cdot \frac{9}{4}$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q1/5

Find $\frac{5}{6} \div \left(-\frac{10}{3}\right)$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q2/5

Which product is positive?

A. $(0.5)(-8)$

B. $0 \cdot 8$

C. $(-0.5)(-8)$

D. $(-0.5)(8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q3/5

True or false?

The quotient of two nonzero rational numbers is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q4/5

True or false?

$\frac{3}{4} \div 0 = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSN-RN.B.3

FORWARD 1

Find exact lengths, perimeters, and areas with irrational side lengths • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

A square has vertices (0, 0), (2, 1), (1, 3), and (−1, 2). Find its exact side length and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q1/5

A rectangle has vertices (0, 0), (3, 3), (1, 5), and (−2, 2). Find its exact perimeter.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q2/5

What is the exact perimeter of the triangle with vertices (0, 0), (4, 0), and (0, 4)?

A. $4\sqrt{2}$

B. 16

C. $8 + 2\sqrt{2}$

D. $8 + 4\sqrt{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q3/5

True or false?

A square with side $\sqrt{5}$ has a rational area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q4/5

True or false?

A square with side $\sqrt{5}$ has a rational perimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q5/5

SET 69 • HSN-RN.B.3

FORWARD 2

Factor sums of squares over the complex numbers • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 9$ over the complex numbers.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q1/5

Expand $(x + 2i)(x - 2i)$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q2/5

Which is a factorization of $x^2 + 25$?

A. $(x + 5i)(x - 5i)$

B. $(x + 5)(x - 5)$

C. $(x + 5i)^2$

D. $(x - 5i)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q3/5

True or false?

$$x^2 + 16 = (x + 4i)(x - 4i)$$

A. True

B. False

True or false?

$x^2 + 4$ cannot be factored, even with complex numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q5/5

SET 70 • HSN-RN.B.3 • NUMBER & QUANTITY

Prove that sums and products of two rational numbers are rational

MAIN	Algebra I main question	**HSN-RN.B.3**
BACKWARD 1	Write quotients of integers as rational numbers	**7.NS.A.2.b**
BACKWARD 2	Divide fractions	**6.NS.A.1**
FORWARD 1	Show that sums and products of complex numbers are complex n…	**HSN-CN.A.2**
FORWARD 2	Show that rational expressions are closed under addition and mult…	**HSA-APR.D.7**

SET 70 • HSN-RN.B.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why the sum of two rational numbers a/b and c/d (with integers a, b, c, d, and b ≠ 0, d ≠ 0) is rational.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q1/5

Prove that the product of two rational numbers a/b and c/d (with integers a, b, c, d, and $b \neq 0$, $d \neq 0$) is rational.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q2/5

In the proof that (a/b)(c/d) is rational, why is bd ≠ 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q3/5

Which statement shows that $(\frac{2}{5})(-\frac{15}{4})$ is rational?

A. It can be written as a decimal with a calculator.

B. It equals $-30/20 = -3/2$, a quotient of integers with a nonzero denominator.

C. It is negative.

D. Its first factor has an even numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q4/5

True or false?

The difference of two rational numbers is always rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q5/5

SET 70 • HSN-RN.B.3

BACKWARD 1

Write quotients of integers as rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Write $-\frac{3}{4}$ as a quotient of integers in two different ways.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q1/5

Is $-12 \div 5$ a rational number? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q2/5

Which quotient is undefined?

A. $-7 \div 2$

B. $0 \div 7$

C. $7 \div 0$

D. $7 \div (-1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q3/5

True or false?

For integers p and q with $q \neq 0$,
$-(p/q) = (-p)/q = p/(-q)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q4/5

True or false?

Some integers are not rational numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSN-RN.B.3

BACKWARD 2

Divide fractions • Grade 6 • The Number System • 6.NS.A.1

Find $\frac{3}{4} \div \frac{1}{8}$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q1/5

Find $\frac{2}{3} \div \frac{5}{6}$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q2/5

Which number equals $\frac{5}{2} \div \frac{1}{4}$?

A. $\frac{5}{8}$

B. $\frac{1}{10}$

C. 10

D. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{2} \div \frac{1}{4} = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q4/5

True or false?

$\frac{3}{5} \div 3 = \frac{9}{5}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSN-RN.B.3

FORWARD 1

Show that sums and products of complex numbers are complex numbers • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2

Add. Write the answer in the form a + bi.

$(3 - 2i) + (-1 + 6i)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q1/5

Multiply. Write the answer in the form a + bi.

$(2 + i)(3 + 2i)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q2/5

Which shows that the product of two complex numbers is a complex number?

A. $i^2 = 1$

B. $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$, and $ac - bd$ and $ad + bc$ are real numbers

C. $(a + bi)(c + di) = ac + bdi$

D. The product of two complex numbers is always real

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q3/5

True or false?

The sum of two complex numbers is a complex number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q4/5

True or false?

(1 + i) + (1 − i) is not a complex number because it has no i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q5/5

SET 70 • HSN-RN.B.3

FORWARD 2

Show that rational expressions are closed under addition and multiplication • High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Assume $x \neq 0$ and $x \neq -1$. Write $1 \div x + 2 \div (x + 1)$ as one fraction.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q1/5

Assume $x \neq 2$ and $x \neq -3$. Multiply and simplify: $(x \div (x - 2)) \cdot ((x - 2) \div (x + 3))$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q2/5

Why is the sum of two rational expressions a rational expression?

A. Over a common denominator, the sum is a polynomial divided by a nonzero polynomial.

B. The denominators always cancel.

C. Rational expressions are always integers.

D. Any sum of fractions equals 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q3/5

True or false?

The quotient of two rational expressions, when the divisor is not 0, is a rational expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q4/5

True or false?

$1 \div x + 1 \div y = 2 \div (x + y)$ for all nonzero x and y with $x + y \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q5/5

SET 4 • HSN-Q.A.1 • NUMBER & QUANTITY

Use units to guide multistep solutions and to choose scales for graphs

MAIN	Algebra I main question	**HSN-Q.A.1**
BACKWARD 1	Convert measurement units with ratio reasoning	**6.RP.A.3.d**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Use density in a model	**HSG-MG.A.2**
FORWARD 2	Solve problems with velocity vectors — nearest related	**HSN-VM.A.3**

SET 4 • HSN-Q.A.1

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1

A car goes 32 miles on each gallon of gas, and gas costs $3.60 per gallon.

Use units to find the cost of gas for a 400-mile trip. Show how the units cancel.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q1/5

Convert 72 kilometers per hour to meters per second.

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q2/5

The formula d = rt gives distance. A rate r is in meters per second and a time t is in minutes.

What must be done before multiplying?

A. Divide r by t.

B. Convert t to seconds (or r to meters per minute).

C. Nothing; the units already match.

D. Convert d to minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q3/5

A graph will show a town's population each year from 1990 to 2020. The populations range from 12,000 to 31,000.

Choose a scale for the vertical axis, and explain your choice.

Scale: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q4/5

True or false?

A speed of 5 miles per hour is about 7.3 feet per second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q5/5

SET 4 • HSN-Q.A.1

BACKWARD 1

Convert measurement units with ratio reasoning • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

How many inches are in 3.5 feet?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q1/5

How many kilograms are in 2,500 grams?

Kilograms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q2/5

How many feet are in 4 yards?

A. 48

B. 36

C. $\frac{4}{3}$

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q3/5

True or false?

1 hour = 3,600 seconds

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q4/5

True or false?

5 km = 500 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSN-Q.A.1

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A walker goes $\frac{3}{4}$ mile in $\frac{1}{4}$ hour.

How many miles per hour is this?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q1/5

Half a pound of cheese costs $3.

What is the price per pound?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q2/5

$\frac{2}{3}$ gallon of paint covers $\frac{1}{3}$ of a fence.

How many gallons cover the whole fence?

A. $\frac{1}{2}$

B. $\frac{2}{9}$

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q3/5

True or false?

Walking $\frac{1}{2}$ mile in $\frac{1}{6}$ hour is a speed of 3 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q4/5

True or false?

Painting $\frac{1}{4}$ of a wall in $\frac{1}{2}$ hour is a rate of $\frac{1}{8}$ wall per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSN-Q.A.1

FORWARD 1

Use density in a model • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2

Copper has a density of 8.96 grams per cubic centimeter.

What is the mass of a 5 cubic-centimeter piece of copper?

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q1/5

A county has 240,000 people living in 600 square miles.

What is its population density?

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q2/5

A 2 cm by 3 cm by 4 cm block has a mass of 60 grams.

What is its density?

A. 0.4 gram per cubic centimeter

B. 60 grams

C. 2.5 grams per cubic centimeter

D. 24 grams per cubic centimeter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q3/5

True or false?

Population density is the number of people divided by the area they live on.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q4/5

True or false?

An object with a volume of 10 cubic centimeters and a density of 3 grams per cubic centimeter has a mass of 3.3 grams.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSN-Q.A.1

FORWARD 2

Solve problems with velocity vectors • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Precalculus standard. Connection: Velocity vectors are rate quantities with units, like the unit rates of the MAIN set, with a direction added.

A boat heads east at 12 km/h across a river whose current flows north at 5 km/h.

What is the boat's resulting speed?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q1/5 • NEAREST RELATED

A plane points north at 300 km/h, and the wind blows west at 40 km/h.

Write its velocity as (east part, north part) in km/h.

Velocity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q2/5 • NEAREST RELATED

Two velocities, (3, 4) and (5, −1), are added.

What is the result?

A. (8, 5)

B. (8, 3)

C. (15, −4)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

An object with velocity (6, 8) meters per second has a speed of 10 meters per second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

Adding the velocities (3, 0) and (0, 4) gives a speed of 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 5 • HSN-Q.A.2 • NUMBER & QUANTITY

Define appropriate quantities for descriptive modeling

MAIN	Algebra I main question	**HSN-Q.A.2**
BACKWARD 1	Use variables to represent quantities in a situation	**6.EE.B.6**
BACKWARD 2	Write and solve px + q = r from a word problem	**7.EE.B.4.a**
FORWARD 1	Use geometric quantities to make design decisions	**HSG-MG.A.3**
FORWARD 2	Recognize surveys, experiments, and observational studies and th…	**HSS-IC.B.3**

SET 5 • HSN-Q.A.2

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2

A school wants to model the total cost of a field trip by bus.

Name the quantities the model needs, with a unit for each.

Quantities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q1/5

A club rents a room for $300 and pays $40 for each member who attends. Its cost model is C = 40n + 300.

Define C and n, with units.

Definitions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q2/5

Which quantity best describes how fast a phone battery runs down?

A. Number of apps on the phone

B. Percent of charge used per hour

C. Width of the screen in inches

D. Mass of the phone in grams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q3/5

Two towns have different populations and different areas.

Define one quantity, with units, that compares how crowded they are.

Quantity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q4/5

True or false?

Kilowatt-hours per month is a sensible quantity for a model of a household's electricity use.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q5/5

SET 5 • HSN-Q.A.2

BACKWARD 1

Use variables to represent quantities in a situation • Grade 6 • Expressions & Equations • 6.EE.B.6

Tickets cost $8 each.

Write an expression for the cost of n tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q1/5

A worker earns $15 per hour plus a $20 bonus.

Write an expression for the pay for h hours.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q2/5

A typist types 30 words per minute.

Which expression gives the number of words typed in m minutes?

A. $\frac{m}{30}$

B. $m + 30$

C. $30m$

D. $\frac{30}{m}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q3/5

True or false?

If w is a number of weeks, 7w is the number of days.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q4/5

True or false?

If d is a number of dollars, d + 5 is 5 times as many dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSN-Q.A.2

BACKWARD 2

Write and solve px + q = r from a word problem • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A gym charges a $25 fee plus $10 per class. Tom paid $95.

How many classes did he take?

Classes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q1/5

A phone costs $50 plus $15 per month. The total so far is $140.

For how many months has it been paid?

Months: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q2/5

A plumber charges a $12 fee plus r dollars per hour. A 3-hour job costs $57.

Which equation fits?

A. $3r + 12 = 57$

B. $3(r + 12) = 57$

C. $12r + 3 = 57$

D. $r + 3 + 12 = 57$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q3/5

True or false?

$x = 6$ is the solution of $4x + 6 = 30$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q4/5

True or false?

$x = 3$ is the solution of $2x - 5 = 11$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSN-Q.A.2

FORWARD 1

Use geometric quantities to make design decisions • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A rectangular garden needs an area of at least 200 square feet and can use at most 60 feet of fence.

Does a 10 ft by 20 ft garden meet both requirements? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q1/5

A box with a square base 10 cm on each side must hold 1,000 cubic centimeters.

How tall must it be?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q2/5

A rectangular pen must have an area of 100 square meters.

Which dimensions use the least fence?

A. 5 m by 20 m

B. 2 m by 50 m

C. 10 m by 10 m

D. 4 m by 25 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q3/5

True or false?

A 6 by 6 square has more area than a 4 by 8 rectangle, even though their perimeters are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q4/5

True or false?

Of all rectangles with a perimeter of 20, the 1 by 9 rectangle has the greatest area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSN-Q.A.2

FORWARD 2

Recognize surveys, experiments, and observational studies and the variables they measure • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Defining the quantities a model needs leads to deciding which variables a study measures and how they are collected.

A researcher randomly assigns 40 plants to two fertilizers and measures each plant's growth in centimeters.

What kind of study is this, and what is the response variable (with units)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q1/5 • NEAREST RELATED

A survey asks 200 randomly chosen students how many hours they sleep on a school night.

What kind of study is this, and what variable does it measure?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which kind of study can show cause and effect?

A. An observational study

B. A poll of volunteers

C. A randomized experiment

D. A survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.3 • Set 5 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

In an observational study, the researchers do not assign treatments.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.3 • Set 5 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A survey of volunteers who answer an online poll is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.3 • Set 5 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 6 • HSN-Q.A.3 • NUMBER & QUANTITY

Choose a level of accuracy that fits the limits of measurement

MAIN	Algebra I main question	**HSN-Q.A.3**
BACKWARD 1	Estimate to check whether an answer is reasonable	**7.EE.B.3**
BACKWARD 2	Round decimals to any place	**5.NBT.A.4**
FORWARD 1	Use volume formulas and report results to a sensible accuracy	**HSG-GMD.A.3**
FORWARD 2	Use the margin of error of a sample survey — nearest related	**HSS-IC.B.4**

SET 6 • HSN-Q.A.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

A ruler measures to the nearest 0.1 cm. A rectangle measures 4.3 cm by 2.7 cm, and a calculator gives its area as 11.61 square cm.

Which report of the area best fits the accuracy of the measurements?

A. 11.610 square cm

B. 11.61 square cm

C. About 10 square cm

D. About 12 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q1/5

A stopwatch reads to the nearest 0.1 second. A runner covers 100 m in 12.4 seconds, and a calculator gives 8.0645161 meters per second.

Report the speed to a sensible accuracy.

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q2/5

A tape measure is marked in centimeters.

Which is the most sensible report of a person's height measured with it?

A. 167.3482 cm

B. 170.00 cm

C. 2 m

D. 167 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q3/5

A bridge is 1,250 m long, measured to the nearest 10 m.

Between which two lengths is its true length?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q4/5

True or false?

Giving more decimal places always makes a measured answer more accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q5/5

SET 6 • HSN-Q.A.3

BACKWARD 1

Estimate to check whether an answer is reasonable • Grade 7 • Expressions & Equations • 7.EE.B.3

Estimate.

4.9×21.2

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q1/5

Is 38.4 ÷ 4.8 = 80 reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q2/5

Which is the best estimate of 19.8% of 503?

A. 100

B. 50

C. 1,000

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q3/5

True or false?

$29.95 × 4 is about $120.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q4/5

True or false?

0.48 × 61 is about 300.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSN-Q.A.3

BACKWARD 2

Round decimals to any place • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 8.0645 to the nearest hundredth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q1/5

Round 11.61 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q2/5

Round 0.0357 to the nearest thousandth.

A. 0.035

B. 0.0357

C. 0.036

D. 0.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q3/5

True or false?

3.149 rounded to the nearest tenth is 3.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q4/5

True or false?

6.95 rounded to the nearest tenth is 6.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSN-Q.A.3

FORWARD 1

Use volume formulas and report results to a sensible accuracy • High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cylinder has a radius of 3.0 cm and a height of 10.0 cm.

Find its volume to the nearest cubic centimeter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q1/5

A sphere has a radius of 5 m.

Find its volume exactly in terms of π, then to the nearest cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q2/5

A cone has a radius of 6 and a height of 4.

What is its volume?

A. 144π

B. 96π

C. 48π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q3/5

True or false?

A cylinder with radius 2 cm and height 2 cm has a volume of 8π cm^3, which is about 25 cm^3 to the nearest cubic centimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q4/5

True or false?

Doubling the radius of a sphere doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSN-Q.A.3

FORWARD 2

Use the margin of error of a sample survey • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: The accuracy a measurement supports leads to the accuracy of an estimate from a sample, given by its margin of error.

A poll reports that 54% of voters support a plan, with a margin of error of 3 percentage points.

What interval of values for the true percent does this give?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.4 • Set 6 • FORWARD 2 • Q1/5 • NEAREST RELATED

A poll reports 54% support with a margin of error of 3 percentage points.

Do the results suggest that a majority supports the plan? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q2/5 • NEAREST RELATED

A poll reports 48% support with a margin of error of 4 percentage points.

Which statement fits?

A. The poll must be wrong.

B. Fewer than half support the plan, for certain.

C. The true percent could be above or below 50%.

D. The true percent is exactly 48%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Larger random samples tend to have smaller margins of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A margin of error of 3 percentage points means the poll is off by exactly 3 points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q5/5 • NEAREST RELATED

Interpret the terms, factors, and coefficients of an expression in context

MAIN	Algebra I main question	**HSA-SSE.A.1.a**
BACKWARD 1	Identify the parts of an expression	**6.EE.A.2.b**
BACKWARD 2	Rewrite an expression to show how quantities are related	**7.EE.A.2**
FORWARD 1	Rewrite a rational expression and interpret its parts	**HSA-APR.D.6**
FORWARD 2	Compose functions and interpret the composite in context	**HSF-BF.A.1.c**

SET 7 • HSA-SSE.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A gym charges $C = 35 + 20m$ dollars for m months of membership.

What do 35 and 20 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

The expression 1.08p gives the total cost, including sales tax, of an item with price p.

What does the factor 1.08 tell you?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

The perimeter of a rectangle is $2(l + w)$.

What does the factor $(l + w)$ represent?

A. The area

B. The length of one side

C. Half the perimeter: one length plus one width

D. Twice the perimeter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

For the expression $3x^2 - 5x + 7$, name its terms, the coefficient of x, and the constant term.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

True or false?

A car's value is $500(0.9)^t$ dollars after t years. The factor 0.9 means the car loses 90% of its value each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 1

Identify the parts of an expression • Grade 6 • Expressions & Equations • 6.EE.A.2.b

How many terms does 4a + 3b − 2 have?

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q1/5

What is the coefficient of y in $7y - 2$?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q2/5

Which description fits 5(x + 2)?

A. A product of two factors, 5 and (x + 2)

B. A sum of three terms

C. A single term with coefficient 2

D. A quotient of two numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q3/5

True or false?

In 8 + 2n, the term 2n has a coefficient of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q4/5

True or false?

The terms of $6(m - 1)$ are 6 and m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 2

Rewrite an expression to show how quantities are related • Grade 7 • Expressions & Equations • 7.EE.A.2

Rewrite a + 0.05a as a single term.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q1/5

A price p is lowered by 20%.

Write the new price as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q2/5

Which expression shows a 15% increase of n?

A. 1.15n

B. 15n

C. n + 15

D. 0.15n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q3/5

True or false?

$p - 0.25p = 0.75p$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q4/5

True or false?

$x + 0.3x = 0.3x^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 1

Rewrite a rational expression and interpret its parts • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

A company's average cost per item for n items is (500 + 3n) ÷ n dollars.

Rewrite it as a sum of two terms, and say what each term means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q1/5

Assume $x \neq 0$. Rewrite $(2x + 5) \div x$ in the form $a + b/x$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q2/5

Assume $x \neq 0$. Which expression is equal to $(x^2 + 3x) \div x$?

A. $x^2 + 3$

B. $x + 3x$

C. $x + 3$

D. $4x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q3/5

True or false?

$$(6x + 4) \div 2 = 3x + 2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q4/5

True or false?

Assume $x \neq 0$. $(x + 6) \div x = 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 2

Compose functions and interpret the composite in context • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

$T(h) = 20 + 5h$ is a temperature in °C after h hours, and $h(t) = t \div 60$ converts t minutes to hours.

Write $T(h(t))$ in simplest form, and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q1/5

$f(x) = x + 3$ and $g(x) = 2x$.

Find $f(g(4))$.

$f(g(4)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q2/5

$f(x) = x^2$ and $g(x) = x - 1$.

Which expression is $f(g(x))$?

A. $x - 1^2$

B. $x^2 - 1$

C. $x^2 - x$

D. $(x - 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q3/5

True or false?

If f(x) = 3x and g(x) = x + 2, then g(f(1)) = 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q4/5

True or false?

f(g(x)) is always equal to g(f(x)).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q5/5

Interpret complicated expressions by viewing one or more of their parts as a single quantity

MAIN	Algebra I main question	**HSA-SSE.A.1.b**
BACKWARD 1	Apply the distributive property to write equivalent expressions	**6.EE.A.3**
BACKWARD 2	Evaluate expressions for given values	**6.EE.A.2.c**
FORWARD 1	Use the formula for the sum of a finite geometric series	**HSA-SSE.B.4**
FORWARD 2	Solve exponential growth models with logarithms	**HSF-LE.A.4**

SET 8 • HSA-SSE.A.1.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

An investment of P dollars is worth $P(1 + r)^n$ dollars after n years, where $r > 0$.

Treating $(1 + r)^n$ as one quantity, explain why the investment grows.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

For which value of x is $10 - (x - 3)^2$ greatest, and what is that greatest value? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

Which expression is never greater than 4, for every real number x?

A. $(x + 1)^2 - 4$

B. $4 - (x + 1)^2$

C. $4(x + 1)^2$

D. $4 + (x + 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

A square picture with side s is surrounded by a frame 2 units wide on every side. The frame's area is $(s + 4)^2 - s^2$.

What do $(s + 4)^2$ and s^2 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

True or false?

For every real number x, $5 + 3(x - 2)^2 \geq 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 1

Apply the distributive property to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$3(2x + 5)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q1/5

Factor using the greatest common factor.

12y + 18

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q2/5

Which expression is equal to $4(a - 3) + 2a$?

A. $6a + 12$

B. $6a - 12$

C. $6a - 3$

D. $4a - 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q3/5

True or false?

$$2(x + y) = 2x + 2y$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q4/5

True or false?

$3(x + 4) = 3x + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 2

Evaluate expressions for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $(x - 3)^2$ when $x = 7$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q1/5

Evaluate $100(1.05)^2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q2/5

Evaluate $10 - (x - 3)^2$ when $x = 1$.

A. 8

B. -6

C. 14

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q3/5

True or false?

When $x = 3$, $(x - 3)^2 = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q4/5

True or false?

When $x = -2$, $x^2 = -4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 1

Use the formula for the sum of a finite geometric series • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Find the sum of the first 6 terms of 3 + 6 + 12 + ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q1/5

Use $S = a(1 - r^n) \div (1 - r)$ to find $2 + 1 + 0.5 + 0.25 + 0.125$.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q2/5

A geometric series has first term 5, ratio 3, and 4 terms.

What is its sum?

A. 405

B. 80

C. 200

D. 121

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q3/5

True or false?

$$1 + 2 + 4 + \ldots + 2^9 = 2^{10} - 1$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q4**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q4/5

True or false?

The formula $a(1 - r^n) \div (1 - r)$ gives the sum of a geometric series with $r = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 2

Solve exponential growth models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

An investment is worth $1{,}000(1.05)^n$ dollars after n years.

Write the number of years to reach $2,000 as an expression with logarithms, then round it to the nearest tenth.

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q1/5

A town of 5,000 people grows 3% per year: $P = 5{,}000(1.03)^t$. When will it reach 8,000? Round to the nearest tenth of a year. (log means base 10.)

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q2/5

Solve $5 \cdot 2^t = 60$. (log means base 10.)

A. $t = \log 60 \div \log 10 \approx 1.78$

B. $t = 3$

C. $t = \log 12 \div \log 2 \approx 3.58$

D. $t = 12 \div 2 = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q3/5

True or false?

$\log(1.05^n) = n \cdot \log 1.05$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q4/5

True or false?

The solution of $2^x = 10$ is $x = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q5/5

Add and subtract polynomials and explain why the result is a polynomial

MAIN	Algebra I main question	**HSA-APR.A.1**
BACKWARD 1	Add and subtract linear expressions	**7.EE.A.1**
BACKWARD 2	Combine like terms to write equivalent expressions	**6.EE.A.4**
FORWARD 1	Use the Remainder Theorem	**HSA-APR.B.2**
FORWARD 2	Rewrite a quotient of polynomials as q(x) + r(x)/b(x)	**HSA-APR.D.6**

SET 9 • HSA-APR.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Add. Write the answer in standard form.

$(4x^2 - 3x + 5) + (2x^2 + 7x - 9)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q1/5

Subtract. Write the answer in standard form.

$(5y^3 + 2y - 1) - (y^3 - 4y^2 + 6y - 3)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q2/5

Which expression is equal to $(3a^2 - 2ab) - (a^2 + 5ab)$?

A. $2a^2 - 7ab$

B. $2a^2 - 3ab$

C. $2a^2 + 3ab$

D. $4a^2 + 3ab$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-M-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q3/5

Explain why the sum of two polynomials is always a polynomial.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q4/5

True or false?

The difference of two polynomials is always a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q5/5

SET 9 • HSA-APR.A.1

BACKWARD 1

Add and subtract linear expressions • Grade 7 • Expressions & Equations • 7.EE.A.1

Add.

$(3x + 4) + (5x - 9)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q1/5

Subtract.

$(7a - 2) - (3a - 6)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q2/5

Subtract.

$(2y + 1) - (y + 5)$

A. $y - 6$

B. $3y - 4$

C. $y + 6$

D. $y - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$$-(x - 3) = -x + 3$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$$(4n + 2) - (n + 2) = 3n + 4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSA-APR.A.1

BACKWARD 2

Combine like terms to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.4

Simplify.

$4m + 2 + m$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q1/5

Are 2(3 + x) and 6 + 2x equivalent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q2/5

Which expression is equal to b + b + b + 4?

A. 7b

B. 3b + 4

C. 4b + 3

D. $b^3 + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q3/5

True or false?

$x + x = 2x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q4/5

True or false?

$5y - y = 4y$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSA-APR.A.1

FORWARD 1

Use the Remainder Theorem • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find the remainder when $x^2 + 3x - 4$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q1/5

Is $x + 1$ a factor of $x^3 + 1$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q2/5

What is the remainder when $p(x) = x^3 - 2x + 5$ is divided by $x - 1$?

A. 8

B. 4

C. -4

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q3/5

True or false?

If $p(3) = 0$, then $x - 3$ is a factor of $p(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q4/5

True or false?

The remainder when p(x) is divided by x + 2 is p(2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSA-APR.A.1

FORWARD 2

Rewrite a quotient of polynomials as q(x) + r(x)/b(x) • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Assume $x \neq -2$. Write $(x^2 + 5x + 7) \div (x + 2)$ in the form $q(x) + r/(x + 2)$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q1/5

Assume $x \neq 0$. Write $(3x + 7) \div x$ in the form $a + b/x$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q2/5

Assume $x \neq 1$. Which expression is equal to $(x^2 - 1) \div (x - 1)$?

A. $x + 1$

B. x

C. $x^2 + 1$

D. $x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q3/5

True or false?

Assume $x \neq 0$. $(2x + 1) \div x = 2 + 1/x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q4/5

True or false?

Assume $x \neq -2$. $(x^2 + 4) \div (x + 2) = x + 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q5/5

Multiply polynomials

MAIN	Algebra I main question	**HSA-APR.A.1**
BACKWARD 1	Apply the distributive property to linear expressions	**6.EE.A.3**
BACKWARD 2	Multiply powers with the same base	**8.EE.A.1**
FORWARD 1	Prove polynomial identities	**HSA-APR.C.4**
FORWARD 2	Expand powers of binomials with the binomial theorem	**HSA-APR.C.5**

SET 10 • HSA-APR.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply. Write the answer in standard form.

$(2x + 3)(x - 5)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q1/5

Multiply. Write the answer in standard form.

$(x - 4)^2$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q2/5

Multiply. Write the answer in standard form.

$(y + 2)(y^2 - 3y + 1)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q3/5

Multiply.

$(3a - 1)(3a + 1)$

A. $9a^2 - 1$

B. $9a^2 - 6a - 1$

C. $6a - 1$

D. $9a^2 + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q4/5

True or false?

$(x + 3)^2 = x^2 + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q5/5

SET 10 • HSA-APR.A.1

BACKWARD 1

Apply the distributive property to linear expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$4(3x - 2)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q1/5

Expand.

$-2(5 - 3a)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q2/5

Expand.

$7(2b + 1)$

A. $14b + 8$

B. $14b + 7$

C. $9b + 8$

D. $14b + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$$5(m - 2) = 5m - 10$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q4/5

True or false?

$$3(2x + 4) = 6x + 4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSA-APR.A.1

BACKWARD 2

Multiply powers with the same base • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$7^2 \cdot 7^3$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

Write as a single power.

$10^4 \cdot 10$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

Write as a single power.

$6^3 \cdot 6^3$

A. 6^6

B. 6^9

C. 12^6

D. 36^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$9 \cdot 9^4 = 9^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

True or false?

$2^3 \cdot 2^3 = 2^9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSA-APR.A.1

FORWARD 1

Prove polynomial identities • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Show that $(a + b)^2 - (a - b)^2 = 4ab$ for all numbers a and b.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q1/5

Use the identity $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$ to factor $x^3 - 8$.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q2/5

Which identity is true for all numbers a and b?

A. $(a - b)^2 = a^2 - b^2$

B. $(a + b)(a - b) = a^2 - b^2$

C. $(a + b)^2 = a^2 + b^2$

D. $a^2 + b^2 = (a + b)(a + b)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q3/5

True or false?

$(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ for all x and y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q4/5

True or false?

$(x + 1)^3 = x^3 + 1$ for all x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSA-APR.A.1

FORWARD 2

Expand powers of binomials with the binomial theorem • High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Expand. Write the answer in standard form.

$(2x - 1)^3$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q1/5

What is the coefficient of the x^3y^2 term in the expansion of $(x + y)^5$?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q2/5

How many terms does the expansion of $(a + b)^7$ have?

A. 128

B. 8

C. 14

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q3/5

True or false?

The number 5 choose 2 equals 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q4/5

True or false?

Every coefficient in the expansion of $(a - b)^4$ is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q5/5

Use structure to factor with a greatest common factor and the difference of squares

MAIN	Algebra I main question	**HSA-SSE.A.2**
BACKWARD 1	Use the greatest common factor and the distributive property	**6.NS.B.4**
BACKWARD 2	Multiply a sum and a difference	**HSA-APR.A.1**
FORWARD 1	Find the zeros of a polynomial from its factors	**HSA-APR.B.3**
FORWARD 2	Factor sums of squares with complex numbers	**HSN-CN.C.8**

SET 11 • HSA-SSE.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor completely.

$6x^3 - 15x^2$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q1/5

Factor completely.

$49y^2 - 25$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q2/5

Factor completely.

$x^4 - 16$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q3/5

Which is the complete factorization of $2x^2 - 18$?

A. $(2x - 6)(x + 3)$

B. $2(x - 3)(x + 3)$

C. $(x - 3)(x + 3)$

D. $2(x^2 - 9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q4/5

True or false?

$x^2 + 9$ can be factored as $(x + 3)(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q5/5

SET 11 • HSA-SSE.A.2

BACKWARD 1

Use the greatest common factor and the distributive property • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q1/5

Use the greatest common factor to write 42 + 28 as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q2/5

What is the greatest common factor of 15 and 40?

A. 5

B. 8

C. 120

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q3/5

True or false?

$18 + 12 = 6(3 + 2)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q4/5

True or false?

The greatest common factor of 9 and 16 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSA-SSE.A.2

BACKWARD 2

Multiply a sum and a difference • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply.

$(x - 7)(x + 7)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q1/5

Multiply.

$(2y + 3)(2y - 3)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q2/5

Multiply.

$(x + 4)(x - 4)$

A. $x^2 - 8$

B. $x^2 + 16$

C. $x^2 - 8x - 16$

D. $x^2 - 16$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-B2-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q3/5

True or false?

$$(a - 5)(a + 5) = a^2 - 25$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q4/5

True or false?

$$(x + 3)(x - 3) = x^2 - 6x - 9$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSA-SSE.A.2

FORWARD 1

Find the zeros of a polynomial from its factors • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

What are the zeros of $p(x) = x(x - 2)(x + 5)$?

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions HSA-APR.B.3 • Set 11 • FORWARD 1 • Q1/5

Factor $x^3 - 9x$ completely, then give all of its zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q2/5

Which polynomial has the zeros -1, 0, and 4?

A. $(x + 1)(x - 4)$

B. $x(x + 1)(x + 4)$

C. $x(x + 1)(x - 4)$

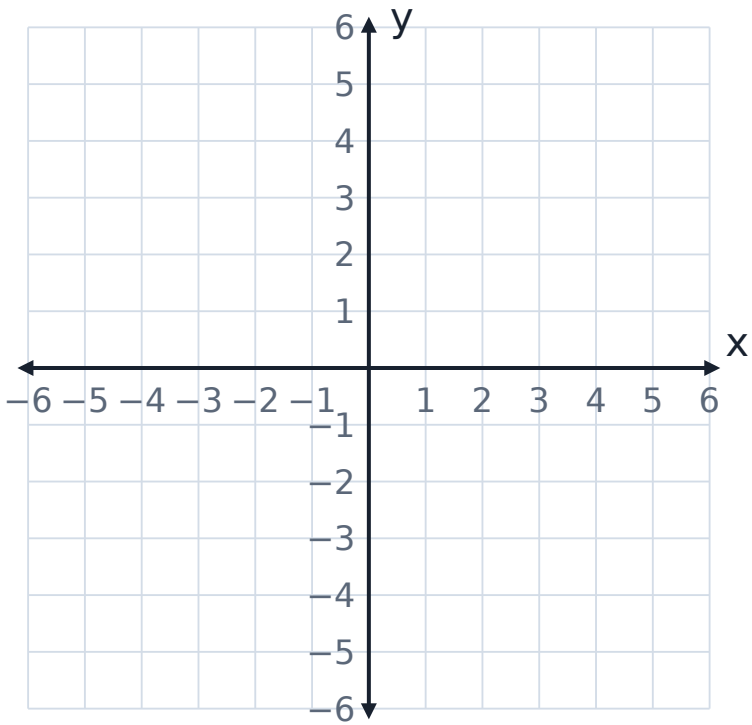
D. $x(x - 1)(x + 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q3/5

Sketch a rough graph of $p(x) = x(x - 2)(x + 5)$. Mark the zeros and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q4/5

True or false?

$p(x) = (x - 3)^2(x + 1)$ has three different zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSA-SSE.A.2

FORWARD 2

Factor sums of squares with complex numbers • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 25$ using complex numbers.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q1/5

Solve $x^2 + 16 = 0$. Give all solutions.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q2/5

Which is the complete factorization of $x^4 - 16$ into linear factors with complex numbers?

A. $(x - 4)(x + 4)$

B. $(x^2 - 4)(x^2 + 4)$

C. $(x - 2)(x + 2)(x - 2i)(x + 2i)$

D. $(x - 2)^2(x + 2)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q3/5

True or false?

$$(x + i)(x - i) = x^2 + 1$$

A. True

B. False

True or false?

$x^2 + 4$ has no linear factors, even with complex numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q5/5

Factor quadratic trinomials

MAIN	Algebra I main question	**HSA-SSE.A.2**
BACKWARD 1	Find factor pairs of a whole number	**4.OA.B.4**
BACKWARD 2	Multiply two binomials	**HSA-APR.A.1**
FORWARD 1	Use factored form to describe the graph of a polynomial function	**HSF-IF.C.7.c**
FORWARD 2	Write a polynomial from its zeros and find zeros by factoring	**HSA-APR.B.3**

SET 12 • HSA-SSE.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 7x + 12$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q1/5

Factor.

$x^2 - 2x - 15$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q2/5

Factor completely.

$2x^2 + 7x + 3$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q3/5

Which is a factor of $3x^2 - 10x - 8$?

A. $3x - 2$

B. $x + 4$

C. $x - 4$

D. $x - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q4/5

True or false?

$x^2 + 5x + 6 = (x + 1)(x + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q5/5

SET 12 • HSA-SSE.A.2

BACKWARD 1

Find factor pairs of a whole number • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factor pairs of 12.

Factor pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q1/5

Two whole numbers multiply to 15 and add to 8.

What are they?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q2/5

Which factor pair of 24 has a sum of 10?

A. 4 and 6

B. 3 and 8

C. 1 and 24

D. 2 and 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q3/5

True or false?

3 and 7 are a factor pair of 21.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q4/5

True or false?

4 and 5 are a factor pair of 25.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSA-SSE.A.2

BACKWARD 2

Multiply two binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply.

$(x + 2)(x + 9)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q1/5

Multiply.

$(x - 6)(x + 1)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q2/5

Multiply.

$(3x + 1)(x + 2)$

A. $3x^2 + 6x + 2$

B. $3x^2 + 7x + 2$

C. $4x^2 + 7x + 2$

D. $3x^2 + 3x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-B2-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q3/5

True or false?

$$(x - 2)(x - 6) = x^2 - 8x + 12$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q4/5

True or false?

$$(x + 1)(x - 4) = x^2 - 3x + 4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSA-SSE.A.2

FORWARD 1

Use factored form to describe the graph of a polynomial function • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

$p(x) = (x + 2)(x - 1)(x - 3)$.

What are its zeros and its y-intercept?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q1/5

Describe the end behavior of $y = x^3 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q2/5

Which describes the end behavior of $y = -x^3$?

A. It rises to the left and falls to the right.

B. It rises on both ends.

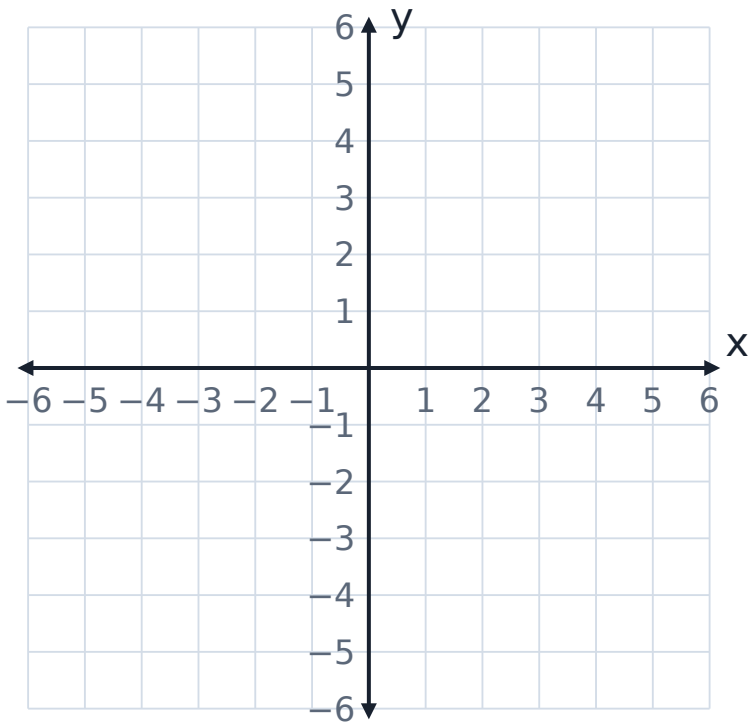
C. It falls to the left and rises to the right.

D. It falls on both ends.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q3/5

Graph
$p(x) = (x + 2)(x - 1)(x - 3)$.
Mark the zeros and the y-intercept, and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q4/5

True or false?

The graph of $y = (x - 1)^2$ crosses the x-axis at $x = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSA-SSE.A.2

FORWARD 2

Write a polynomial from its zeros and find zeros by factoring • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Write a polynomial function of degree 3 whose only zeros are 2, −3, and 5.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q1/5

Factor $x^3 - x^2 - 6x$ completely, then give its zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions HSA-APR.B.3 • Set 12 • FORWARD 2 • Q2/5

What are the zeros of $p(x) = (2x - 1)(x + 4)$?

A. 2 and -4

B. 1 and 4

C. $\frac{1}{2}$ and -4

D. $-\frac{1}{2}$ and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F2-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q3/5

True or false?

If $(x - 7)$ is a factor of $p(x)$, then 7 is a zero of p.

A. True

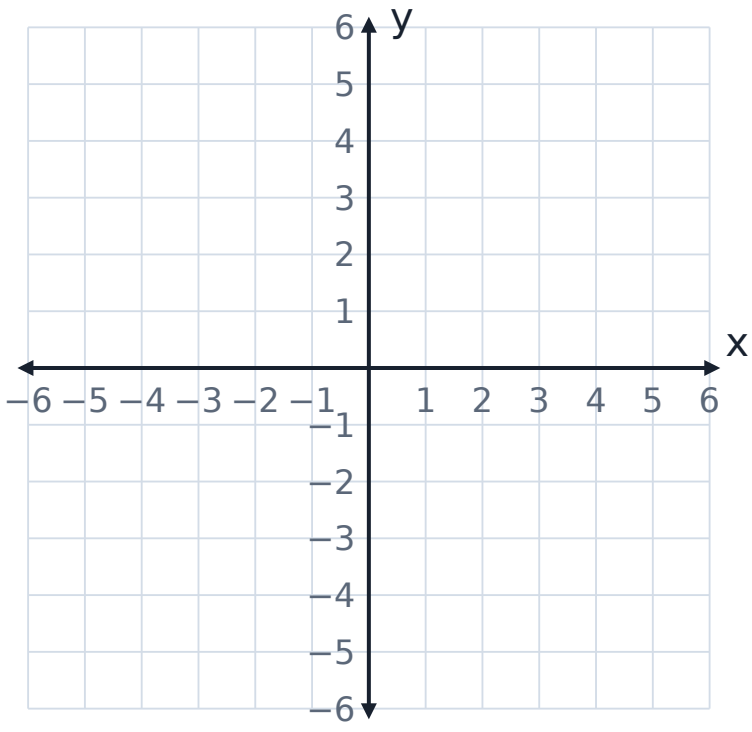
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q4/5

Sketch a rough graph of $p(x) = (x - 1)^2(x + 2)(x - 3)$. Show what the graph does at each zero and its end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q5/5

Use structure to factor perfect squares and expressions in quadratic form

MAIN	Algebra I main question	**HSA-SSE.A.2**
BACKWARD 1	Find squares of whole numbers and decimals	**6.EE.A.1**
BACKWARD 2	Square a binomial	**HSA-APR.A.1**
FORWARD 1	Find the zeros of polynomial functions in quadratic form	**HSF-IF.C.7.c**
FORWARD 2	Factor quadratic expressions with complex numbers	**HSN-CN.C.8**

SET 13 • HSA-SSE.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 10x + 25$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q1/5

Factor completely. (Think of x^2 as one quantity.)

$x^4 - 5x^2 + 4$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q2/5

Factor, and simplify each factor. (Think of x + 1 as one quantity.)

$(x + 1)^2 - 9$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q3/5

Which expression is a perfect square trinomial?

A. $x^2 + 9$

B. $4x^2 - 12x + 9$

C. $4x^2 + 6x + 9$

D. $x^2 - 6x - 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q4/5

True or false?

$9x^2 + 30x + 25 = (3x + 5)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q5/5

SET 13 • HSA-SSE.A.2

BACKWARD 1

Find squares of whole numbers and decimals • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

15^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

What whole number squared is 169?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

Find the value.

2.5^2

A. 5

B. 4.25

C. 62.5

D. 6.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

True or false?

$11^2 = 121$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

True or false?

$6^2 = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSA-SSE.A.2

BACKWARD 2

Square a binomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply.

$(x + 5)^2$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q1/5

Multiply.

$(3y - 2)^2$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q2/5

Multiply.

$(x - 6)^2$

A. $x^2 - 6x + 36$

B. $x^2 - 36$

C. $x^2 + 36$

D. $x^2 - 12x + 36$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-B2-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q3/5

True or false?

$(a + 1)^2 = a^2 + 2a + 1$

A. True

B. False

True or false?

$(x - 3)^2 = x^2 - 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSA-SSE.A.2

FORWARD 1

Find the zeros of polynomial functions in quadratic form • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

Find all real zeros of $f(x) = x^4 - 5x^2 + 4$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q1/5

Find all real zeros of $y = x^4 - 9x^2$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q2/5

How many real zeros does $y = x^4 - 1$ have?

A. 2

B. 1

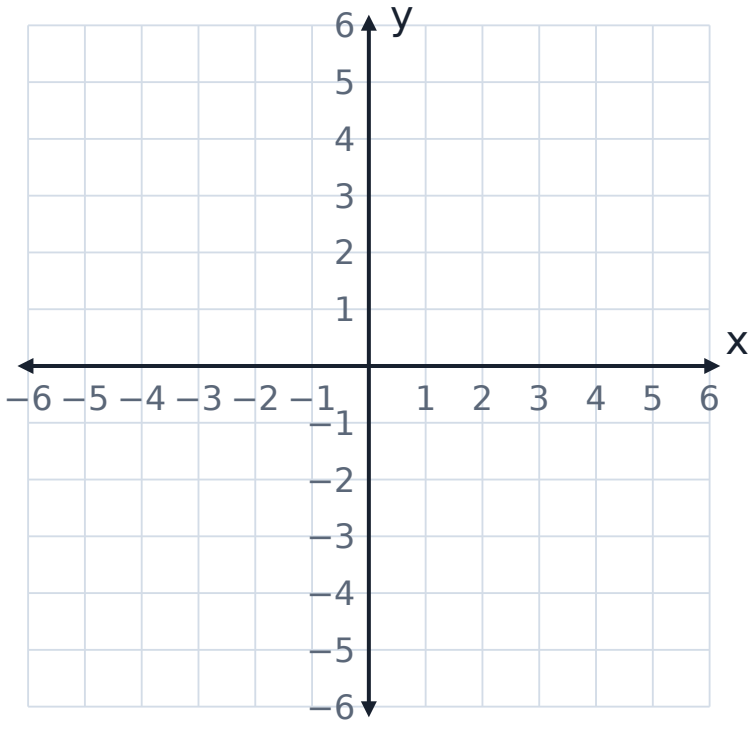
C. 0

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q3/5

Graph $f(x) = x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4)$. Mark the zeros and the y-intercept, and show the end behavior.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q4/5

True or false?

$y = x^4 + 1$ has a real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSA-SSE.A.2

FORWARD 2

Factor quadratic expressions with complex numbers • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 49$ using complex numbers.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q1/5

The solutions of $x^2 + 2x + 5 = 0$ are $-1 + 2i$ and $-1 - 2i$.

Write $x^2 + 2x + 5$ as a product of two linear factors.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q2/5

Multiply.

$(x - 3i)(x + 3i)$

A. $x^2 - 9$

B. $x^2 + 6ix - 9$

C. $x^2 + 9$

D. $x^2 - 9i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q3/5

True or false?

$$x^4 - 1 = (x - 1)(x + 1)(x - i)(x + i)$$

A. True

B. False

True or false?

$$(x + 2i)^2 = x^2 - 4$$

A. True

B. False

Factor a quadratic expression to reveal the zeros of the function it defines

MAIN	Algebra I main question	**HSA-SSE.B.3.a**
BACKWARD 1	Factor trinomials with a leading coefficient of 1	**HSA-SSE.A.2**
BACKWARD 2	Use substitution to decide which values make an equation true	**6.EE.B.5**
FORWARD 1	Find zeros of cubic functions by factoring	**HSA-APR.B.3**
FORWARD 2	Sketch polynomial graphs from their zeros	**HSF-IF.C.7.c**

SET 14 • HSA-SSE.B.3.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

$f(x) = x^2 - x - 12$.

Factor f(x), then give the zeros of f.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q1/5

A ball's height is $h(t) = -16t^2 + 48t$ feet after t seconds.

Factor h(t), and use the factors to say when the ball is at ground level.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q2/5

What are the zeros of $g(x) = 2x^2 + 5x - 3$?

A. $\frac{1}{2}$ and -3

B. $-\frac{1}{2}$ and 3

C. 2 and -3

D. 1 and -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q3/5

Write a quadratic function, in factored form with a leading coefficient of 1, whose zeros are 2 and −5.

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q4/5

True or false?

The zeros of $f(x) = (x - 6)(x + 1)$ are 6 and -1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q5/5

SET 14 • HSA-SSE.B.3.a

BACKWARD 1

Factor trinomials with a leading coefficient of 1 • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 9x + 20$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q1/5

Factor.

$x^2 - 9x + 14$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q2/5

Factor.

$x^2 + x - 6$

A. $(x - 6)(x + 1)$

B. $(x - 3)(x + 2)$

C. $(x + 3)(x - 2)$

D. $(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q3/5

True or false?

$$x^2 - 4x - 21 = (x - 7)(x + 3)$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 8x + 15 = (x + 3)(x + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSA-SSE.B.3.a

BACKWARD 2

Use substitution to decide which values make an equation true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is $x = 4$ a solution of $x^2 = 16$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q1/5

Which of the values 1, 2, and 3 make $x^2 - 5x + 6 = 0$ true?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q2/5

Which value makes $3x - 6 = 0$ true?

A. 3

B. -2

C. 2

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q3/5

True or false?

$x = 5$ makes $x(x - 5) = 0$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q4/5

True or false?

$x = -2$ makes $x + 2 = 4$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSA-SSE.B.3.a

FORWARD 1

Find zeros of cubic functions by factoring • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Find the zeros of $f(x) = x^3 + 2x^2 - 15x$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q1/5

$g(x) = (x - 1)(x^2 - 4)$.

List all the zeros of g.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q2/5

Which function has a double zero at $x = 2$?

A. $f(x) = (x - 2)^2(x + 1)$

B. $f(x) = 2x(x - 1)$

C. $f(x) = (x + 2)^2(x - 1)$

D. $f(x) = (x - 2)(x + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q3/5

True or false?

$f(x) = x^3 - x$ has three real zeros.

A. True

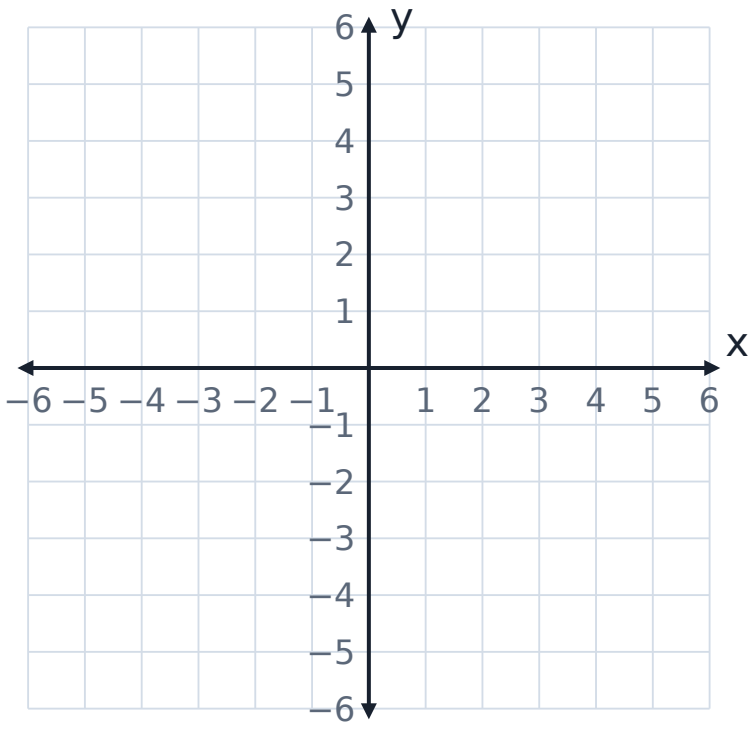
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q4/5

Sketch a rough graph of $f(x) = (x - 2)^2(x + 1)$. Mark the zeros and the y-intercept, and show what the graph does at each zero. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSA-SSE.B.3.a

FORWARD 2

Sketch polynomial graphs from their zeros • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

Describe where $f(x) = (x + 1)(x - 2)(x - 4)$ is above the x-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q1/5

A cubic function has zeros -3, 0, and 3 and a positive leading coefficient.

Describe its end behavior.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q2/5

How many turning points can the graph of a cubic function have, at most?

A. 2

B. 1

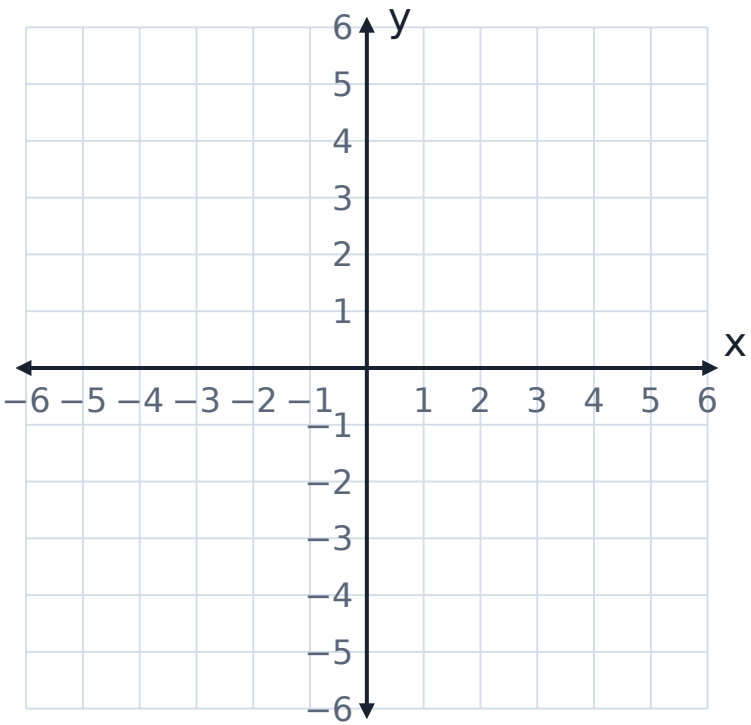
C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q3/5

Graph
$f(x) = (x + 1)(x - 2)(x - 4)$.
Mark the zeros and the y-intercept, and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q4/5

True or false?

A polynomial function with a negative leading coefficient always falls on both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q5/5

Complete the square to reveal the maximum or minimum value of a quadratic function

MAIN	Algebra I main question	**HSA-SSE.B.3.b**
BACKWARD 1	Recognize and complete perfect square trinomials	**HSA-SSE.A.2**
BACKWARD 2	Describe where a graph rises, falls, and reaches its highest point	**8.F.B.5**
FORWARD 1	Complete the square to find the center and radius of a circle	**HSG-GPE.A.1**
FORWARD 2	Derive the equations of ellipses and hyperbolas from their foci	**HSG-GPE.A.3**

SET 15 • HSA-SSE.B.3.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Write $f(x) = x^2 + 6x + 5$ in vertex form by completing the square. Show the steps.

$f(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q1/5

$g(x) = x^2 - 10x + 30$.

Complete the square to write g in vertex form, then give the minimum value of g and where it happens. Show the steps.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q2/5

A ball's height is $h(t) = -t^2 + 8t + 2$ meters after t seconds.

Complete the square (write h in vertex form) to find the greatest height and when it happens. Show the steps.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q3/5

Which is $x^2 - 4x + 7$ in vertex form?

A. $(x - 4)^2 + 7$

B. $(x - 2)^2 + 7$

C. $(x - 2)^2 + 3$

D. $(x + 2)^2 + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q4/5

True or false?

The minimum value of $(x + 1)^2 - 6$ is -6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q5/5

SET 15 • HSA-SSE.B.3.b

BACKWARD 1

Recognize and complete perfect square trinomials • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 14x + 49$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q1/5

What number c makes $x^2 - 8x + c$ a perfect square trinomial?

$c =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q2/5

Which number completes $x^2 + 12x + ___$ to a perfect square trinomial?

A. 36

B. 6

C. 24

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q3/5

True or false?

$$x^2 - 2x + 1 = (x - 1)^2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 6x + 6$ is a perfect square trinomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q5/5

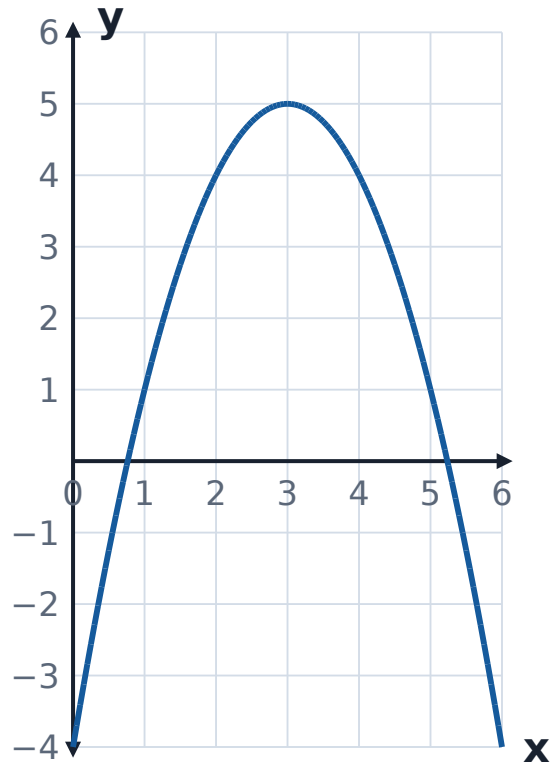
SET 15 • HSA-SSE.B.3.b

BACKWARD 2

Describe where a graph rises, falls, and reaches its highest point • Grade 8 • Functions • 8.F.B.5

Where is the highest point of the graph?

Point: _____



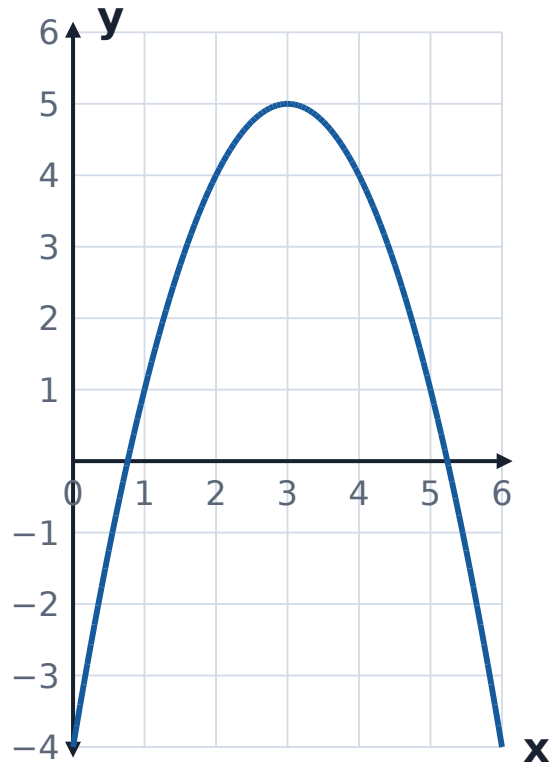
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q1/5

For which x-values shown is the graph rising?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q2/5

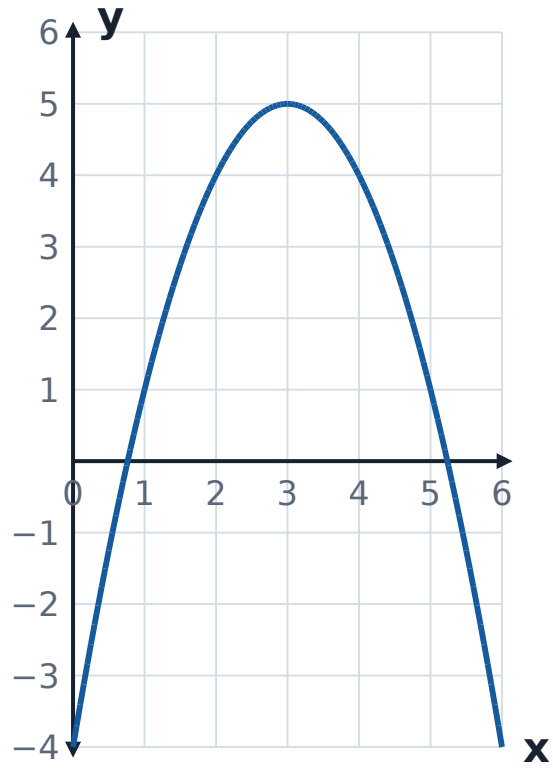
For which x-values shown is the graph falling?

A. From x = 3 to x = 6

B. It never falls.

C. From x = 0 to x = 3

D. From x = 0 to x = 6



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

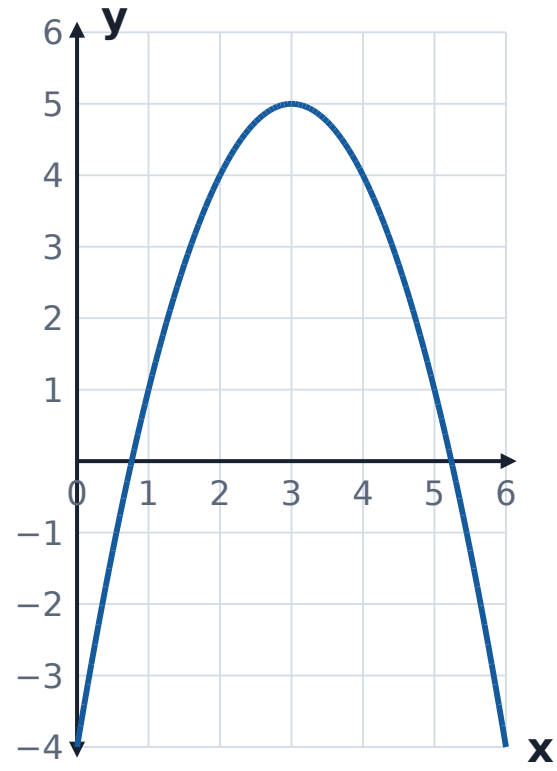
Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q3/5

True or false?

The graph has the same height at $x = 2$ and at $x = 4$.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

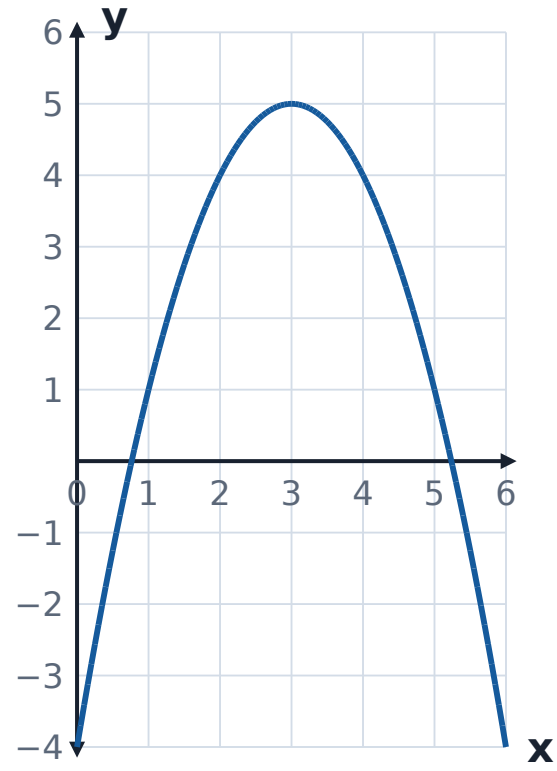
Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q4/5

True or false?

The graph is a straight line.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSA-SSE.B.3.b

FORWARD 1

Complete the square to find the center and radius of a circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 3 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q1/5

Write an equation of the circle with center $(-1, 4)$ and radius 3 in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q2/5

Find the center and radius of $x^2 + y^2 + 8x = 9$.

A. Center (4, 0), radius 5

B. Center (0, 0), radius 9

C. Center (−4, 0), radius 3

D. Center (−4, 0), radius 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q3/5

True or false?

The circle $(x - 2)^2 + (y + 1)^2 = 4$ has a radius of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q4/5

True or false?

The circle $x^2 + y^2 = 25$ has its center at (5, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSA-SSE.B.3.b

FORWARD 2

Derive the equations of ellipses and hyperbolas from their foci • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

An ellipse has foci $(4, 0)$ and $(-4, 0)$, and the sum of the distances from any point on it to the foci is 10.

Write its equation in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q1/5

A hyperbola has foci (5, 0) and (−5, 0), and the absolute value of the difference of the distances from any point on it to the foci is 8.

Write its equation in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q2/5

The ellipse $x^2 \div 25 + y^2 \div 9 = 1$ has foci (4, 0) and (−4, 0).

Show that (3, 2.4) is on the ellipse and that its distances to the two foci add to 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q3/5

True or false?

$x^2/25 + y^2/9 = 1$ is the equation of an ellipse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q4/5

True or false?

For every point on an ellipse, the difference of the distances to the two foci is constant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q5/5

Use the properties of exponents to transform exponential expressions

MAIN	Algebra I main question	**HSA-SSE.B.3.c**
BACKWARD 1	Use the meaning of rational exponents and the power-of-a-power…	**HSN-RN.A.1**
BACKWARD 2	Solve percent increase and decrease problems	**7.RP.A.3**
FORWARD 1	Solve exponential equations with logarithms	**HSF-LE.A.4**
FORWARD 2	Use the sum of a geometric series in a growth model	**HSA-SSE.B.4**

SET 16 • HSA-SSE.B.3.c

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c

An account grows by 6% each year: $A = 500(1.06)^t$ after t years.

Rewrite the expression to show the monthly growth factor, rounded to four decimal places.

$A =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q1/5

Use the properties of exponents to explain why $1.21^{t} = 1.1^{2t}$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q2/5

A population doubles every 5 years: $P = 300 \cdot 2^{t/5}$ after t years.

Find the yearly growth factor to three decimal places, and the yearly percent growth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q3/5

Which expression is equal to 8^t?

A. 2^{t+3}

B. 3^{2t}

C. 4^{2t}

D. 2^{3t}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q4/5

True or false?

In $y = 100(0.5)^{t/2}$, the amount is cut in half every 2 time units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q5/5

SET 16 • HSA-SSE.B.3.c

BACKWARD 1

Use the meaning of rational exponents and the power-of-a-power property • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

Find the value.

$4^{1/2}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q1/5

Write $(1.1^{2})^{t}$ as a single power of 1.1.

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q2/5

Find the value.

$9^{1/2}$

A. $\frac{1}{9}$

B. 4.5

C. 3

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q3/5

True or false?

$(2^3)^t = 2^{3t}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q4/5

True or false?

$5^{1/2} = 2.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSA-SSE.B.3.c

BACKWARD 2

Solve percent increase and decrease problems • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Increase $500 by 6%.

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q1/5

Decrease 80 by 25%.

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q2/5

What do you multiply by to increase an amount by 4%?

A. 0.04

B. 1.4

C. 1.04

D. 0.96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q3/5

True or false?

Decreasing an amount by 10% is the same as multiplying it by 0.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q4/5

True or false?

An increase of 5% followed by a decrease of 5% gives back the original amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSA-SSE.B.3.c

FORWARD 1

Solve exponential equations with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $500(1.06)^t = 800$. Round t to the nearest tenth.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q1/5

A population is $P = 300 \cdot 2^{t/5}$ after t years. When is P = 1,000? Round to the nearest tenth. (log means base 10.)

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q2/5

Solve $10^{2x} = 50$. (log means base 10.)

A. $x = 2.5$

B. $x = (\log 50) \div 2 \approx 0.85$

C. $x = 25$

D. $x = \log 25 \approx 1.40$

True or false?

If $3^x = 3^5$, then $x = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q4/5

True or false?

The solution of $1.06^t = 2$ is $t = 2 \div 1.06$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSA-SSE.B.3.c

FORWARD 2

Use the sum of a geometric series in a growth model • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Deposits of \$100, \$100(1.005), \$100(1.005)^2, ..., \$100(1.005)^{11} are added. Use $S = a(1 - r^n) \div (1 - r)$ to find the total, to the nearest dollar.

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q1/5

Find the sum 5 + 10 + 20 + ... + 640.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q2/5

What is 1 + 3 + 9 + 27 + 81?

A. 81

B. 121

C. 243

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q3/5

True or false?

The series 2 + 6 + 18 + 54 has a common ratio of 3.

A. True

B. False

True or false?

The sum of the first n terms of a geometric series with first term a and ratio $r \neq 1$ is $a \cdot r^n$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q5/5

Explain each step in solving an equation and justify the solution method

MAIN	Algebra I main question	**HSA-REI.A.1**
BACKWARD 1	Use substitution to decide whether a number makes an equation…	**6.EE.B.5**
BACKWARD 2	Solve linear equations by expanding and collecting like terms	**8.EE.C.7.b**
FORWARD 1	Justify the steps of a proof about lines and angles	**HSG-CO.C.9**
FORWARD 2	Solve radical and rational equations and identify extraneous solut..	**HSA-REI.A.2**

SET 17 • HSA-REI.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

Solve $3(x - 4) + 5 = 2x + 7$. Give a reason for each step.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q1/5

A student solves $5x - 8 = 12$.

Step 1: $5x = 20$

Step 2: $x = 4$

Which property justifies Step 1?

A. Distributive property

B. Addition property of equality: add 8 to both sides

C. Subtraction property of equality: subtract 8 from both sides

D. Multiplication property of equality: multiply both sides by 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q2/5

A student wrote:

$4(x + 2) = 20$

$4x + 2 = 20$

$4x = 18$

$x = 4.5$

Which step has the error? What is the correct solution?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q3/5

True or false?

If $a = b$, then $a \div c = b \div c$ for every number c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q4/5

True or false?

Subtracting 2x from both sides of $2(x + 1) = 2x + 2$ gives $2 = 2$, so every real number is a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q5/5

SET 17 • HSA-REI.A.1

BACKWARD 1

Use substitution to decide whether a number makes an equation true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is $x = 3$ a solution of $4x - 5 = 7$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q1/5

Which number from 1, 2, 3, 4 makes 2x + 1 = 9 true?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q2/5

Which value of y makes $15 - y = 9$ true?

A. 6

B. -6

C. 9

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q3/5

True or false?

$x = 5$ is a solution of $3x = 15$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q4/5

True or false?

$n = 2$ is a solution of $n + 7 = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSA-REI.A.1

BACKWARD 2

Solve linear equations by expanding and collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve $2(x + 3) = 14$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q1/5

Solve $5y - 2y + 4 = 19$.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q2/5

Solve $3(n - 2) = n + 8$.

A. $n = 7$

B. $n = 5$

C. $n = 3$

D. $n = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q3/5

True or false?

The solution of $0.5x + 1 = 4$ is $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q4/5

True or false?

The solution of $4(x - 1) = 12$ is $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSA-REI.A.1

FORWARD 1

Justify the steps of a proof about lines and angles • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9

Angles 1 and 2 are vertical angles. Angle 1 and angle 3 form a linear pair, and so do angle 2 and angle 3.

Explain why $m\angle 1 = m\angle 2$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

In a proof, the line $m\angle A + 40° = 90°$ is followed by $m\angle A = 50°$.

Which reason justifies that step?

A. Definition of a linear pair

B. Vertical angles are congruent

C. Addition property of equality

D. Subtraction property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5

Two angles form a linear pair. One measures (3x + 10)° and the other measures (2x + 20)°.

Find x and both angle measures.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

True or false?

Vertical angles have equal measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

True or false?

If two angles have equal measures, they must be vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSA-REI.A.1

FORWARD 2

Solve radical and rational equations and identify extraneous solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $\sqrt{(x + 3)} = 5$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q1/5

Solve $\sqrt{(2x + 3)} = x$. Name any extraneous solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q2/5

Assume $x \neq 0$. Solve $6 \div x = 3$.

A. $x = \frac{1}{2}$

B. $x = 3$

C. $x = 18$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q3/5

True or false?

$x = 4$ is a solution of $\sqrt{x} = -2$.

A. True

B. False

True or false?

Squaring both sides of an equation can produce a value that does not satisfy the original equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q5/5

Solve linear equations in one variable, including equations with letter coefficients

MAIN	Algebra I main question	**HSA-REI.B.3**
BACKWARD 1	Name the property that justifies a step	**HSA-REI.A.1**
BACKWARD 2	Solve two-step equations from word problems	**7.EE.B.4.a**
FORWARD 1	Prove and apply theorems about triangles	**HSG-CO.C.10**
FORWARD 2	Solve rational equations that lead to linear equations	**HSA-REI.A.2**

SET 18 • HSA-REI.B.3

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $\frac{2}{3}x - 5 = \frac{1}{4}x + 5$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q1/5

Solve $0.4(3x - 5) = 2.2x + 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q2/5

How many solutions does $5x - 2(x + 4) = 3x + 1$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q3/5

Solve $ax + b = c$ for x, where $a \neq 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q4/5

True or false?

The equation $2(x - 3) = 2x - 6$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q5/5

SET 18 • HSA-REI.B.3

BACKWARD 1

Name the property that justifies a step • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

Name the property used to go from $3x + 4 = 19$ to $3x = 15$.

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q1/5

Name the property used to go from $\frac{x}{5} = 3$ to $x = 15$.

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q2/5

Which step keeps the same solutions as 2x = 10?

A. Divide both sides by 2

B. Square the right side only

C. Add 2 to the left side only

D. Multiply both sides by 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q3/5

True or false?

Adding the same number to both sides of an equation keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q4/5

True or false?

Multiplying both sides of an equation by 0 keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSA-REI.B.3

BACKWARD 2

Solve two-step equations from word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A taxi charges $3 plus $2 per mile. A ride cost $19.

How many miles was the ride?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q1/5

Solve $-4x + 9 = 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q2/5

Solve $\frac{1}{2}(x + 6) = 10$.

A. $x = 14$

B. $x = 17$

C. $x = 2$

D. $x = 26$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q3/5

True or false?

The solution of $6x - 4 = 20$ is $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q4/5

True or false?

The solution of $2x + 5 = -1$ is $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSA-REI.B.3

FORWARD 1

Prove and apply theorems about triangles • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10

Prove that the angles of any triangle ABC add to 180°. Start by drawing the line through B parallel to AC.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q1/5

Explain why an exterior angle of a triangle equals the sum of the two remote interior angles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q2/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C. (Draw the bisector of ∠A, meeting BC at D.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q3/5

D and E are the midpoints of AB and AC in triangle ABC, and BC = 18.

What does the midsegment theorem let you conclude?

A. DE ⊥ BC and DE = 9

B. DE ∥ BC and DE = 18

C. DE = 9, but DE need not be parallel to BC

D. DE ∥ BC and DE = 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q4/5

True or false?

The three medians of a triangle meet at one point, and that point is $\frac{2}{3}$ of the way from each vertex to the midpoint of the opposite side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSA-REI.B.3

FORWARD 2

Solve rational equations that lead to linear equations • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Assume $x \neq 0$. Solve $3 \div x + 2 = 5 \div x$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q1/5

Assume $x \neq 2$. Solve $(x + 1) \div (x - 2) = 4$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q2/5

Assume $x \neq 3$. Solve $x \div (x - 3) = 3 \div (x - 3)$.

A. $x = 0$

B. No solution: $x = 3$ is extraneous

C. $x = 3$

D. Every real number except 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q3/5

True or false?

$x = 1$ is a solution of $2x \div (x + 1) = 1$.

A. True

B. False

True or false?

The solution of $6 \div (x + 1) = 2$ is $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q5/5

Solve linear inequalities and compound inequalities in one variable

MAIN	Algebra I main question	**HSA-REI.B.3**
BACKWARD 1	Solve two-step inequalities from word problems	**7.EE.B.4.b**
BACKWARD 2	Write and graph simple inequalities	**6.EE.B.8**
FORWARD 1	Use inequalities to meet constraints in a design problem	**HSG-MG.A.3**
FORWARD 2	Write a margin-of-error interval as a compound inequality — neare…	**HSS-IC.B.4**

SET 19 • HSA-REI.B.3

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $5 - 3x \leq 17$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q1/5

Solve $-2 < 3x + 4 \leq 13$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q2/5

The number line shows the solution set of which inequality?



A. $x + 5 > 7$

B. $4x + 3 < 2x + 7$

C. $3x \leq 6$

D. $-2x < -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q3/5

Solve $\frac{x}{4} - 1 > \frac{x}{2} + 2$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q4/5

True or false?

The solution of $-4x > 8$ is $x > -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q5/5

SET 19 • HSA-REI.B.3

BACKWARD 1

Solve two-step inequalities from word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.b

Solve $3x + 2 > 14$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q1/5

A phone plan costs $20 plus $5 per gigabyte g. Lena can spend at most $45.

Write and solve an inequality for g.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q2/5

Solve $-2x + 1 \geq 9$.

A. $x \leq -4$

B. $x \geq 5$

C. $x \leq 4$

D. $x \geq -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q3/5

True or false?

$x = 6$ is a solution of $2x - 3 > 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q4/5

True or false?

Dividing both sides of an inequality by a positive number reverses the inequality sign.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSA-REI.B.3

BACKWARD 2

Write and graph simple inequalities • Grade 6 • Expressions & Equations • 6.EE.B.8

A rider must be at least 48 inches tall.

Write an inequality for the rider's height h.

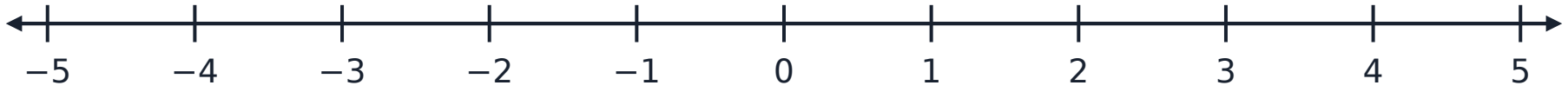
Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q1/5

Graph $x < 3$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q2/5

Which number is a solution of $x > -2$?

A. -2

B. -3

C. -5

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

The inequality $x > 5$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

On a number line, $x \leq 2$ is shown with an open circle at 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSA-REI.B.3

FORWARD 1

Use inequalities to meet constraints in a design problem • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A rectangular garden is 12 ft long. Its perimeter can be at most 50 ft.

Write and solve an inequality for the width w.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q1/5

A box with a 10 in by 10 in base must hold at least 600 in^3.

Write and solve an inequality for the height h.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q2/5

A poster 15 in wide must have an area of at least 300 in^2.

Which lengths L work?

A. $L \geq 285$ inches

B. $L \geq 20$ inches

C. $L \geq 4.5$ inches

D. $L \leq 20$ inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q3/5

True or false?

A rectangle 8 cm long and w cm wide has an area of at most 40 cm² exactly when $0 < w \leq 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q4/5

True or false?

A square with side s cm has a perimeter less than 20 cm exactly when $s < 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSA-REI.B.3

FORWARD 2

Write a margin-of-error interval as a compound inequality • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: A margin-of-error interval is written as a compound inequality like those solved in the MAIN set.

A survey estimates that 54% of voters support a plan, with a margin of error of 3 percentage points.

Write the interval for the true proportion p as a compound inequality.

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.4 • Set 19 • FORWARD 2 • Q1/5 • NEAREST RELATED

A poll reports 62% with a margin of error of 4 percentage points.

Is 60% inside the interval? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.4 • Set 19 • FORWARD 2 • Q2/5 • NEAREST RELATED

The interval $0.40 \leq p \leq 0.48$ is an estimate plus or minus a margin of error.

What are the estimate and the margin of error?

A. 0.44 and 0.08

B. 0.40 and 0.08

C. 0.44 and 0.04

D. 0.48 and 0.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

An estimate of 70% with a margin of error of 5 percentage points gives $65\% \leq p \leq 75\%$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

If the interval is $0.30 \leq p \leq 0.36$, the margin of error is 0.06.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.4 • Set 19 • FORWARD 2 • Q5/5 • NEAREST RELATED

Create linear equations and inequalities in one variable and use them to solve problems

MAIN	Algebra I main question	**HSA-CED.A.1**
BACKWARD 1	Solve linear equations and inequalities	**HSA-REI.B.3**
BACKWARD 2	Write and solve one-step equations	**6.EE.B.7**
FORWARD 1	Write and solve equations from similar triangles	**HSG-SRT.B.5**
FORWARD 2	Write and solve an equation for an expected value — nearest relat.	**HSS-MD.A.2**

SET 20 • HSA-CED.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

A plumber charges a $65 call fee plus $48 per hour. A bill is $257.

Write an equation for the number of hours h, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q1/5

The sum of three consecutive integers is 141.

Write an equation and find the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q2/5

Maya has $180 and saves $15 each week. She wants at least $400.

Write and solve an inequality for the number of weeks w. How many whole weeks must she save?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q3/5

A rental costs $40 per day plus a one-time $25 fee.

Which inequality gives the number of days d you can rent for at most $265?

A. $40(d + 25) \leq 265$

B. $40d + 25 \geq 265$

C. $40d + 25 \leq 265$

D. $25d + 40 \leq 265$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q4/5

True or false?

Plan A costs 30 + 0.10m dollars and Plan B costs 0.25m dollars for m minutes. Plan A costs less when $m < 200$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q5/5

SET 20 • HSA-CED.A.1

BACKWARD 1

Solve linear equations and inequalities • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $7x - 4 = 3x + 20$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q1/5

Solve $2(5 - x) \geq 16$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q2/5

Solve 0.25t + 3 = 7.

A. t = 1

B. t = 40

C. t = 2.5

D. t = 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q3/5

True or false?

$x = -3$ is a solution of $4 - x > 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q4/5

True or false?

The solution of $12 - 3x < 0$ is $x < 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSA-CED.A.1

BACKWARD 2

Write and solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve x + 17 = 42.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q1/5

Tickets cost $9 each. Sam spent $63.

Write an equation and find the number of tickets t.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q2/5

Solve 4.5 = 1.5y.

A. y = 6.75

B. y = 6

C. y = 3

D. y = 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q3/5

True or false?

If $n - 8 = 15$, then $n = 23$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q4/5

True or false?

If 6p = 3, then p = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSA-CED.A.1

FORWARD 1

Write and solve equations from similar triangles • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

A 6-ft person casts a 4-ft shadow at the same time a tree casts a 22-ft shadow.

Write a proportion and find the tree's height.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q1/5

Triangle ABC ~ triangle DEF with AB = 8, BC = 12, and DE = 10.

Find EF.

EF = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q2/5

Triangle PQR ~ triangle STU, with PQ = 5, ST = 15, QR = x + 2, and TU = 2x + 9.

What is x?

A. x = 1

B. x = 4.5

C. x = 3

D. x = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q3/5

True or false?

If two triangles are similar, their corresponding side lengths are proportional.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q4/5

True or false?

In similar triangles with a ratio of 2 : 3, a side of 10 corresponds to a side of 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSA-CED.A.1

FORWARD 2

Write and solve an equation for an expected value • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Writing and solving a linear equation from a situation is applied to an expected-value condition.

A game pays $10 with probability 0.2 and $0 otherwise.

What is the expected payout?

Expected payout: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q1/5 • NEAREST RELATED

A game pays x dollars with probability 0.25 and $0 otherwise.

Write and solve an equation for x so that the expected payout is $3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q2/5 • NEAREST RELATED

A random variable is 1 with probability 0.5, 2 with probability 0.3, and 4 with probability 0.2.

What is its expected value?

A. 7

B. 1.9

C. 2

D. 2.33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

The expected value of one roll of a fair number cube numbered 1 to 6 is 3.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A game costs $5 to play. It pays $20 with probability 0.2 and $0 otherwise. The expected net gain is $4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q5/5 • NEAREST RELATED

Rearrange formulas to highlight a quantity of interest

MAIN	Algebra I main question	**HSA-CED.A.4**
BACKWARD 1	Solve linear equations with letter coefficients	**HSA-REI.B.3**
BACKWARD 2	Evaluate formulas for given values	**6.EE.A.2.c**
FORWARD 1	Rearrange the arc-length formula $s = r\theta$	**HSG-C.B.5**
FORWARD 2	Solve exponential formulas for the exponent with logarithms	**HSF-LE.A.4**

SET 21 • HSA-CED.A.4

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

Solve $V = \pi r^2 h$ for h. ($r > 0$)

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q1/5

Solve $C = \frac{5}{9}(F - 32)$ for F.

F = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q2/5

Solve $A = P(1 + rt)$ for t. ($P \neq 0$, $r \neq 0$)

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q3/5

Ohm's law is $V = IR$. Solve for R. ($I \neq 0$)

A. $R = \frac{I}{V}$

B. $R = V - I$

C. $R = VI$

D. $R = \frac{V}{I}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q4/5

True or false?

Solving $y = mx + b$ for x ($m \neq 0$) gives $x = y \div m - b$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q5/5

SET 21 • HSA-CED.A.4

BACKWARD 1

Solve linear equations with letter coefficients • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $3x + k = 10$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q1/5

Solve $ax = bx + c$ for x, where $a \neq b$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q2/5

Solve $x \div m = n$ for x. ($m \neq 0$)

A. $x = m + n$

B. $x = mn$

C. $x = n \div m$

D. $x = m \div n$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q3/5

True or false?

If $2x - p = q$, then $x = (p + q) \div 2$.

A. True

B. False

True or false?

If $kx = 5$ and $k \neq 0$, then $x = 5k$.

A. True

B. False

SET 21 • HSA-CED.A.4

BACKWARD 2

Evaluate formulas for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use A = lw to find A when l = 9 and w = 4.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q1/5

Use d = rt to find d when r = 55 and t = 3.

d = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q2/5

Use P = 2l + 2w to find P when l = 7 and w = 3.

A. 10

B. 21

C. 20

D. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q3/5

True or false?

$V = s^3$ gives $V = 27$ when $s = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q4/5

True or false?

$A = \frac{1}{2}bh$ gives $A = 40$ when $b = 10$ and $h = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSA-CED.A.4

FORWARD 1

Rearrange the arc-length formula s = rθ • High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5

A circle has radius 6 cm. An arc has length 9 cm.

Use $s = r\theta$ to find the central angle θ in radians.

$\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q1/5

Solve $s = r\theta$ for r. Then find r when $s = 10\pi$ and $\theta = \pi \div 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q2/5

The area of a sector is $A = \frac{1}{2}r^2\theta$. Solve for θ. ($r > 0$)

A. $\theta = A - \frac{1}{2}r^2$

B. $\theta = 2Ar^2$

C. $\theta = A \div (2r^2)$

D. $\theta = 2A \div r^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q3/5

True or false?

An arc of length $2\pi r$ has a central angle of 2π radians.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q4/5

True or false?

If $s = r\theta$, then $\theta = r \div s$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSA-CED.A.4

FORWARD 2

Solve exponential formulas for the exponent with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $A = Pe^{rt}$ for t. (A, P, r > 0)

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q1/5

$500 grows to $1,000 with continuous interest at r = 0.05.

Find t to the nearest tenth of a year.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q2/5

Solve $2^x = y$ for x. ($y > 0$)

A. $x = y^2$

B. $x = \sqrt{y}$

C. $x = y \div 2$

D. $x = \log_2 y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q3/5

True or false?

If $10^k = 1{,}000$, then $k = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q4/5

True or false?

Solving $N = 5 \cdot 3^t$ for t gives $t = \log_3 N - 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q5/5

Complete the square to solve quadratic equations and derive the quadratic formula

MAIN	Algebra I main question	**HSA-REI.B.4.a**
BACKWARD 1	Complete the square to write a quadratic in vertex form	**HSA-SSE.B.3.b**
BACKWARD 2	Solve $x^2 = p$ and $x^3 = p$ with square and cube roots	**8.EE.A.2**
FORWARD 1	Complete the square to find the center and radius of a circle	**HSG-GPE.A.1**
FORWARD 2	Complete the square to write the equation of an ellipse	**HSG-GPE.A.3**

SET 22 • HSA-REI.B.4.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

Solve $x^2 + 6x - 7 = 0$ by completing the square.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q1/5

Rewrite $x^2 - 10x + 18 = 0$ in the form $(x - p)^2 = q$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q2/5

Rewrite $2x^2 + 12x + 5 = 0$ in the form $(x - p)^2 = q$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q3/5

Which number completes the square?

$x^2 + 8x + ___$

A. 8

B. 16

C. 4

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q4/5

True or false?

Completing the square on $ax^2 + bx + c = 0$ ($a \neq 0$) gives $(x + b \div (2a))^2 = (b^2 - 4ac) \div (4a^2)$, which leads to the quadratic formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q5/5

SET 22 • HSA-REI.B.4.a

BACKWARD 1

Complete the square to write a quadratic in vertex form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Write $x^2 + 4x + 1$ in the form $(x - h)^2 + k$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q1/5

Write $x^2 - 6x + 11$ in the form $(x - h)^2 + k$, and give its minimum value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q2/5

Which expression is equal to $x^2 + 2x + 5$?

A. $(x + 2)^2 + 1$

B. $(x + 1)^2 + 5$

C. $(x + 1)^2 + 4$

D. $(x - 1)^2 + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q3/5

True or false?

$$(x - 4)^2 = x^2 - 8x + 16$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 10x = (x + 5)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSA-REI.B.4.a

BACKWARD 2

Solve $x^2 = p$ and $x^3 = p$ with square and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve $x^2 = 49$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q1/5

Solve $x^3 = -64$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q2/5

What are the solutions of $x^2 = 81$?

A. 9 and −9

B. 9 only

C. −9 only

D. 40.5 and −40.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{64} = 8$ and $\sqrt[3]{64} = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q4/5

True or false?

The only solution of $x^2 = 25$ is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSA-REI.B.4.a

FORWARD 1

Complete the square to find the center and radius of a circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q1/5

Write an equation of the circle with center (2, −5) and radius 6 in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q2/5

What is the center of the circle $x^2 + 8x + y^2 = 9$?

A. $(-8, 0)$

B. $(-4, 0)$

C. $(4, 0)$

D. $(0, -4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q3/5

True or false?

The circle $(x - 2)^2 + y^2 = 16$ has radius 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q4/5

True or false?

The circle $x^2 + (y + 5)^2 = 9$ has center (0, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSA-REI.B.4.a

FORWARD 2

Complete the square to write the equation of an ellipse • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Write $x^2 + 4y^2 - 6x - 7 = 0$ in standard form, and give the center of the ellipse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q1/5

Derive the equation of the set of points whose distances to the foci (3, 0) and (−3, 0) add to 10.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q2/5

The ellipse $(x - 1)^2 \div 25 + (y + 3)^2 \div 9 = 1$ has a horizontal major axis.

How long is the major axis?

A. 6

B. 5

C. 25

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q3/5

True or false?

Completing the square on $x^2 - 8x$ gives $(x - 4)^2 - 16$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q4/5

True or false?

The ellipse $x^2 \div 4 + y^2 \div 9 = 1$ is wider than it is tall.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q5/5

Derive the quadratic formula by completing the square

MAIN	Algebra I main question	**HSA-REI.B.4.a**
BACKWARD 1	Complete the square in expressions	**HSA-SSE.B.3.b**
BACKWARD 2	Solve literal equations for a variable	**HSA-CED.A.4**
FORWARD 1	Use the quadratic formula when the discriminant is negative	**HSN-CN.C.7**
FORWARD 2	Count the complex solutions of quadratic equations	**HSN-CN.C.9**

SET 71 • HSA-REI.B.4.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

Derive the quadratic formula. Start from $ax^2 + bx + c = 0$ with $a \neq 0$, and complete the square to solve for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q1/5

The equation $x^2 + (b \div a)x = -c \div a$ comes from $ax^2 + bx + c = 0$. What must be added to both sides to complete the square, and what does the left side become?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q2/5

After completing the square,
$(x + b \div (2a))^2 = (b^2 - 4ac) \div (4a^2)$. Explain how taking square roots of both sides gives the ± and the denominator 2a.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q3/5

In the derivation of the quadratic formula, why must $a \neq 0$?

A. Because the discriminant must be positive.

B. Because b must be positive.

C. Because c must be 0.

D. The first step divides by a, and if $a = 0$ the equation is not quadratic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q4/5

True or false?

The derivation shows that when $b^2 - 4ac < 0$, the equation has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q5/5

SET 71 • HSA-REI.B.4.a

BACKWARD 1

Complete the square in expressions • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Complete the square.

$x^2 + 12x + 7$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q1/5

Complete the square.

$x^2 - 5x$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q2/5

What number completes the square for $x^2 + \frac{2}{3}x$?

A. $\frac{1}{9}$

B. $\frac{2}{3}$

C. $\frac{4}{9}$

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q3/5

True or false?

$$x^2 + bx + (b \div 2)^2 = (x + b \div 2)^2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q4/5

True or false?

To complete the square on $2x^2 + 8x$, add 16 to the expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSA-REI.B.4.a

BACKWARD 2

Solve literal equations for a variable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

Solve $ax + b = 0$ for x. ($a \neq 0$)

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q1/5

Solve $(x + p)^2 = q$ for x. ($q \geq 0$)

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q2/5

Solve $y = kx^2$ for x, where $k > 0$ and $y \geq 0$.

A. $x = \pm\sqrt{(ky)}$

B. $x = \sqrt{y} \div k$

C. $x = \pm\sqrt{(y \div k)}$

D. $x = y \div k$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q3/5

True or false?

Solving $A = \pi r^2$ for r (with $r > 0$) gives $r = \sqrt{(A \div \pi)}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q4/5

True or false?

Solving $2x - c = d$ for x gives $x = (d - c) \div 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSA-REI.B.4.a

FORWARD 1

Use the quadratic formula when the discriminant is negative • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7

Use the quadratic formula to solve $x^2 + 6x + 13 = 0$. Write the solutions as $a + bi$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q1/5

Solve $2x^2 - 2x + 5 = 0$. Write the solutions as $a + bi$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q2/5

What is the discriminant of $3x^2 + x + 2$?

A. 25

B. -23

C. -25

D. 23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q3/5

True or false?

When $b^2 - 4ac < 0$, the quadratic formula gives two complex solutions that are conjugates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q4/5

True or false?

The solutions of $x^2 + 9 = 0$ are 3 and -3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSA-REI.B.4.a

FORWARD 2

Count the complex solutions of quadratic equations • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9

How many complex solutions, counted with multiplicity, does every quadratic equation have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q1/5

$x^2 - 6x + 9 = 0$ has one real solution. How is it counted by the Fundamental Theorem of Algebra?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q2/5

A quadratic equation with real coefficients has the solution 2 + i. What is its other solution?

A. $-2 - i$

B. $2 - i$

C. $-2 + i$

D. 2 + i again

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q3/5

True or false?

Every quadratic equation with real coefficients has two solutions in the complex numbers, counted with multiplicity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q4/5

True or false?

A quadratic equation can have three different complex solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q5/5

Solve quadratic equations by inspection, square roots, and factoring

MAIN	Algebra I main question	**HSA-REI.B.4.b**
BACKWARD 1	Factor quadratic expressions	**HSA-SSE.A.2**
BACKWARD 2	Solve the linear equations that come from each factor	**HSA-REI.B.3**
FORWARD 1	Find the zeros of a polynomial from its factors	**HSA-APR.B.3**
FORWARD 2	Use the Fundamental Theorem of Algebra to count complex zeros	**HSN-CN.C.9**

SET 23 • HSA-REI.B.4.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve by factoring.

$x^2 - 5x - 24 = 0$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q1/5

Solve $3(x - 2)^2 = 75$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q2/5

Solve $2x^2 = 7x$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q3/5

Solve $6x^2 + x - 2 = 0$.

A. $x = 2$ or $x = -1$

B. $x = -\frac{2}{3}$ or $x = \frac{1}{2}$

C. $x = \frac{2}{3}$ or $x = -\frac{1}{2}$

D. $x = -2$ or $x = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q4/5

True or false?

A student divides both sides of $x^2 = 4x$ by x and gets $x = 4$. This finds every solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q5/5

SET 23 • HSA-REI.B.4.b

BACKWARD 1

Factor quadratic expressions • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor completely.

$x^2 + 2x - 15$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q1/5

Factor completely.

$4x^2 - 9$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q2/5

Factor $x^2 - 9x + 20$.

A. $(x - 4)(x + 5)$

B. $(x - 2)(x - 10)$

C. $(x - 4)(x - 5)$

D. $(x + 4)(x + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q3/5

True or false?

$3x^2 - 12x = 3x(x - 4)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q4/5

True or false?

$$x^2 + 9 = (x + 3)(x - 3)$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSA-REI.B.4.b

BACKWARD 2

Solve the linear equations that come from each factor • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $2x + 7 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q1/5

Solve $3x - 4 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q2/5

Solve $5 - 2x = 0$.

A. $x = 3$

B. $x = -2.5$

C. $x = 2.5$

D. $x = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q3/5

True or false?

If $x - 6 = 0$, then $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q4/5

True or false?

If $4x = 0$, then $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSA-REI.B.4.b

FORWARD 1

Find the zeros of a polynomial from its factors • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Find all zeros of $p(x) = (x - 2)(x + 1)(x - 5)$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q1/5

Find all zeros of $p(x) = x^3 - 4x$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q2/5

Which polynomial has zeros -3, 0, and 4?

A. $x(x + 3)(x - 4)$

B. $(x + 3)(x - 4)$

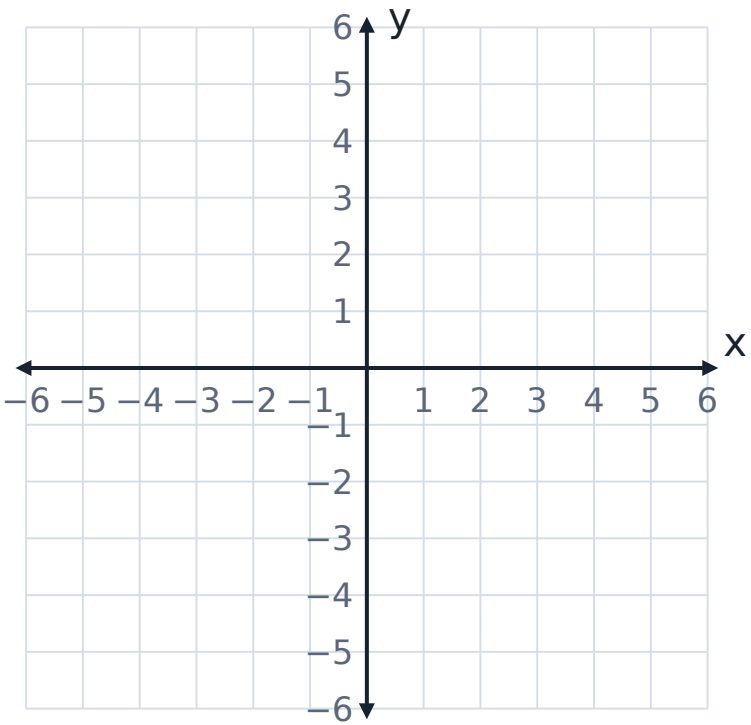
C. $x(x - 3)(x + 4)$

D. $x(x - 3)(x - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q3/5

Sketch a rough graph of $p(x) = -(x + 3)(x - 1)(x - 4)$. Mark the zeros and the y-intercept, and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q4/5

True or false?

$p(x) = x^2(x - 3)$ has three different zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSA-REI.B.4.b

FORWARD 2

Use the Fundamental Theorem of Algebra to count complex zeros • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9

How many complex zeros, counted with multiplicity, does a polynomial of degree 5 have?

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q1/5

$p(x) = x^3 - x^2 + 4x - 4 = (x - 1)(x^2 + 4)$.

Find all complex zeros.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q2/5

A polynomial with real coefficients has zeros 3 and 2i.

Which number must also be a zero?

A. 3i

B. 2

C. −3

D. −2i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q3/5

True or false?

$x^2 + 1 = (x + i)(x - i)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q4/5

True or false?

Every polynomial of degree 3 has three real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q5/5

Solve quadratic equations with the quadratic formula and use the discriminant

MAIN	Algebra I main question	**HSA-REI.B.4.b**
BACKWARD 1	Complete the square to solve a quadratic equation	**HSA-REI.B.4.a**
BACKWARD 2	Estimate square roots with rational approximations	**8.NS.A.2**
FORWARD 1	Solve right-triangle problems that lead to quadratic equations	**HSG-SRT.C.8**
FORWARD 2	Solve quadratic equations with complex solutions	**HSN-CN.C.7**

SET 24 • HSA-REI.B.4.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $2x^2 - 3x - 4 = 0$ with the quadratic formula.

Give exact answers in simplest radical form.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q1/5

Solve $x^2 + 4x - 1 = 0$. Round each solution to the nearest hundredth.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q2/5

Use the discriminant to find how many real solutions $3x^2 - 2x + 5 = 0$ has.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q3/5

Solve $x^2 + 2x + 7 = 0$ over the real numbers.

A. No real solution

B. $x = -1 + \sqrt{6}$ or $x = -1 - \sqrt{6}$

C. $x = -7$ or $x = 1$

D. $x = 1 + \sqrt{6}$ or $x = 1 - \sqrt{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q4/5

True or false?

The equation $4x^2 - 12x + 9 = 0$ has exactly one real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q5/5

SET 24 • HSA-REI.B.4.b

BACKWARD 1

Complete the square to solve a quadratic equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

Solve $x^2 - 4x = 5$ by completing the square.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q1/5

Solve $x^2 + 2x - 2 = 0$ by completing the square. Give exact answers.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q2/5

Which equation results from completing the square on $x^2 - 6x + 4 = 0$?

A. $(x + 3)^2 = 5$

B. $(x - 3)^2 = 5$

C. $(x - 3)^2 = 13$

D. $(x - 6)^2 = 32$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-B1-Q3**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q3/5

True or false?

Adding 25 to both sides of $x^2 + 10x = 3$ gives $(x + 5)^2 = 28$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q4/5

True or false?

Completing the square on $x^2 + 6x = 1$ gives $(x + 3)^2 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSA-REI.B.4.b

BACKWARD 2

Estimate square roots with rational approximations • Grade 8 • The Number System • 8.NS.A.2

Between which two consecutive whole numbers is $\sqrt{41}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q1/5

Round $\sqrt{5}$ to the nearest hundredth.

$\sqrt{5} \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q2/5

Which is the best estimate of $\sqrt{30}$?

A. 6.2

B. 5.0

C. 15

D. 5.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{2} \approx 1.41$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q4/5

True or false?

$\sqrt{90}$ is between 8 and 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSA-REI.B.4.b

FORWARD 1

Solve right-triangle problems that lead to quadratic equations • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A right triangle's legs are x and x + 7, and its hypotenuse is 13.

Write an equation and find the lengths of the legs.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q1/5

A 10-ft ladder leans against a wall and makes a 70° angle with the ground.

How high up the wall does it reach, to the nearest tenth of a foot?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q2/5

A rectangle's diagonal is 10, and its length is 2 more than its width.

What is the width?

A. -8

B. 6

C. 8

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q3/5

True or false?

A ramp rises 1 ft over a horizontal run of 12 ft. Its angle with the ground is about 4.8°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q4/5

True or false?

A right triangle with legs 1 and 1 has a hypotenuse of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSA-REI.B.4.b

FORWARD 2

Solve quadratic equations with complex solutions • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 - 4x + 13 = 0$. Write the solutions in the form $a + bi$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q1/5

Solve $x^2 + 25 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q2/5

What are the solutions of $x^2 + 2x + 5 = 0$?

A. $1 + 2i$ and $1 - 2i$

B. $-1 + 4i$ and $-1 - 4i$

C. $-2 + 4i$ and $-2 - 4i$

D. $-1 + 2i$ and $-1 - 2i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q3/5

True or false?

If a quadratic equation with real coefficients has a negative discriminant, its two solutions are complex conjugates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q4/5

True or false?

The solutions of $x^2 + 2x + 2 = 0$ are $1 + i$ and $1 - i$.

A. True

B. False

Create quadratic and exponential equations in one variable and use them to solve problems

MAIN	Algebra I main question	**HSA-CED.A.1**
BACKWARD 1	Solve quadratic equations	**HSA-REI.B.4.b**
BACKWARD 2	Use properties of integer exponents	**8.EE.A.1**
FORWARD 1	Use volume formulas to write and solve equations for a dimension	**HSG-GMD.A.3**
FORWARD 2	Solve exponential models with logarithms	**HSF-LE.A.4**

SET 25 • HSA-CED.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

A ball's height is $h = -16t^2 + 48t + 4$ feet after t seconds.

Write and solve an equation to find when the ball is 36 feet high.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q1/5

A rectangle's length is 5 cm more than its width. Its area is 84 cm².

Write an equation and find the dimensions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q2/5

A culture of 200 bacteria doubles every hour.

Write an equation for the number of hours t until there are 6,400 bacteria, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q3/5

A $2,000 investment earns 5% interest, added once at the end of each year.

Which inequality gives the whole numbers of years n after which it is worth at least $3,000?

A. $2{,}000(1.5)^n \geq 3{,}000$

B. $2{,}000(0.05)^n \geq 3{,}000$

C. $2{,}000 + 0.05n \geq 3{,}000$

D. $2{,}000(1.05)^n \geq 3{,}000$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q4/5

True or false?

A square garden of side x ft has a 2-ft path around it, and the garden and path cover 144 ft². Then $(x + 4)^2 = 144$, so $x = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q5/5

SET 25 • HSA-CED.A.1

BACKWARD 1

Solve quadratic equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $t^2 - 3t + 2 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q1/5

Solve $w^2 + 5w = 84$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q2/5

Solve $(x + 4)^2 = 144$.

A. $x = 12$ or $x = -12$

B. $x = 140$ or $x = -148$

C. $x = 8$ only

D. $x = 8$ or $x = -16$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q3/5

True or false?

$x = -2$ is a solution of $x^2 + 5x + 6 = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q4/5

True or false?

The only solution of $x^2 = 3x$ is $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSA-CED.A.1

BACKWARD 2

Use properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Write 32 as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q1/5

Simplify $2^3 \cdot 2^4$. Write the answer as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q2/5

Which power of 2 equals 64?

A. 2^{32}

B. 2^{8}

C. 2^{5}

D. 2^{6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q3/5

True or false?

$3^4 = 81$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q4/5

True or false?

$2^{-3} = \frac{1}{8}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSA-CED.A.1

FORWARD 1

Use volume formulas to write and solve equations for a dimension • High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cylinder has height 10 cm and volume 250π cm³.

Write an equation and find its radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q1/5

A square pyramid has height 9 in and volume 300 in³.

Find the side length of its base.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q2/5

A cone has volume 12π and height 4.

What is its radius?

A. 9

B. 6

C. $\sqrt{3}$

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q3/5

True or false?

A sphere of radius 3 has volume 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q4/5

True or false?

Doubling the radius of a cylinder doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSA-CED.A.1

FORWARD 2

Solve exponential models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $200 \cdot 2^{t} = 5{,}000$. Give the exact answer using logarithms, then round it to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q1/5

Solve $2{,}000(1.05)^n = 3{,}000$. Round to the nearest tenth.

$n \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q2/5

Solve $10^x = 500$. (log means base 10.)

A. $x = 2.5$

B. $x = 5{,}000$

C. $x = 50$

D. $x = \log 500 \approx 2.70$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q3/5

True or false?

The solution of $e^x = 7$ is $x = \ln 7$. (ln means base e.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q4/5

True or false?

The solution of $5^x = 30$ is $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q5/5

Show that replacing an equation by its sum with a multiple of the other keeps the solutions of a system

MAIN	Algebra I main question	**HSA-REI.C.5**
BACKWARD 1	Justify steps with the properties of equality	**HSA-REI.A.1**
BACKWARD 2	Recognize the solution of a system as a point on both lines	**8.EE.C.8.a**
FORWARD 1	Write a system as a matrix equation AX = B and use row operatio..	**HSA-REI.C.8**
FORWARD 2	Use determinants and inverse matrices	**HSN-VM.C.10**

SET 26 • HSA-REI.C.5

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

System: $x + y = 7$ and $x - y = 1$.

Replace the second equation by the sum of the two equations. Write the new system, solve it, and show that the solution also satisfies the original system.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q1/5

Suppose (a, b) is a solution of the system $2x + y = 5$ and $x - 3y = 6$.

Explain why (a, b) is also a solution of $(2x + y) + 2(x - 3y) = 5 + 2(6)$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q2/5

System: $3x + 2y = 12$ and $x - y = -1$.

Which system has the same solution?

A. $3x + 2y = 12$ and $4x + y = 13$

B. $3x + 2y = 12$ and $2x + 3y = 11$

C. $3x + 2y = 12$ and $5x = 14$

D. $3x + 2y = 12$ and $5x = 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q3/5

True or false?

Replacing one equation of a system by the sum of that equation and 3 times the other equation can change the solutions of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q4/5

True or false?

In the system $2x + 3y = 7$ and $4x - 3y = 5$, adding the equations eliminates y and gives $6x = 12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q5/5

SET 26 • HSA-REI.C.5

BACKWARD 1

Justify steps with the properties of equality • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

If a = b and c = d, explain why a + c = b + d.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q1/5

Name the property used to go from $x = 5$ to $2x = 10$.

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q2/5

If $p = q$, which equation must also be true?

A. $4p = q$

B. $p = q + 4$

C. $p + 4 = q + 4$

D. $p + 4 = q$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q3/5

True or false?

If a = b, then 5a = 5b.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q4/5

True or false?

If $a = b$ and $c = d$, then $a - c = b + d$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSA-REI.C.5

BACKWARD 2

Recognize the solution of a system as a point on both lines • Grade 8 • Expressions & Equations • 8.EE.C.8.a

Is (2, 5) a solution of the system $y = 2x + 1$ and $y = x + 3$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q1/5

The lines $y = x + 1$ and $y = -x + 5$ meet at $(2, 3)$.

What is the solution of the system?

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q2/5

Which point is a solution of both $x + y = 6$ and $x - y = 2$?

A. (2, 4)

B. (4, 2)

C. (3, 3)

D. (5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q3/5

True or false?

The solution of a system of two linear equations is the point where their graphs intersect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q4/5

True or false?

(1, 1) is a solution of the system y = 2x and y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSA-REI.C.5

FORWARD 1

Write a system as a matrix equation AX = B and use row operations • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8

Write the system $2x + y = 7$ and $x - 3y = 0$ as a matrix equation $AX = B$. Give A, X, and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q1/5

In the matrix equation $AX = B$, A has rows $[1\ 2]$ and $[3\ -1]$, X is the column $[x, y]$, and B is the column $[5, 1]$.

Write the system it represents.

System: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q2/5

On the rows [1 2 | 5] and [3 −1 | 1], replace row 2 by (row 2) − 3 × (row 1).

What is the new row 2?

A. [0 −7 | 14]

B. [2 −3 | −4]

C. [0 −7 | −14]

D. [0 5 | 16]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q3/5

True or false?

Adding a multiple of one row of an augmented matrix to another row does the same thing as replacing that equation by its sum with a multiple of the other equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q4/5

True or false?

The coefficient matrix of $x + y = 3$ and $2x - y = 0$ has the rows [1 1] and [2 1].

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSA-REI.C.5

FORWARD 2

Use determinants and inverse matrices • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10

Find the determinant of the matrix with rows [3 1] and [5 2].

Determinant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q1/5

Show that the matrix with rows [2 −1] and [−5 3] is the inverse of the matrix with rows [3 1] and [5 2].

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q2/5

Which matrix has no inverse?

A. Rows [2 4] and [1 2]

B. Rows [1 0] and [0 1]

C. Rows [2 1] and [1 1]

D. Rows [3 1] and [5 2]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q3/5

True or false?

If the coefficient matrix of a system of two linear equations has an inverse, the system has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q4/5

True or false?

The inverse of the matrix with rows [2 0] and [0 4] has rows [−2 0] and [0 −4].

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q5/5

Prove that replacing an equation by its sum with a multiple of the other gives an equivalent system, in both directions

MAIN	Algebra I main question	**HSA-REI.C.5**
BACKWARD 1	Use the properties of equality to justify a step	**HSA-REI.A.1**
BACKWARD 2	Solve simple systems by adding equations	**8.EE.C.8.b**
FORWARD 1	Use row operations on an augmented matrix as elimination steps	**HSA-REI.C.8**
FORWARD 2	Solve the same system with an inverse matrix	**HSA-REI.C.9**

SET 72 • HSA-REI.C.5

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

System A: $x + 2y = 7$ and $3x - y = 7$. System B: $x + 2y = 7$ and $7x = 21$, where $7x = 21$ is the first equation plus 2 times the second.

Show that every solution of A solves B and every solution of B solves A. Then give the solution.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q1/5

Let P = c and Q = d be the two equations of a system (P and Q are expressions in x and y). Prove that the system P = c and Q + kP = d + kc has exactly the same solutions, for any number k.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q2/5

Why must the new system keep one of the original equations unchanged? Give an example.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q3/5

System: $2x + y = 5$ and $x - y = 1$. Which step can produce a system with DIFFERENT solutions?

A. Replacing both equations by their sum, $3x = 6$, alone

B. Multiplying the first equation by 3

C. Writing the equations in the other order

D. Replacing the second equation by the sum of the two equations

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q4/5

True or false?

If (2, 1) solves $x + y = 3$ and $x - y = 1$, then (2, 1) also solves $(x + y) + 4(x - y) = 3 + 4(1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q5/5

SET 72 • HSA-REI.C.5

BACKWARD 1

Use the properties of equality to justify a step • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

If P = c and Q = d are true, explain why P + Q = c + d is true.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q1/5

If P + 3Q = c + 3d and Q = d, explain why P = c.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q2/5

Which property justifies going from $x - y = 1$ to $4x - 4y = 4$?

A. Addition property of equality

B. Commutative property

C. Substitution of y

D. Multiplication property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q3/5

True or false?

Subtracting equal quantities from both sides of a true equation gives a true equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q4/5

True or false?

If a + b = 10 and a = 4, then b = 14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSA-REI.C.5

BACKWARD 2

Solve simple systems by adding equations • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Solve by adding the equations.

$x + y = 9$

$x - y = 3$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q1/5

Solve.

$2x + y = 11$

$y = 3$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q2/5

Solve.

$x + 2y = 8$

$x = 2$

A. (2, 5)

B. (8, 0)

C. (2, 3)

D. (3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q3/5

True or false?

Adding $x + y = 5$ and $x - y = 1$ gives $2x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q4/5

True or false?

The system $x + y = 4$ and $2x + 2y = 8$ has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSA-REI.C.5

FORWARD 1

Use row operations on an augmented matrix as elimination steps • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8

Write the augmented matrix of the system $x + 2y = 7$ and $3x - y = 7$.

Rows: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q1/5

Apply R2 → 2R2 + R1 to the rows [1 2 | 7] and [3 −1 | 7]. What is the new second row, and which equation does it represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q2/5

Which row operation matches "replace equation 2 by itself minus 3 times equation 1"?

A. $R2 \to R1$

B. $R2 \to R2 - 3R1$

C. $R2 \to 3R2 - R1$

D. $R1 \to R1 - 3R2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q3/5

True or false?

A row operation R2 → R2 + kR1 does not change the solutions of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q4/5

True or false?

Replacing R2 by 0 · R2 + R1 keeps the solutions of every system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q5/5

SET 72 • HSA-REI.C.5

FORWARD 2

Solve the same system with an inverse matrix • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9

For $x + 2y = 7$ and $3x - y = 7$, the coefficient matrix A has rows [1 2] and [3 −1]. Find the determinant of A.

Determinant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q1/5

A^{-1} is $(1 \div (-7))$ times the matrix with rows $[-1\ -2]$ and $[-3\ 1]$. Use $X = A^{-1}B$, with B the column $[7, 7]$, to solve the system.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q2/5

If the determinant of the coefficient matrix of a 2-by-2 linear system is 0, which is true?

A. $X = A^{-1}B$ gives the solution.

B. The system has exactly one solution.

C. The system has no solution or infinitely many solutions.

D. The two equations must be identical.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q3/5

True or false?

Solving $AX = B$ with $X = A^{-1}B$ gives the same solution as elimination.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q4/5

True or false?

For all 2-by-2 matrices A and B, $A^{-1}B = BA^{-1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q5/5

Solve systems of linear equations exactly by substitution and elimination

MAIN	Algebra I main question	**HSA-REI.C.6**
BACKWARD 1	Eliminate a variable by adding a multiple of an equation	**HSA-REI.C.5**
BACKWARD 2	Solve simple systems of linear equations algebraically	**8.EE.C.8.b**
FORWARD 1	Use slopes and systems to find where parallel or perpendicular li..	**HSG-GPE.B.5**
FORWARD 2	Multiply a matrix by a column to check a solution	**HSN-VM.C.8**

SET 27 • HSA-REI.C.6

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Solve by substitution.

$y = 3x - 5$

$2x + y = 15$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q1/5

Solve by elimination.

$3x + 4y = 10$

$5x - 2y = 34$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q2/5

Solve.

$\frac{1}{2}x + \frac{1}{3}y = 4$

$x - y = 3$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q3/5

Solve.

$2x - 3y = -1$

$4x + y = 19$

A. $(-4, -3)$

B. $(4, 3)$

C. $(3, 4)$

D. $(5, -1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q4/5

True or false?

The system $2x + y = 5$ and $4x + 2y = 9$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q5/5

SET 27 • HSA-REI.C.6

BACKWARD 1

Eliminate a variable by adding a multiple of an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

Add the equations $x + 2y = 9$ and $3x - 2y = 3$.

What equation results, and what is x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q1/5

System: $2x + y = 8$ and $6x - 5y = 8$.

What multiple of the first equation should you subtract from the second to eliminate x?

Multiple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q2/5

System: $x + y = 5$ and $2x + 3y = 12$.

Which equation can replace the second equation without changing the solution?

A. $x = 2$

B. $y = 7$

C. $y = 2$

D. $3x + 4y = 12$

True or false?

Multiplying one equation of a system by a nonzero number does not change the solutions of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q4/5

True or false?

Adding the equations $x + y = 4$ and $x - y = 2$ gives $2x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSA-REI.C.6

BACKWARD 2

Solve simple systems of linear equations algebraically • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Solve.

$y = x + 2$

$y = 3x$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q1/5

Solve.

$x + y = 10$

$y = 4$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q2/5

Solve.

$y = 2x$

$x + y = 12$

A. (4, 8)

B. (6, 6)

C. (8, 4)

D. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q3/5

True or false?

The system $y = 2x + 3$ and $y = 2x - 1$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q4/5

True or false?

The solution of $y = x$ and $y = -x + 6$ is $(2, 4)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSA-REI.C.6

FORWARD 1

Use slopes and systems to find where parallel or perpendicular lines meet • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Line ℓ is $y = 2x + 1$. Write an equation of the line through $(5, 1)$ that is perpendicular to ℓ, and find where the two lines meet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q1/5

Are the lines $2x + 3y = 6$ and $3x - 2y = 4$ perpendicular? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q2/5

Which line is parallel to $y = 3x - 2$ and passes through $(0, 5)$?

A. $y = 5x + 3$

B. $y = -3x + 5$

C. $y = 3x + 5$

D. $y = -\frac{1}{3}x + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations

HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q3/5

True or false?

The lines $y = 4x + 1$ and $y = 4x - 7$ have no point in common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q4/5

True or false?

The lines $y = 2x$ and $y = -2x$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSA-REI.C.6

FORWARD 2

Multiply a matrix by a column to check a solution • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8

Multiply the matrix with rows [2 1] and [1 −3] by the column [4, 1].

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q1/5

The matrix A has rows [2 1] and [1 −3].

Use multiplication by A to decide whether x = 4, y = 1 solves 2x + y = 9 and x − 3y = 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q2/5

What is the product of the matrix with rows [1 2] and [3 4] and the column [1, 1]?

A. The column [3, 7]

B. The column [2, 6]

C. The column [10]

D. The column [4, 6]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q3/5

True or false?

Multiplying a 2-by-2 matrix by a 2-by-1 column gives a 2-by-1 column.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q4/5

True or false?

For all 2-by-2 matrices A and B, AB = BA.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q5/5

Solve systems of linear equations in context, including systems with no solution or infinitely many solutions

MAIN	Algebra I main question	**HSA-REI.C.6**
BACKWARD 1	Write and solve an equation for one quantity in context	**HSA-CED.A.1**
BACKWARD 2	Solve real-world problems with two linear equations	**8.EE.C.8.c**
FORWARD 1	Use matrices to organize data in context	**HSN-VM.C.6**
FORWARD 2	Solve a system in context with an inverse matrix	**HSA-REI.C.9**

SET 28 • HSA-REI.C.6

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

A theater sold 340 tickets. Adult tickets cost $12 and student tickets cost $7. Sales totaled $3,255.

Write a system and find how many of each kind were sold.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q1/5

How many solutions does the system $6x - 4y = 10$ and $3x - 2y = 5$ have? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q2/5

A chemist mixes a 10% acid solution with a 30% acid solution to make 20 liters of a 25% acid solution.

How many liters of each are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q3/5

For which value of k does the system $y = 3x + 2$ and $6x - 2y = k$ have infinitely many solutions?

A. $k = 2$

B. $k = -2$

C. $k = 4$

D. $k = -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q4/5

True or false?

A system of two linear equations in two variables can have exactly two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q5/5

SET 28 • HSA-REI.C.6

BACKWARD 1

Write and solve an equation for one quantity in context • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

A gym has 340 members. There are 40 more adults than teens.

Write and solve an equation for the number of teens t.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q1/5

Pens cost $2 and notebooks cost $5. Ana buys 3 notebooks and some pens and spends $27.

Write and solve an equation for the number of pens p.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q2/5

A 20-liter mix has x liters of a 10% acid solution and the rest a 30% acid solution.

Which expression gives the liters of acid?

A. $0.40(20)$

B. $0.10x + 0.30x$

C. $0.10x + 0.30(20 - x)$

D. $10x + 30(20 - x)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q3/5

True or false?

25 coins are all dimes and quarters. If d coins are dimes, then 25 − d coins are quarters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q4/5

True or false?

"3 more than 5 times a number is 48" gives $5n + 3 = 48$, and its solution is $n = 10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSA-REI.C.6

BACKWARD 2

Solve real-world problems with two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.c

Two numbers have a sum of 30 and a difference of 8.

Find the numbers.

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q1/5

Plan A costs $10 plus $2 per hour. Plan B costs $4 per hour.

After how many hours do they cost the same?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q2/5

3 pencils and 2 erasers cost $1.30. 1 pencil and 2 erasers cost $0.70.

What does one pencil cost?

A. $0.60

B. $0.20

C. $0.35

D. $0.30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q3/5

True or false?

Two cars leave the same place at the same time in the same direction, at 50 mi/h and 60 mi/h. After 2 hours the faster car is 20 miles ahead.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q4/5

True or false?

One company charges $y = 2x + 5$ dollars and another charges $y = 2x + 8$ dollars for x units. They charge the same for 3 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSA-REI.C.6

FORWARD 1

Use matrices to organize data in context • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6

Friday: 120 adult and 90 student tickets. Saturday: 150 adult and 60 student tickets.

Write a 2-by-2 matrix with one row per day (adult first).

Rows: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q1/5

Adult tickets cost $12 and student tickets $7.

Multiply the matrix with rows [120 90] and [150 60] by the column [12, 7]. What do the entries mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q2/5

A matrix has rows [5 2] and [3 4]. Each row is a store and each column is a product.

What does the entry in row 2, column 1 represent?

A. Store 2's amount of product 1

B. Store 1's amount of product 2

C. Store 2's amount of product 2

D. The total of product 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q3/5

True or false?

A matrix with 3 rows and 2 columns is a 3-by-2 matrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q4/5

True or false?

Any two matrices can be added, whatever their dimensions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSA-REI.C.6

FORWARD 2

Solve a system in context with an inverse matrix • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9

Write the system $a + s = 340$ and $12a + 7s = 3{,}255$ as a matrix equation $AX = B$. Give A, X, and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q1/5

The matrix with rows [1 1] and [12 7] has determinant −5.

Does the ticket system have exactly one solution? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q2/5

The inverse of the matrix with rows [1 1] and [12 7] is $(1 \div (-5))$ times which matrix?

A. Rows [7 −1] and [−12 1]

B. Rows [7 1] and [12 1]

C. Rows [−7 1] and [12 −1]

D. Rows [1 −1] and [−12 7]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q3/5

True or false?

If the coefficient matrix of a system has determinant 0, the system does not have exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q4/5

True or false?

The determinant of the matrix with rows [2 3] and [4 6] is 24.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q5/5

Solve a system of a linear equation and a quadratic equation algebraically

MAIN	Algebra I main question	**HSA-REI.C.7**
BACKWARD 1	Solve quadratic equations by factoring	**HSA-REI.B.4.b**
BACKWARD 2	Solve systems of linear equations	**HSA-REI.C.6**
FORWARD 1	Decide whether points lie on a circle	**HSG-GPE.B.4**
FORWARD 2	Find complex solutions when a line and a parabola do not meet	**HSN-CN.C.7**

SET 29 • HSA-REI.C.7

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7

Solve the system.

$y = x^2 - 3$

$y = 2x$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q1/5

Solve the system.

$y = x^2 + 4x + 1$

$y = x - 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q2/5

How many real solutions does the system $y = x^2 + 2$ and $y = x$ have?

A. Infinitely many

B. One

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q3/5

Solve the system.

$x^2 + y^2 = 25$

$y = x + 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q4/5

True or false?

The line $y = 2x - 1$ meets the parabola $y = x^2$ at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q5/5

SET 29 • HSA-REI.C.7

BACKWARD 1

Solve quadratic equations by factoring • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $x^2 - 2x - 3 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q1/5

Solve $x^2 + x - 12 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q2/5

Solve $x^2 + 3x + 2 = 0$.

A. $x = -1$ or $x = 2$

B. $x = 1$ or $x = 2$

C. $x = 1$ or $x = -2$

D. $x = -1$ or $x = -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q3/5

True or false?

$x^2 - x + 2 = 0$ has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q4/5

True or false?

The solutions of $x^2 = 2x$ are $x = 2$ and $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSA-REI.C.7

BACKWARD 2

Solve systems of linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Solve.

$y = 2x$

$y = -x + 9$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q1/5

Solve.

$x + y = 1$

$y = x + 1$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q2/5

Solve.

$2x + y = 7$

$x - y = 2$

A. $(4, -1)$

B. $(3, 1)$

C. $(1, 3)$

D. $(2, 3)$

True or false?

Substitution and elimination give the same solution of a system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q4/5

True or false?

(2, 1) is a solution of the system $x + y = 3$ and $x - y = 3$.

A. True

B. False

SET 29 • HSA-REI.C.7

FORWARD 1

Decide whether points lie on a circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A circle is centered at the origin and passes through (0, 5).

Does the point (3, −4) lie on the circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q1/5

Find the points where the line $y = 3$ meets the circle $x^2 + y^2 = 25$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q2/5

A circle has center (1, 2) and radius 5.

Which point is on the circle?

A. (1, 6)

B. (4, 6)

C. (3, 3)

D. (6, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q3/5

True or false?

The line $y = 6$ does not meet the circle $x^2 + y^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q4/5

True or false?

The point (2, 2) lies on the circle $x^2 + y^2 = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q5/5

SET 29 • HSA-REI.C.7

FORWARD 2

Find complex solutions when a line and a parabola do not meet • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7

Substituting $y = 2x$ into $y = x^2 + 2$ gives $x^2 - 2x + 2 = 0$.

Solve it over the complex numbers.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q1/5

Solve $x^2 + 4 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q2/5

Solve $x^2 - 6x + 10 = 0$.

A. $3 + i$ and $3 - i$

B. $6 + i$ and $6 - i$

C. $3 + 2i$ and $3 - 2i$

D. $-3 + i$ and $-3 - i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q3**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q3/5

True or false?

When a line and a parabola do not intersect, the equation from substitution has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q4/5

True or false?

$i^3 = i$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q5**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q5/5

Understand the graph of an equation as the set of all its solutions

MAIN	Algebra I main question	**HSA-REI.D.10**
BACKWARD 1	Use tables and graphs to relate two variables	**6.EE.C.9**
BACKWARD 2	Understand a function's graph as its input-output pairs	**8.F.A.1**
FORWARD 1	Derive a parabola from its focus and directrix	**HSG-GPE.A.2**
FORWARD 2	Use the unit circle as the set of points (cos θ, sin θ)	**HSF-TF.A.2**

SET 30 • HSA-REI.D.10

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

Is $(3, -2)$ on the graph of $2x + 5y = -4$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q1/5

The graph of $y = x^2 - 2x - 3$ is shown.

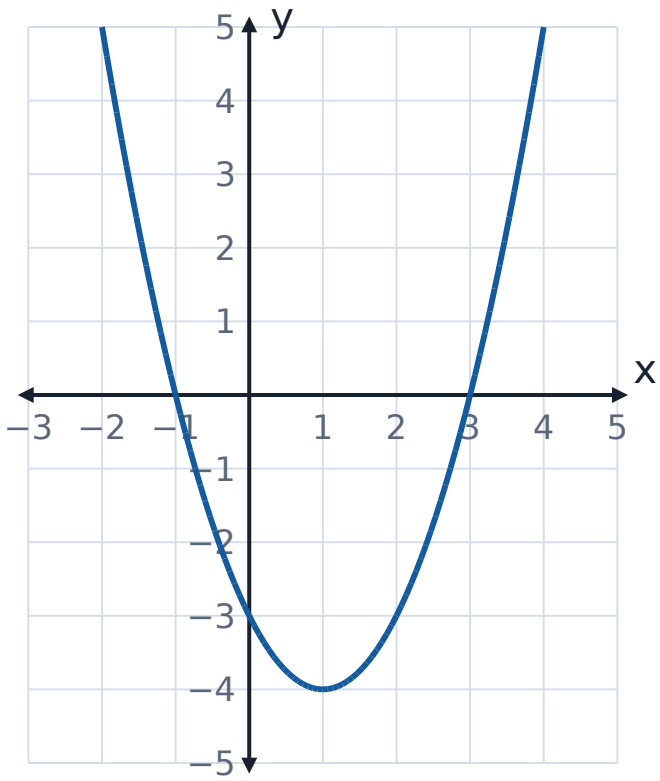
Which point is NOT a solution of the equation?

A. (2, 3)

B. (3, 0)

C. (0, −3)

D. (1, −4)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q2/5

The point (k, 7) is on the graph of $y = 3x - 2$.

Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q3/5

The graph of an equation in x and y passes through (4, 1) and (−2, 5).

What does that tell you about the equation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q4/5

True or false?

A point that is not on the graph of $y = x^2$ can still be a solution of $y = x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q5/5

SET 30 • HSA-REI.D.10

BACKWARD 1

Use tables and graphs to relate two variables • Grade 6 • Expressions & Equations • 6.EE.C.9

$y = 3x$. Find y for $x = 0, 1, 2,$ and 3.

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q1/5

A car travels at 40 miles per hour.

Write an equation for the distance d after t hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q2/5

Which point is on the graph of y = x + 4?

A. (2, 6)

B. (1, 3)

C. (4, 0)

D. (6, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q3/5

True or false?

In y = 5x, y is the dependent variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q4/5

True or false?

The point (2, 8) is on the graph of $y = 3x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSA-REI.D.10

BACKWARD 2

Understand a function's graph as its input-output pairs • Grade 8 • Functions • 8.F.A.1

The function f pairs each x with $2x - 1$.

What output does it pair with $x = 4$?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q1/5

Is {(1, 2), (2, 3), (1, 5)} a function? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q2/5

Which point is on the graph of $y = x^2$?

A. $(9, -3)$

B. $(-3, -9)$

C. $(-3, 9)$

D. $(3, 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q3/5

True or false?

A function assigns exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q4/5

True or false?

The vertical line $x = 2$ is the graph of a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSA-REI.D.10

FORWARD 1

Derive a parabola from its focus and directrix • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

A parabola is the set of points equally far from the focus $(0, 2)$ and the directrix $y = -2$.

Derive its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q1/5

Is (4, 2) on the parabola with focus (0, 2) and directrix $y = -2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q2/5

Where is the focus of the parabola $y = x^2 \div 4$?

A. (0, 4)

B. $(0, \frac{1}{4})$

C. (1, 0)

D. (0, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q3/5

True or false?

Every point on a parabola is the same distance from its focus as from its directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q4/5

True or false?

The vertex of a parabola lies on its directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q5/5

SET 30 • HSA-REI.D.10

FORWARD 2

Use the unit circle as the set of points (cos θ, sin θ) • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

Explain why the point $(\cos \theta, \sin \theta)$ lies on the circle $x^2 + y^2 = 1$ for every angle θ.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q1/5

Which point on the unit circle corresponds to $\theta = \pi \div 2$?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q2/5

Which point is on the unit circle $x^2 + y^2 = 1$?

A. (0.5, 0.5)

B. (1, 1)

C. (0.6, 0.8)

D. (0.8, 0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q3/5

True or false?

The point $(-1, 0)$ on the unit circle corresponds to an angle of π radians.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q4/5

True or false?

The point (cos 0, sin 0) is (0, 1).

A. True

B. False

Explain why intersections of y = f(x) and y = g(x) solve f(x) = g(x), and approximate solutions with graphs and tables

MAIN	Algebra I main question	**HSA-REI.D.11**
BACKWARD 1	Check whether points are on a graph	**HSA-REI.D.10**
BACKWARD 2	Read the solution of a system from the graphs of two lines	**8.EE.C.8.a**
FORWARD 1	Solve exponential equations exactly with logarithms	**HSF-LE.A.4**
FORWARD 2	Solve trigonometric equations with graphs and technology	**HSF-TF.B.7**

SET 31 • HSA-REI.D.11

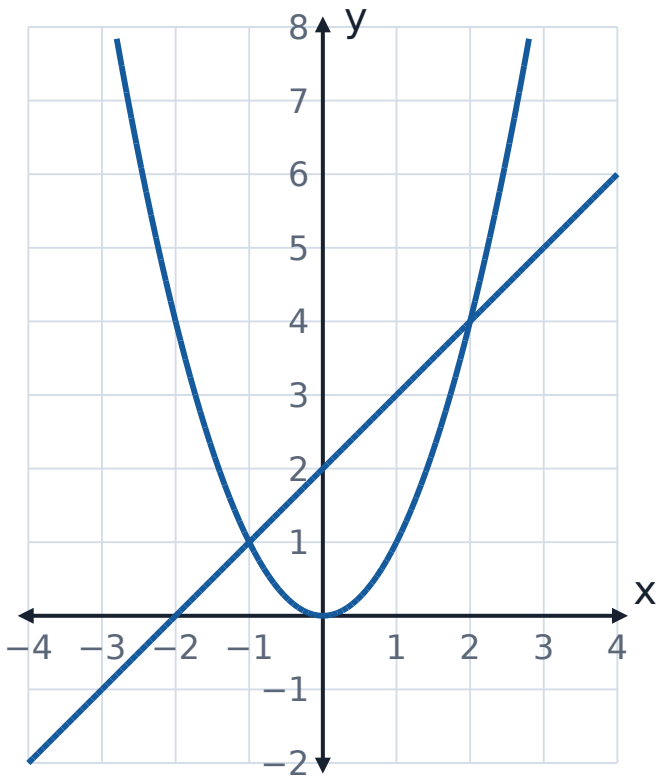
MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

The graphs of $f(x) = x^2$ and $g(x) = x + 2$ are shown.

Use the graphs to solve $x^2 = x + 2$.

Solutions: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q1/5

Use a table of values to find the solution of $2^x = 3x$ between $x = 3$ and $x = 4$, to the nearest tenth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q2/5

The table shows $f(x) = |x - 1|$ and $g(x) = 3 - x$.

For which x is $f(x) = g(x)$?

x	0	1	2	3	4
f(x)	1	0	1	2	3
g(x)	3	2	1	0	−1

A. $x = 2$ **B.** $x = 3$ **C.** $x = 0$ **D.** $x = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q3/5

Explain why the x-coordinates of the points where the graphs of $y = f(x)$ and $y = g(x)$ meet are the solutions of $f(x) = g(x)$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q4/5

True or false?

The graphs of $y = 3x + 1$ and $y = 3x - 4$ never meet, so $3x + 1 = 3x - 4$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q5/5

SET 31 • HSA-REI.D.11

BACKWARD 1

Check whether points are on a graph • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

Is (2, 4) on both $y = x^2$ and $y = x + 2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q1/5

The point $(-1, y)$ is on the graph of $y = x + 2$.

Find y.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q2/5

Which point is on the graph of $y = |x - 1|$?

A. $(-2, -3)$

B. $(-2, 3)$

C. $(3, -2)$

D. $(1, 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q3/5

True or false?

If (a, b) is on the graph of y = f(x), then b = f(a).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q4/5

True or false?

(0, 1) is on the graph of $y = 2^x + 1$.

A. True

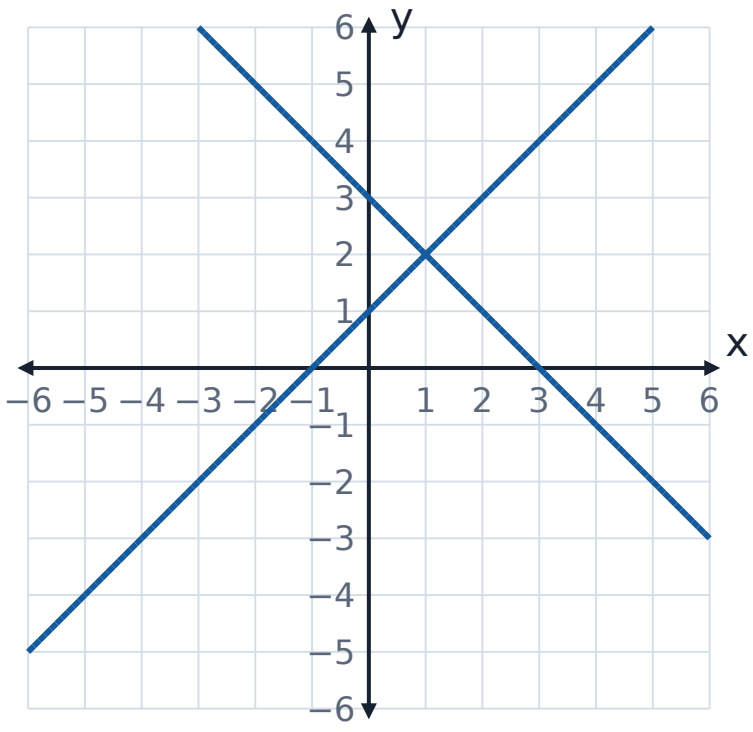
B. False

SET 31 • HSA-REI.D.11

BACKWARD 2

Read the solution of a system from the graphs of two lines • Grade 8 • Expressions & Equations • 8.EE.C.8.a

Use the graph to solve the system $y = x + 1$ and $y = -x + 3$.



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q1/5

At what x-value do the graphs of $y = 2x - 3$ and $y = 5$ meet?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q2/5

The graphs of two linear equations are the same line.

How many solutions does the system have?

A. One

B. Infinitely many

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q3/5

True or false?

Two lines with different slopes meet at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q4/5

True or false?

Two lines with the same slope always meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSA-REI.D.11

FORWARD 1

Solve exponential equations exactly with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

The graphs of $y = 2^x$ and $y = 10$ meet at one point.

Find its x-coordinate exactly using logarithms, and to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q1/5

Solve $5e^{x} = 20$ exactly.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q2/5

A graph shows $y = 3^x$ meeting $y = 15$ at about $x = 2.5$.

Which expression is the exact solution of $3^x = 15$? (log means base 10.)

A. $15 \div 3$

B. $\log 5$

C. $\log 15 \div \log 3$

D. $\sqrt{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q3/5

True or false?

$\log_2 8 = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q4/5

True or false?

The solution of $10^x = 0.01$ is $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSA-REI.D.11

FORWARD 2

Solve trigonometric equations with graphs and technology • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7

Solve $\sin x = \frac{1}{2}$ for $0 \leq x \leq \pi$. Give exact answers.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q1/5

Use a calculator to solve $\cos x = 0.3$ for $0 \leq x \leq \pi$, to the nearest hundredth of a radian.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q2/5

How many solutions does $\sin x = 2$ have?

A. Two

B. One

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q3/5

True or false?

The graphs of $y = \sin x$ and $y = 0.5$ meet at infinitely many points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q4/5

True or false?

For $0 \leq x \leq \pi$, the equation $\cos x = 0$ has two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q5/5

Solve linear-quadratic systems by graphing and check the solutions algebraically

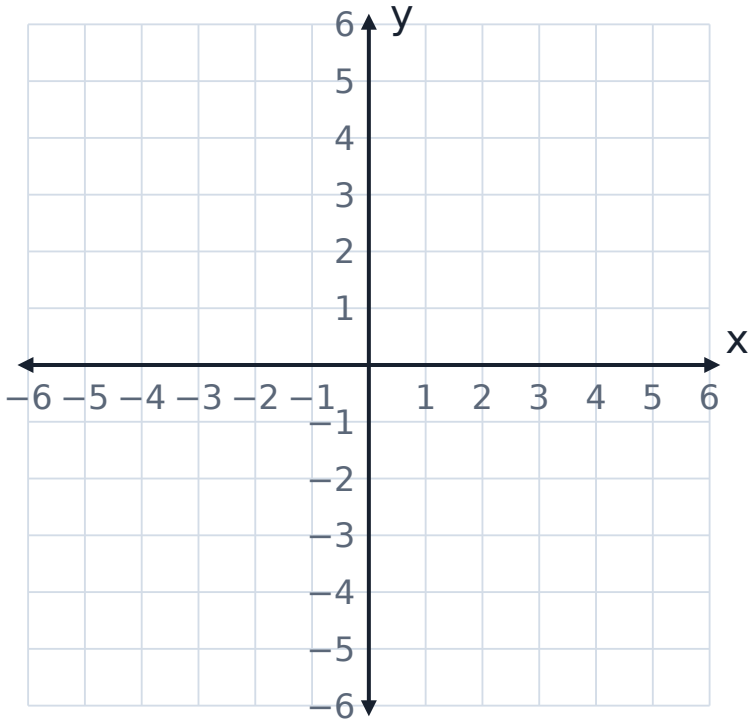
MAIN	Algebra I main question	**HSA-REI.C.7**
BACKWARD 1	Read solutions of f(x) = g(x) from intersections	**HSA-REI.D.11**
BACKWARD 2	Check whether a point is on a graph	**HSA-REI.D.10**
FORWARD 1	Find where a line meets a circle with coordinates	**HSG-GPE.B.4**
FORWARD 2	Find where a line meets an ellipse	**HSG-GPE.A.3**

SET 73 • HSA-REI.C.7

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7

Graph $y = x^2 - 2$ and $y = x$ on the grid. Read the solutions of the system from the intersections, then check them algebraically.



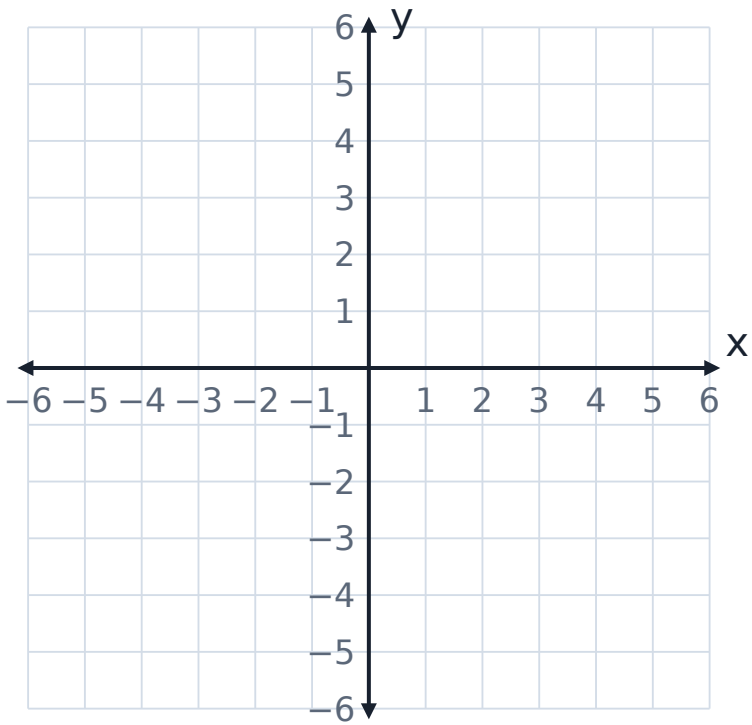
Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q1/5

Graph the circle $x^2 + y^2 = 25$ and the line $y = x + 1$. Read the intersection points, then check one of them algebraically.



Points: _____

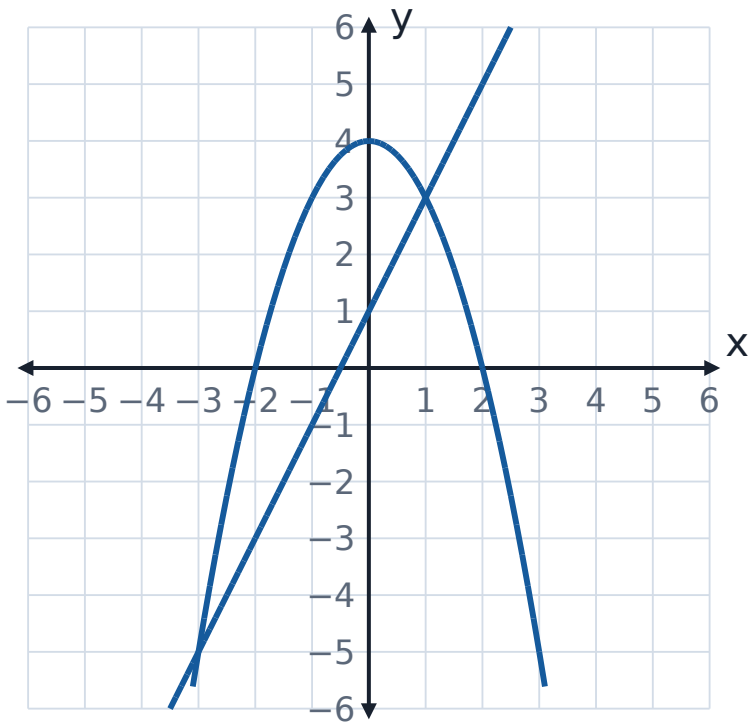
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q2/5

The graphs of $y = -x^2 + 4$ and $y = 2x + 1$ are shown.

Read the solutions of the system from the graph, then verify them algebraically.



Solutions: _____

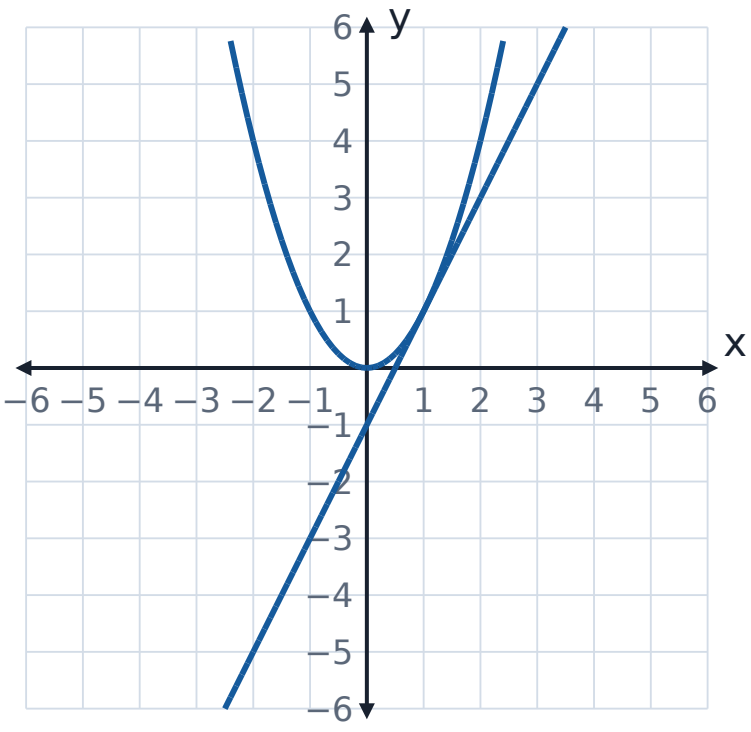
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q3/5

The graphs of $y = x^2$ and $y = 2x - 1$ are shown.

How many solutions does the system have?



A. One

B. Two

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q4/5

True or false?

The line $y = 5$ and the circle $x^2 + y^2 = 9$ intersect at two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q5/5

SET 73 • HSA-REI.C.7

BACKWARD 1

Read solutions of f(x) = g(x) from intersections • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

The graphs of $y = x^2$ and $y = 4$ meet at two points.

What are the solutions of $x^2 = 4$?

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q1/5

The graphs of $y = 2x + 1$ and $y = x^2 + 1$ meet at (0, 1) and (2, 5). Which x-values solve $2x + 1 = x^2 + 1$? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q2/5

The graphs of $y = f(x)$ and $y = g(x)$ meet only at $(3, 7)$. Which statement is true?

A. $f(x) = g(x)$ for every x

B. $f(7) = g(7) = 3$

C. $f(3) = 3$

D. $f(3) = g(3) = 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q3/5

True or false?

If the graphs of $y = f(x)$ and $y = g(x)$ never meet, then $f(x) = g(x)$ has no real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q4/5

True or false?

The solutions of f(x) = g(x) are the y-coordinates of the intersection points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSA-REI.C.7

BACKWARD 2

Check whether a point is on a graph • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

Is (3, 4) on the circle $x^2 + y^2 = 25$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q1/5

Is (2, 3) on the parabola $y = x^2 - 2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q2/5

Which point is on both $y = x$ and $y = x^2 - 2$?

A. (2, 2)

B. (−2, −2)

C. (1, 1)

D. (0, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q3/5

True or false?

A point is a solution of a system exactly when it lies on the graphs of both equations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q4/5

True or false?

(0, 5) is on both the circle $x^2 + y^2 = 25$ and the line $y = x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSA-REI.C.7

FORWARD 1

Find where a line meets a circle with coordinates • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Find the points where the line $y = 7 - x$ meets the circle $x^2 + y^2 = 25$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q1/5

Show that the line $y = 5$ touches the circle $x^2 + y^2 = 25$ at exactly one point.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q2/5

How many points do the line $y = 2x$ and the circle $x^2 + y^2 = 20$ share?

A. One

B. Four

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q3/5

True or false?

The line $x = 6$ does not meet the circle $x^2 + y^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q4/5

True or false?

The point (1, 5) lies on the circle $x^2 + y^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q5/5

SET 73 • HSA-REI.C.7

FORWARD 2

Find where a line meets an ellipse • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Find where the line $y = 2$ meets the ellipse $x^2 \div 16 + y^2 \div 4 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q1

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q1/5

Find where the x-axis meets the ellipse $x^2 \div 9 + y^2 \div 16 = 1$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q2/5

Where does the line $x = 0$ meet the ellipse $x^2 \div 25 + y^2 \div 9 = 1$?

A. (0, 5) and (0, −5)

B. (5, 0) and (−5, 0)

C. Nowhere

D. (0, 3) and (0, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F2-Q3**

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q3/5

True or false?

A line can meet an ellipse in at most two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q4

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q4/5

True or false?

The line $y = 4$ meets the ellipse $x^2 \div 25 + y^2 \div 9 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q5

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q5/5

Solve systems of linear equations by graphing, including approximate solutions

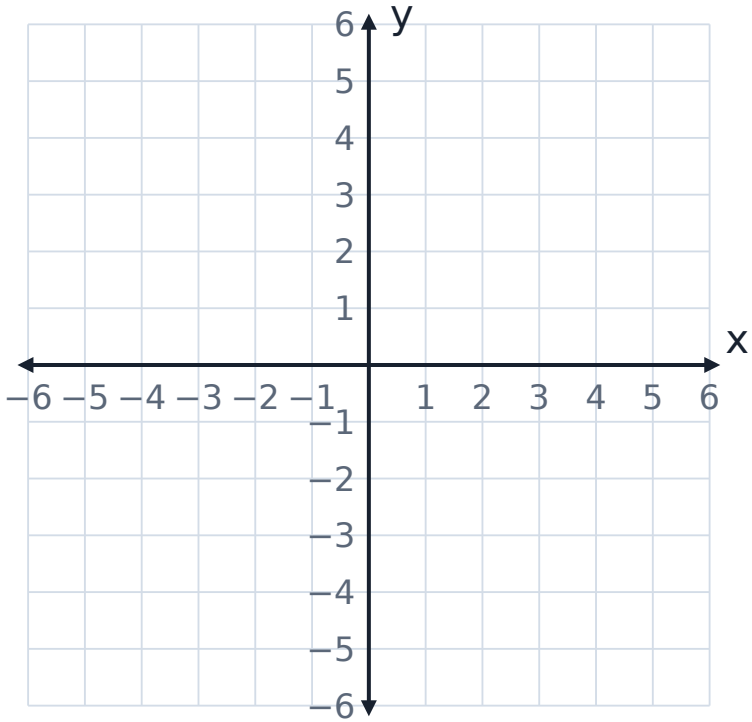
MAIN	Algebra I main question	**HSA-REI.C.6**
BACKWARD 1	Find the solution of a system where two lines cross	**8.EE.C.8.a**
BACKWARD 2	Decide whether points are on a line	**HSA-REI.D.10**
FORWARD 1	Graph parallel and perpendicular lines and find where they meet	**HSG-GPE.B.5**
FORWARD 2	Check a graphical estimate with an inverse matrix	**HSA-REI.C.9**

SET 84 • HSA-REI.C.6

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Graph $y = 2x - 1$ and $y = -x + 5$ on the grid. What is the solution of the system?



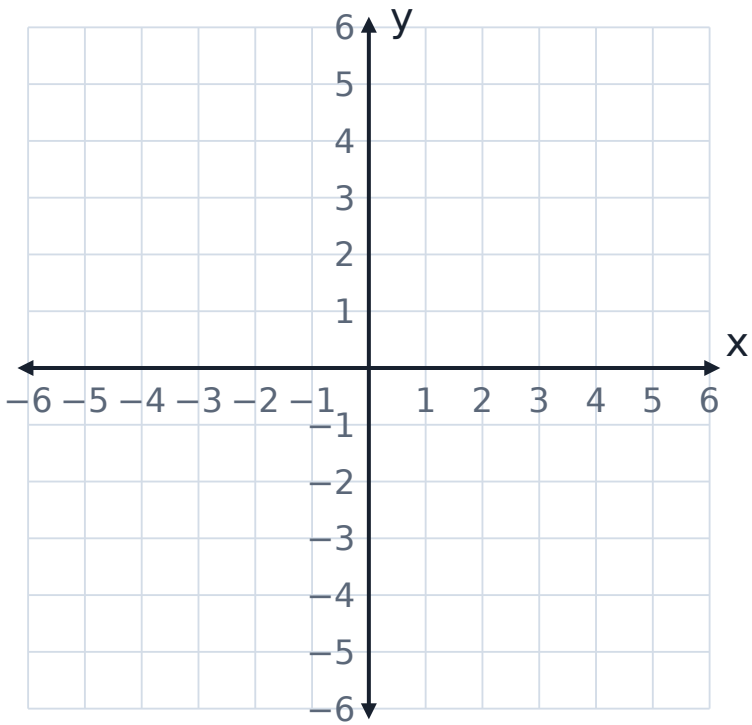
Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q1/5

Graph $y = 0.5x + 2$ and $y = -2x + 4$. Estimate the solution to the nearest half unit, then solve algebraically to check.



Answer: _____

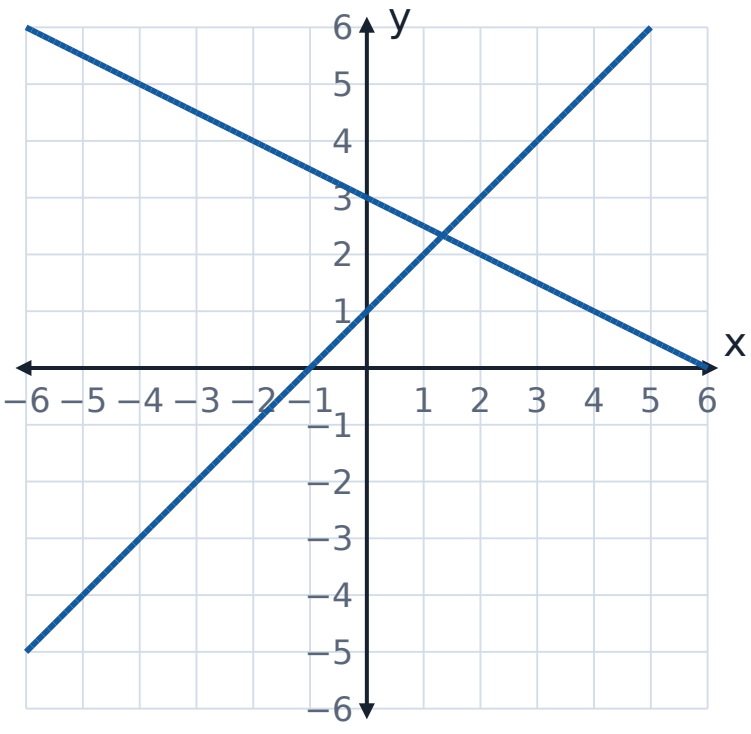
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q2/5

The graphs of $y = x + 1$ and $y = -0.5x + 3$ are shown.

Estimate the solution of the system to the nearest tenth, and explain how you made the estimate.



Answer: _____

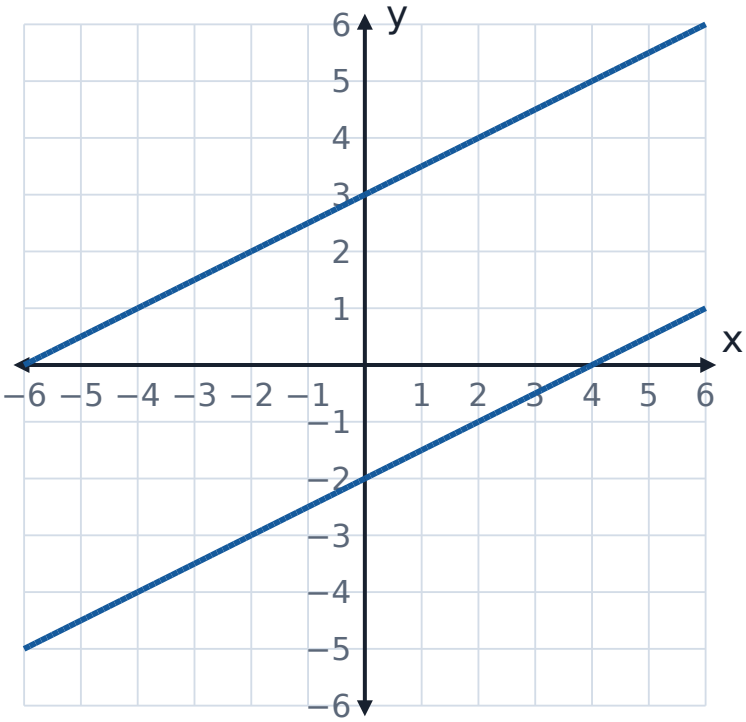
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q3/5

The graphs of the two equations of a system are shown.

How many solutions does the system have?



A. Two

B. One

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q4/5

True or false?

If the graphs of two linear equations lie on top of each other, the system has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q5

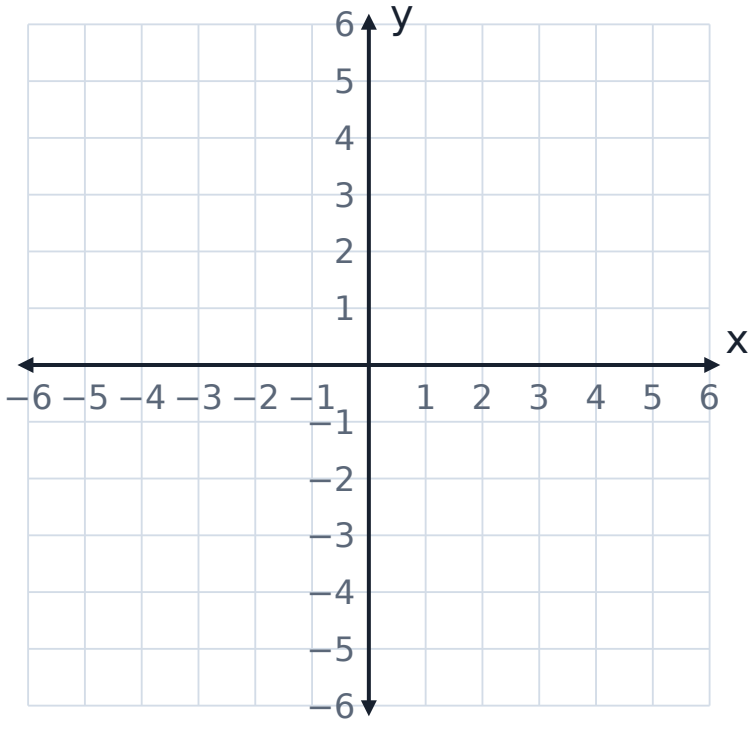
High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q5/5

SET 84 • HSA-REI.C.6

BACKWARD 1

Find the solution of a system where two lines cross • Grade 8 • Expressions & Equations • 8.EE.C.8.a

Graph $y = x$ and $y = 4 - x$, and mark the point where they meet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q1/5

Two lines meet at $(-1, 3)$. What is the solution of their system?

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q2/5

The graphs of two lines cross at (4, −2). Which equations could they be?

A. $y = x - 6$ and $y = x + 2$

B. $y = x - 6$ and $y = -x + 2$

C. $y = x + 6$ and $y = -x + 2$

D. $y = 2x$ and $y = -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q3/5

True or false?

A point that lies on both lines is a solution of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q4/5

True or false?

Two lines with different slopes can have no point in common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSA-REI.C.6

BACKWARD 2

Decide whether points are on a line • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

Is (2, 3) on the line $y = 2x - 1$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q1/5

Is (2, 3) on the line $y = -x + 4$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q2/5

Which point is on the line $y = 0.5x + 2$?

A. (2, 3)

B. (3, 2)

C. (0, 0.5)

D. (4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q3/5

True or false?

The graph of a linear equation in two variables is the set of all its solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q4/5

True or false?

(0, 4) is on the graph of $y = -2x + 2$.

A. True

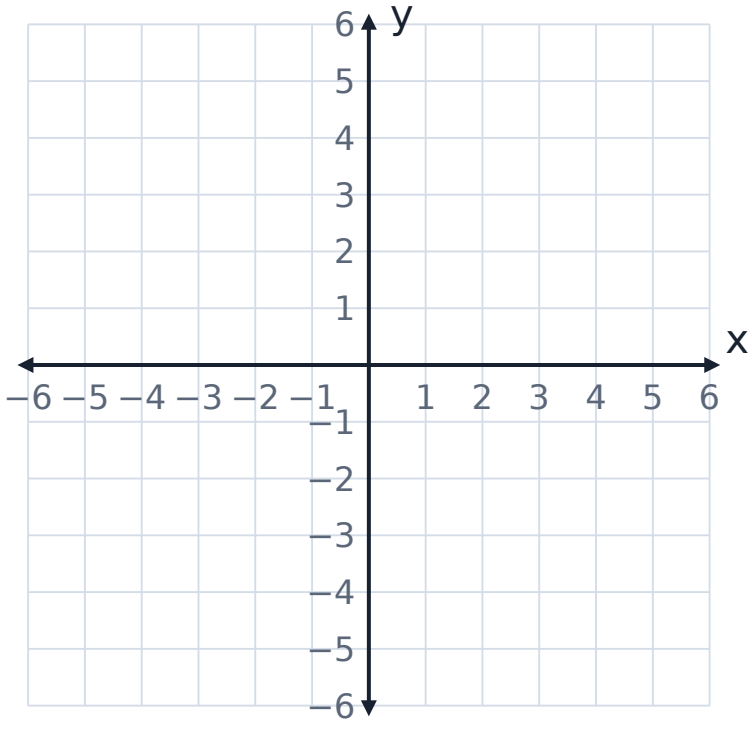
B. False

SET 84 • HSA-REI.C.6

FORWARD 1

Graph parallel and perpendicular lines and find where they meet • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Graph y = 2x + 1 and the line through (4, 1) that is perpendicular to it. Mark the point where they meet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q1/5

Two lines have slopes 3 and $-\frac{1}{3}$. What angle do they form?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q2/5

Which system has no solution?

A. $y = 3x + 1$ and $y = -3x + 1$

B. $y = 3x + 1$ and $y = x + 1$

C. $y = 3x + 1$ and $y = 3x - 4$

D. $y = 3x$ and $y = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q3/5

True or false?

The lines $y = -x + 2$ and $y = x - 2$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q4/5

True or false?

The lines $y = 2x + 5$ and $y = 2x + 5$ have no points in common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q5/5

SET 84 • HSA-REI.C.6

FORWARD 2

Check a graphical estimate with an inverse matrix • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9

The system $y = 0.5x + 2$ and $y = -2x + 4$ is written $-0.5x + y = 2$ and $2x + y = 4$. Find the determinant of the matrix A with rows $[-0.5\ 1]$ and $[2\ 1]$.

Determinant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q1/5

A^{-1} is $(1 \div (-2.5))$ times the matrix with rows $[1\ -1]$ and $[-2\ -0.5]$. Use $X = A^{-1}B$, with B the column $[2, 4]$, to find the exact solution.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q2/5

A graph shows two lines crossing near (1.3, 2.3). The inverse-matrix solution is (4/3, 7/3). Which statement is true?

A. The two results contradict each other.

B. The system has two solutions.

C. The graph is exact; the matrix method is approximate.

D. The graph gives an estimate; the matrix method gives the exact solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q3/5

True or false?

If the coefficient matrix of a 2-by-2 linear system has a nonzero determinant, the system has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q4/5

True or false?

If the graphs of two lines coincide, the determinant of the coefficient matrix is not 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q5/5

Graph the solutions of linear inequalities and systems of linear inequalities in two variables as half-planes

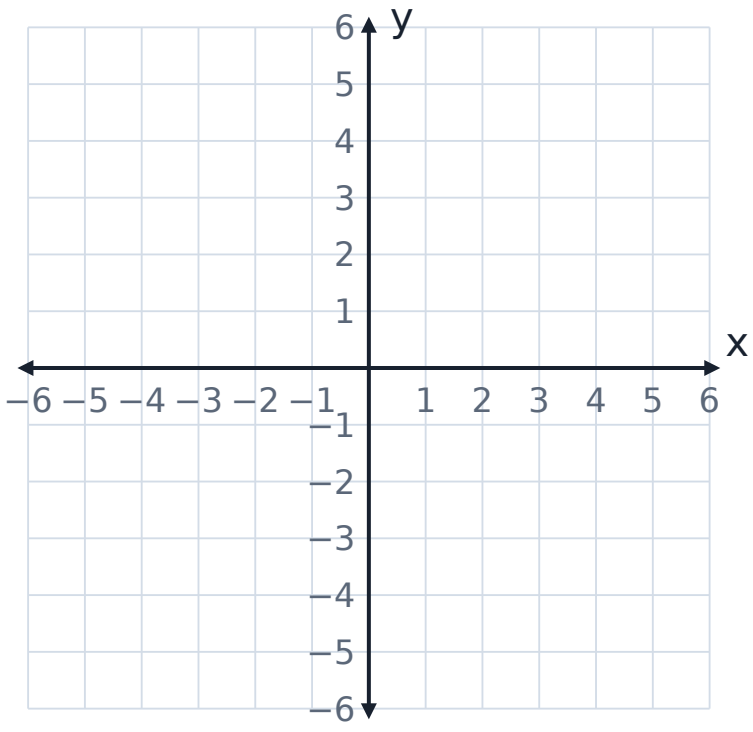
MAIN	Algebra I main question	**HSA-REI.D.12**
BACKWARD 1	Solve an inequality in two variables for y	**HSA-REI.B.3**
BACKWARD 2	Plot points and find distances in the coordinate plane	**6.NS.C.8**
FORWARD 1	Find the area of a region described by inequalities	**HSG-GPE.B.7**
FORWARD 2	Use zeros to find where a polynomial is positive or negative	**HSA-APR.B.3**

SET 32 • HSA-REI.D.12

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

Graph $y < 2x - 1$ on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q1/5

Is (4, 1) a solution of the system $y \leq -x + 6$ and $y > \frac{1}{2}x - 2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q2/5

A graph has a solid boundary line through (0, 2) and (3, 0), and the side that contains the origin is shaded.

Which inequality does it show?

A. $2x + 3y < 6$

B. $2x + 3y \geq 6$

C. $3x + 2y \leq 6$

D. $2x + 3y \leq 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

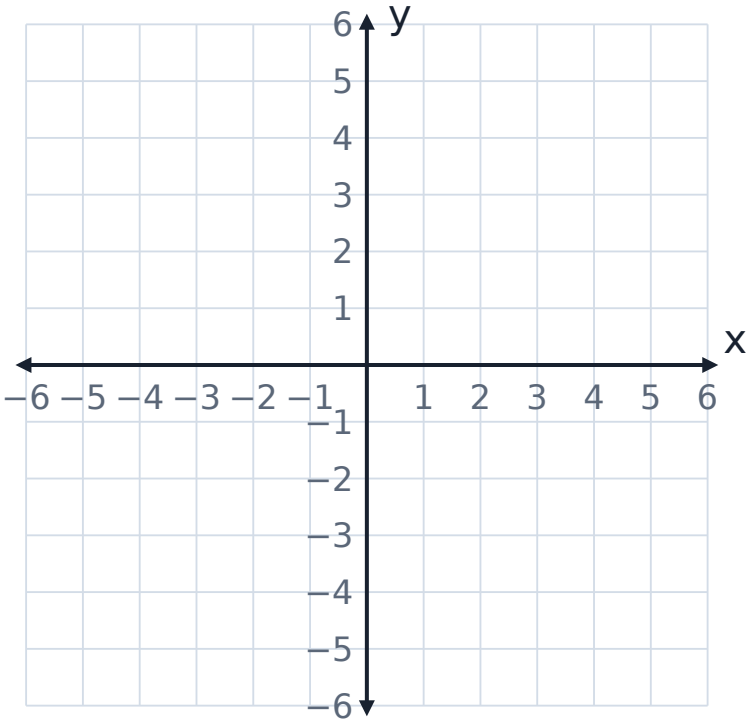
ID S32-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q3/5

Graph the system $y \geq x - 2$ and $x + y < 4$ on the coordinate plane.

Name one point that is a solution of the system.

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q4/5

True or false?

Points on the boundary line $y = 3x$ are solutions of $y > 3x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q5/5

SET 32 • HSA-REI.D.12

BACKWARD 1

Solve an inequality in two variables for y • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $2x + 3y \leq 6$ for y.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q1/5

Solve $x - 2y > 4$ for y.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q2/5

Solve $-y \geq x - 5$ for y.

A. $y \geq x + 5$

B. $y \geq -x + 5$

C. $y \leq -x + 5$

D. $y \leq x - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q3/5

True or false?

Solving $4x + y < 3$ for y gives $y < -4x + 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q4/5

True or false?

Solving $-3y < 9$ for y gives $y < -3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSA-REI.D.12

BACKWARD 2

Plot points and find distances in the coordinate plane • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(2, -3)$ and $(2, 5)$.

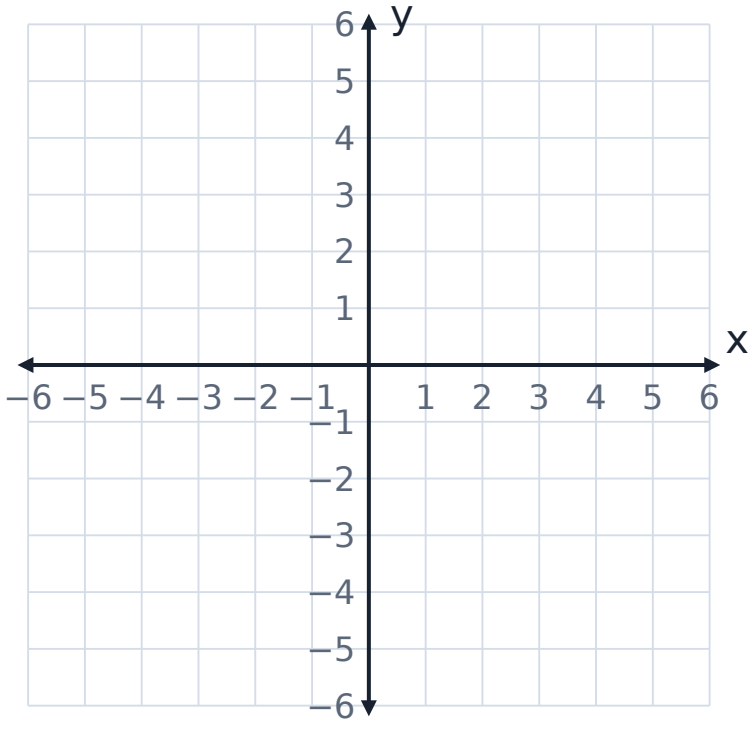
Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

Plot the points $(-3, 2)$ and $(4, -1)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

Which point is in Quadrant III?

A. (2, −5)

B. (−2, −5)

C. (2, 5)

D. (−2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

True or false?

The points (3, 4) and (3, −4) are reflections of each other across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

True or false?

The distance between $(-4, 1)$ and $(6, 1)$ is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSA-REI.D.12

FORWARD 1

Find the area of a region described by inequalities • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

The system $x \geq 0$, $y \geq 0$, and $x + y \leq 6$ describes a triangle.

Find its vertices and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q1/5

Find the perimeter of the rectangle with vertices (1, 1), (5, 1), (5, 4), and (1, 4).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q2/5

The region $0 \leq x \leq 4$ and $0 \leq y \leq 2x$ is a triangle.

What is its area?

A. 32

B. 16

C. 8

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q3/5

True or false?

The triangle with vertices (0, 0), (3, 0), and (0, 4) has perimeter 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q4/5

True or false?

The square with vertices (0, 0), (2, 0), (2, 2), and (0, 2) has area 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSA-REI.D.12

FORWARD 2

Use zeros to find where a polynomial is positive or negative • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

$p(x) = (x - 1)(x - 4)$.

For which x is $p(x) < 0$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q1/5

$p(x) = x(x + 2)(x - 3)$.

Is $p(1)$ positive or negative? Explain using its factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q2/5

For which x is $(x + 3)(x - 2) > 0$?

A. $x < -3$ only

B. $x > 2$ only

C. $x < -3$ or $x > 2$

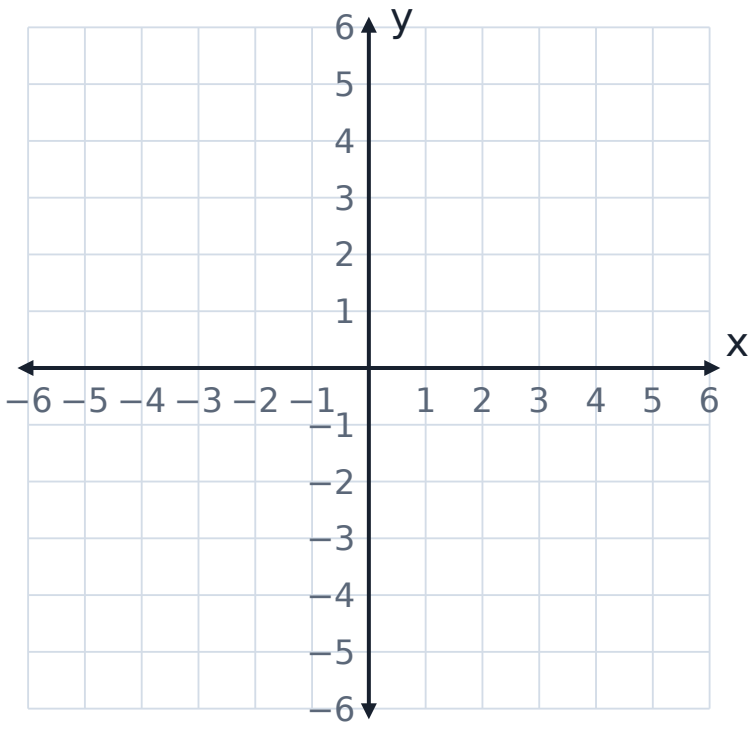
D. $-3 < x < 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q3/5

Sketch a rough graph of $y = x(x + 2)(x - 3)$, and mark the x-values where $y < 0$. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q4/5

True or false?

$(x - 5)(x + 1) \geq 0$ for every real number x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q5/5

Create equations in two or more variables and graph them on labeled axes

MAIN	Algebra I main question	**HSA-CED.A.2**
BACKWARD 1	Write and solve an equation in one variable	**HSA-CED.A.1**
BACKWARD 2	Write the equation of a linear function	**8.F.B.4**
FORWARD 1	Write the equation of a circle from a situation	**HSG-GPE.A.1**
FORWARD 2	Write and interpret polynomial models	**HSF-IF.C.7.c**

SET 33 • HSA-CED.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

A savings account has $150 and grows by $25 each week.

Write an equation for the balance B after w weeks.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

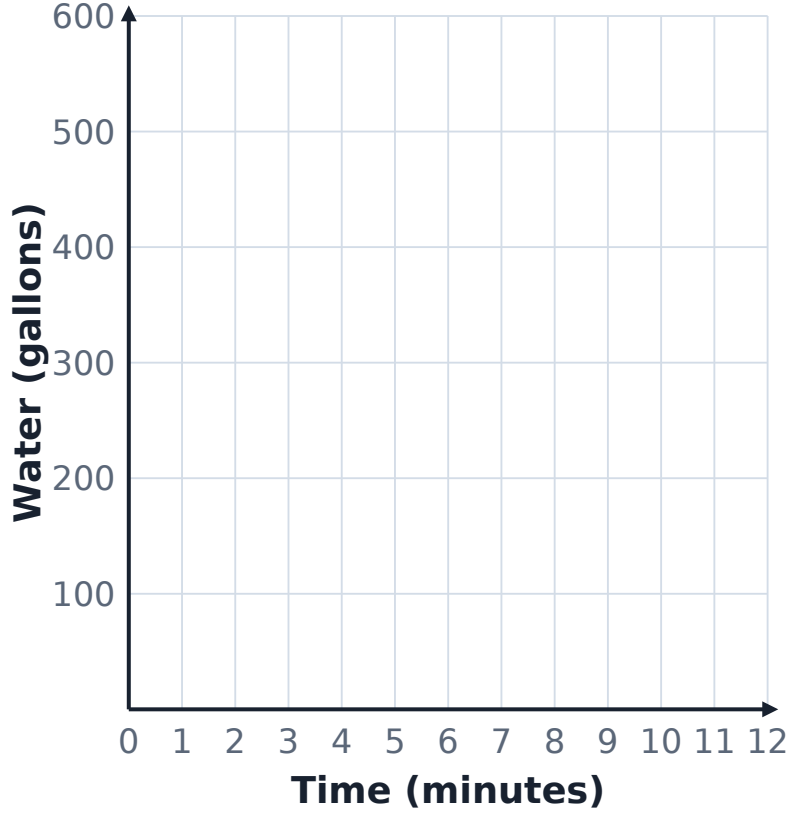
ID S33-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q1/5

A pool holds 600 gallons and drains at 50 gallons per minute.

Write an equation for the gallons G left after t minutes, and graph it for $0 \leq t \leq 12$.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q2/5

A rectangle's length is 3 more than twice its width w.

Write an equation for its area A.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q3/5

A town has 8,000 people, and its population grows 3% each year.

Which equation gives the population P after t years?

A. $P = 8{,}000(0.03)^t$

B. $P = 8{,}000 + 0.03t$

C. $P = 8{,}000(1.03)^t$

D. $P = 8{,}000(1.3)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q4/5

True or false?

Student tickets cost $4 and adult tickets cost $9. The equation 4s + 9a = 300 relates the numbers of tickets when sales total $300.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q5/5

SET 33 • HSA-CED.A.2

BACKWARD 1

Write and solve an equation in one variable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

Solve 150 + 25w = 400 for w.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q1/5

A pool has 600 gallons and drains at 50 gallons per minute.

Write and solve an equation for when it is empty.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q2/5

A plan costs $12 per month plus a $30 fee.

Which equation gives the number of months m for a total of $186?

A. $12m - 30 = 186$

B. $30m + 12 = 186$

C. $12(m + 30) = 186$

D. $12m + 30 = 186$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q3/5

True or false?

$w = 2.5$ is a solution of $w(2w + 3) = 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q4/5

True or false?

Solving $4s + 9(20) = 300$ gives $s = 40$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSA-CED.A.2

BACKWARD 2

Write the equation of a linear function • Grade 8 • Functions • 8.F.B.4

A line passes through (0, 5) and (2, 11).

Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q1/5

A plumber charges $50 plus $30 per hour.

Write a function for the cost C(h).

C(h) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q2/5

A table has x-values 0, 1, 2 and y-values 4, 7, 10.

Which equation fits the table?

A. $y = 4x + 3$

B. $y = x + 4$

C. $y = 3x$

D. $y = 3x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q3/5

True or false?

The rate of change of $y = -2x + 7$ is -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q4/5

True or false?

The initial value of $y = 6x - 1$ is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSA-CED.A.2

FORWARD 1

Write the equation of a circle from a situation • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

A sprinkler at (3, −2) waters everything within 4 meters.

Write an equation of the boundary of the watered region in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q1/5

A cell tower at the origin reaches 10 km. Is a phone at (6, 7) within range? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q2/5

Which equation describes all the points 5 units from (0, 0)?

A. $x^2 - y^2 = 25$

B. $x^2 + y^2 = 25$

C. $x + y = 5$

D. $x^2 + y^2 = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q3/5

True or false?

The circle $(x - 1)^2 + (y - 1)^2 = 4$ passes through $(3, 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q4/5

True or false?

The circle $x^2 + y^2 = 9$ passes through (3, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSA-CED.A.2

FORWARD 2

Write and interpret polynomial models • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

An open box is made from a 10 in by 8 in sheet by cutting a square of side x from each corner.

Write an equation for its volume V.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q1/5

An open box is made from a 10 in by 8 in sheet by cutting a square of side x from each corner and folding up the sides. Its volume is $V = x(10 - 2x)(8 - 2x)$.

Which values of x make sense in the situation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q2/5

What are the zeros of $V = x(10 - 2x)(8 - 2x)$?

A. 5 and 4 only

B. 0, 10, and 8

C. 0, −5, and −4

D. 0, 5, and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q3/5

True or false?

The graph of $y = x^3 - x$ crosses the x-axis three times.

A. True

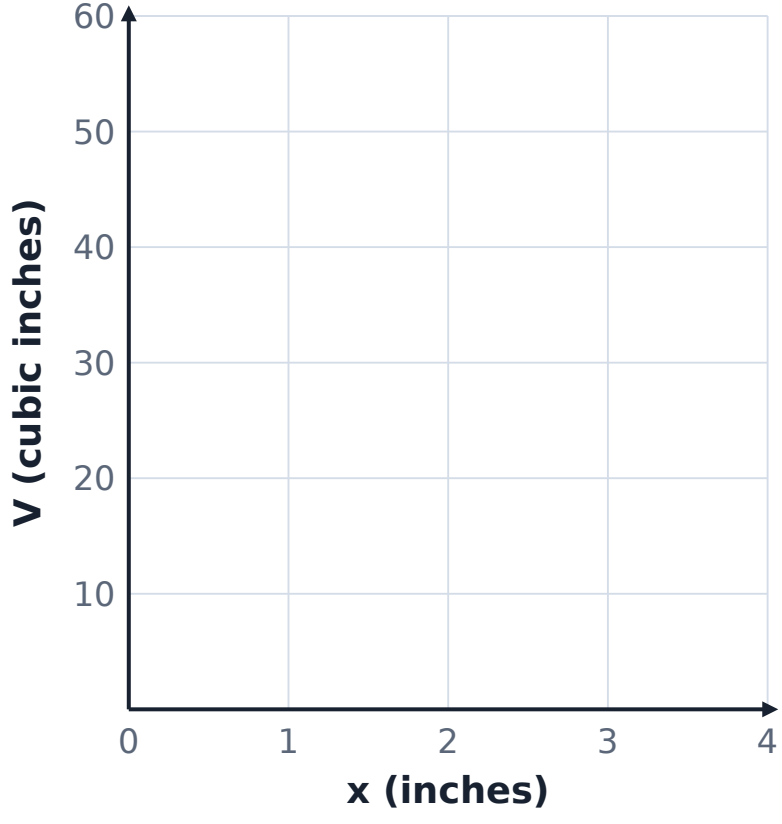
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q4/5

An open box is made from a 10 in by 8 in sheet by cutting a square of side x from each corner. Its volume is $V = x(10 - 2x)(8 - 2x)$. Graph V for $0 \leq x \leq 4$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q5/5

Represent constraints with equations and inequalities and decide whether solutions are viable

MAIN	Algebra I main question	**HSA-CED.A.3**
BACKWARD 1	Test points in a system of inequalities	**HSA-REI.D.12**
BACKWARD 2	Solve a system of linear equations	**HSA-REI.C.6**
FORWARD 1	Use constraints to optimize a design	**HSG-MG.A.3**
FORWARD 2	Use expected values to compare decisions — nearest related	**HSS-MD.B.7**

SET 34 • HSA-CED.A.3

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

A bakery makes m muffins and b loaves. A muffin uses 1 cup of flour and a loaf uses 3 cups; there are 60 cups. A muffin takes 2 minutes and a loaf takes 5 minutes; there are 140 minutes.

Write the system of constraints.

Constraints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q1/5

A bakery makes m muffins and b loaves. A muffin uses 1 cup of flour and a loaf uses 3 cups; there are 60 cups. A muffin takes 2 minutes and a loaf takes 5 minutes; there are 140 minutes.

Is making 40 muffins and 10 loaves possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q2/5

A club needs at least $500 from selling shirts at $15 and hats at $10, and it can sell at most 45 items.

Write the constraints for s shirts and h hats.

Constraints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q3/5

A club sells shirts for $15 and hats for $10. It needs at least $500 and can sell at most 45 items.

Which combination meets both constraints?

A. 15 shirts and 25 hats

B. 25 shirts and 15 hats

C. 30 shirts and 20 hats

D. 20 shirts and 10 hats

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q4/5

True or false?

If x counts people, x = 12.5 can be a viable solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q5/5

SET 34 • HSA-CED.A.3

BACKWARD 1

Test points in a system of inequalities • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

Is (10, 10) a solution of $x + 3y \leq 60$ and $2x + 5y \leq 140$? Explain.

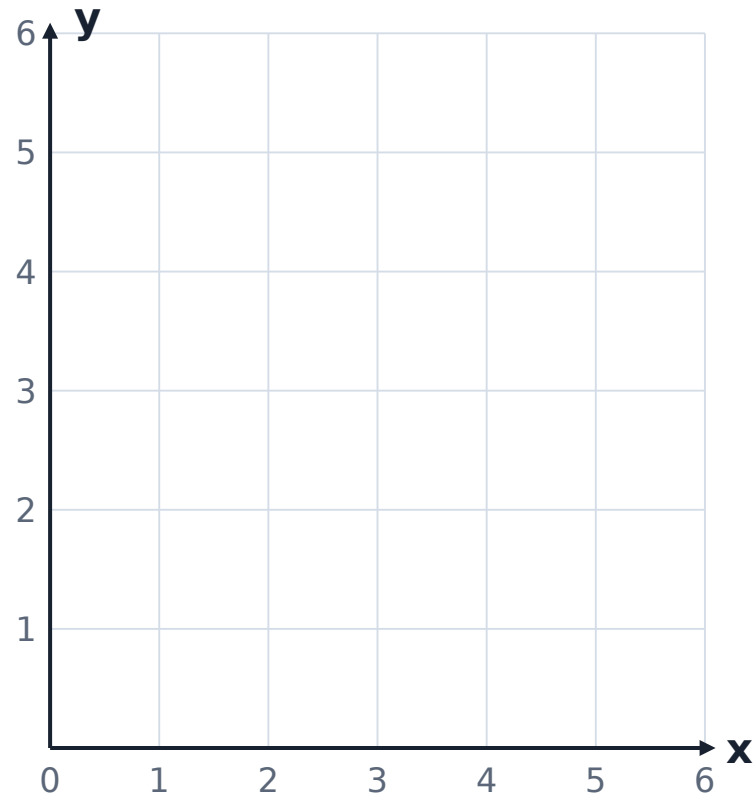
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q1/5

Graph the system $x + y \leq 5$, $x \geq 0$, and $y \geq 0$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q2/5

Which point is in the solution region of $y \geq 2x$ and $y \leq 6$?

A. (4, 1)

B. (1, 7)

C. (3, 5)

D. (1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q3/5

True or false?

The point (0, 0) satisfies $x \geq 0$ and $y \geq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q4/5

True or false?

The point (2, 5) satisfies $x + y < 7$.

A. True

B. False

SET 34 • HSA-CED.A.3

BACKWARD 2

Solve a system of linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Solve.

$x + y = 45$

$15x + 10y = 500$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q1/5

Solve.

$2x + y = 12$

$x + y = 8$

Solution: _____

Solve.

$x + 3y = 60$

$x + y = 40$

A. (30, 10)

B. (15, 15)

C. (10, 30)

D. (20, 20)

True or false?

(10, 35) is a solution of $x + y = 45$.

A. True

B. False

True or false?

The system $x + y = 10$ and $x + y = 12$ has one solution.

A. True

B. False

SET 34 • HSA-CED.A.3

FORWARD 1

Use constraints to optimize a design • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A feasible region has vertices (0, 0), (60, 0), and (0, 20).

Find the greatest value of P = 2m + 7b on the region and where it occurs.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q1/5

A rectangular pen uses at most 40 m of fence.

What dimensions give the greatest area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q2/5

The cost $C = 3x + 2y$ is checked at the vertices (0, 6), (2, 2), and (6, 0) of a region.

Where is the cost least?

A. (2, 2)

B. (0, 0)

C. (6, 0)

D. (0, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q3/5

True or false?

For a linear expression on a polygonal region, the greatest value occurs at a vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q4/5

True or false?

Of all rectangles with area 36, a square has the smallest perimeter, so its side is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q5/5

FORWARD 2

Use expected values to compare decisions • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Choosing viable options under constraints extends to choosing between options by expected value.

Plan A earns $100 with probability 0.6 and $0 otherwise.
Plan B earns $70 for sure.

Which plan has the greater expected value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q1/5 • NEAREST RELATED

Guessing on a 4-choice question earns 1 point if right and loses 0.25 point if wrong.

What is the expected score from guessing?

Expected score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q2/5 • NEAREST RELATED

A $5 warranty pays $100 if the item breaks (probability 0.04) and $0 otherwise.

What is the expected net value of buying it?

A. $-\$5$

B. $\$1$

C. $\$4$

D. $-\$1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A choice with the higher expected value can still lose money on a single trial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

If a game has an expected value of 0, a player cannot lose money playing it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 35 • HSF-IF.A.1 • INTERPRETING FUNCTIONS

Understand a function as a rule that assigns exactly one output to each input, with a domain and a range

MAIN	Algebra I main question	**HSF-IF.A.1**
BACKWARD 1	Decide whether a relation is a function	**8.F.A.1**
BACKWARD 2	Check whether points are on the graph of an equation	**HSA-REI.D.10**
FORWARD 1	Describe transformations of the plane as functions of points	**HSG-CO.A.2**
FORWARD 2	Restrict a domain to make a function invertible	**HSF-BF.B.4.d**

SET 35 • HSF-IF.A.1

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1

Does the relation $\{(-2, 4), (0, 1), (2, 4), (3, -1)\}$ define y as a function of x? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q1/5

Give the domain and the range of $f(x) = x^2 + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q2/5

Which equation does NOT define y as a function of x?

A. $y = x^2$

B. $x = y^2$

C. $y = 2^x$

D. $y = |x|$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q3/5

$f(x) = \sqrt{(x - 3)}$. Over the real numbers, what is the domain of f?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q4/5

True or false?

If f is a function and f(2) = 5, then f(2) can also equal 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q5/5

SET 35 • HSF-IF.A.1

BACKWARD 1

Decide whether a relation is a function • Grade 8 • Functions • 8.F.A.1

A table pairs x = 1, 2, 3, 2 with y = 4, 5, 6, 7. Is y a function of x? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q1/5

Each button on a vending machine gives exactly one kind of snack. Is the snack a function of the button? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q2/5

Which set of ordered pairs is a function?

A. {(5, 1), (5, 2), (6, 3)}

B. {(0, 0), (0, 1)}

C. {(1, 3), (2, 3), (3, 3)}

D. {(1, 2), (1, 3), (2, 4)}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q3/5

True or false?

Two different inputs of a function can have the same output.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q4/5

True or false?

A function can pair the input 4 with two different outputs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSF-IF.A.1

BACKWARD 2

Check whether points are on the graph of an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

Is $(-2, 5)$ on the graph of $y = x^2 + 1$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q1/5

The point (a, 10) is on the graph of y = 2x + 4.

Find a.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q2/5

Which point is on the graph of $y = \sqrt{x}$?

A. (9, −3)

B. (−9, 3)

C. (9, 3)

D. (3, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q3/5

True or false?

The graph of $y = 4$ is the set of all points with y-coordinate 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q4/5

True or false?

(1, 3) is on the graph of $y = x^2 + 3$.

A. True

B. False

SET 35 • HSF-IF.A.1

FORWARD 1

Describe transformations of the plane as functions of points • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2

The translation T maps (x, y) to (x + 3, y − 2).

Find T(4, 1).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q1/5

A reflection across the y-axis maps (x, y) to which point?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q2/5

Which transformation does NOT preserve distance?

A. $(x, y) \rightarrow (x + 5, y)$

B. $(x, y) \rightarrow (-x, y)$

C. $(x, y) \rightarrow (y, x)$

D. $(x, y) \rightarrow (2x, y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q3/5

True or false?

A transformation of the plane is a function: each input point has exactly one image.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q4/5

True or false?

A rotation of 90° counterclockwise about the origin maps (1, 0) to (0, −1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q5/5

SET 35 • HSF-IF.A.1

FORWARD 2

Restrict a domain to make a function invertible • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d

Explain why $f(x) = x^2$, with all real numbers as its domain, has no inverse function.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q1/5

$f(x) = (x - 1)^2$. Restrict f to the largest interval that contains $x = 3$ on which f is one-to-one, and write the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q2/5

Which function is one-to-one on all real numbers?

A. $f(x) = x^2 - 4$

B. $f(x) = x^3$

C. $f(x) = |x|$

D. $f(x) = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^2$ with domain $x \geq 0$ has the inverse $f^{-1}(x) = \sqrt{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q4/5

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSF-IF.A.2 • INTERPRETING FUNCTIONS

Use function notation, evaluate functions, and interpret statements in function notation in context

MAIN	Algebra I main question	**HSF-IF.A.2**
BACKWARD 1	Evaluate expressions for given values	**6.EE.A.2.c**
BACKWARD 2	Solve linear equations	**HSA-REI.B.3**
FORWARD 1	Evaluate compositions of functions	**HSF-BF.A.1.c**
FORWARD 2	Read values of an inverse function	**HSF-BF.B.4.c**

SET 36 • HSF-IF.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$f(x) = 2x^2 - 3x + 1$.

Find $f(-2)$.

$f(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q1/5

$h(t) = -16t^2 + 64t$ gives a ball's height in feet t seconds after it is hit.

Find h(1), and explain what it means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q2/5

$g(x) = 5x - 7$.

Find the value of x for which $g(x) = 18$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q3/5

C(n) is the cost in dollars of n tickets.

What does C(4) = 60 mean?

A. 4 tickets cost $60.

B. C equals 4 times 60.

C. Each ticket costs $60.

D. 60 tickets cost $4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q4/5

True or false?

If $f(x) = x^2$, then $f(a + 1) = a^2 + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q5/5

SET 36 • HSF-IF.A.2

BACKWARD 1

Evaluate expressions for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $2x^2 - 3x + 1$ when $x = 3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q1/5

Evaluate $-16t^2 + 64t$ when $t = 2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q2/5

Evaluate $3(a - 4)^2$ when $a = 6$.

A. 144

B. 12

C. 6

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q3/5

True or false?

When $x = -1$, $x^2 - x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q4/5

True or false?

When $n = 5$, $2n^2 = 100$.

A. True

B. False

SET 36 • HSF-IF.A.2

BACKWARD 2

Solve linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $5x - 7 = 18$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q1/5

Solve $3(x + 2) = 2x - 4$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q2/5

Solve 0.5x + 3 = 11.

A. x = 7

B. x = 28

C. x = 16

D. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q3/5

True or false?

$x = -2$ is the solution of $4x + 9 = 1$.

A. True

B. False

True or false?

The solution of $7 - x = 10$ is $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSF-IF.A.2

FORWARD 1

Evaluate compositions of functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

$f(x) = x + 4$ and $g(x) = 3x$.

Find $f(g(2))$.

$f(g(2)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q1/5

$f(x) = x^2$ and $g(x) = x - 3$.

Write $f(g(x))$.

$f(g(x)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q2/5

$f(x) = 2x$ and $g(x) = x + 1$.

Find $g(f(5))$.

A. 11

B. 10

C. 6

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q3/5

True or false?

If $f(x) = x - 1$ and $g(x) = x + 1$, then $f(g(x)) = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q4/5

True or false?

If $f(x) = 2x$ and $g(x) = x^2$, then $f(g(3)) = 36$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSF-IF.A.2

FORWARD 2

Read values of an inverse function • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

The function f has domain {1, 2, 3, 4}, with f(1) = 3, f(2) = 5, f(3) = 7, and f(4) = 9. Its outputs are all different, so f has an inverse.

Find $f^{-1}(7)$.

$f^{-1}(7) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q1/5

$f(x) = 2x + 1$ is invertible.

Find $f^{-1}(11)$.

$f^{-1}(11) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q2/5

f is a one-to-one function with $f(4) = 10$.

Which statement is true?

A. $f^{-1}(10) = 4$

B. $f^{-1}(4) = 10$

C. $f(10) = 4$

D. $f^{-1}(10) = \frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q3/5

True or false?

If f is a one-to-one function, the graph of f^{-1} is the reflection of the graph of f across the line y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q4/5

True or false?

$f^{-1}(x)$ means $1 \div f(x)$.

A. True

B. False

SET 37 • HSF-IF.A.3 • INTERPRETING FUNCTIONS

Recognize that sequences are functions whose domain is a set of integers

MAIN	Algebra I main question	**HSF-IF.A.3**
BACKWARD 1	Evaluate functions in function notation	**HSF-IF.A.2**
BACKWARD 2	Generate two numerical patterns and relate their terms	**5.OA.B.3**
FORWARD 1	Repeat a dilation and describe the lengths — nearest related	**HSG-SRT.A.1**
FORWARD 2	Find the sum of a finite geometric series	**HSA-SSE.B.4**

SET 37 • HSF-IF.A.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

A sequence is defined by $f(1) = 3$ and $f(n) = f(n - 1) + 4$ for $n \geq 2$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q1/5

A sequence is defined by $f(1) = 2$, $f(2) = 3$, and $f(n) = f(n - 1) + 2f(n - 2)$ for $n \geq 3$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q2/5

Write an explicit formula for the sequence 5, 10, 20, 40, …, with f(1) = 5.

f(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q3/5

A sequence f(n) starts with f(1).

What is its domain?

A. The positive integers 1, 2, 3, ...

B. The numbers from 0 to 1

C. All real numbers

D. All integers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q4/5

True or false?

The graph of the sequence f(n) = 2n + 1 for n = 1, 2, 3, ... is a solid line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q5/5

SET 37 • HSF-IF.A.3

BACKWARD 1

Evaluate functions in function notation • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$f(n) = 3n - 1$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q1/5

$f(n) = 2^n$.

Find f(5).

f(5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q2/5

$f(n) = n^2 + 1$.

Find $f(3)$.

A. 7

B. 10

C. 16

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q3/5

True or false?

If $f(n) = 10 - n$, then $f(10) = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q4/5

True or false?

If f(n) = 4n, then f(3) = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSF-IF.A.3

BACKWARD 2

Generate two numerical patterns and relate their terms • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 6.

Write the first 5 terms of each pattern.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q1/5

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 8.

Write ordered pairs (A term, B term) for the first 4 terms. What relationship do you see?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q2/5

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 6.

How do their third terms compare?

A. A's is 3 times B's.

B. B's is 4 more than A's.

C. B's is 3 times A's.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q3/5

True or false?

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10. Each term of B is twice the matching term of A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q4/5

True or false?

Pattern A starts at 1 and adds 2. Pattern B starts at 2 and adds 4. Each term of B is 4 times the matching term of A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSF-IF.A.3

FORWARD 1

Repeat a dilation and describe the lengths • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Geometry standard. Connection: Repeated dilations produce a geometric sequence of lengths, a function defined on the positive integers.

A segment of length 3 is dilated by scale factor 2, then the image is dilated by 2 again, and so on.

Write the length after n dilations.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q1/5 • NEAREST RELATED

A dilation with center (0, 0) and scale factor 3 maps (2, −1) to which point?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q2/5 • NEAREST RELATED

A dilation with scale factor $\frac{1}{2}$ maps a segment of length 10 to a segment of what length?

A. 20

B. 9.5

C. 10

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A dilation maps a line that does not pass through the center to a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

A dilation with scale factor 2 doubles the area of a figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 37 • HSF-IF.A.3

FORWARD 2

Find the sum of a finite geometric series • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Find the sum of the first 5 terms of 2, 6, 18, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q1/5

Write a formula for the sum of the first n terms of 5, 10, 20, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q2/5

What is 1 + 4 + 16 + 64?

A. 256

B. 85

C. 84

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q3/5

True or false?

The sum of the first 4 terms of 1, 2, 4, 8, ... is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q4/5

True or false?

A geometric series with first term 4, ratio 1, and 5 terms has sum 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q5/5

SET 38 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Interpret key features of graphs and tables of functions in context

MAIN	Algebra I main question	**HSF-IF.B.4**
BACKWARD 1	Describe a function qualitatively from a situation	**8.F.B.5**
BACKWARD 2	Use vertex form to find a maximum or minimum	**HSA-SSE.B.3.b**
FORWARD 1	Identify key features of polynomial functions	**HSF-IF.C.7.c**
FORWARD 2	Identify amplitude, midline, and period of a periodic model	**HSF-TF.B.5**

SET 38 • HSF-IF.B.4

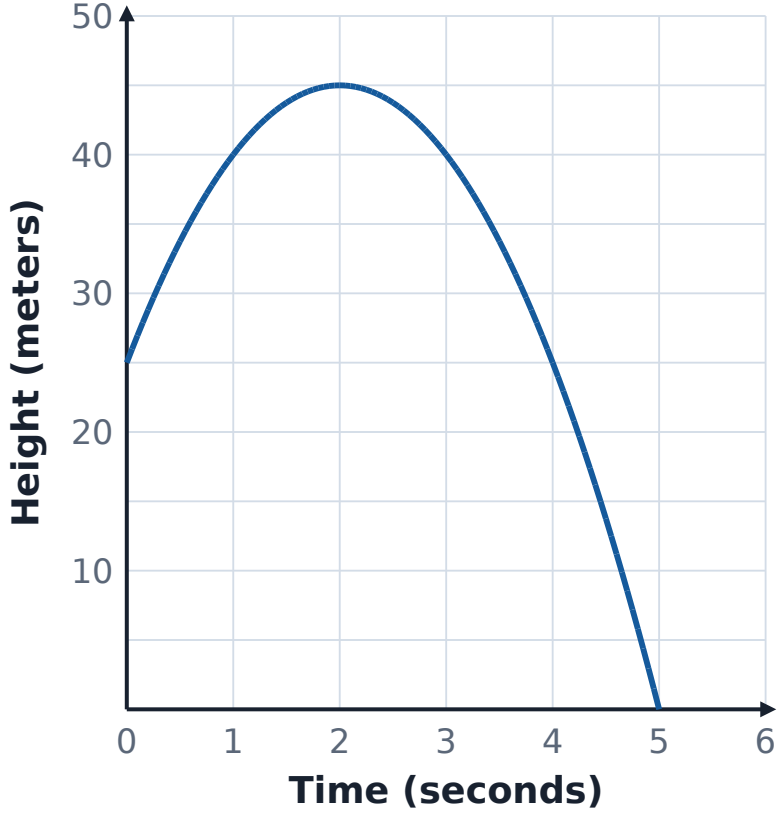
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony.

What is the maximum height, and when does it occur?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

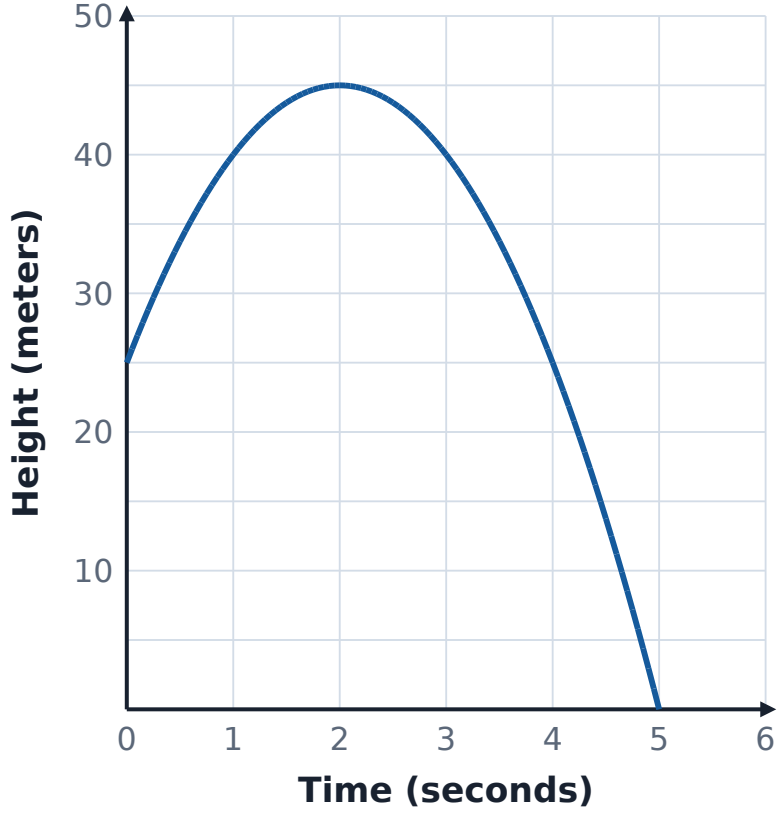
ID S38-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q1/5

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony (t = 0 is the moment it is thrown).

What is the h-intercept of the graph, and what does it mean?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

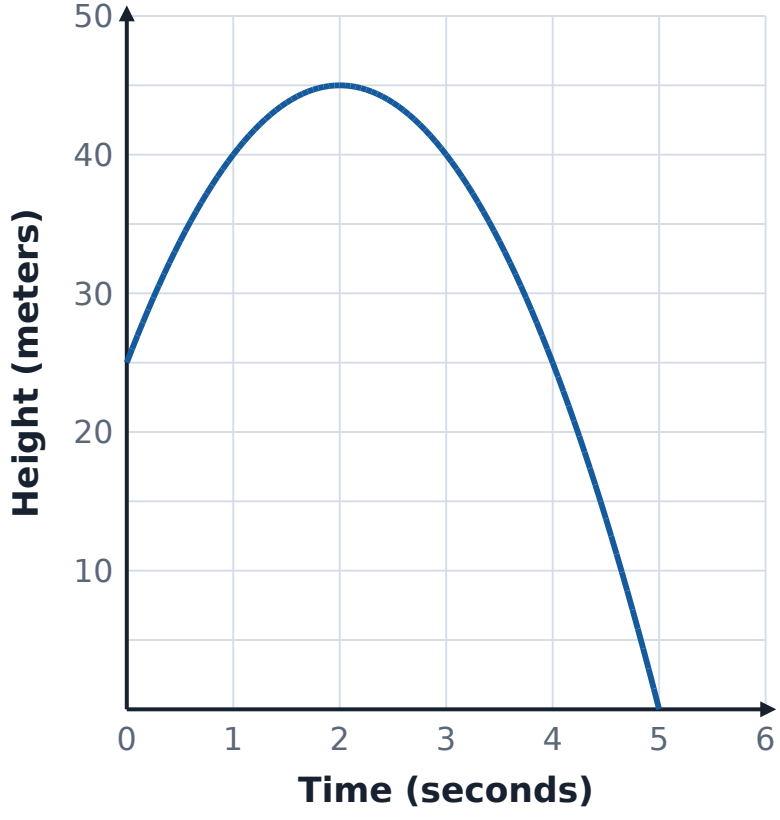
ID S38-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q2/5

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony.

When does the ball reach the ground? Over what interval is its height increasing?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

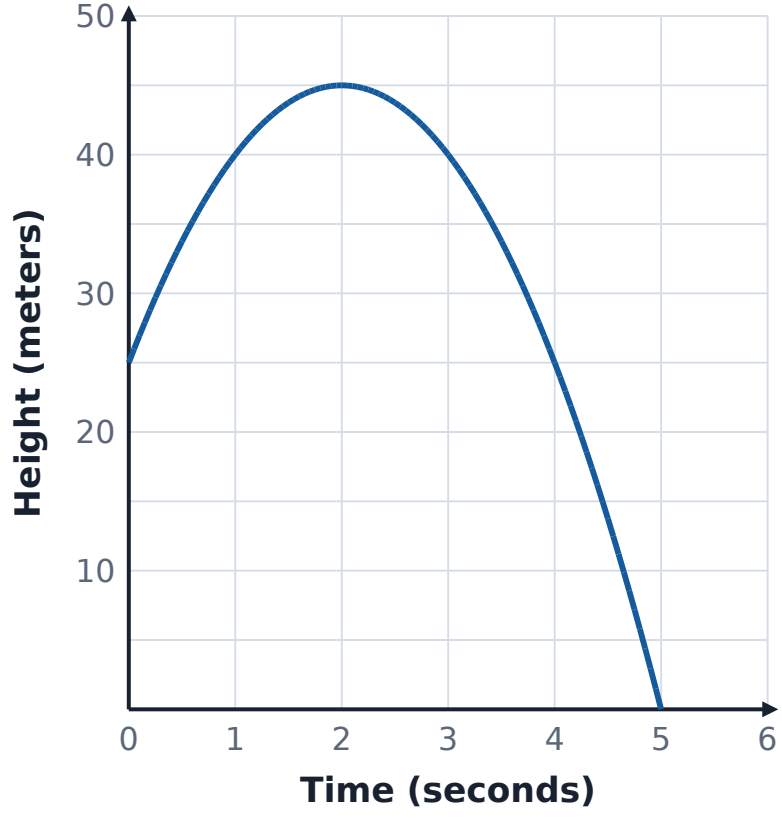
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q3/5

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony.

Over what interval is the ball's height decreasing?

A. $0 < t < 2$ **B.** $0 < t < 5$

C. $2 < t < 5$ **D.** $25 < t < 45$



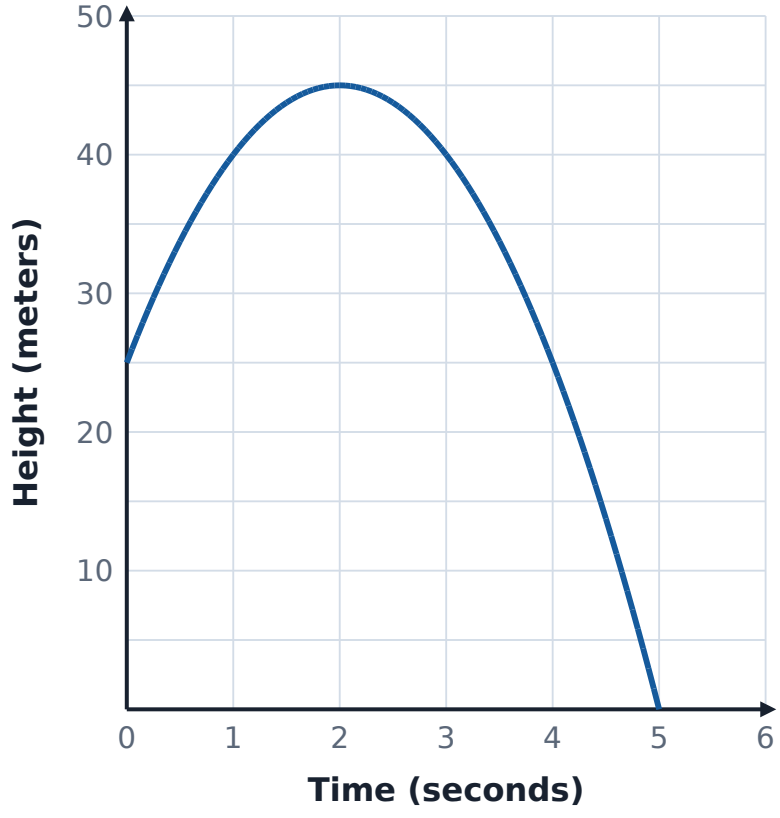
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q4/5

True or false?

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony. The ball is 25 meters high twice: at t = 0 and at t = 4.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q5/5

SET 38 • HSF-IF.B.4

BACKWARD 1

Describe a function qualitatively from a situation • Grade 8 • Functions • 8.F.B.5

A car speeds up, drives at a constant speed, then slows to a stop.

Describe the graph of its speed over time.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q1/5

A graph of water level is horizontal from t = 2 to t = 5.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q2/5

Which situation has a graph that first increases and then decreases?

A. Savings with a steady deposit each week

B. The height of a ball thrown up that comes back down

C. The distance a parked car travels

D. The height of a burning candle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q3/5

True or false?

On a distance-time graph, a steeper segment means a greater speed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q4/5

True or false?

A decreasing graph always means the quantity is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSF-IF.B.4

BACKWARD 2

Use vertex form to find a maximum or minimum • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Write $-5t^2 + 20t + 25$ in the form $a(t - h)^2 + k$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q1/5

What is the maximum value of $-(x - 3)^2 + 7$, and for what x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q2/5

What is the minimum value of $2(x + 1)^2 - 8$?

A. -8

B. -1

C. 2

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q3/5

True or false?

$-5(t - 2)^2 + 45$ is greatest when $t = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q4/5

True or false?

The minimum value of $(x - 4)^2 + 1$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSF-IF.B.4

FORWARD 1

Identify key features of polynomial functions • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

Describe the end behavior of $f(x) = -2x^3 + x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q1/5

Find the zeros of $f(x) = x(x - 2)(x + 3)$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q2/5

Which function rises on the right and falls on the left?

A. $f(x) = -x^4$

B. $f(x) = x^3 - 4x$

C. $f(x) = x^2$

D. $f(x) = -x^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q3/5

True or false?

A polynomial function of degree 4 has at most 4 real zeros.

A. True

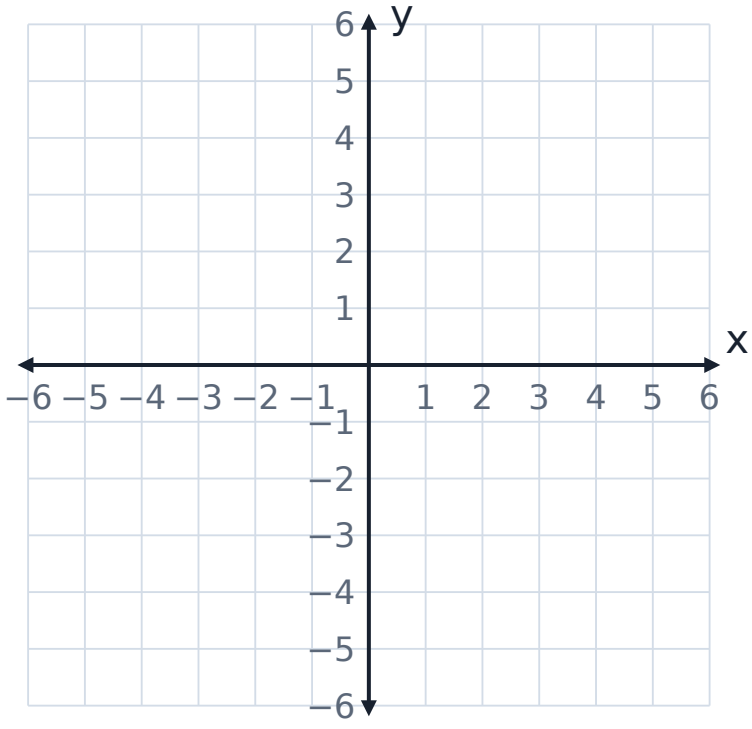
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q4/5

Graph $f(x) = -x^2(x - 3)$. Mark the zeros and show what the graph does at each zero and at the ends. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSF-IF.B.4

FORWARD 2

Identify amplitude, midline, and period of a periodic model • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel seat's height is $h(t) = 30 - 25\cos(\pi t \div 5)$ meters after t minutes.

What are the greatest and least heights?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q1/5

What is the period of $h(t) = 30 - 25\cos(\pi t \div 5)$?

Period: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q2/5

What is the midline of $y = 4\sin x + 2$?

A. $y = 4$

B. $y = 2$

C. $y = 6$

D. $y = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q3/5

True or false?

The amplitude of $y = 3\sin x$ is 3.

A. True

B. False

True or false?

The function $y = \sin x$ has period π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q5/5

SET 78 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Sketch graphs that show key features given in a verbal description

MAIN	Algebra I main question	**HSF-IF.B.4**
BACKWARD 1	Sketch a graph from a description	**8.F.B.5**
BACKWARD 2	Write and graph equations on labeled axes	**HSA-CED.A.2**
FORWARD 1	Sketch polynomial graphs from a description of their features	**HSF-IF.C.7.c**
FORWARD 2	Sketch periodic graphs from a description	**HSF-TF.B.5**

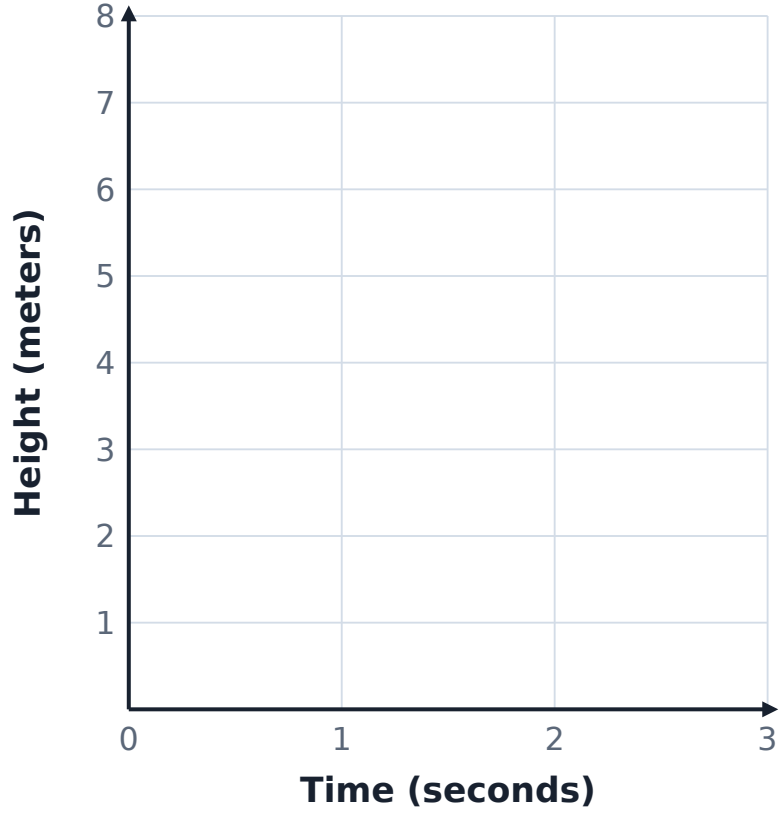
SET 78 • HSF-IF.B.4

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

A ball is thrown up from a 2-m platform. It reaches its highest point, 7 m, after 1 second, and it lands on the ground after about 2.2 seconds.

Sketch its height against time. Label the starting point, the maximum, and the landing time.



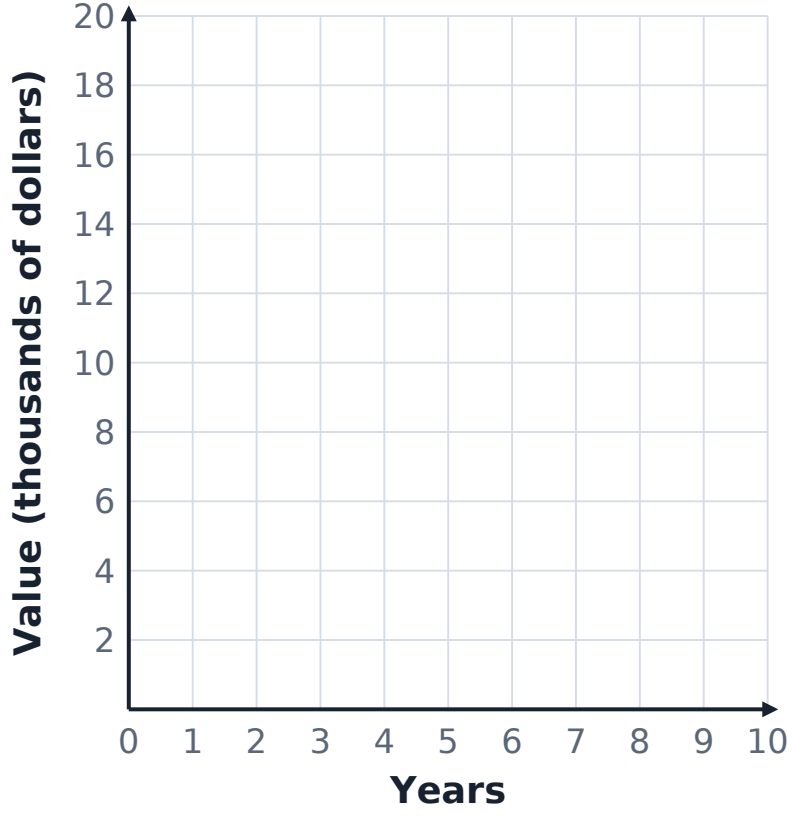
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q1/5

A car is worth $20,000 when new, and it loses 15% of its value each year, never reaching $0.

Sketch its value for 0 to 10 years. Show the starting value and what happens as time goes on.



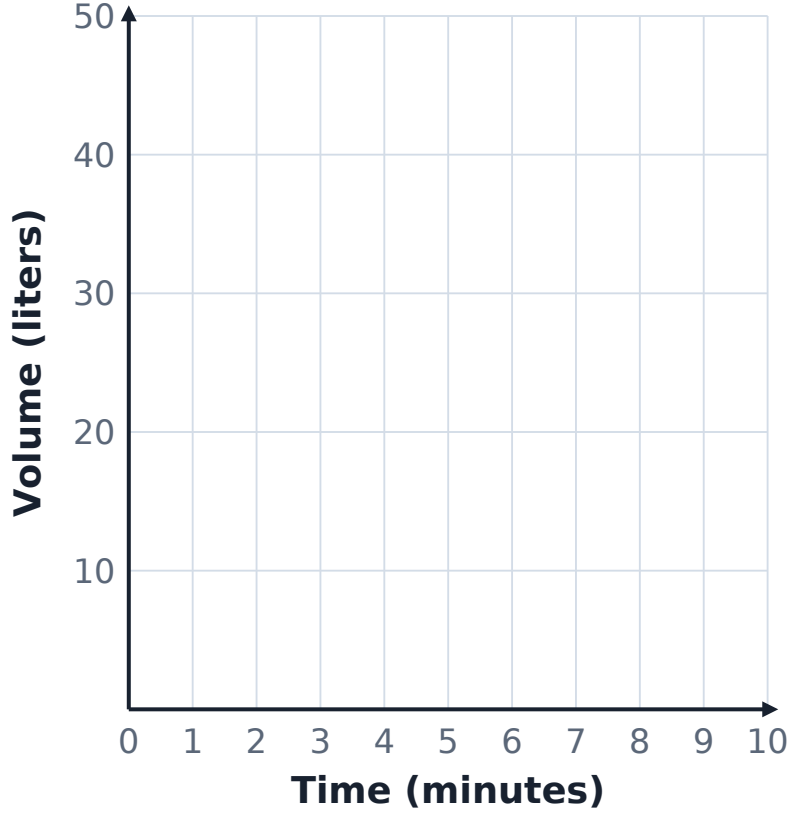
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q2/5

A 50-liter tank drains at a constant rate until it is empty after 5 minutes, stays empty for 2 minutes, and then refills at a constant rate to 30 liters in 3 minutes.

Sketch the volume over the 10 minutes.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q3/5

A town's population grew slowly at first and then faster and faster.

Which graph shape fits?

A. Increasing and leveling off

B. Increasing and curving upward

C. A horizontal line

D. Decreasing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q4/5

True or false?

A graph of the temperature of a cup of coffee left on a table falls at a constant rate until it reaches 0 °C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q5/5

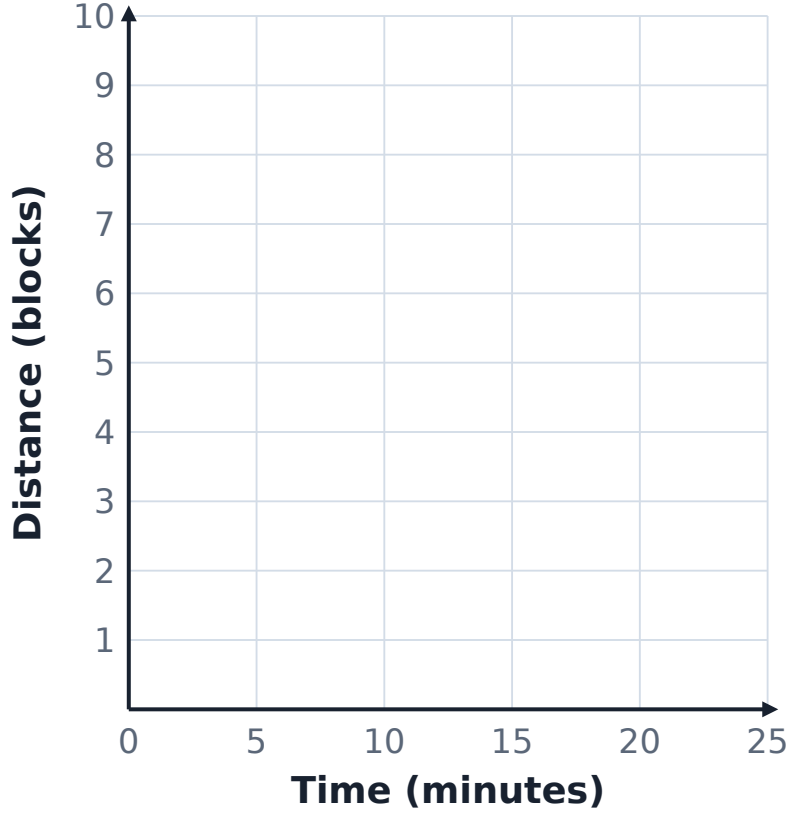
SET 78 • HSF-IF.B.4

BACKWARD 1

Sketch a graph from a description • Grade 8 • Functions • 8.F.B.5

You walk away from home at a steady pace for 10 minutes, rest for 5 minutes, then walk home faster in 10 minutes.

Sketch your distance from home over 25 minutes.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q1

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q1/5

A graph of a plant's height rises steeply and then levels off. Describe what happened to the plant.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q2

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q2/5

Which situation has a graph that stays constant (horizontal) for a while?

A. A rising balloon's height

B. A burning candle's height

C. A parked car's distance from home

D. A falling ball's height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q3

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q3/5

True or false?

A graph that rises and then falls has a highest point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q4

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q4/5

True or false?

A horizontal segment on a speed-time graph means the car is stopped.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q5

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSF-IF.B.4

BACKWARD 2

Write and graph equations on labeled axes • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

Write an equation for the area A of a square with side s. What kind of graph does it have for s ≥ 0?

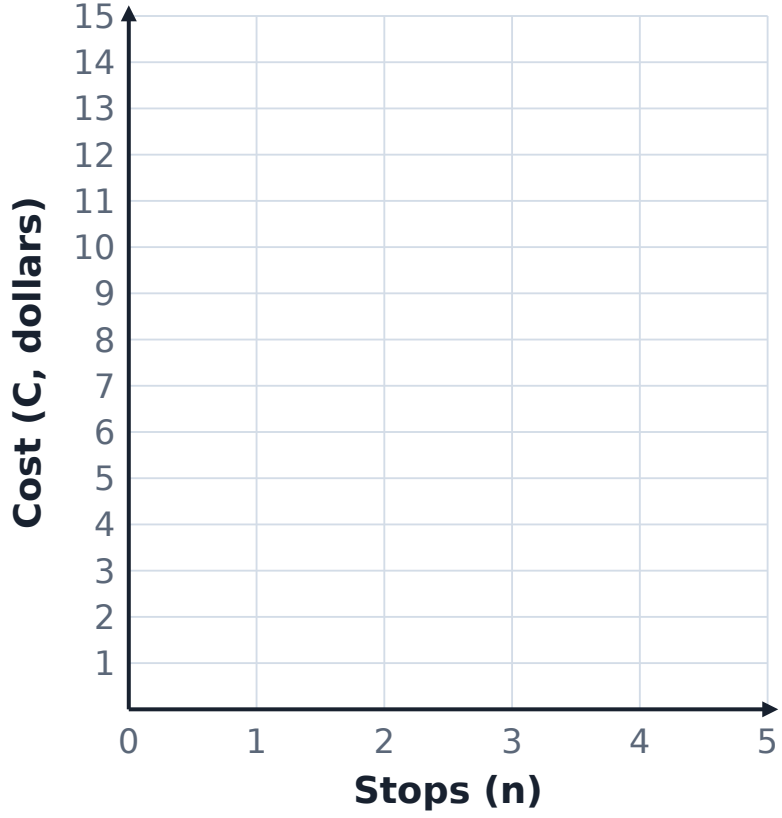
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q1/5

Graph C = 5 + 2n, the cost in dollars of a ride with n stops, for n = 0 to 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q2/5

A $1,200 laptop loses $200 of value each year. Which equation gives its value V after t years?

A. $V = 1{,}200 - 200t$

B. $V = 200 - 1{,}200t$

C. $V = 1{,}200 + 200t$

D. $V = 1{,}200(0.8)^{t}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q3/5

True or false?

An equation in two variables can be graphed on a coordinate plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q4/5

True or false?

The graph of d = 50t is a curve.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q5

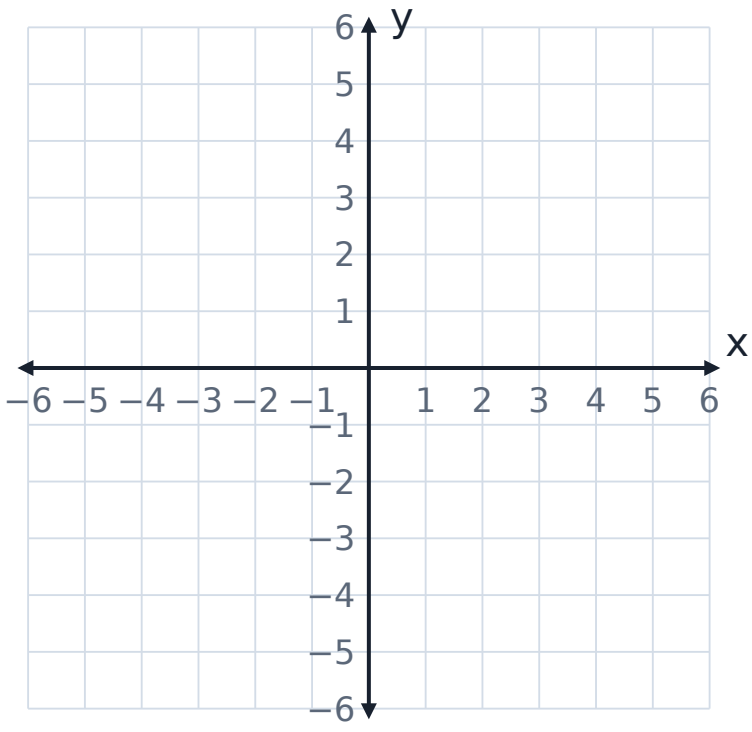
High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q5/5

SET 78 • HSF-IF.B.4

FORWARD 1

Sketch polynomial graphs from a description of their features • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

Sketch a cubic function that falls to the left, rises to the right, and has zeros at −2, 1, and 4. (The y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q1/5

A polynomial graph touches the x-axis at $x = 3$ and turns around there. What can you say about the factor $x - 3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q2/5

Which polynomial function has a graph that falls on both ends?

A. $y = -x^4 + 3x$

B. $y = x^3$

C. $y = -x^3$

D. $y = x^4 - 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q3/5

True or false?

The graph of a degree-5 polynomial with a positive leading coefficient rises to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q4/5

True or false?

The graph of a cubic function can rise on both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q5/5

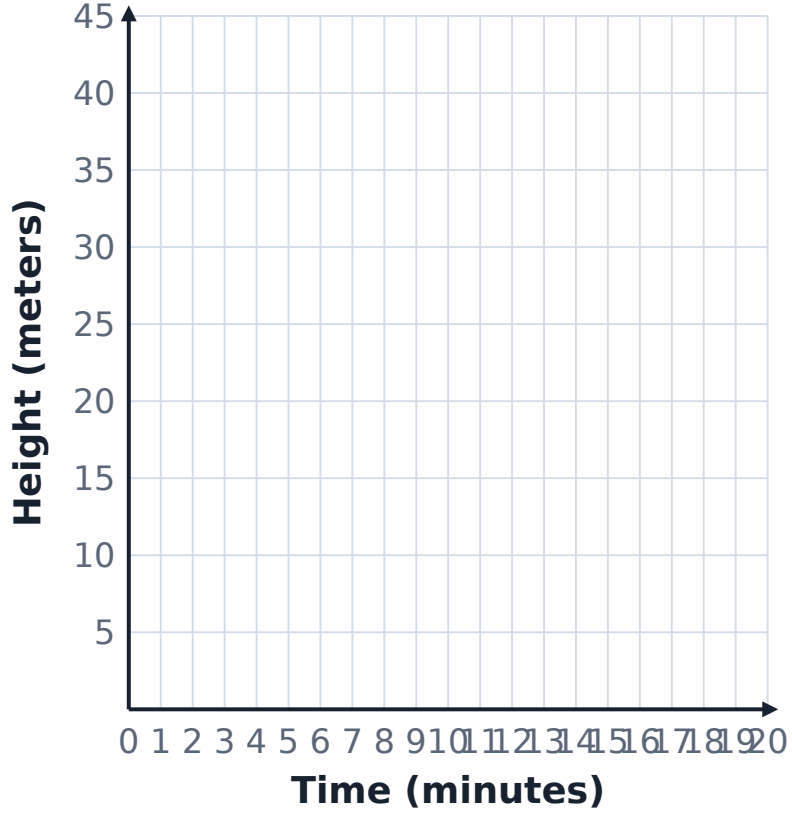
SET 78 • HSF-IF.B.4

FORWARD 2

Sketch periodic graphs from a description • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel seat starts at its lowest height, 2 m, reaches its highest height, 42 m, after 5 minutes, and repeats every 10 minutes.

Sketch its height over 20 minutes.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q1/5

Daylight lasts 15 hours in late June and 9 hours in late December. Describe a graph of daylight hours over a year.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q2/5

Which feature of a periodic graph is the time for one full cycle?

A. The maximum

B. The midline

C. The amplitude

D. The period

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q3/5

True or false?

The midline of a periodic graph is halfway between its maximum and minimum values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q4/5

True or false?

A tide graph that repeats every 12.4 hours has amplitude 12.4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q5/5

SET 39 • HSF-IF.B.5 • INTERPRETING FUNCTIONS

Relate the domain of a function to the quantity it describes

MAIN	Algebra I main question	**HSF-IF.B.5**
BACKWARD 1	Decide whether a solution is viable	**HSA-CED.A.3**
BACKWARD 2	Find where a quadratic model equals 0	**HSA-REI.B.4.b**
FORWARD 1	Use trigonometric ratios as functions of acute angles — nearest r..	**HSG-SRT.C.6**
FORWARD 2	Extend trigonometric functions to all real numbers with the unit ci…	**HSF-TF.A.2**

SET 39 • HSF-IF.B.5

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

$n(k) = 12k$ gives the number of eggs in k full cartons. The store has 30 cartons.

What is a reasonable domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q1/5

$h(t) = -16t^2 + 64t$ gives a ball's height in feet from the moment it is hit until it lands.

What is the domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q2/5

A taxi fare is $C(m) = 3 + 2.5m$ dollars for m miles, with a limit of 40 miles per ride.

What is the domain?

A. All real numbers

B. $m \geq 3$

C. $3 \leq m \leq 103$

D. $0 \leq m \leq 40$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q3/5

A rectangle has perimeter 20. Its area is $A(w) = w(10 - w)$, where w is the width.

What is the domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q4/5

True or false?

P(n) = 50n gives the number of seats on n buses. n = 2.5 is in a reasonable domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q5/5

SET 39 • HSF-IF.B.5

BACKWARD 1

Decide whether a solution is viable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

Solving a problem gives 7.4 buses needed to carry every student.

What is the viable answer? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q1/5

The number of tickets t must satisfy $5t \leq 60$ and $t \geq 0$.

What whole numbers of tickets are viable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q2/5

Which value is NOT viable for the number of people in a room?

A. 40

B. 0

C. 12

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q3/5

True or false?

A length of 0.25 meter can be a viable solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q4/5

True or false?

$w(10 - w) = 24$ has solutions $w = 4$ and $w = 6$, but only one of them is a viable width.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSF-IF.B.5

BACKWARD 2

Find where a quadratic model equals 0 • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $-16t^2 + 64t = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q1/5

Solve $w(10 - w) = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q2/5

Solve $-16t^2 + 64t + 80 = 0$.

A. $t = 4$ or $t = -2$

B. $t = 5$ or $t = -1$

C. $t = 5$ only

D. $t = -5$ or $t = 1$

True or false?

$t = 4$ is a solution of $t^2 - 4t = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q4/5

True or false?

$x^2 + 9 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSF-IF.B.5

FORWARD 1

Use trigonometric ratios as functions of acute angles • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Geometry standard. Connection: A trigonometric ratio defined by a right triangle has a domain limited to acute angles, a domain set by the situation.

Right triangle ABC has its right angle at C, with AC = 4, BC = 3, and AB = 5.

Find sin A.

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q1/5 • NEAREST RELATED

For which angle measures θ is sin θ defined as a ratio of sides of a right triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q2/5 • NEAREST RELATED

In a right triangle, $\cos A = \frac{12}{13}$.

What is sin A?

A. $\frac{5}{13}$

B. $\frac{12}{5}$

C. $\frac{1}{13}$

D. $\frac{13}{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In a right triangle, $\sin A = \cos B$ when A and B are the two acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

For an acute angle θ, $\sin\theta$ can be greater than 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 39 • HSF-IF.B.5

FORWARD 2

Extend trigonometric functions to all real numbers with the unit circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

Find cos π.

cos π = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q1/5

Find sin(3π ÷ 2).

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q2/5

With the unit circle, what is the domain of $f(\theta) = \sin \theta$?

A. All real numbers

B. $0 < \theta < \pi \div 2$

C. $\theta \geq 0$

D. $0 \leq \theta \leq 2\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q3/5

True or false?

A negative radian measure gives a clockwise rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q4/5

True or false?

sin θ is defined only for acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q5/5

SET 40 • HSF-IF.B.6 • INTERPRETING FUNCTIONS

Calculate, interpret, and estimate the average rate of change of a function over an interval

MAIN	Algebra I main question	**HSF-IF.B.6**
BACKWARD 1	Find the rate of change of a linear function	**8.F.B.4**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Build a periodic model from amplitude, period, and midline	**HSF-TF.B.5**
FORWARD 2	Analyze rational functions: rates of change and asymptotes	**HSF-IF.C.7.d**

SET 40 • HSF-IF.B.6

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2 + 2x$.

Find the average rate of change from $x = 1$ to $x = 4$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q1/5

A town had 12,000 people in 2010 and 15,600 in 2022.

Find the average rate of change, with units.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q2/5

The table gives the distance d a cart has traveled after t seconds.

Over which 2-second interval was its average speed greatest? Give that speed.

t (seconds)	0	2	4	6
d (meters)	0	30	100	150

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q3/5

What is the average rate of change of $g(x) = 3^x$ from $x = 0$ to $x = 2$?

A. 4

B. 3

C. 8

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q4/5

True or false?

A linear function has the same average rate of change over every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q5/5

SET 40 • HSF-IF.B.6

BACKWARD 1

Find the rate of change of a linear function • Grade 8 • Functions • 8.F.B.4

Find the rate of change of the line through (2, 5) and (6, 17).

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q1/5

A tank had 50 liters at t = 0 and 20 liters at t = 10 minutes, draining steadily.

What is the rate of change?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q2/5

What is the rate of change of $y = 7 - 4x$?

A. 7

B. -4

C. $-\frac{4}{7}$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q3/5

True or false?

A negative rate of change means the quantity is decreasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q4/5

True or false?

The rate of change from (0, 3) to (2, 3) is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSF-IF.B.6

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A runner goes $\frac{3}{4}$ mile in $\frac{1}{8}$ hour.

What is the speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q1/5

A car uses 12 gallons to go 300 miles.

How many miles per gallon?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q2/5

A recipe uses $\frac{2}{3}$ cup of sugar for $\frac{1}{3}$ of a batch.

How much sugar is used per batch?

A. $\frac{1}{2}$ cup

B. 1 cup

C. 2 cups

D. $\frac{2}{9}$ cup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q3/5

True or false?

120 km in 1.5 hours is 80 km per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q4/5

True or false?

$18 for 4 pounds is $4 per pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSF-IF.B.6

FORWARD 1

Build a periodic model from amplitude, period, and midline • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel has diameter 40 m, its center is 25 m above the ground, and it turns once every 8 minutes. A rider starts at the lowest point.

Write a model $h(t) = A\cos(Bt) + D$ for the rider's height after t minutes.

$h(t) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q1/5

$h(t) = -20\cos(\pi t \div 4) + 25$ gives a rider's height in meters after t minutes.

Give the amplitude, the period, and the midline.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q2/5

Daily high temperatures in a town vary from a low of 4 °C to a high of 28 °C over a 365-day year.

Which values fit a sinusoidal model?

A. Amplitude 24, midline 16, period 365 days

B. Amplitude 16, midline 12, period 365 days

C. Amplitude 12, midline 14, period 182.5 days

D. Amplitude 12, midline 16, period 365 days

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q3/5

True or false?

A model $y = A\cos(Bt) + D$ with period 12 hours has $B = \pi \div 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q4/5

True or false?

A tide that rises from 1 m to 5 m and falls back again has amplitude 4 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSF-IF.B.6

FORWARD 2

Analyze rational functions: rates of change and asymptotes • High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d

$f(x) = 12 \div x$.

Find the average rate of change from $x = 2$ to $x = 6$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q1/5

Give the vertical and horizontal asymptotes of $f(x) = 1 \div (x - 3) + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q2/5

Where is $f(x) = (x + 1) \div (x - 4)$ undefined?

A. $x = -4$

B. $x = 1$

C. $x = -1$

D. $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q3/5

True or false?

The graph of $y = 1 \div x$ gets closer to the x-axis as x gets large.

A. True

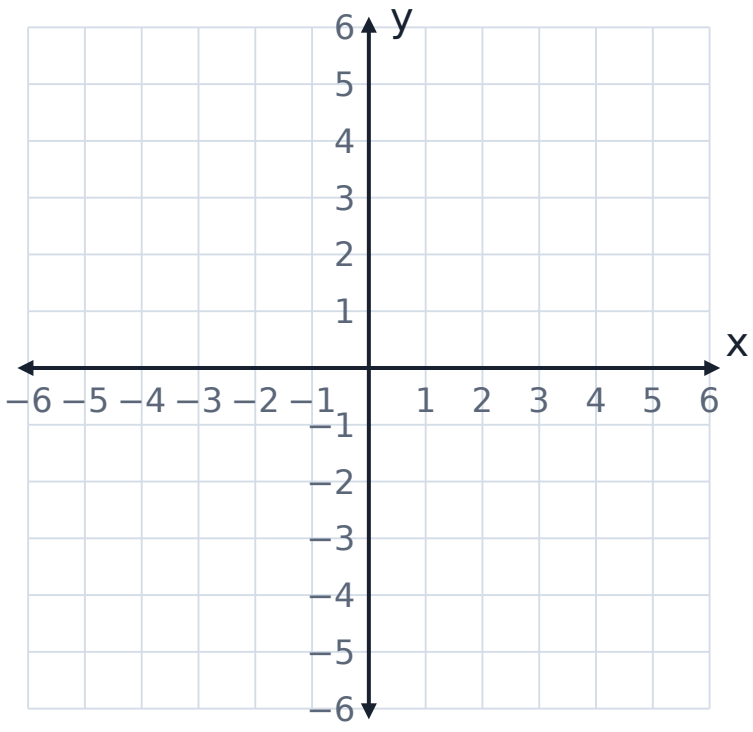
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q4/5

Graph f(x) = 1 ÷ (x − 2) + 1. Draw both asymptotes as lines and mark the intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q5/5

SET 79 • HSF-IF.B.6 • INTERPRETING FUNCTIONS

Estimate the average rate of change of a function from its graph

MAIN	Algebra I main question	**HSF-IF.B.6**
BACKWARD 1	Find a rate of change from two points	**8.F.B.4**
BACKWARD 2	Read a unit rate from a graph	**7.RP.A.2.b**
FORWARD 1	Estimate average rates of change from a polynomial graph	**HSF-IF.C.7.c**
FORWARD 2	Estimate average rates of change from a rational-function graph	**HSF-IF.C.7.d**

SET 79 • HSF-IF.B.6

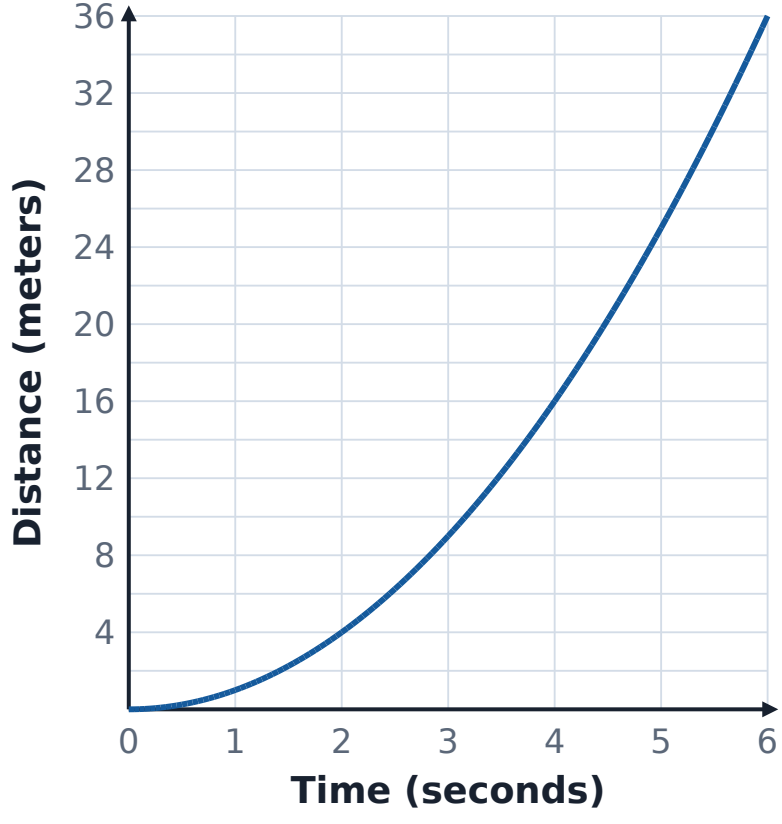
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

The graph shows the distance d, in meters, that a cart has rolled after t seconds.

Estimate the average rate of change from t = 1 to t = 3, with units.

Rate: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

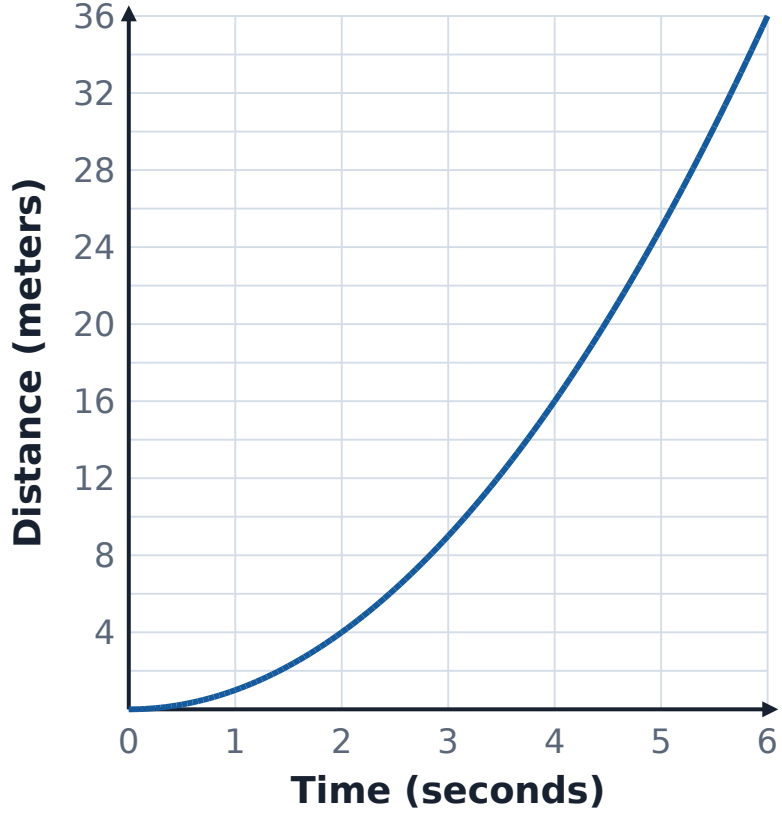
ID S79-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q1/5

The graph shows the distance d, in meters, that a cart has rolled after t seconds.

Estimate the average rate of change from t = 3 to t = 5. Is the cart speeding up?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

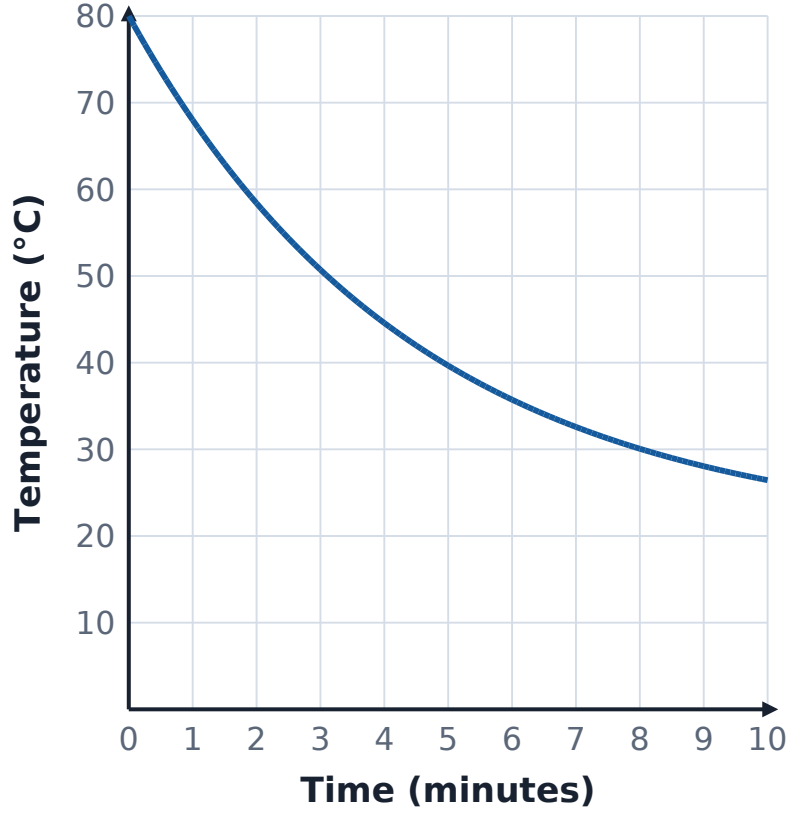
ID S79-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q2/5

The graph shows the temperature of a cup of coffee.

Estimate the average rate of change of the temperature over the first 5 minutes, with units.

Rate: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q3/5

The graph shows the temperature of a cup of coffee.

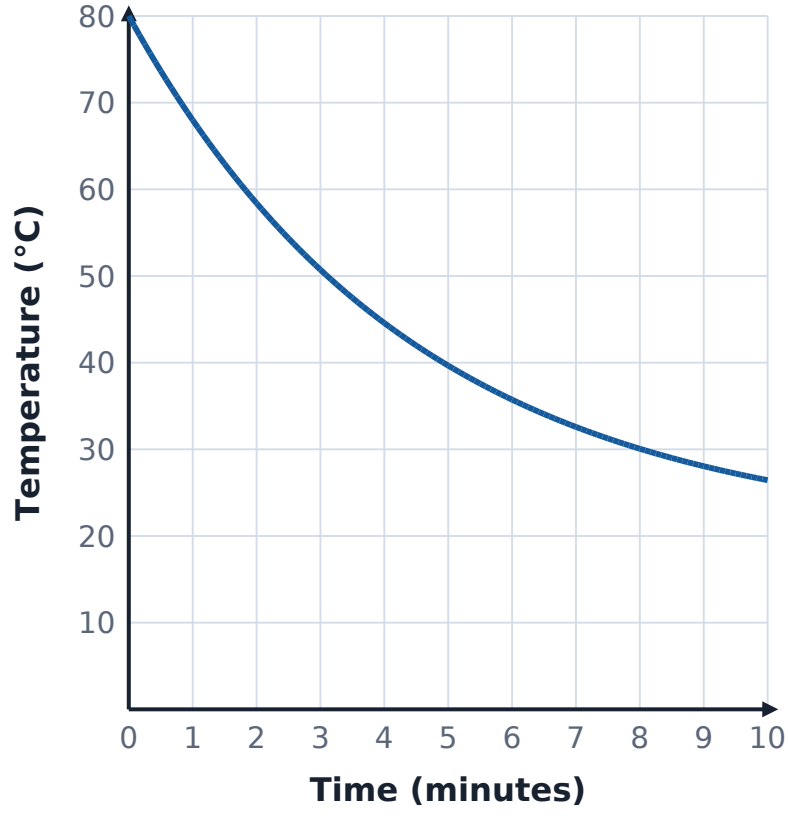
Over which 2-minute interval is the coffee cooling fastest on average?

A. 4 to 6 minutes

B. 8 to 10 minutes

C. 0 to 2 minutes

D. 6 to 8 minutes



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

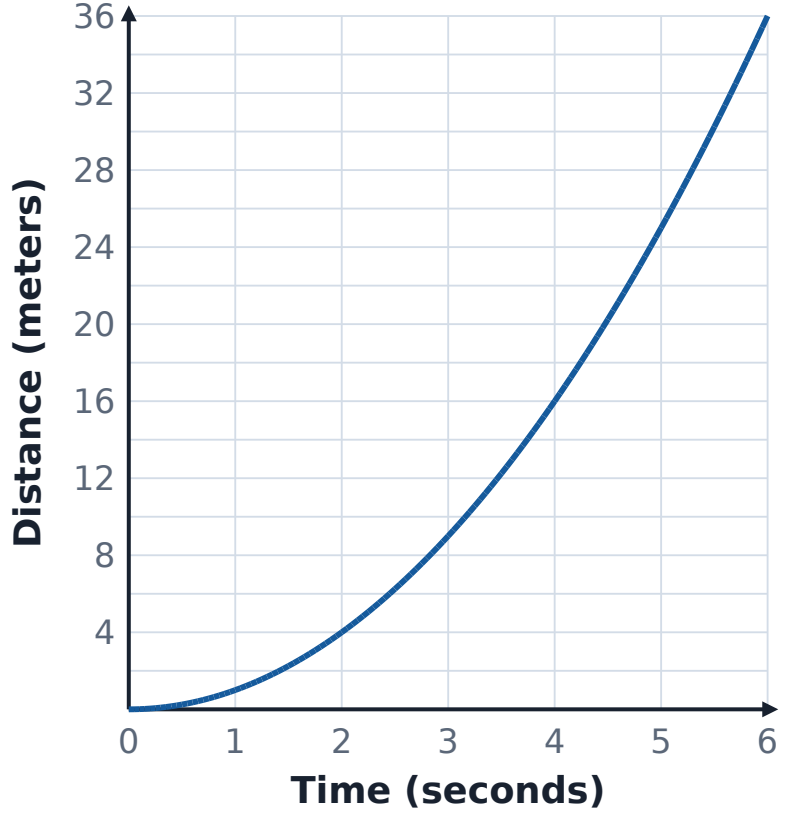
ID S79-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q4/5

True or false?

The average rate of change of d from t = 0 to t = 6 equals the slope of the segment joining (0, 0) and (6, 36) on the graph.

A. True **B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q5/5

SET 79 • HSF-IF.B.6

BACKWARD 1

Find a rate of change from two points • Grade 8 • Functions • 8.F.B.4

A line passes through (2, 10) and (6, 30). Find its rate of change.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q1/5

A straight-line graph of a hike shows 3 km at 1 hour and 9 km at 3 hours. What is the rate of change, with units?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q2/5

What is the slope of the line through (0, 50) and (5, 25)?

A. 5

B. -5

C. 0.2

D. -25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q3/5

True or false?

The slope between two points on a line is (change in y) ÷ (change in x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q4/5

True or false?

The line through (1, 4) and (3, 4) has rate of change 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSF-IF.B.6

BACKWARD 2

Read a unit rate from a graph • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The graph of a proportional relationship passes through (4, 12). What is the unit rate?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q1/5

A proportional cost graph passes through (0, 0) and (5, 7.5). What is the cost per item?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q2/5

A proportional graph of miles against gallons passes through (3, 90). What is the unit rate?

A. 3 miles per gallon

B. 270 miles per gallon

C. 90 miles per gallon

D. 30 miles per gallon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q3/5

True or false?

On a proportional graph, the point (1, r) shows the unit rate r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q4/5

True or false?

A proportional graph passes through (2, 5), so its unit rate is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q5/5

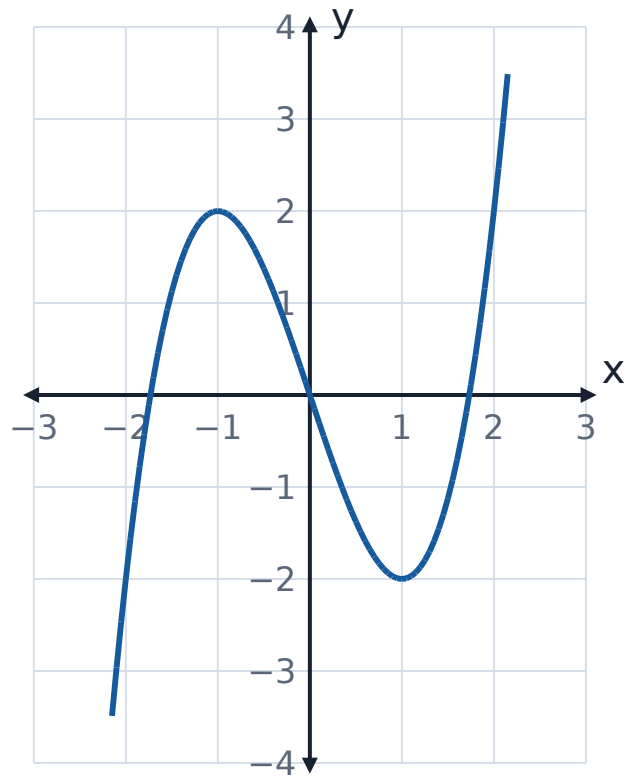
SET 79 • HSF-IF.B.6

FORWARD 1

Estimate average rates of change from a polynomial graph • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

The graph shows $f(x) = x^3 - 3x$.

Estimate the average rate of change from $x = -1$ to $x = 1$.



Rate: _____

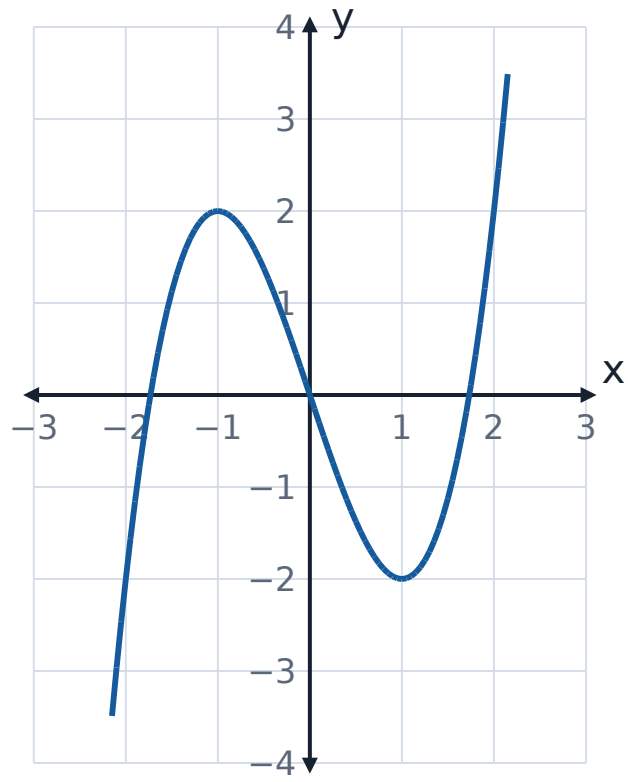
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q1/5

The graph shows $f(x) = x^3 - 3x$.

Estimate the average rate of change from $x = 1$ to $x = 2$.



Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q2/5

The graph shows $f(x) = x^3 - 3x$.

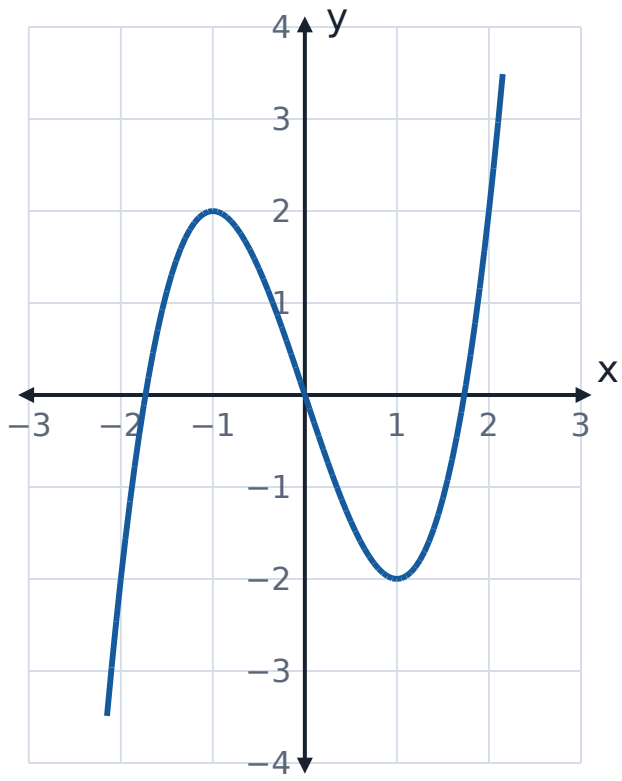
On which interval is f increasing?

A. $-1 < x < 1$

B. $1 < x < 2$

C. $-1 < x < 0$

D. $0 < x < 1$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q3

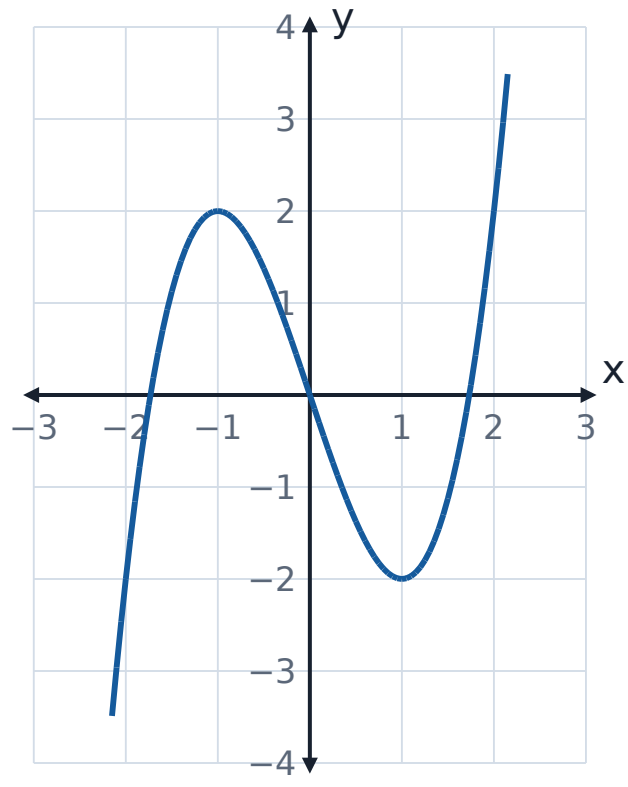
High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q3/5

True or false?

The average rate of change of $f(x) = x^3 - 3x$ from $x = -2$ to $x = 2$ is 1.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q4/5

True or false?

A positive average rate of change on an interval means the function increases at every point of the interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q5/5

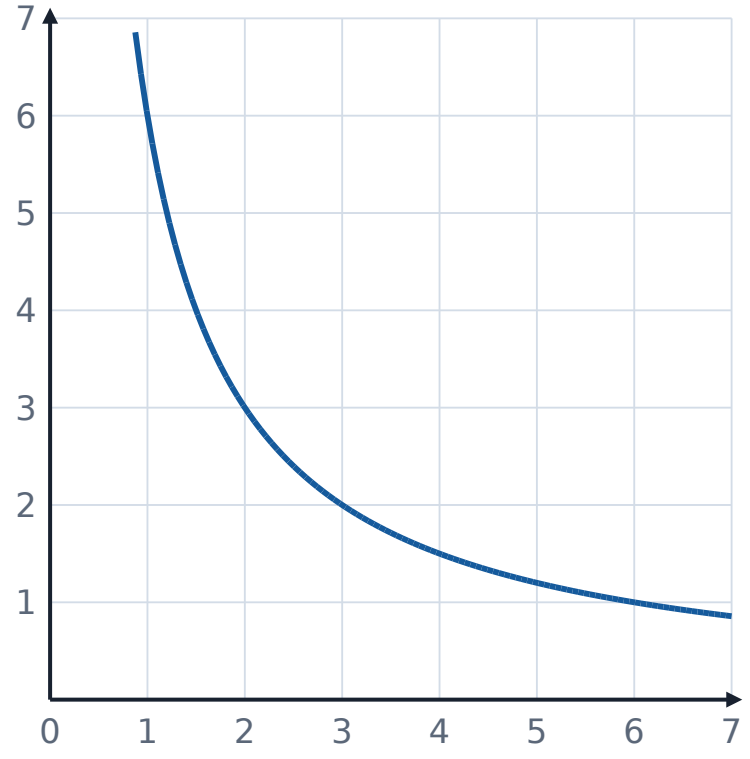
SET 79 • HSF-IF.B.6

FORWARD 2

Estimate average rates of change from a rational-function graph • High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d

The graph shows f(x) = 6 ÷ x.

Estimate the average rate of change from x = 1 to x = 3.



Rate: _____

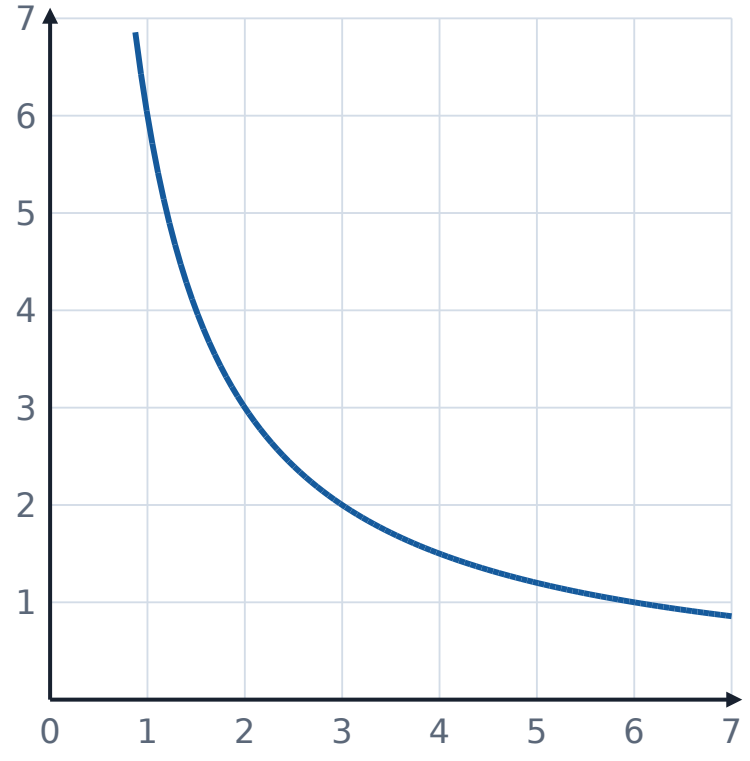
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q1/5

The graph shows f(x) = 6 ÷ x.

Estimate the average rate of change from x = 3 to x = 6.



Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q2/5

The graph shows f(x) = 6 ÷ x.

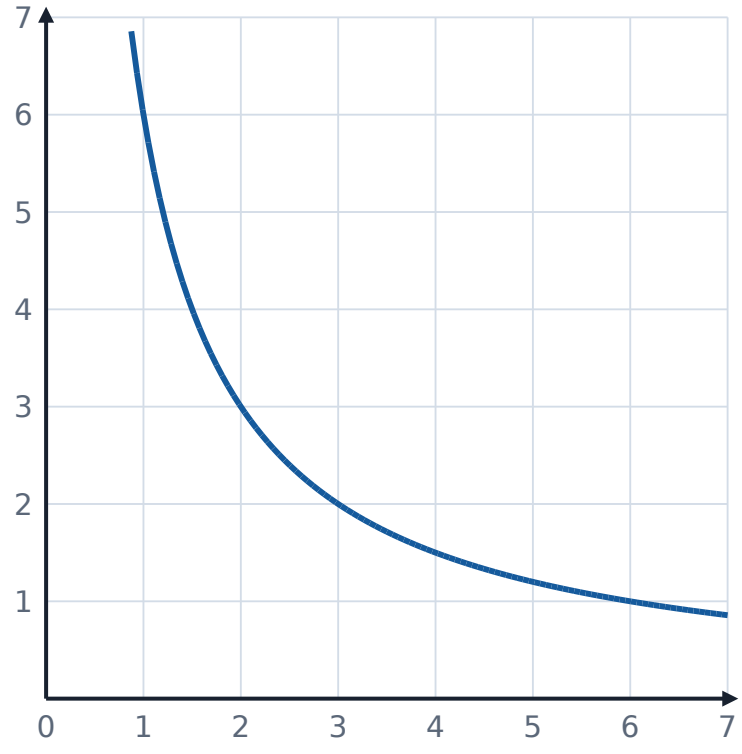
What happens to its average rate of change on intervals farther to the right?

A. It gets closer to 0.

B. It becomes positive.

C. It stays −2.

D. It gets more negative.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q3/5

True or false?

The graph of $y = 6 \div x$ has the horizontal asymptote $y = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q4/5

True or false?

The graph of $y = 6 \div x$ passes through (0, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q5/5

SET 41 • HSF-IF.C.7.a • INTERPRETING FUNCTIONS

Graph linear functions and show their intercepts

MAIN	Algebra I main question	**HSF-IF.C.7.a**
BACKWARD 1	Find slope and write y = mx + b	**8.EE.B.6**
BACKWARD 2	Write linear equations in two variables from a situation	**HSA-CED.A.2**
FORWARD 1	Find a point that partitions a segment in a given ratio	**HSG-GPE.B.6**
FORWARD 2	Find and verify inverses of linear functions	**HSF-BF.B.4.b**

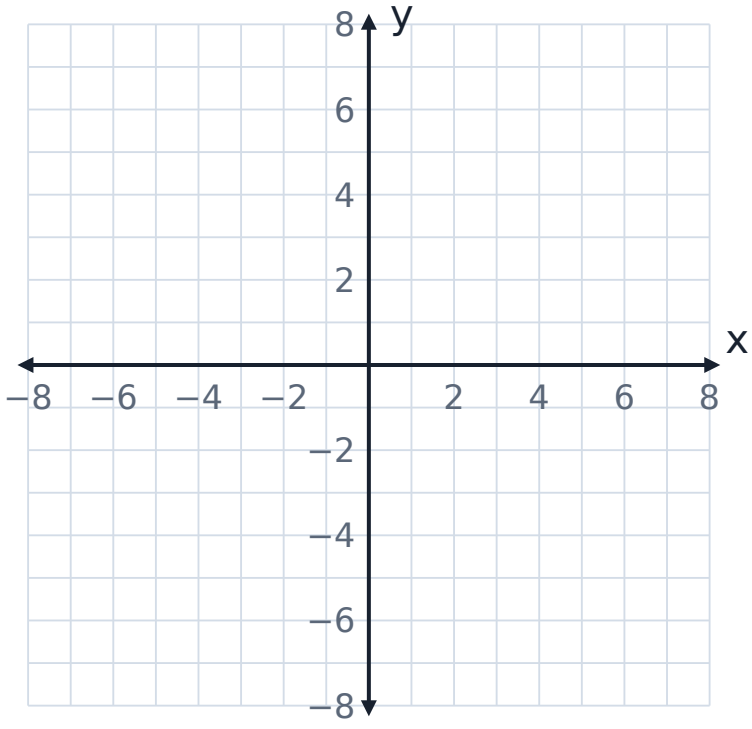
SET 41 • HSF-IF.C.7.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

Graph $f(x) = -\frac{2}{3}x + 4$ on the coordinate plane.

Give the x-intercept and the y-intercept.



Intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q1/5

Find the x- and y-intercepts of $3x - 5y = 15$.

Intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q2/5

Write the equation of the line through $(-2, 7)$ and $(4, -5)$ in slope-intercept form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q3/5

Which describes the graph of $y = -3$?

A. A line with slope 1 through $(-3, -3)$

B. A vertical line through $(-3, 0)$

C. A line with slope -3 through the origin

D. A horizontal line through $(0, -3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q4/5

True or false?

The graph of $2x + y = 6$ has x-intercept 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q5/5

SET 41 • HSF-IF.C.7.a

BACKWARD 1

Find slope and write y = mx + b • Grade 8 • Expressions & Equations • 8.EE.B.6

Find the slope of the line through (1, 2) and (3, 8).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q1/5

A line has slope 2 and y-intercept −1.

Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q2/5

What is the y-intercept of $y = 5x - 4$?

A. $\frac{4}{5}$

B. 4

C. 5

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q3/5

True or false?

Any two distinct points on a nonvertical line give the same slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q4/5

True or false?

The line $y = 4x$ passes through $(1, 5)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSF-IF.C.7.a

BACKWARD 2

Write linear equations in two variables from a situation • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

A gym charges $25 to join plus $15 per month.

Write an equation for the cost C after m months.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q1/5

A candle 20 cm tall burns 2 cm per hour.

Write an equation for its height h after t hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q2/5

Ana earns $8 per hour plus $5 in tips.

Which equation gives her earnings E for h hours?

A. E = 8h + 5

B. E = 5h + 8

C. E = 13h

D. E = 8(h + 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q3/5

True or false?

d = 60t gives the distance traveled at 60 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q4/5

True or false?

$y = 100 - 4x$ describes a quantity that grows by 4 each step.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSF-IF.C.7.a

FORWARD 1

Find a point that partitions a segment in a given ratio • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the midpoint of the segment from $(-2, 3)$ to $(6, 7)$.

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q1/5

Find the point $\frac{1}{4}$ of the way from A(0, 0) to B(8, 12).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q2/5

Point P divides the segment from A(1, 2) to B(7, 11) in the ratio 1 : 2 (AP : PB).

Find P.

A. (5, 8)

B. (3, 5)

C. (4, 6.5)

D. (2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q3/5

True or false?

The midpoint of (0, 0) and (10, 4) is (5, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q4/5

True or false?

The point $\frac{1}{3}$ of the way from (0, 0) to (9, 6) is (6, 4).

A. True

B. False

SET 41 • HSF-IF.C.7.a

FORWARD 2

Find and verify inverses of linear functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

Find the inverse of $f(x) = 3x - 6$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q1/5

Show that f(x) = 2x + 1 and g(x) = (x − 1) ÷ 2 are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q2/5

Which is the inverse of $f(x) = x \div 4 + 1$?

A. $f^{-1}(x) = 4x + 1$

B. $f^{-1}(x) = 4(x + 1)$

C. $f^{-1}(x) = 4x - 4$

D. $f^{-1}(x) = x \div 4 - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q3/5

True or false?

If $f(g(x)) = x$ and $g(f(x)) = x$ for every x, then f and g are inverses.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q4/5

True or false?

The inverse of $f(x) = x + 5$ is $f^{-1}(x) = x + 5$.

A. True

B. False

SET 42 • HSF-IF.C.7.a • INTERPRETING FUNCTIONS

Graph quadratic functions and show intercepts, maxima, and minima

MAIN	Algebra I main question	**HSF-IF.C.7.a**
BACKWARD 1	Factor to find the zeros of a quadratic	**HSA-SSE.B.3.a**
BACKWARD 2	Interpret key features of a function	**HSF-IF.B.4**
FORWARD 1	Relate a parabola's equation to its focus and directrix	**HSG-GPE.A.2**
FORWARD 2	Use factors to describe the graph of a polynomial	**HSF-IF.C.7.c**

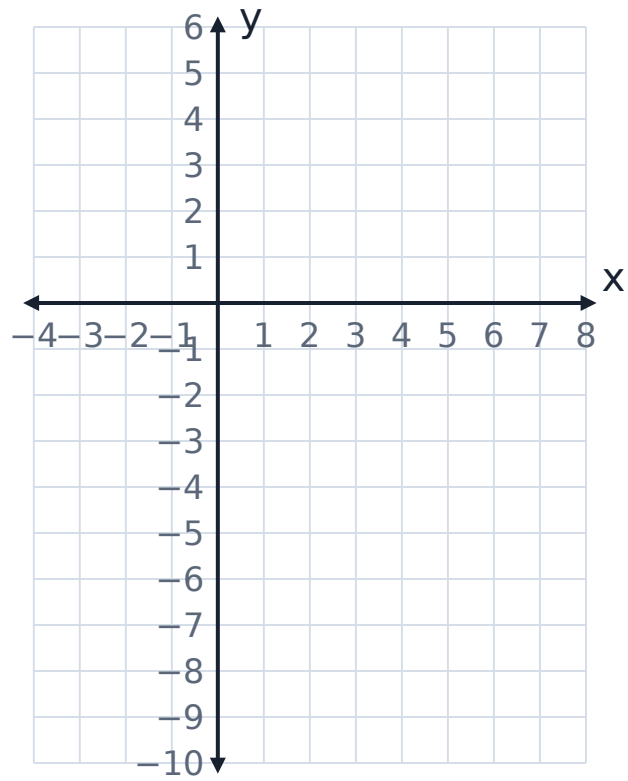
SET 42 • HSF-IF.C.7.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

Graph $f(x) = x^2 - 4x - 5$.

Give the vertex, the x-intercepts, and the y-intercept.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q1/5

Find the vertex and the axis of symmetry of $y = 2x^2 + 8x + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q2/5

Does $y = -3x^2 + 6x - 1$ have a maximum or a minimum? Find it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q3/5

What are the x-intercepts of $y = (x + 3)(x - 1)$?

A. (3, 0) and (−1, 0)

B. (−3, 0) and (1, 0)

C. (−3, 0) and (−1, 0)

D. (0, −3) only

True or false?

The graph of $y = x^2 + 4$ has no x-intercepts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q5/5

SET 42 • HSF-IF.C.7.a

BACKWARD 1

Factor to find the zeros of a quadratic • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

Factor $x^2 - 4x - 5$, and give its zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q1/5

Find the zeros of $2x^2 - 8$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q2/5

What are the zeros of $x^2 + 6x + 8$?

A. 2 and −4

B. −2 and −4

C. 2 and 4

D. −2 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q3/5

True or false?

The zeros of $(x - 7)(x + 2)$ are 7 and -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q4/5

True or false?

The zeros of $x^2 - 9x$ are 0 and -9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSF-IF.C.7.a

BACKWARD 2

Interpret key features of a function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

A table has x = −2, −1, 0, 1, 2 and y = 5, 2, 1, 2, 5.

Describe its symmetry and its minimum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q1/5

A profit function P(x) is positive for $10 < x < 50$, where x is the number of items sold.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q2/5

The highest point of a graph is (3, 40).

Which statement describes it?

A. A maximum of 40 at x = 3

B. A maximum of 3 at x = 40

C. A minimum of 3 at x = 40

D. A y-intercept of 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q3/5

True or false?

A function is increasing on an interval when its outputs get larger as its inputs get larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q4/5

True or false?

The x-intercepts of a graph are where the output is greatest.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSF-IF.C.7.a

FORWARD 1

Relate a parabola's equation to its focus and directrix • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

Find the focus and directrix of $y = \frac{1}{8}x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q1/5

Derive the equation of the parabola with focus $(0, -1)$ and directrix $y = 1$ from the distance definition.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q2/5

A parabola has focus (0, 3) and directrix $y = -3$.

Where is its vertex?

A. (0, 0)

B. (0, 3)

C. (0, −3)

D. (3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q3/5

True or false?

The parabola $x^2 = 12y$ opens upward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q4/5

True or false?

The parabola with focus (0, 1) and directrix $y = -1$ is $y = x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSF-IF.C.7.a

FORWARD 2

Use factors to describe the graph of a polynomial • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

$f(x) = (x + 1)(x - 2)(x - 4)$.

Find the zeros and the y-intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q1/5

At most how many turning points can the graph of a degree-3 polynomial have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q2/5

$f(x) = (x - 3)^2(x + 1)$.

What does the graph do at $x = 3$?

A. It touches the x-axis and turns around.

B. It is undefined.

C. It has a vertical asymptote.

D. It crosses the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^3 - 8$ has exactly one real zero.

A. True

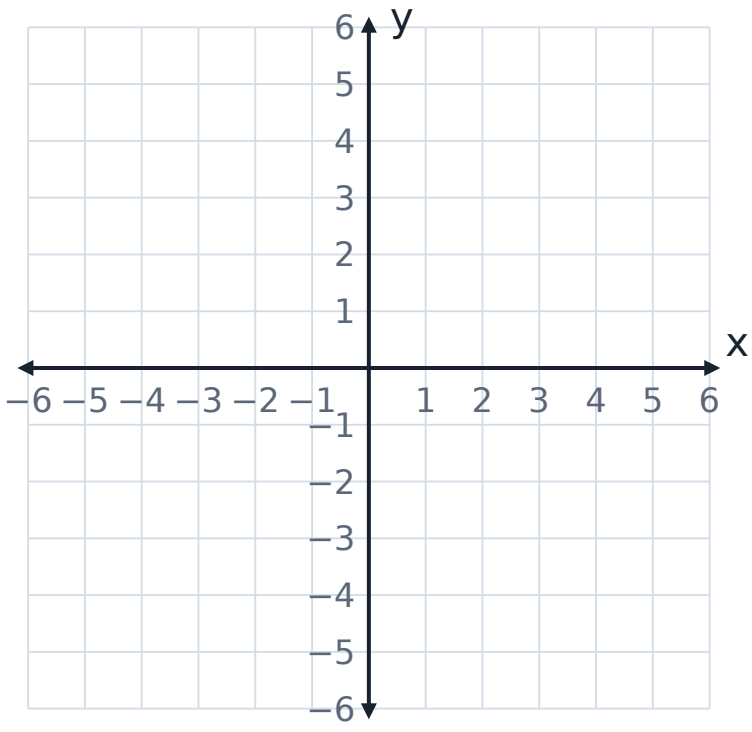
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q4/5

Graph $f(x) = (x + 2)(x - 1)^2$. Mark the zeros and the y-intercept, and show what the graph does at each zero. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q5/5

SET 43 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph absolute value functions and show their key features

MAIN	Algebra I main question	**HSF-IF.C.7.b**
BACKWARD 1	Find the domain and range of a function	**HSF-IF.A.1**
BACKWARD 2	Understand absolute value as distance from 0	**6.NS.C.7.c**
FORWARD 1	Solve radical equations and check for extraneous solutions	**HSA-REI.A.2**
FORWARD 2	Read values of inverses of power and root functions	**HSF-BF.B.4.c**

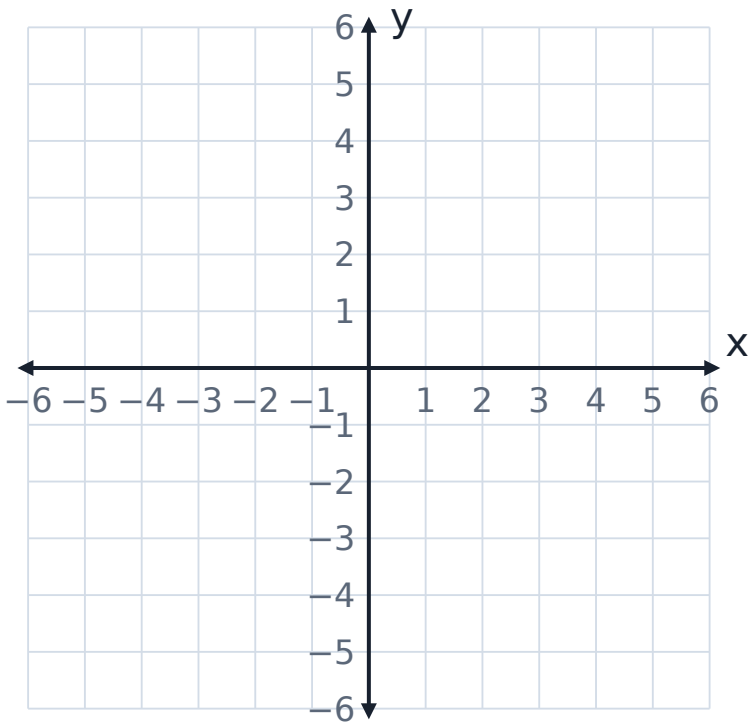
SET 43 • HSF-IF.C.7.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

Graph $f(x) = |x - 2| - 3$.

Give the vertex and the x-intercepts.



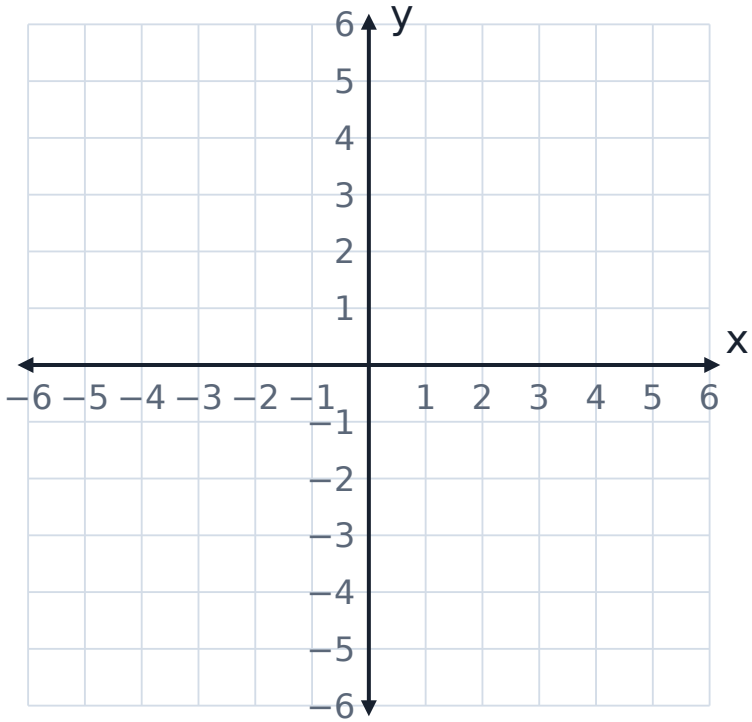
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q1/5

Graph $g(x) = -2|x + 1| + 4$, and label its vertex and its x-intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q2/5

Describe how the graph of $y = |x + 3| + 1$ is related to the graph of $y = |x|$, and give its range.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q3/5

Which function's graph is a V opening downward with vertex (2, 5)?

A. $y = |x - 2| + 5$

B. $y = -|x + 2| + 5$

C. $y = -|x - 5| + 2$

D. $y = -|x - 2| + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q4/5

True or false?

The graph of $y = 3|x|$ is narrower than the graph of $y = |x|$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q5/5

SET 43 • HSF-IF.C.7.b

BACKWARD 1

Find the domain and range of a function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1

Over the real numbers, what is the domain of $f(x) = \sqrt{(x - 5)}$?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q1/5

What is the range of f(x) = |x|?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q2/5

Which value is NOT in the domain of $g(x) = \sqrt{(6 - x)}$?

A. 7

B. -3

C. 0

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q3/5

True or false?

Every real number is in the domain of $f(x) = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q4/5

True or false?

The range of $f(x) = x^2$ is all real numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSF-IF.C.7.b

BACKWARD 2

Understand absolute value as distance from 0 • Grade 6 • The Number System • 6.NS.C.7.c

What is $|-9|$?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q1/5

Which is farther from 0, −7 or 5? Explain with absolute value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q2/5

Which value equals $|3 - 8|$?

A. 5

B. -5

C. 11

D. -11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q3/5

True or false?

$|-4| = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q4/5

True or false?

$|0| = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSF-IF.C.7.b

FORWARD 1

Solve radical equations and check for extraneous solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $\sqrt{(x - 2)} = 4$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q1/5

Solve $\sqrt{(x + 6)} = x$. Name any extraneous solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q2/5

Solve $\sqrt[3]{(x + 1)} = 2$.

A. $x = 7$

B. $x = 3$

C. $x = 9$

D. $x = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q3/5

True or false?

$\sqrt{x} + 3 = 0$ has no real solution.

A. True

B. False

True or false?

$x = 16$ is a solution of $\sqrt{x} = -4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSF-IF.C.7.b

FORWARD 2

Read values of inverses of power and root functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

$g(x) = x^3$.

Find $g^{-1}(27)$.

$g^{-1}(27) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q1/5

The point (4, 2) is on the graph of $f(x) = \sqrt{x}$.

Which point must be on the graph of f^{-1}?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q2/5

$f(x) = x^2$ with domain $x \geq 0$.

What is $f^{-1}(49)$?

A. -7

B. 24.5

C. 2,401

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q3/5

True or false?

The inverse of $f(x) = x^3$ is $f^{-1}(x) = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q4/5

True or false?

f is a one-to-one function. If (3, 10) is on the graph of f, then (3, 10) must be on the graph of f^{-1}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph square root and cube root functions

MAIN	Algebra I main question	**HSF-IF.C.7.b**
BACKWARD 1	Find domains and ranges of root functions	**HSF-IF.A.1**
BACKWARD 2	Evaluate square roots and cube roots	**8.EE.A.2**
FORWARD 1	Solve radical equations and check the solutions	**HSA-REI.A.2**
FORWARD 2	Find inverses of square root and cube root functions	**HSF-BF.B.4.c**

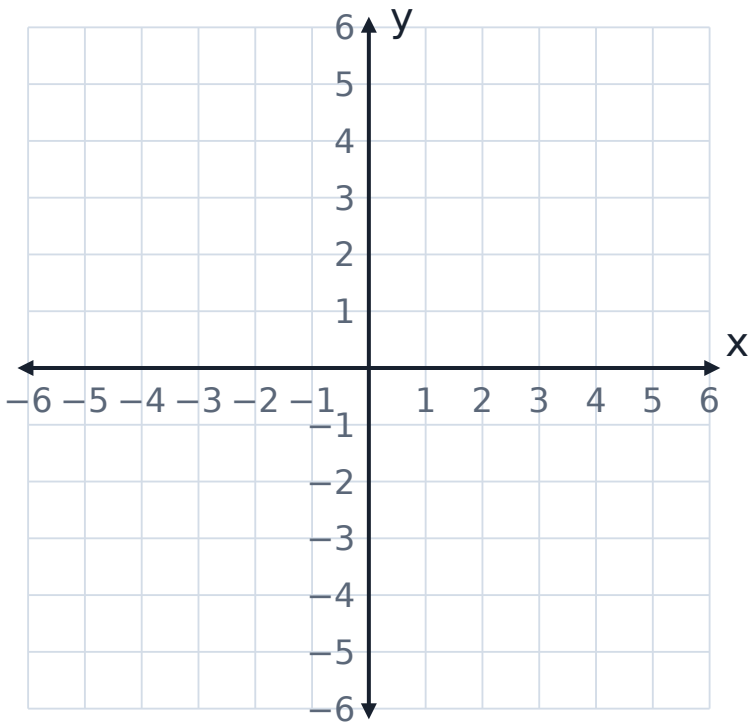
SET 74 • HSF-IF.C.7.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

Graph $f(x) = \sqrt{(x + 4)}$.

Give its domain and range.



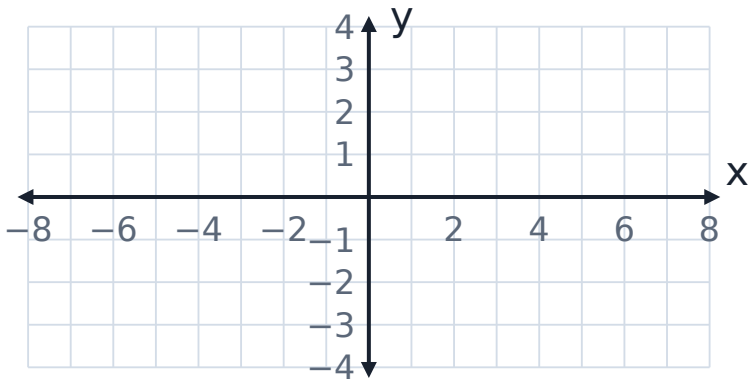
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q1/5

Graph $g(x) = \sqrt[3]{x}$ for $-8 \le x \le 8$. Plot the points where x is a perfect cube, then draw the curve.



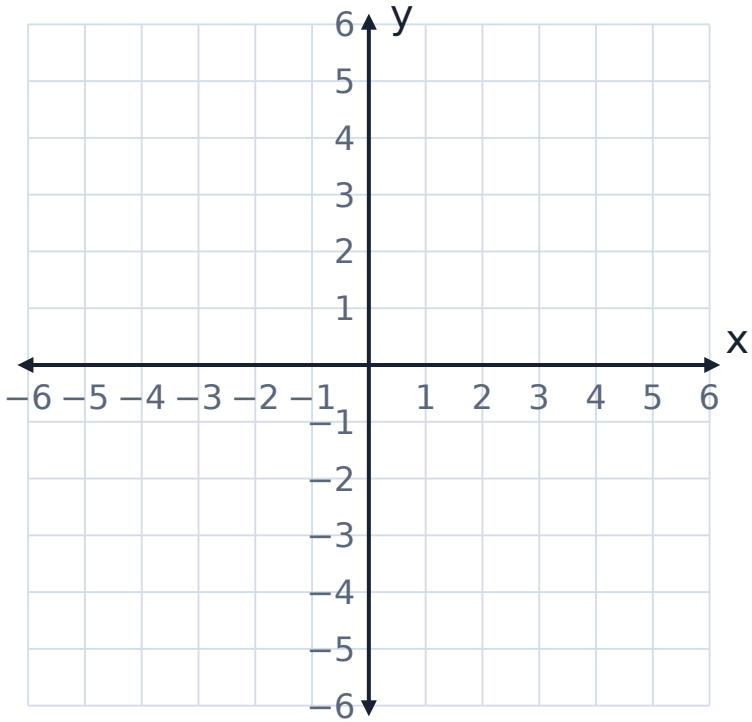
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q2/5

Graph $h(x) = -\sqrt{x} + 3$.

Give its starting point and its range.



Answer: _____

Which describes the graph of $y = \sqrt[3]{x}$?

A. It starts at (0, 0) and goes only to the right.

B. It has the horizontal asymptote $y = 0$.

C. It passes through $(-8, -2)$, (0, 0), and (8, 2).

D. It is V-shaped with vertex (0, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q4/5

True or false?

The graph of $y = \sqrt{(x - 1)}$ starts at $(-1, 0)$.

A. True

B. False

SET 74 • HSF-IF.C.7.b

BACKWARD 1

Find domains and ranges of root functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1

Over the real numbers, what is the domain of $g(x) = \sqrt{(2x - 6)}$?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q1/5

What is the range of $h(x) = \sqrt{x} + 2$?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q2/5

What is the domain of $y = \sqrt[3]{(x - 5)}$?

A. All real numbers

B. $x \leq 5$

C. $x > 5$

D. $x \geq 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q3/5

True or false?

The range of $y = -\sqrt{x}$ is $y \leq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q4/5

True or false?

$x = -1$ is in the domain of $y = \sqrt{x}$ over the real numbers.

A. True

B. False

SET 74 • HSF-IF.C.7.b

BACKWARD 2

Evaluate square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Evaluate $\sqrt{81}$ and $\sqrt[3]{(-27)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q1/5

Solve $x^3 = 125$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q2/5

What is $\sqrt[3]{8} + \sqrt{4}$?

A. 6

B. 12

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{0.25} = 0.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q4/5

True or false?

$\sqrt[3]{(-8)}$ is not a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.b

FORWARD 1

Solve radical equations and check the solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $\sqrt{(x + 4)} = 3$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q1/5

Solve $\sqrt[3]{(2x)} = -2$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q2/5

Solve $\sqrt{(x + 2)} = x$.

A. $x = -1$ only

B. $x = 2$ or $x = -1$

C. No solution

D. $x = 2$ only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q3/5

True or false?

$\sqrt{(x - 3)} = -2$ has no real solution.

A. True

B. False

True or false?

Cubing both sides of $\sqrt[3]{x} = 4$ gives $x = 12$.

A. True

B. False

SET 74 • HSF-IF.C.7.b

FORWARD 2

Find inverses of square root and cube root functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

$f(x) = \sqrt{x}$ for $x \geq 0$. Find $f^{-1}(5)$.

$f^{-1}(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q1/5

$g(x) = \sqrt[3]{x}$. Find $g^{-1}(-2)$.

$g^{-1}(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q2/5

Which is the inverse of $f(x) = \sqrt{(x - 1)}$, for $x \geq 1$?

A. $f^{-1}(x) = \sqrt{(x + 1)}$

B. $f^{-1}(x) = x^2 + 1$, for $x \geq 0$

C. $f^{-1}(x) = x^2 - 1$

D. $f^{-1}(x) = (x + 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q3/5

True or false?

The graph of $y = \sqrt{x}$ $(x \geq 0)$ is the reflection of the graph of $y = x^2$ $(x \geq 0)$ across the line $y = x$.

A. True

B. False

True or false?

$g(x) = \sqrt[3]{x}$ has no inverse function.

A. True

B. False

SET 75 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph piecewise-defined and step functions

MAIN	Algebra I main question	**HSF-IF.C.7.b**
BACKWARD 1	Evaluate piecewise-defined functions	**HSF-IF.A.2**
BACKWARD 2	Order numbers and describe intervals on a number line	**6.NS.C.7.a**
FORWARD 1	Graph the probability distribution of a discrete random variable —..	**HSS-MD.A.1**
FORWARD 2	Read inverse values of piecewise and step functions	**HSF-BF.B.4.c**

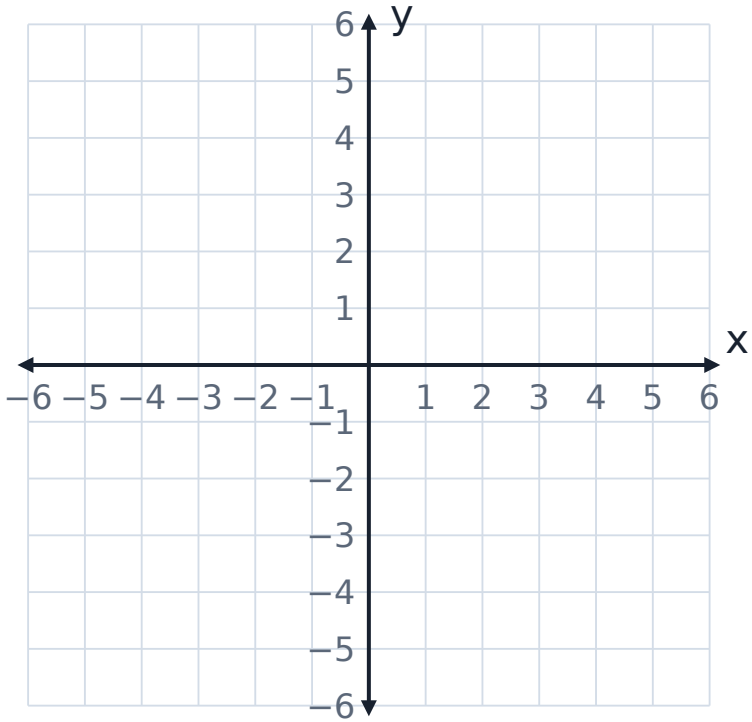
SET 75 • HSF-IF.C.7.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

$f(x) = x + 1$ for $x < 2$, and
$f(x) = 5 - x$ for $x \geq 2$.

Graph f. Then find f(0), f(2), and f(4).



Answer: _____

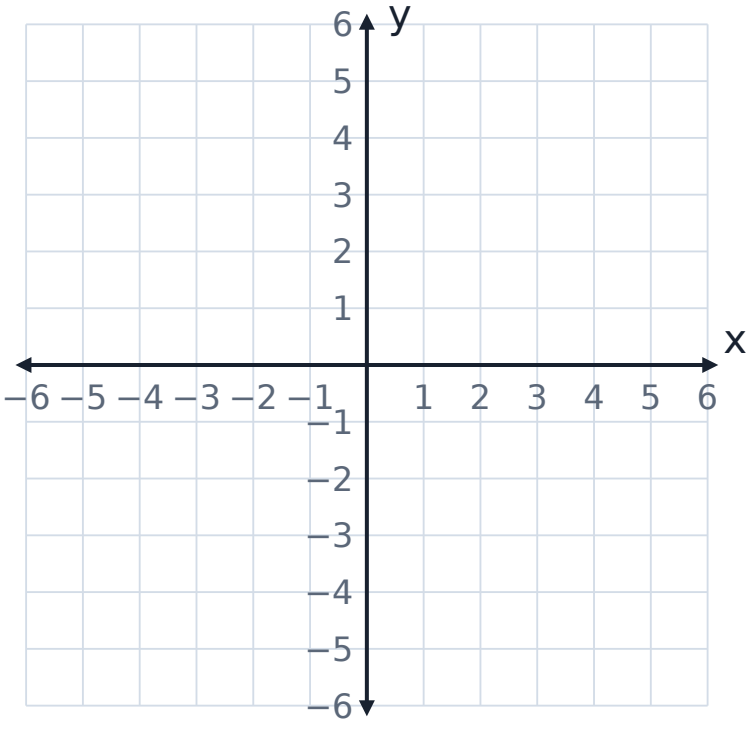
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q1/5

$g(x) = -1$ for $x < 0$, $g(x) = 2$ for $0 \leq x < 3$, and $g(x) = x - 1$ for $x \geq 3$.

Graph g. Use open and closed dots at the breaks.

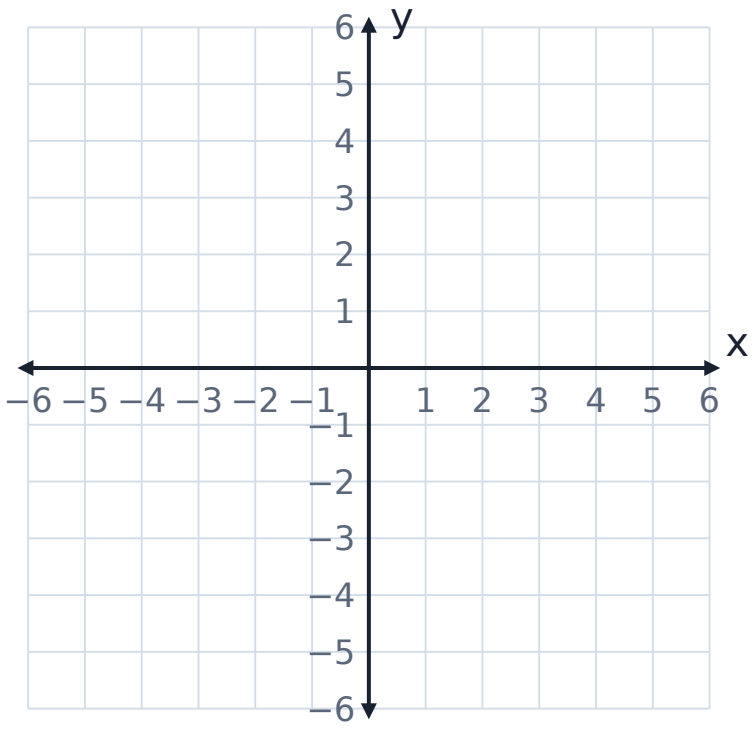


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q2/5

Graph the greatest integer function $y = \lfloor x \rfloor$ (the greatest integer less than or equal to x) for $-2 \leq x < 3$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q3/5

A parking garage charges $3 for each hour or part of an hour.

What is the charge for 2.25 hours?

A. $6

B. $6.75

C. $7.50

D. $9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q4/5

True or false?

For the greatest integer function, $\lfloor 2.7 \rfloor = 3$.

A. True

B. False

SET 75 • HSF-IF.C.7.b

BACKWARD 1

Evaluate piecewise-defined functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$f(x) = 2x - 1$ for $x < 3$, and $f(x) = 8$ for $x \geq 3$.

Find $f(3)$ and $f(1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q1/5

$g(x) = |x|$ for $x \leq 0$, and $g(x) = x^2$ for $x > 0$.

Find $g(-4)$ and $g(4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q2/5

$h(x) = 3$ for $x < 1$, and $h(x) = 2x$ for $x \geq 1$. What is $h(1)$?

A. 2

B. 5

C. 3

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q3/5

True or false?

$p(x) = 5$ for $x \leq 2$, and $p(x) = x$ for $x > 2$. Then $p(2) = 5$.

A. True

B. False

True or false?

$\lfloor -1.5 \rfloor = -1$

A. True

B. False

SET 75 • HSF-IF.C.7.b

BACKWARD 2

Order numbers and describe intervals on a number line • Grade 6 • The Number System • 6.NS.C.7.a

Is −3 to the left or to the right of −1 on a number line? Which number is greater?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q1/5

Write an inequality for "x is at least 0 and less than 3".

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q2/5

Which statement is true?

A. $-4 > -2$

B. $-4 < -2$

C. $0 < -1$

D. $-2 < -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q3/5

True or false?

-5 is less than -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q4/5

True or false?

On a number line, 2 is to the left of −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q5/5

FORWARD 1

Graph the probability distribution of a discrete random variable • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: A probability distribution assigns a value to each separate outcome, like a function defined piece by piece, and its bar graph is read the same way.

A fair coin is flipped twice, and X is the number of heads. Give P(X = 0), P(X = 1), and P(X = 2).

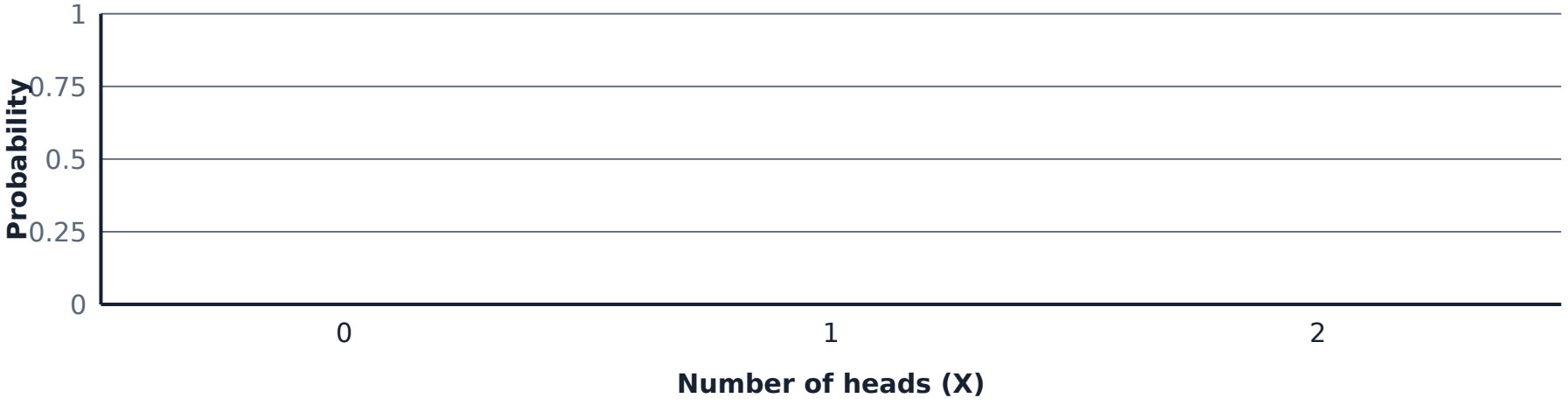
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q1/5 • NEAREST RELATED

A fair coin is flipped twice, and X is the number of heads. Graph the probability distribution of X as a bar graph.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions HSS-MD.A.1 • Set 75 • FORWARD 1 • Q2/5 • NEAREST RELATED

X is the result of one roll of a fair number cube numbered 1 to 6. What is $P(X \geq 5)$?

A. $\frac{5}{6}$

B. $\frac{1}{6}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

The probabilities of a probability distribution add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

For X = the number of heads in 2 fair coin flips, $P(X = 1) = \frac{1}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 75 • HSF-IF.C.7.b

FORWARD 2

Read inverse values of piecewise and step functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

$f(x) = 2x$ for $x < 0$, and $f(x) = x^2$ for $x \geq 0$. This f is one-to-one. Find $f^{-1}(9)$ and $f^{-1}(-4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q1/5

$g(x) = x + 1$ for $x < 2$, and $g(x) = 2x - 1$ for $x \geq 2$. This g is one-to-one. Find $g^{-1}(7)$.

$g^{-1}(7) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q2/5

Why does the greatest integer function $y = \lfloor x \rfloor$ have no inverse function?

A. Many inputs share each output, such as $\lfloor 2.1 \rfloor = \lfloor 2.9 \rfloor = 2$.

B. It is a linear function.

C. It is not defined for negative x.

D. Its outputs are not integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q3/5

True or false?

A piecewise function whose pieces are increasing and whose output ranges do not overlap can have an inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q4/5

True or false?

If f is one-to-one and $f(3) = 5$, then $f^{-1}(3) = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q5/5

SET 44 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph exponential functions and show intercepts and end behavior

MAIN	Algebra I main question	**HSF-IF.C.7.e**
BACKWARD 1	Evaluate integer exponents	**8.EE.A.1**
BACKWARD 2	Evaluate exponential functions	**HSF-IF.A.2**
FORWARD 1	Solve exponential equations with logarithms	**HSF-LE.A.4**
FORWARD 2	Understand logarithms as inverses of exponential functions	**HSF-BF.B.5**

SET 44 • HSF-IF.C.7.e

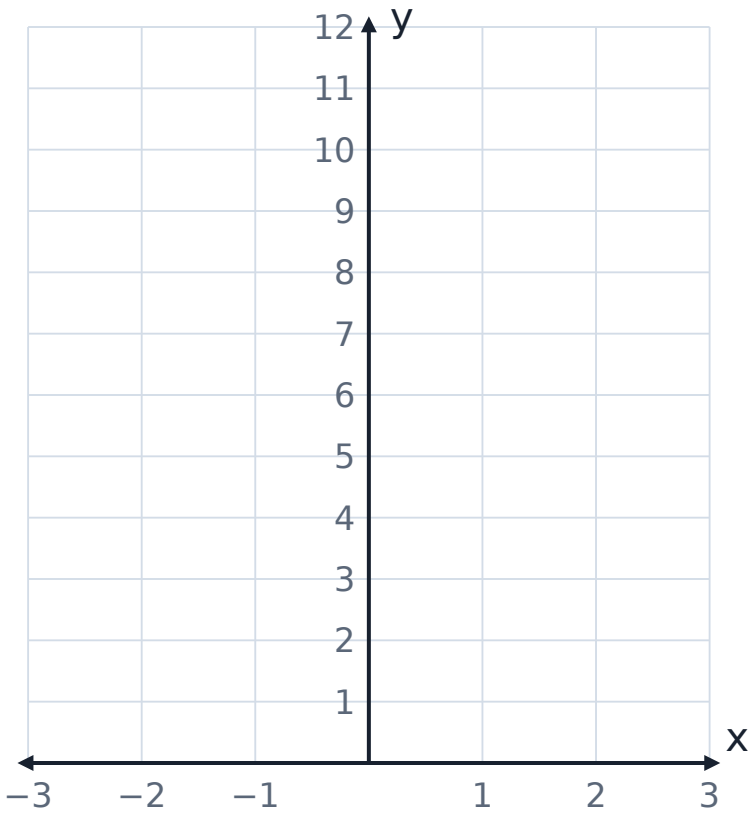
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

Graph $f(x) = 3 \cdot 2^x$ for $-2 \leq x \leq 2$.

Give its y-intercept and its horizontal asymptote.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q1/5

$g(x) = 50(0.8)^x$.

Is this growth or decay? Give its y-intercept, and say what happens to g(x) as x gets large.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q2/5

$h(x) = 2^x - 4$.

Find the y-intercept, the x-intercept, and the horizontal asymptote.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q3/5

Which function's graph rises from left to right and passes through (0, 5)?

A. $y = 1.2(5)^x$

B. $y = 0.8(5)^x$

C. $y = 5(0.8)^x$

D. $y = 5(1.2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q4/5

True or false?

The graph of $y = 4(0.5)^x$ crosses the x-axis.

A. True

B. False

SET 44 • HSF-IF.C.7.e

BACKWARD 1

Evaluate integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Evaluate 2^{-2}.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q1/5

Evaluate $3 \cdot 2^{-1}$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q2/5

Which number equals $(0.5)^{-1}$?

A. -0.5

B. 0.5

C. 2

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q3/5

True or false?

$5^0 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q4/5

True or false?

$4^{-1} = \frac{1}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSF-IF.C.7.e

BACKWARD 2

Evaluate exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$f(x) = 50(0.8)^x$.

Find f(2).

f(2) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q1/5

$g(x) = 2^x - 4$.

Find $g(3)$.

$g(3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q2/5

$h(x) = 3 \cdot 2^x$.

Find $h(-1)$.

A. -1.5

B. 6

C. -6

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q3/5

True or false?

If $f(x) = 10^x$, then $f(3) = 1{,}000$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q4/5

True or false?

If $f(x) = 3^x$, then $f(2) = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSF-IF.C.7.e

FORWARD 1

Solve exponential equations with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $3 \cdot 2^x = 48$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q1/5

Solve $50(0.8)^x = 20$. Round to the nearest tenth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q2/5

Solve $2^x = 12$. (log means base 10.)

A. $x = \log 6$

B. $x = \log 12 \div \log 2 \approx 3.58$

C. $x = 3.5$

D. $x = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q3/5

True or false?

log 1 = 0 (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q4/5

True or false?

log 100 = 2 (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSF-IF.C.7.e

FORWARD 2

Understand logarithms as inverses of exponential functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

Write the inverse of $f(x) = 2^x$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q1/5

Explain why $\log_3 81 = 4$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q2/5

The graph of $y = \log_2 x$ is the reflection of the graph of $y = 2^x$ across which line?

A. $x = 0$

B. $y = 0$

C. $y = x$

D. $y = -x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q3/5

True or false?

The domain of $y = \log x$ is $x > 0$. (log means base 10.)

A. True

B. False

True or false?

$\log_2(2^5) = 2$

A. True

B. False

SET 45 • HSF-IF.C.8.a • INTERPRETING FUNCTIONS

Factor and complete the square to show zeros, extreme values, and symmetry, and interpret them

MAIN	Algebra I main question	**HSF-IF.C.8.a**
BACKWARD 1	Complete the square	**HSA-SSE.B.3.b**
BACKWARD 2	Use the greatest common factor	**6.NS.B.4**
FORWARD 1	Prove polynomial identities	**HSA-APR.C.4**
FORWARD 2	Complete the square to find the center of a hyperbola	**HSG-GPE.A.3**

SET 45 • HSF-IF.C.8.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a

$f(x) = x^2 - 8x + 12$.

Write f in factored form and in vertex form. Give the zeros, the vertex, and the axis of symmetry.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q1/5

A company's profit is $P(x) = -2x^2 + 120x - 1{,}000$ dollars when it charges x dollars.

Complete the square to find the price that gives the most profit, and the greatest profit.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q2/5

A rock's height is $h(t) = -16t^2 + 32t + 48$ feet after t seconds.

Factor to find when it hits the ground.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q3/5

Which form of $f(x) = x^2 + 6x + 5$ shows the vertex directly?

A. $(x + 1)(x + 5)$

B. $x^2 + 6x + 5$

C. $(x + 3)^2 - 4$

D. $x(x + 6) + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q4/5

True or false?

The axis of symmetry of $y = (x - 1)(x - 7)$ is $x = 4$.

A. True

B. False

SET 45 • HSF-IF.C.8.a

BACKWARD 1

Complete the square • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Complete the square.

$x^2 - 8x + 12$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q1/5

Complete the square.

$x^2 + 3x$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q2/5

Which expression equals $-(x^2 - 6x)$?

A. $-(x - 3)^2 - 9$

B. $-(x - 3)^2 + 9$

C. $(x - 3)^2 - 9$

D. $-(x + 3)^2 + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q3/5

True or false?

$$x^2 + 2x + 10 = (x + 1)^2 + 9$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q4/5

True or false?

$(x - 5)^2 - 3$ has its minimum at $x = -5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSF-IF.C.8.a

BACKWARD 2

Use the greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Use the greatest common factor to write 16 + 40 as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q1/5

Find the greatest common factor of 32 and 48.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q2/5

Which expression equals 36 + 27 and uses the greatest common factor?

A. 9(4 + 3)

B. 3(12 + 6)

C. 6(6 + 27)

D. 9(4 + 27)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q3/5

True or false?

The greatest common factor of 12 and 18 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q4/5

True or false?

$20 + 50 = 5(4 + 5)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSF-IF.C.8.a

FORWARD 1

Prove polynomial identities • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Show that $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ by expanding both sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q1/5

Use $x = 2$ and $y = 1$ in $x^2 - y^2$, $2xy$, and $x^2 + y^2$ to make a Pythagorean triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q2/5

Which equation is an identity (true for all numbers a and b)?

A. $a^3 + b^3 = (a + b)^3$

B. $a^3 - b^3 = (a - b)^3$

C. $a^2 + b^2 = (a + b)^2$

D. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q3/5

True or false?

$$(a + b)^2 - (a - b)^2 = 4ab$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q4/5

True or false?

$(x + 1)^3 = x^3 + 1$

A. True

B. False

SET 45 • HSF-IF.C.8.a

FORWARD 2

Complete the square to find the center of a hyperbola • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Complete the square to find the center of the hyperbola $x^2 - 4y^2 - 6x - 8y - 11 = 0$.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q1/5

The hyperbola $x^2 \div 16 - y^2 \div 9 = 1$ has foci $(5, 0)$ and $(-5, 0)$.

Show that $(5, 2.25)$ is on the hyperbola and that the absolute difference of its distances to the foci is 8.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q2/5

Which equation is a hyperbola?

A. $y = x^2 \div 4$

B. $x^2 \div 4 + y^2 \div 9 = 1$

C. $x^2 \div 4 - y^2 \div 9 = 1$

D. $x^2 + y^2 = 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q3/5

True or false?

For a hyperbola, the absolute value of the difference of the distances from any of its points to the two foci is constant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q4/5

True or false?

The hyperbola $x^2 - y^2 = 1$ passes through $(0, 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q5/5

SET 46 • HSF-IF.C.8.b • INTERPRETING FUNCTIONS

Use the properties of exponents to interpret growth and decay rates

MAIN	Algebra I main question	**HSF-IF.C.8.b**
BACKWARD 1	Read growth and decay from an exponential function	**HSF-IF.C.7.e**
BACKWARD 2	Solve percent increase and decrease problems	**7.RP.A.3**
FORWARD 1	Sum a geometric series in a savings context	**HSA-SSE.B.4**
FORWARD 2	Find times in exponential models with logarithms	**HSF-LE.A.4**

SET 46 • HSF-IF.C.8.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

A savings balance is $y = 800(1.035)^t$ dollars after t years.

Is this growth or decay, and what is the percent rate of change per year?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q1/5

A loan balance grows by 1.5% each month:

$B = 2{,}000(1.015)^{12t}$, where t is in years.

Rewrite it to find the yearly growth factor (to four decimal places) and the yearly percent rate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q2/5

The value of a machine is $V = 45{,}000(0.88)^t$ dollars after t years.

What is the yearly percent decrease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q3/5

A forest's area is $A = 600(1.3)^{t/20}$ hectares after t years.

By what percent does A grow every 20 years?

A. 1.3%

B. 13%

C. 3%

D. 30%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q4/5

True or false?

$y = 100(0.5)^{t/8}$ halves every 8 units of t.

A. True

B. False

SET 46 • HSF-IF.C.8.b

BACKWARD 1

Read growth and decay from an exponential function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

Is $y = 30(1.5)^x$ increasing or decreasing? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q1/5

What is the y-intercept of $y = 7(0.9)^x$?

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q2/5

Which function is decreasing?

A. $y = 2(1.6)^x$

B. $y = 0.6(1.2)^x$

C. $y = 2(0.6)^x$

D. $y = 0.6(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q3/5

True or false?

The graph of $y = 3^x$ has the horizontal asymptote $y = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q4/5

True or false?

$y = 5(1.1)^x$ has y-intercept 1.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSF-IF.C.8.b

BACKWARD 2

Solve percent increase and decrease problems • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A $40 shirt is marked up 15%.

What is the new price?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q1/5

A town grows from 2,000 to 2,300 people.

What is the percent increase?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q2/5

A price drops from $80 to $60.

What is the percent decrease?

A. 25%

B. 20%

C. 33%

D. 75%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q3/5

True or false?

Multiplying by 1.06 increases a number by 6%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q4/5

True or false?

Multiplying by 0.85 decreases a number by 85%.

A. True

B. False

SET 46 • HSF-IF.C.8.b

FORWARD 1

Sum a geometric series in a savings context • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

You save $100 the first month and 2% more each month after that.

Write the total for 12 months as a sum formula, and find it to the nearest dollar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q1/5

Find $4 + 4(1.5) + 4(1.5)^2 + 4(1.5)^3$.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q2/5

Which expression is the sum of the first n terms of $a + ar + ar^2 + \ldots$ $(r \neq 1)$?

A. $a \div (1 - r)$

B. $a(1 - r) \div (1 - r^n)$

C. $a(1 - r^n) \div (1 - r)$

D. nar

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q3/5

True or false?

$$1 + 0.5 + 0.25 + 0.125 = 1.875$$

A. True

B. False

True or false?

The sum of 10 terms of $3 + 3 + 3 + \ldots$ is 3^{10}.

A. True

B. False

SET 46 • HSF-IF.C.8.b

FORWARD 2

Find times in exponential models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$y = 100(0.5)^{t/8}$.

Find t when y = 10. Round to the nearest tenth.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q1/5

Find the continuous rate r for which $e^{10r} = 2$. Round to the nearest thousandth.

$r \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q2/5

To the nearest tenth, how long does $y = (1.05)^t$ take to double?

A. 20

B. 7.2

C. 10.5

D. 14.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q3/5

True or false?

If $y = 2^{t/3}$, then y doubles every 3 units of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q4/5

True or false?

$\log(2^{10}) = 20$ (log means base 10.)

A. True

B. False

SET 47 • HSF-IF.C.9 • INTERPRETING FUNCTIONS

Compare properties of two functions represented in different ways

MAIN	Algebra I main question	**HSF-IF.C.9**
BACKWARD 1	Interpret key features in context	**HSF-IF.B.4**
BACKWARD 2	Compute average rates of change	**HSF-IF.B.6**
FORWARD 1	Compare periodic functions given in different forms	**HSF-TF.B.5**
FORWARD 2	Compare normal distributions by mean and standard deviation —…	**HSS-ID.A.4**

SET 47 • HSF-IF.C.9

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9

f(x) = 3x + 2. The table shows g.

Which function has the greater value at x = 3? Which has the greater average rate of change from x = 0 to x = 3?

x	0	1	2	3
g(x)	1	4	9	16

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q1/5

Function A is $y = -2(x - 3)^2 + 10$. Function B is a parabola that opens down with vertex (1, 12).

Which has the greater maximum value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q2/5

Function f has y-intercept 4 and increases by 5 for each 1-unit increase in x. $g(x) = 5x - 1$.

Compare their slopes and y-intercepts.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q3/5

$P(x) = 1.08^x$ grows 8% per year. The table shows Q.

Which grows faster?

x	0	1	2
Q(x)	1	1.1	1.21

A. P: its graph is a line.

B. Q: it grows 10% per year.

C. P: it grows 8% per year and Q grows 1%.

D. They grow at the same rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q4/5

True or false?

$f(x) = x^2$, and g is a parabola opening up with vertex $(0, -4)$. The two functions have the same minimum value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q5/5

SET 47 • HSF-IF.C.9

BACKWARD 1

Interpret key features in context • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

T(h) is the temperature h hours after midnight, and its minimum is T(6) = 4.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q1/5

A graph of savings S(m) is horizontal from m = 3 to m = 5.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q2/5

A function's graph is an unbroken curve. It is positive for $0 < x < 8$ and negative for $x > 8$.

What is true at $x = 8$?

A. The output is greatest.

B. The output is 8.

C. The output is 0.

D. The function is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q3/5

True or false?

A y-intercept gives the starting value of a quantity that changes over time.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q4/5

True or false?

An x-intercept tells the starting value of a quantity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSF-IF.C.9

BACKWARD 2

Compute average rates of change • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = 2^x$.

Find the average rate of change from $x = 1$ to $x = 3$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q1/5

A savings account had $100 at week 0 and $180 at week 4.

What is the average rate of change?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q2/5

$g(x) = x^2 - 1$.

What is the average rate of change from $x = -1$ to $x = 2$?

A. 1

B. -1

C. 3

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q3/5

True or false?

The average rate of change of $y = 5$ is 0 on every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q4/5

True or false?

The average rate of change of $y = x^2$ from $x = 0$ to $x = 2$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSF-IF.C.9

FORWARD 1

Compare periodic functions given in different forms • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

$f(x) = 3\sin(2x)$. Function g has amplitude 4 and period π.

Which has the greater amplitude? Do they have the same period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q1/5

Tide A rises to 12 ft and falls to 2 ft. Tide B is $h(t) = 6 + 4\sin t$.

Which has the greater amplitude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q2/5

What is the period of $y = \sin(4x)$?

A. $\pi \div 2$

B. $\pi \div 4$

C. 2π

D. 4π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q3/5

True or false?

$y = 2\cos x$ and $y = 2\sin x$ have the same amplitude and the same period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q4/5

True or false?

$y = 5\sin x$ has a maximum value of 10.

A. True

B. False

SET 47 • HSF-IF.C.9

FORWARD 2

Compare normal distributions by mean and standard deviation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Comparing two functions by their key features leads to comparing two normal curves by their center and spread.

Test A scores are normal with mean 70 and standard deviation 5. Test B scores are normal with mean 70 and standard deviation 10.

Which test's scores are more spread out?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 47 • FORWARD 2 • Q1/5 • NEAREST RELATED

Heights are normal with mean 64 in and standard deviation 3 in.

About what percent are between 61 in and 67 in?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q2/5 • NEAREST RELATED

Scores are normal with mean 100 and standard deviation 15.

About what percent of scores are above 130?

A. About 16%

B. About 34%

C. About 2.5%

D. About 5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A normal distribution is symmetric about its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 47 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

In a normal distribution, about 95% of the data lie within 1 standard deviation of the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 47 • FORWARD 2 • Q5/5 • NEAREST RELATED

Write an explicit expression or a recursive process for a function from a context

MAIN	Algebra I main question	**HSF-BF.A.1.a**
BACKWARD 1	Evaluate a sequence from a recursive or explicit rule	**HSF-IF.A.3**
BACKWARD 2	Write equations in two or more variables	**HSA-CED.A.2**
FORWARD 1	Build a composite function from a context	**HSF-BF.A.1.c**
FORWARD 2	Find and interpret the inverse of a model	**HSF-BF.B.4.b**

SET 48 • HSF-BF.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a

A phone plan has a $60 activation fee and costs $35 per month.

Write an explicit expression for the total cost C(n) after n months, and a recursive rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q1/5

A tree is 4 ft tall and grows 1.5 ft per year.

Write a function for its height h(t) after t years.

h(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q2/5

A ball is dropped from 10 m. Each bounce reaches 60% of the height of the bounce before.

Write a function for the height h(n) of the nth bounce.

h(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q3/5

A table shows n = 1, 2, 3, 4 and a(n) = 7, 11, 15, 19.

Which recursive process fits?

A. $a(1) = 4,\ a(n) = a(n - 1) + 7$

B. $a(1) = 7,\ a(n) = 4a(n - 1)$

C. $a(1) = 7,\ a(n) = a(n - 1) + 4$

D. $a(n) = 7n + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q4/5

True or false?

The steps "start at 500 and multiply by 1.03 each year" give f(t) = 500 + 1.03t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q5/5

SET 48 • HSF-BF.A.1.a

BACKWARD 1

Evaluate a sequence from a recursive or explicit rule • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

$f(1) = 2$ and $f(n) = 3f(n - 1)$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q1/5

$f(n) = 10 - 2n$ for $n \geq 1$.

List the first four terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q2/5

$f(1) = 5$ and $f(n) = f(n - 1) - 3$.

What is $f(3)$?

A. -1

B. 11

C. 2

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q3/5

True or false?

The terms f(1), f(2), f(3) of a sequence are its outputs for the inputs 1, 2, 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q4/5

True or false?

$f(n) = n^2$ gives the terms 1, 2, 3, 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSF-BF.A.1.a

BACKWARD 2

Write equations in two or more variables • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

A car rental costs $45 per day plus $0.20 per mile.

Write an equation for the cost C for d days and m miles.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q1/5

Write an equation for the perimeter P of a rectangle with length l and width w.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q2/5

A $900 laptop loses $150 of value each year.

Which equation gives its value V after t years?

A. $V = 900 - 150t$

B. $V = 150 - 900t$

C. $V = 900(150)^t$

D. $V = 900 + 150t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q3/5

True or false?

$A = s^2$ gives the area of a square with side s.

A. True

B. False

True or false?

y = 12x gives the number of feet in x inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSF-BF.A.1.a

FORWARD 1

Build a composite function from a context • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

An oil spill's radius is $r(t) = 2t$ meters after t hours, and the area of a circle is $A(r) = \pi r^2$.

Write $A(r(t))$ and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q1/5

A store takes 20% off and then $5 off: $d(x) = 0.8x$ and $c(x) = x - 5$.

Find $c(d(50))$.

$c(d(50)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q2/5

With $d(x) = 0.8x$ and $c(x) = x - 5$, which expression is $d(c(x))$?

A. $0.8x - 4$

B. $0.8x + 4$

C. $0.8x - 5$

D. $x - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F1-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q3/5

True or false?

A store takes 20% off and then $5 off: $d(x) = 0.8x$ and $c(x) = x - 5$. For these functions, $c(d(50))$ and $d(c(50))$ are different.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q4/5

True or false?

If $f(x) = 3x$ and $g(x) = x + 2$, then $f(g(x)) = 3x + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSF-BF.A.1.a

FORWARD 2

Find and interpret the inverse of a model • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

$C(n) = 60 + 35n$ is the total cost after n months.

Find the inverse and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q1/5

Show that f(x) = 5x − 2 and g(x) = (x + 2) ÷ 5 are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q2/5

$F = 1.8C + 32$ converts Celsius to Fahrenheit.

Which formula is its inverse?

A. $C = F \div 1.8 - 32$

B. $C = 1.8F + 32$

C. $C = (F + 32) \div 1.8$

D. $C = (F - 32) \div 1.8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q3/5

True or false?

If f is invertible and $f(2) = 9$, then $f^{-1}(9) = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q4/5

True or false?

The inverse of $f(x) = 2x$ is $f^{-1}(x) = -2x$.

A. True

B. False

Build quadratic functions that model situations

MAIN	Algebra I main question	**HSF-BF.A.1.a**
BACKWARD 1	Write equations in two variables for areas and perimeters	**HSA-CED.A.2**
BACKWARD 2	Interpret the parts of an expression in context	**HSA-SSE.A.1.a**
FORWARD 1	Use quadratic models to make the best design choice	**HSG-MG.A.3**
FORWARD 2	Compose functions to build quadratic models	**HSF-BF.A.1.c**

SET 80 • HSF-BF.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a

A farmer has 100 m of fence for three sides of a rectangular pen; a barn wall forms the fourth side. Each side perpendicular to the wall is x m long.

Write the area A(x) of the pen.

A(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q1/5

A theater sells 300 tickets at $20 each. For each $1 increase in price, it sells 10 fewer tickets.

Write the revenue R(n) after n price increases of $1.

R(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q2/5

A stone is thrown upward at 20 m/s from the top of a 5-m wall. Using $h(t) = -4.9t^2 + v_0t + h_0$, where v_0 is the starting upward speed and h_0 is the starting height, write the stone's height h(t).

$h(t) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q3/5

A table has x = 0, 1, 2, 3 and y = 3, 5, 11, 21.

Which function fits the table?

A. $y = 3(2)^x$

B. $y = 2x + 3$

C. $y = 2x^2 + 3$

D. $y = x^2 + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q4/5

True or false?

If each of n people shakes hands once with every other person, the number of handshakes is $h(n) = n(n - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q5/5

SET 80 • HSF-BF.A.1.a

BACKWARD 1

Write equations in two variables for areas and perimeters • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

A rectangle has length 2w and width w. Write an equation for its perimeter P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q1/5

A triangle has base b and height 2b. Write an equation for its area A.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q2/5

A rectangle has perimeter 40 and width w. Which equation gives its area A?

A. $A = w^2 + 20$

B. $A = w(40 - w)$

C. $A = 20w$

D. $A = w(20 - w)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q3/5

True or false?

The area of a circle as a function of its radius is $A = \pi r^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q4/5

True or false?

If a rectangle's length is its width w plus 2, then its area is w + 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q5/5

SET 80 • HSF-BF.A.1.a

BACKWARD 2

Interpret the parts of an expression in context • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

In $R = (20 + n)(300 - 10n)$, what does the factor $300 - 10n$ represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q1/5

In $A = x(100 - 2x)$ for a pen against a wall, what does $100 - 2x$ represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q2/5

In $h = -4.9t^2 + 20t + 5$, what does 5 represent?

A. The starting upward speed

B. The starting height

C. The effect of gravity

D. The time in the air

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q3/5

True or false?

In $-4.9t^2 + 20t + 5$, the coefficient of t is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q4/5

True or false?

The constant term of $2w^2 + 3w$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSF-BF.A.1.a

FORWARD 1

Use quadratic models to make the best design choice • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

For a pen with area $A(x) = x(100 - 2x)$, which x gives the greatest area, and what is that area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q1/5

For $R(n) = (20 + n)(300 - 10n)$, which number of \$1 increases gives the most revenue? What are the ticket price and the revenue?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q2/5

A 24-m rope is laid out as a rectangle. Which dimensions give the greatest area?

A. 2 m by 10 m

B. 12 m by 12 m

C. 6 m by 6 m

D. 4 m by 8 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q3/5

True or false?

Of all rectangles with perimeter 40, the square has the greatest area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q4/5

True or false?

For $A(x) = x(100 - 2x)$, the area is greatest when $x = 50$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q5/5

SET 80 • HSF-BF.A.1.a

FORWARD 2

Compose functions to build quadratic models • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

Ticket demand is $q(p) = 500 - 10p$ tickets at a price of p dollars, and the price after n increases is $p(n) = 20 + n$.

Write $q(p(n))$ and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q1/5

A square's area is $A(s) = s^2$, and its side grows with time: $s(t) = 2 + 0.5t$.

Write $A(s(t))$ in standard form.

$A(s(t)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q2/5

$f(x) = x^2$ and $g(x) = x + 3$. Which is $f(g(x))$ in standard form?

A. $x^2 + 3x$

B. $x^2 + 9$

C. $x^2 + 3$

D. $x^2 + 6x + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q3/5

True or false?

If $f(x) = x^2$ and $g(x) = 2x$, then $g(f(3)) = 18$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q4/5

True or false?

If $f(x) = x^2$ and $g(x) = 2x$, then $f(g(3)) = 18$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q5/5

Combine standard function types using arithmetic operations

MAIN	Algebra I main question	**HSF-BF.A.1.b**
BACKWARD 1	Add, subtract, and multiply polynomials	**HSA-APR.A.1**
BACKWARD 2	Evaluate and interpret functions	**HSF-IF.A.2**
FORWARD 1	Rewrite rational expressions such as average cost	**HSA-APR.D.6**
FORWARD 2	Graph and interpret an average-cost rational function	**HSF-IF.C.7.d**

SET 49 • HSF-BF.A.1.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

Revenue is R(x) = 25x and cost is C(x) = 400 + 9x dollars for x items.

Write the profit function P(x) in simplest form and find P(50).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q1/5

A ball's height is $h(t) = -4.9t^2 + 30t + 2$ meters after t seconds.

Name the three simple functions added together, and what each one describes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q2/5

$f(x) = x^2 + 1$ and $g(x) = 2x - 3$.

Find $(f - g)(x)$ in simplest form, and find $(f \cdot g)(2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q3/5

The temperature of tea is $T(t) = 20 + 70(0.9)^t$ °C after t minutes.

Which description fits this combination of functions?

A. A linear function with slope 0.9

B. An exponential function that grows 90% per minute

C. A constant room temperature of 20 °C plus a difference of $70(0.9)^t$ that decays

D. A constant temperature of 90 °C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q4/5

True or false?

If $f(x) = 3x$ and $g(x) = x^2$, then $(f + g)(2) = 10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q5/5

SET 49 • HSF-BF.A.1.b

BACKWARD 1

Add, subtract, and multiply polynomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Subtract.

$25x - (400 + 9x)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q1/5

Add.

$(x^2 + 1) + (2x - 3)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q2/5

Multiply.

$(x^2 + 1)(2x - 3)$

A. $2x^3 + 2x - 3$

B. $2x^2 + 2x - 3$

C. $2x^3 - 3x^2 - 3$

D. $2x^3 - 3x^2 + 2x - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-B1-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q3/5

True or false?

The product of two polynomials is a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q4/5

True or false?

$(x + 2) - (x - 2) = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSF-BF.A.1.b

BACKWARD 2

Evaluate and interpret functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$P(x) = 16x - 400$.

Find x so that $P(x) = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q1/5

$T(t) = 20 + 70(0.9)^t$.

Find $T(0)$.

$T(0) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q2/5

R(x) is the revenue in dollars from selling x items.

What does R(10) = 250 mean?

A. Selling 250 items brings in $10.

B. Each item costs $10.

C. The profit is $250.

D. Selling 10 items brings in $250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q3/5

True or false?

If C(x) = 400 + 9x, then C(0) = 400.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q4/5

True or false?

If $h(t) = -4.9t^2 + 30t + 2$, then $h(0) = 30$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSF-BF.A.1.b

FORWARD 1

Rewrite rational expressions such as average cost • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Write the average cost $A(x) = (400 + 9x) \div x$ as a constant plus a fraction.

$A(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q1/5

Assume $x \neq -1$. Write $(x^2 + 3x + 5) \div (x + 1)$ in the form $q(x) + r \div (x + 1)$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q2/5

Assume $x \neq 0$. Which expression equals $(6x^2 + 4x) \div (2x)$?

A. $3x + 4x$

B. $3x^2 + 2$

C. $6x + 2$

D. $3x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q3/5

True or false?

$(x^2 - 1) \div (x - 1) = x + 1$ for $x \neq 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q4/5

True or false?

$(x + 8) \div x = 9$ for $x \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSF-BF.A.1.b

FORWARD 2

Graph and interpret an average-cost rational function • High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d

$A(x) = 9 + 400 \div x$ is the average cost per item, in dollars, for x items.

Give its horizontal asymptote and what it means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q1/5

$A(x) = 9 + 400 \div x$ is the average cost per item, in dollars, for x items.

Find A(100).

A(100) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q2/5

What is the vertical asymptote of $y = 400 \div x + 9$?

A. $x = 400$

B. $x = 9$

C. $y = 9$

D. $x = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q3/5

True or false?

As x increases, 9 + 400 ÷ x decreases toward 9.

A. True

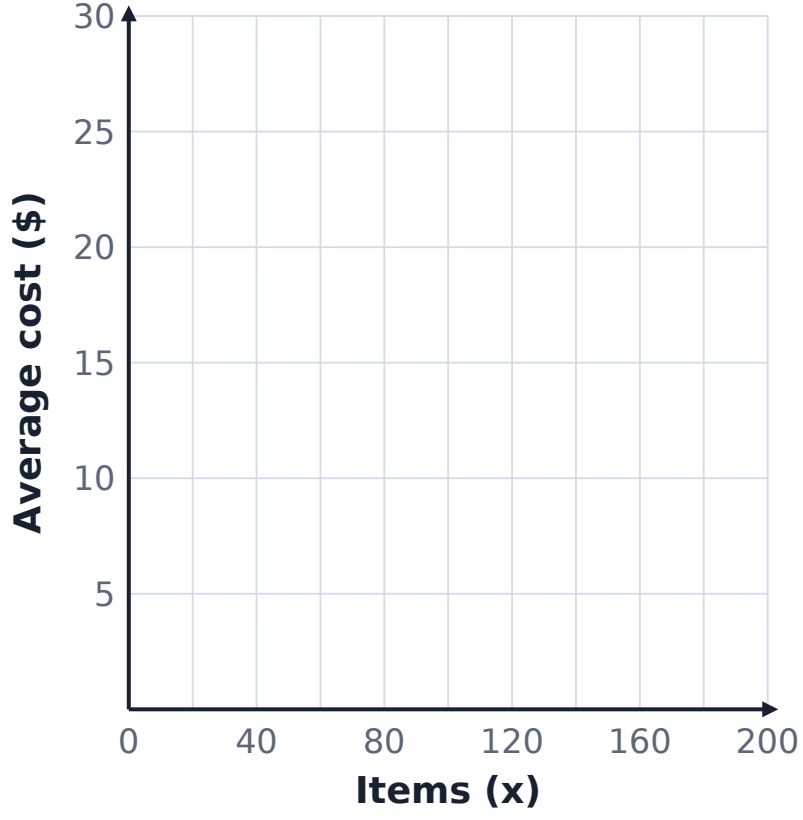
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q4/5

$A(x) = 9 + 400 \div x$ is the average cost per item, in dollars, for x items. Graph A for $20 \leq x \leq 200$, and draw its horizontal asymptote.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q5/5

Write arithmetic and geometric sequences recursively and with an explicit formula

MAIN	Algebra I main question	**HSF-BF.A.2**
BACKWARD 1	Evaluate recursively defined sequences	**HSF-IF.A.3**
BACKWARD 2	Generate a pattern from a rule and describe its features	**4.OA.C.5**
FORWARD 1	Sum geometric sequences	**HSA-SSE.B.4**
FORWARD 2	Use logarithms to find a term number	**HSF-LE.A.4**

SET 50 • HSF-BF.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.2

Write an explicit formula and a recursive formula for 7, 4, 1, −2, ...

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q1/5

Write an explicit formula and a recursive formula for 3, 12, 48, 192, ...

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q2/5

An arithmetic sequence has a(3) = 14 and a(8) = 39.

Find a(1) and the common difference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q3/5

A geometric sequence of real numbers has a(1) = 2 and a(4) = 54.

What is the common ratio?

A. 3

B. 18

C. 27

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q4/5

True or false?

The sequence 2, 4, 8, 14, ... is arithmetic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q5/5

SET 50 • HSF-BF.A.2

BACKWARD 1

Evaluate recursively defined sequences • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

$f(1) = 4$ and $f(n) = f(n - 1) + 6$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q1/5

$f(1) = 1$ and $f(n) = 2f(n - 1) + 1$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q2/5

$f(n) = 2n - 1$.

What is $f(10)$?

A. 20

B. 21

C. 19

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q3/5

True or false?

A recursive formula defines each term after the starting term(s) using one or more earlier terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q4/5

True or false?

$f(1) = 3$ and $f(n) = 2f(n - 1)$ give 3, 5, 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSF-BF.A.2

BACKWARD 2

Generate a pattern from a rule and describe its features • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

Rule: start at 5 and add 5.

Write the first 6 terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q1/5

Rule: start at 1 and add 4.

Explain why every term is odd.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q2/5

Rule: start at 2 and multiply by 2.

What is the 5th term?

A. 10

B. 16

C. 64

D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q3/5

True or false?

With the rule "start at 3 and add 3," every term is a multiple of 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q4/5

True or false?

With the rule "start at 10 and add 10," the 4th term is 50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSF-BF.A.2

FORWARD 1

Sum geometric sequences • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Find the sum of the first 6 terms of 3, 12, 48, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q1/5

A job pays $1 on day 1, and the pay doubles each day.

What is the total pay for 10 days?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q2/5

What is 5 + 10 + 20 + 40 + 80?

A. 160

B. 80

C. 155

D. 150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q3/5

True or false?

The formula $a(1 - r^n) \div (1 - r)$ works for $r = 0.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q4/5

True or false?

$$2 + 6 + 18 + 54 = 2(3^4 - 1)$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSF-BF.A.2

FORWARD 2

Use logarithms to find a term number • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$a(n) = 3 \cdot 4^{n-1}$.

What is the smallest n for which a(n) > 10,000?

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

A sequence starts at 5 and grows by 20% per term: $a(n) = 5(1.2)^{n-1}$. Which term is the first to exceed 100? (log means base 10.)

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

Solve $5 \cdot 2^n = 600$. (log means base 10.)

A. n = 7

B. n = log 600 ÷ log 2 ≈ 9.23

C. n = log 120 ÷ log 2 ≈ 6.91

D. n = 120 ÷ 2 = 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q3/5

True or false?

If $4^k = 64$, then k = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

True or false?

If $3^n = 30$, then n = 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

Model situations with arithmetic and geometric sequences and translate between recursive and explicit forms

MAIN	Algebra I main question	**HSF-BF.A.2**
BACKWARD 1	Evaluate sequences from recursive and explicit rules	**HSF-IF.A.3**
BACKWARD 2	Write a function or a recursive process from a situation	**HSF-BF.A.1.a**
FORWARD 1	Sum geometric sequences that model situations	**HSA-SSE.B.4**
FORWARD 2	Use logarithms to find when a sequence reaches a value	**HSF-LE.A.4**

SET 81 • HSF-BF.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.2

A gym charges $50 to join and $30 each month. a(n) is the total paid after n months, with a(1) = 80.

Write a recursive rule and an explicit rule for a(n).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q1/5

A ball bounces to 80% of its previous height. Its first bounce reaches 5 m.

Write an explicit rule and a recursive rule for the height h(n) of the nth bounce.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q2/5

Translate the explicit rule $a(n) = 7 + 4(n - 1)$, for $n \geq 1$, into a recursive rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q3/5

Which explicit rule matches $a(1) = 3$ and $a(n) = 2a(n - 1)$ for $n \geq 2$?

A. $a(n) = 3 \cdot 2^{n-1}$

B. $a(n) = 2 \cdot 3^{n-1}$

C. $a(n) = 3 \cdot 2^{n}$

D. $a(n) = 3 + 2(n - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q4/5

True or false?

The recursive rule $a(n) = a(n - 1) - 6$ needs a starting value, such as $a(1) = 50$, to define a sequence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q5/5

SET 81 • HSF-BF.A.2

BACKWARD 1

Evaluate sequences from recursive and explicit rules • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

$f(1) = 50$ and $f(n) = f(n - 1) - 6$ for $n \geq 2$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q1/5

$f(n) = 5(0.8)^{n-1}$.

Find f(3).

f(3) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q2/5

$f(1) = 2$ and $f(n) = 3f(n - 1) + 1$ for $n \geq 2$.

What is $f(3)$?

A. 7

B. 22

C. 19

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q3/5

True or false?

The domain of the sequence $a(n) = 2n + 1$, $n \geq 1$, is the positive integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q4/5

True or false?

$f(n) = n^2$ gives the sequence 1, 3, 5, 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSF-BF.A.2

BACKWARD 2

Write a function or a recursive process from a situation • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a

A savings jar starts with $20 and gains $5 each week.

Write a function for the amount A after w weeks.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q1/5

A culture of 100 cells triples every day.

Write a function for the number of cells N after d days.

N = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q2/5

Which recursive process means "start at 10 and multiply by 1.1 each year"?

A. $a(0) = 10, a(n) = a(n - 1) + 1.1$

B. $a(n) = 10 + 1.1n$

C. $a(0) = 1.1, a(n) = 10a(n - 1)$

D. $a(0) = 10, a(n) = 1.1a(n - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q3/5

True or false?

Adding the same amount at each step gives a linear function of the step number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q4/5

True or false?

Multiplying by 2 at each step, starting at 1, gives f(n) = 2n.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSF-BF.A.2

FORWARD 1

Sum geometric sequences that model situations • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

A ball's bounce heights are 5, 4, 3.2, ... meters (each 80% of the one before). Find the total of the first 5 bounce heights, to the nearest hundredth.

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q1/5

You deposit $100 the first month and 3% more each month after that. Find the total deposited in 12 months, to the nearest cent.

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q2/5

Which expression is the sum of the first 10 terms of $5(0.8)^{n-1}$?

A. $5(1 - 0.8^{10}) \div 0.2$

B. $5(0.8)^{10}$

C. $10(5)(0.8)$

D. $5 \div 0.2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q3/5

True or false?

The sum of the first n bounce heights 5, 4, 3.2, ... is always less than 25.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q4/5

True or false?

For a geometric series with ratio 2, adding one more term always doubles the sum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSF-BF.A.2

FORWARD 2

Use logarithms to find when a sequence reaches a value • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$h(n) = 5(0.8)^{n-1}$ is the height of the nth bounce, in meters. Which bounce is the first below 1 m? (log means base 10.)

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q1/5

$a(n) = 1{,}000(1.05)^n$ is a savings balance after n years. After how many whole years does it first exceed $2,000? (log means base 10.)

Years: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q2/5

Solve $3^n = 50$. (log means base 10.)

A. n = log 3 ÷ log 50

B. n = 50 ÷ 3

C. n = log 50 ÷ log 3 ≈ 3.56

D. n = 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q3/5

True or false?

$\log(0.8^n) = n \log 0.8$, and log 0.8 is negative. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q4/5

True or false?

If $2^n = 10$, then n is a whole number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q5/5

Identify the effect of replacing f(x) by f(x) + k, kf(x), f(kx), and f(x + k), and recognize even and odd functions

MAIN	Algebra I main question	**HSF-BF.B.3**
BACKWARD 1	Graph absolute value and square root functions	**HSF-IF.C.7.b**
BACKWARD 2	Read key features of quadratic graphs	**HSF-IF.C.7.a**
FORWARD 1	Describe a sequence of rigid transformations	**HSG-CO.A.5**
FORWARD 2	Transform sine and cosine graphs	**HSF-TF.B.5**

SET 51 • HSF-BF.B.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

$g(x) = (x - 4)^2 + 1$.

Describe how to get the graph of g from the graph of $y = x^2$.

Answer: _____

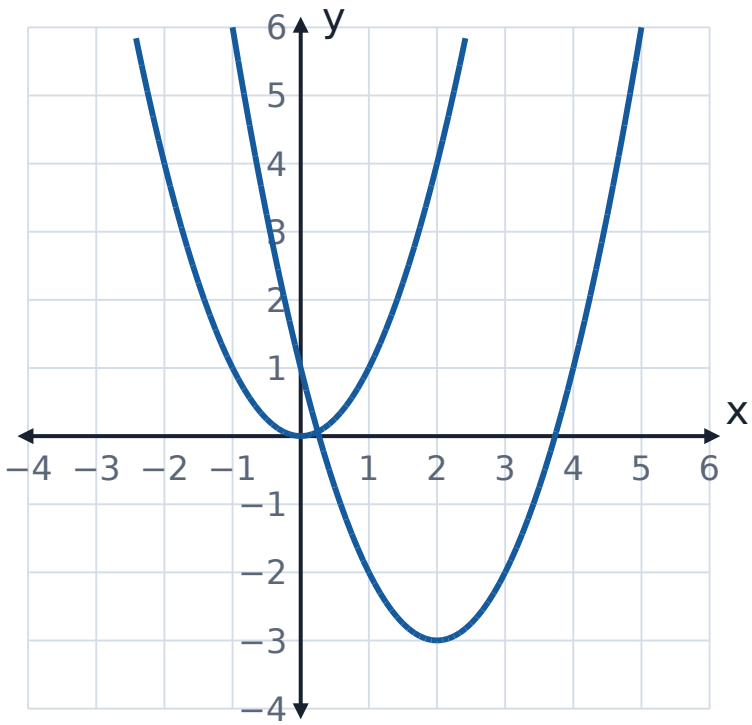
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q1/5

The graph shows $y = x^2$ and a translated graph $y = g(x)$.

Write $g(x)$.



$g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q2/5

$f(x) = x^2$. Compare $g(x) = 3f(x)$ and $h(x) = f(3x)$.

Which is a vertical stretch and which is a horizontal compression? Are g and h the same function?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q3/5

$f(x) = x^3$.

Is f even, odd, or neither?

A. Odd: $f(-x) = -f(x)$

B. Even: $f(-x) = f(x)$

C. Both even and odd

D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q4/5

True or false?

The graph of $y = f(x + 2)$ is the graph of $y = f(x)$ shifted 2 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q5/5

SET 51 • HSF-BF.B.3

BACKWARD 1

Graph absolute value and square root functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

What is the vertex of $y = |x + 3| - 2$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q1/5

Give two points on the graph of $y = \sqrt{(x - 1)}$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q2/5

What is the vertex of $y = -|x - 1| + 4$?

A. (1, 4)

B. (1, −4)

C. (4, 1)

D. (−1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q3/5

True or false?

$y = |x|$ has its minimum at (0, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q4/5

True or false?

The domain of $y = \sqrt{(x + 2)}$ is $x \geq 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSF-BF.B.3

BACKWARD 2

Read key features of quadratic graphs • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

Give the vertex of $y = (x - 2)^2 - 3$, and say whether it is a maximum or a minimum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q1/5

Find the x-intercepts of $y = x^2 - 9$.

x-intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q2/5

What is the y-intercept of $y = 2x^2 - 5x + 7$?

A. 7

B. 0

C. -5

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q3/5

True or false?

The graph of $y = -x^2 + 4$ opens downward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q4/5

True or false?

The vertex of $y = x^2 + 6$ is $(6, 0)$.

A. True

B. False

SET 51 • HSF-BF.B.3

FORWARD 1

Describe a sequence of rigid transformations • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5

Describe a transformation that maps the triangle with vertices (0, 0), (2, 0), (0, 1) to the triangle with vertices (3, 2), (5, 2), (3, 3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q1/5

The point (2, 5) is reflected across the x-axis and then translated 4 units left.

Where does it end up?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q2/5

Which single transformation maps every (x, y) to (−x, −y)?

A. A translation

B. A rotation of 180° about the origin

C. A reflection across the y-axis

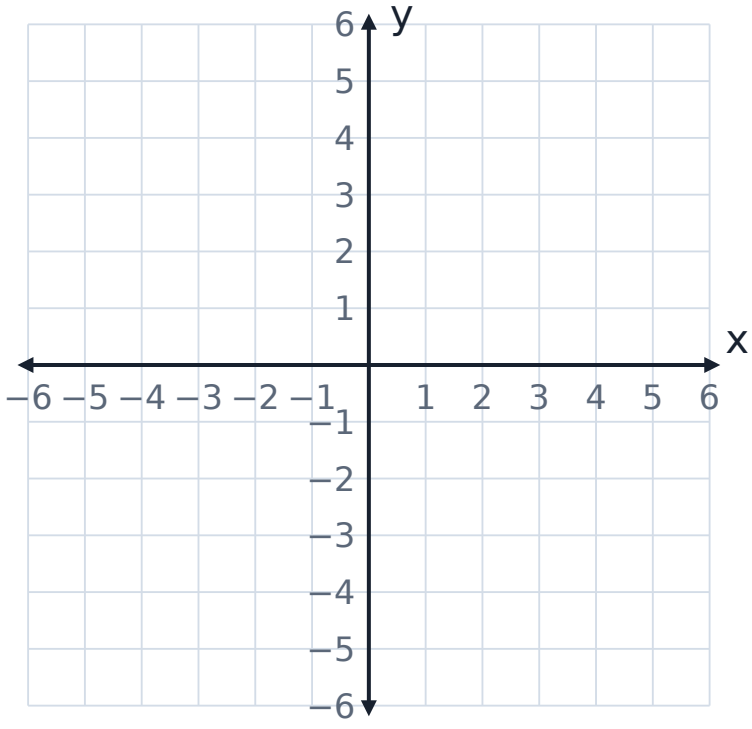
D. A reflection across the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q3/5

Triangle ABC has A(1, 1), B(4, 1), and C(1, 3). Reflect it across the y-axis, then translate the image 2 units down. Draw the final image and label its vertices.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q4/5

True or false?

Translating and then reflecting always gives the same result as reflecting and then translating.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSF-BF.B.3

FORWARD 2

Transform sine and cosine graphs • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

How is the graph of $y = \sin x + 3$ related to the graph of $y = \sin x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q1/5

Write a sine function with amplitude 2, midline $y = 1$, and period 2π whose graph passes through $(0, 1)$ and is rising there.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q2/5

How is $y = 4\sin x$ obtained from $y = \sin x$?

A. A shift 4 units right

B. A horizontal stretch by a factor of 4

C. A vertical stretch by a factor of 4

D. A shift 4 units up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q3/5

True or false?

$y = -\sin x$ is the reflection of $y = \sin x$ across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q4/5

True or false?

$y = \sin x - 2$ has midline $y = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q5/5

Experiment with transformations of graphs and recognize even and odd functions from graphs and equations

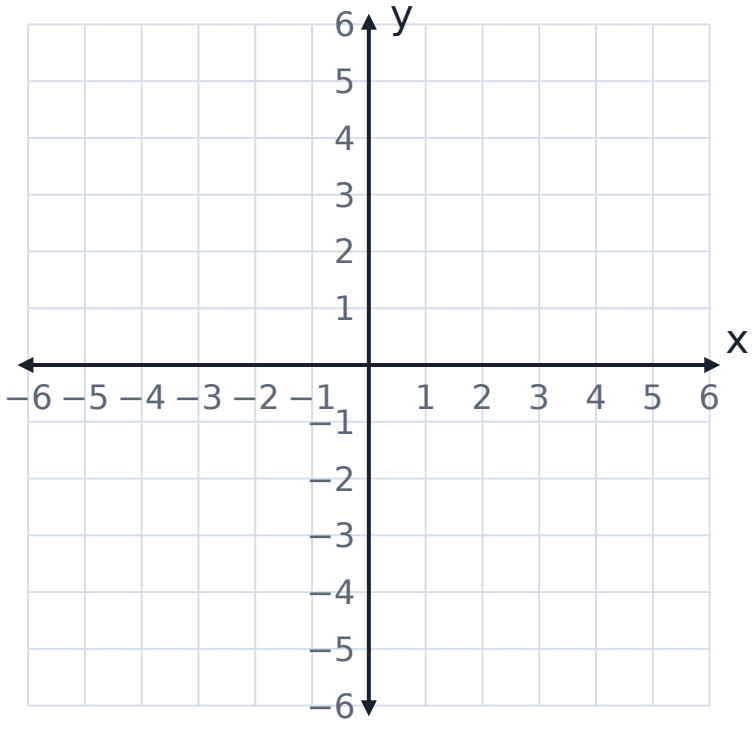
MAIN	Algebra I main question	**HSF-BF.B.3**
BACKWARD 1	Graph quadratic functions	**HSF-IF.C.7.a**
BACKWARD 2	Graph absolute value and root functions	**HSF-IF.C.7.b**
FORWARD 1	Draw rigid transformations of figures	**HSG-CO.A.5**
FORWARD 2	Transform sine and cosine graphs and recognize their symmetry	**HSF-TF.B.5**

SET 82 • HSF-BF.B.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

Graph $f(x) = x^2$ and $g(x) = (x - 3)^2 - 2$ on the same grid. Label both, and describe how g is related to f.

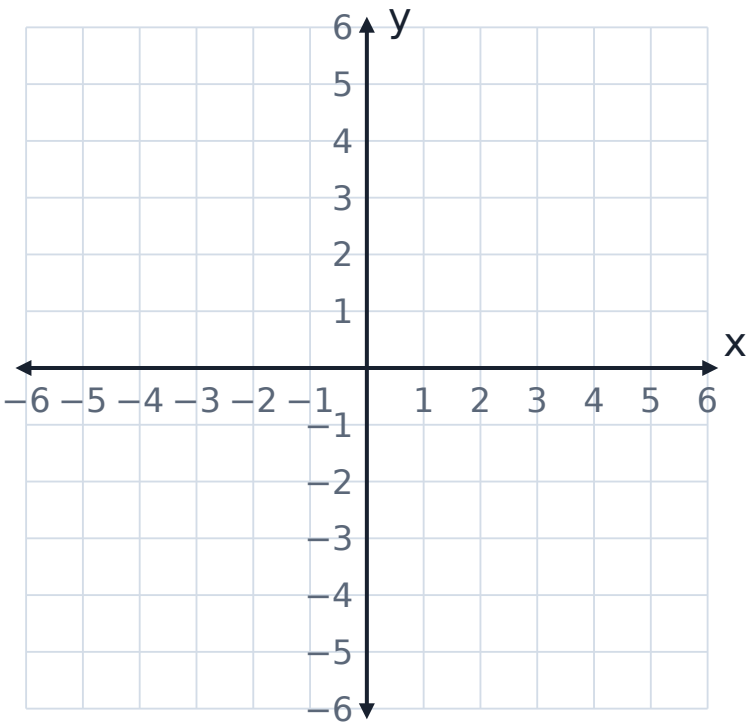


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q1/5

$f(x) = |x|$. Graph $g(x) = 2f(x)$ and $h(x) = f(2x)$ for $-3 \leq x \leq 3$, using a table or graphing technology. Are g and h the same function?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q2/5

Use a table or graphing technology to compare $f(x) = \sqrt{x}$ with $g(x) = \sqrt{(x - 4)} + 1$ at $x = 4$, 5, and 8. Describe the transformation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q3/5

The graph of f is shown.

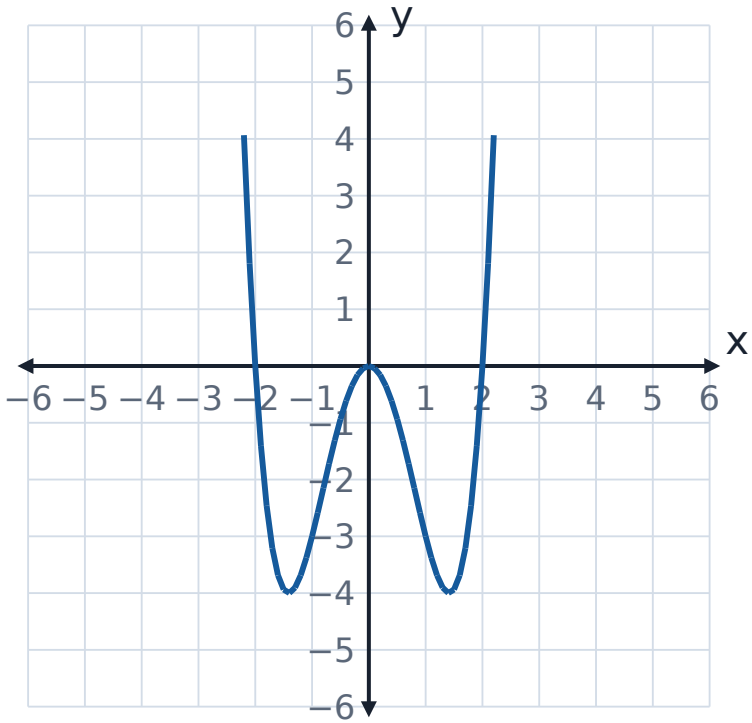
Is f even, odd, or neither?

A. Both even and odd

B. Even

C. Odd

D. Neither



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q4/5

True or false?

$f(x) = x^3 + x$ is an odd function, because $f(-x) = -f(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q5

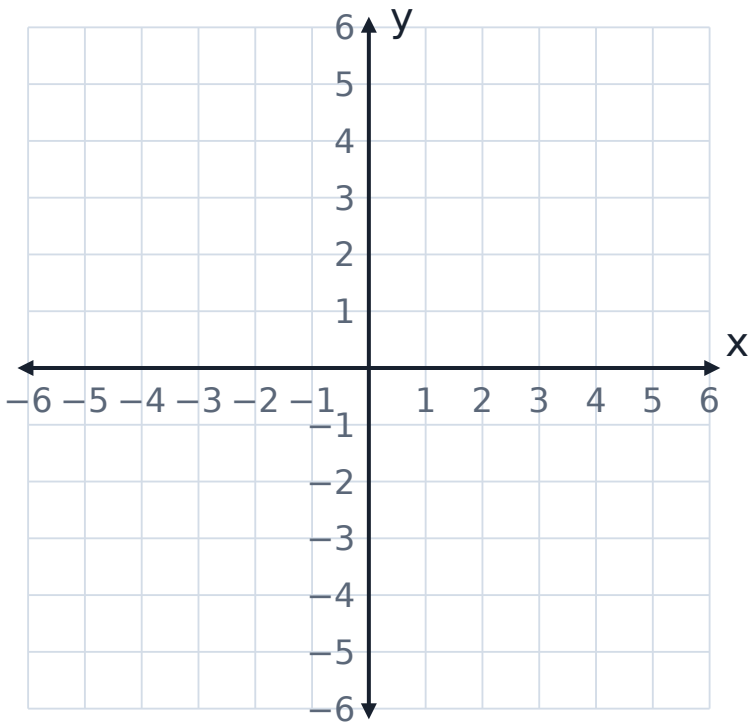
High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q5/5

SET 82 • HSF-BF.B.3

BACKWARD 1

Graph quadratic functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

Graph $y = x^2 - 4$, and mark its vertex and x-intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q1/5

What is the vertex of $y = (x + 1)^2 + 2$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q2/5

What is the axis of symmetry of $y = x^2 - 6x + 1$?

A. $x = 6$

B. $x = 3$

C. $x = -3$

D. $x = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q3/5

True or false?

$y = -x^2 + 1$ has a maximum value of 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q4/5

True or false?

The x-intercepts of $y = x^2 + 1$ are 1 and -1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSF-BF.B.3

BACKWARD 2

Graph absolute value and root functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

What is the vertex of $y = |x - 4| + 1$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q1/5

Where does the graph of $y = \sqrt{(x + 2)} - 3$ start?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q2/5

Which function's graph is V-shaped?

A. $y = \sqrt[3]{(x + 1)}$

B. $y = |x + 1|$

C. $y = \sqrt{(x + 1)}$

D. $y = (x + 1)^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q3/5

True or false?

The graph of $y = \sqrt[3]{x}$ passes through $(-1, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q4/5

True or false?

The graph of $y = |x| - 2$ has vertex $(2, 0)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q5

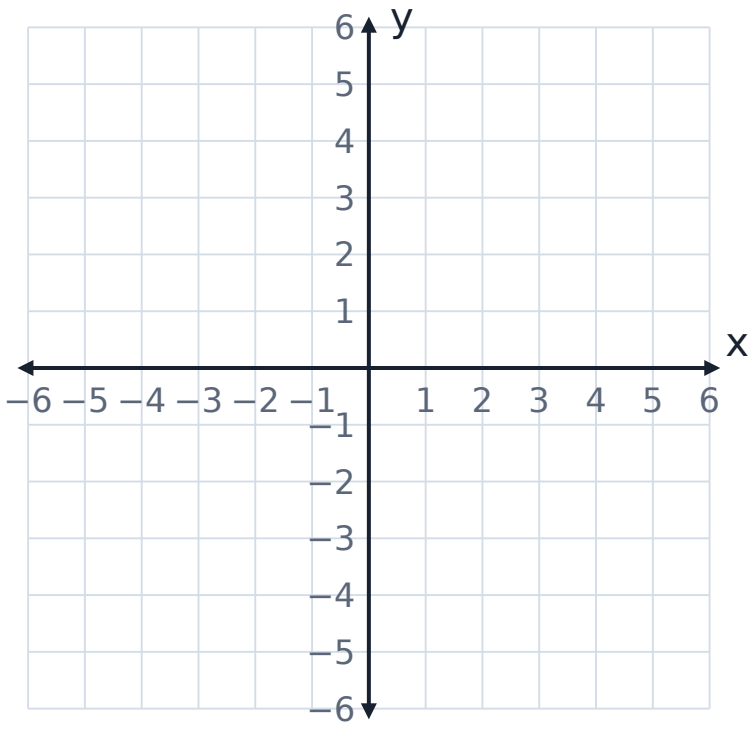
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q5/5

SET 82 • HSF-BF.B.3

FORWARD 1

Draw rigid transformations of figures • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5

Translate triangle A(0, 0), B(2, 0), C(0, 3) by (x, y) → (x − 3, y + 1). Draw and label the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q1/5

The point (4, −1) is rotated 90° counterclockwise about the origin. Where does it go?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q2/5

Which sequence of transformations maps (x, y) to $(-x, y - 2)$?

A. Reflect across the x-axis, then translate 2 units down

B. Rotate 180°, then translate 2 units up

C. Reflect across the y-axis, then translate 2 units down

D. Translate 2 units left, then reflect across the y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q3/5

True or false?

A translation followed by another translation is a translation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q4/5

True or false?

Reflecting a figure across the same line twice moves it to a new position.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q5

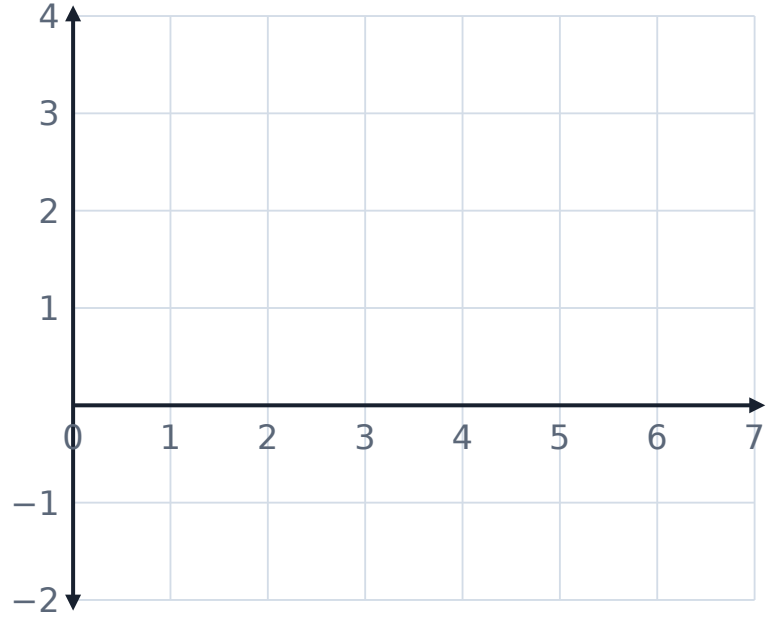
High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q5/5

SET 82 • HSF-BF.B.3

FORWARD 2

Transform sine and cosine graphs and recognize their symmetry • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

Sketch $y = 2\sin x + 1$ for $0 \le x \le 2\pi$ (about 6.28).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q1/5

Describe how the graph of $y = \sin(x - \pi \div 2)$ is related to the graph of $y = \sin x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q2/5

Which function has period π?

A. $y = \sin x + 2$

B. $y = \sin(x + 2)$

C. $y = 2\sin x$

D. $y = \sin(2x)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q3/5

True or false?

$y = -\cos x$ is the reflection of $y = \cos x$ across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q4/5

True or false?

$y = \cos x$ is an odd function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q5/5

Solve f(x) = c for a simple function f and write an expression for the inverse

MAIN	Algebra I main question	**HSF-BF.B.4.a**
BACKWARD 1	Rearrange a formula for another variable	**HSA-CED.A.4**
BACKWARD 2	Use cube roots to solve $x^3 = p$	**8.EE.A.2**
FORWARD 1	Verify inverses by composition	**HSF-BF.B.4.b**
FORWARD 2	Restrict a domain to produce an inverse	**HSF-BF.B.4.d**

SET 52 • HSF-BF.B.4.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

$f(x) = 3x^3 - 1$. Solve $f(x) = 80$, and write an expression for the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q1/5

$f(x) = (2x + 3) \div (x - 4)$, for $x \neq 4$. Write an expression for the inverse.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q2/5

$F(C) = 1.8C + 32$ gives degrees Fahrenheit.

Write the inverse $C(F)$, and find C when $F = 77$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q3/5

$f(x) = 4x - 7$.

What is $f^{-1}(x)$?

A. $(x - 7) \div 4$

B. $(x + 7) \div 4$

C. $x \div 4 + 7$

D. $4x + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q4/5

True or false?

If $f(x) = x \div 3 + 2$, then $f^{-1}(x) = 3x - 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q5/5

SET 52 • HSF-BF.B.4.a

BACKWARD 1

Rearrange a formula for another variable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

Solve $y = 4x - 7$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q1/5

Solve $P = 2l + 2w$ for l.

$l =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q2/5

Solve $d = rt$ for t. ($r \neq 0$)

A. $t = r \div d$

B. $t = d - r$

C. $t = d \div r$

D. $t = dr$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q3/5

True or false?

Solving $y = x^3$ for x gives $x = \sqrt[3]{y}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q4/5

True or false?

Solving $y = 5x$ for x gives $x = 5y$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSF-BF.B.4.a

BACKWARD 2

Use cube roots to solve $x^3 = p$ • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve $x^3 = 27$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q1/5

Evaluate $\sqrt[3]{125}$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q2/5

What is $\sqrt[3]{64}$?

A. 21.3

B. 4

C. 8

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q3/5

True or false?

$\sqrt[3]{(-27)} = -3$

A. True

B. False

True or false?

$\sqrt{16} = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSF-BF.B.4.a

FORWARD 1

Verify inverses by composition • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

Show that $f(x) = 2x^3$ and $g(x) = \sqrt[3]{(x \div 2)}$ are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q1/5

$f(x) = (x + 1) \div (x - 1)$.

Find $f(f(3))$.

$f(f(3)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q2/5

Which function is its own inverse?

A. $f(x) = 1 \div x$

B. $f(x) = 2x$

C. $f(x) = x + 1$

D. $f(x) = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F1-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q3/5

True or false?

If f(g(x)) = x on the domain of g and g(f(x)) = x on the domain of f, then g is the inverse of f.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q4/5

True or false?

$f(x) = x + 3$ and $g(x) = x - 3$ are not inverses, because $f(g(1)) = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSF-BF.B.4.a

FORWARD 2

Restrict a domain to produce an inverse • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d

$f(x) = x^2 + 2$. Restrict f to the largest interval that contains $x = 1$ on which f is one-to-one, and give the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q1/5

Explain why f(x) = |x| has no inverse on all real numbers.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q2/5

$f(x) = (x + 4)^2$.

What is the largest interval containing $x = 0$ on which f is one-to-one?

A. All real numbers

B. $-8 \leq x \leq 0$

C. $x \geq 0$

D. $x \geq -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^2$ restricted to $x \leq 0$ has the inverse $f^{-1}(x) = -\sqrt{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q4/5

True or false?

$f(x) = x^2 + 1$ is one-to-one on all real numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q5/5

Prove that linear functions grow by equal differences and exponential functions grow by equal factors over equal intervals

MAIN	Algebra I main question	**HSF-LE.A.1.a**
BACKWARD 1	Tell linear from nonlinear functions	**8.F.A.3**
BACKWARD 2	Compare average rates of change over intervals	**HSF-IF.B.6**
FORWARD 1	Derive the formula for the sum of a finite geometric series	**HSA-SSE.B.4**
FORWARD 2	Use logarithms to turn equal factors into equal differences	**HSF-BF.B.5**

SET 53 • HSF-LE.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

$f(x) = 3x + 5$. Show that f grows by equal differences over equal intervals: find $f(x + h) - f(x)$ in simplest form, and explain why it does not depend on x.

$f(x + h) - f(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q1/5

$g(x) = 4 \cdot 2^x$. Show that g grows by equal factors over equal intervals: find $g(x + h) \div g(x)$ in simplest form, and explain why it does not depend on x.

$g(x + h) \div g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q2/5

A table shows x = 0, 2, 4, 6 and y = 5, 11, 17, 23.

Is the relationship linear or exponential? Explain using equal intervals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q3/5

A table shows x = 0, 1, 2, 3 and y = 80, 40, 20, 10.

Which description is correct?

A. Linear: y decreases by 40 each time

B. Exponential: y is multiplied by $\frac{1}{2}$ each time x increases by 1

C. Linear: y decreases by 20 each time

D. Neither linear nor exponential

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q4/5

True or false?

A table with y-values 2, 5, 10, 17 for x = 0, 1, 2, 3 is linear.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q5/5

SET 53 • HSF-LE.A.1.a

BACKWARD 1

Tell linear from nonlinear functions • Grade 8 • Functions • 8.F.A.3

Is $y = x^2 + 1$ linear? Use the points (0, 1), (1, 2), and (2, 5) to explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q1/5

Is $y = 4 - 2x$ linear? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q2/5

Which function is linear?

A. $y = 2^x$

B. $y = x^2$

C. $y = 0.5x - 3$

D. $y = 1 \div x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q3/5

True or false?

The graph of a linear function is a straight line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q4/5

True or false?

The area of a square, $A = s^2$, is a linear function of s.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSF-LE.A.1.a

BACKWARD 2

Compare average rates of change over intervals • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = 3x + 5$.

Find the average rate of change from $x = 0$ to $x = 10$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q1/5

$g(x) = 4 \cdot 2^x$.

Find the average rate of change from $x = 0$ to $x = 1$, and from $x = 1$ to $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q2/5

What is the average rate of change of $5 \cdot 3^x$ from $x = 0$ to $x = 2$?

A. 15

B. 20

C. 40

D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q3/5

True or false?

An exponential growth function has a larger average rate of change on later intervals of the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q4/5

True or false?

The average rate of change of 2^x from $x = 0$ to $x = 3$ is 2.

A. True

B. False

SET 53 • HSF-LE.A.1.a

FORWARD 1

Derive the formula for the sum of a finite geometric series • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Let $S = 1 + 2 + 4 + \ldots + 2^9$. Show that $2S - S = 2^{10} - 1$, and find S.

$S =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q1/5

Use $S = a(1 - r^n) \div (1 - r)$ to find the sum of the first 6 terms of 1, 3, 9, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q2/5

When deriving the sum formula, why do we multiply S by r?

A. To double the sum

B. So that $S - rS$ cancels all but two terms

C. To make every term equal

D. To make r equal 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q3/5

True or false?

For $r \neq 1$, $a + ar + \ldots + ar^{n-1} = a(1 - r^n) \div (1 - r)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q4/5

True or false?

$1 + 3 + 9 + 27 = 3^4$

A. True

B. False

SET 53 • HSF-LE.A.1.a

FORWARD 2

Use logarithms to turn equal factors into equal differences • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

$g(x) = 4 \cdot 2^x$.

Show that $\log_2 g(x) = x + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q1/5

$y = 10^x$.

What is log y? (log means base 10.)

log y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q2/5

The values y = 1, 10, 100, 1,000 occur at x = 0, 1, 2, 3.

What are the values of log y? (log means base 10.)

A. 0, 1, 2, 3

B. 0, 10, 20, 30

C. 1, 2, 3, 4

D. 1, 10, 100, 1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q3/5

True or false?

For $a > 0$ and $b > 0$, taking the logarithm of $a \cdot b^x$ gives a linear function of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q4/5

True or false?

$\log(100 \cdot 10^x) = 100x$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q5/5

Recognize situations in which a quantity changes at a constant rate per unit interval

MAIN	Algebra I main question	**HSF-LE.A.1.b**
BACKWARD 1	Find the constant of proportionality	**7.RP.A.2.b**
BACKWARD 2	Write a linear function from a rate and an initial value	**8.F.B.4**
FORWARD 1	Relate arc length and sector area to the central angle — nearest r…	**HSG-C.B.5**
FORWARD 2	Use radian measure as arc length on a circle — nearest related	**HSF-TF.A.1**

SET 54 • HSF-LE.A.1.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b

A pool fills at a constant rate. At 2 p.m. it holds 1,500 gallons; at 5 p.m. it holds 3,300 gallons.

What is the rate, and how much water was in it at noon?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q1/5

A plant is 6 cm tall on day 0 and grows 0.8 cm each day.

How tall is it on day 15?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q2/5

A taxi charges the same amount for every mile.

Explain why the fare for the miles driven is a linear function of the miles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q3/5

Which situation is modeled by a linear function?

A. A cell population doubles each hour.

B. A tank loses 4 liters each minute.

C. A loan grows 4% each month.

D. A car loses 15% of its value each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q4/5

True or false?

A savings account that earns the same dollar amount of interest each year grows linearly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q5/5

SET 54 • HSF-LE.A.1.b

BACKWARD 1

Find the constant of proportionality • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

$y = 2.5x$ gives the cost of x pounds of apples.

What is the constant of proportionality?

Constant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q1/5

A proportional table has x = 2, 4, 6 and y = 7, 14, 21.

What is the unit rate?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q2/5

A cyclist rides 12 miles in 3 hours at a constant rate.

What is the constant of proportionality in miles per hour?

A. 15

B. 36

C. $\frac{1}{4}$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q3/5

True or false?

In $y = 9x$, y increases by 9 whenever x increases by 1.

A. True

B. False

True or false?

A proportional relationship through (5, 20) has constant of proportionality 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSF-LE.A.1.b

BACKWARD 2

Write a linear function from a rate and an initial value • Grade 8 • Functions • 8.F.B.4

A line passes through (2, 1,500) and (5, 3,300).

Find its slope and its y-intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q1/5

A phone battery is at 80% and drops 5 percentage points per hour.

Write a function B(t) for the charge after t hours.

B(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q2/5

Which function has rate of change 0.8 and initial value 6?

A. $y = 6x + 0.8$

B. $y = 0.8x + 6$

C. $y = 6.8x$

D. $y = 6(0.8)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q3/5

True or false?

A function with initial value 10 and rate of change -2 is $y = 10 - 2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q4/5

True or false?

The line through (0, 4) and (2, 10) has rate of change 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSF-LE.A.1.b

FORWARD 1

Relate arc length and sector area to the central angle • High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Geometry standard. Connection: Arc length grows by a constant amount (the radius) for each radian, a constant-rate situation.

In a circle of radius 5 cm, by how much does the arc length increase for each 1-radian increase in the central angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q1/5 • NEAREST RELATED

A circle has radius 10.

Find the arc lengths for central angles of 0.5, 1, and 1.5 radians.

Arc lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q2/5 • NEAREST RELATED

The area of a sector is $A = \frac{1}{2}r^2\theta$, and $r = 4$.

By how much does A increase when θ increases by 1 radian?

A. 4

B. 2

C. 16

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

For a fixed radius, arc length is a linear function of the central angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

Doubling the radius while keeping the angle the same doubles the area of a sector.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 54 • HSF-LE.A.1.b

FORWARD 2

Use radian measure as arc length on a circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Algebra II standard. Connection: Radian measure is arc length per unit of radius, continuing the constant-rate arc length of FORWARD 1.

A wheel of radius 0.3 m turns through 4π radians.

How far does a point on its rim travel? Give the exact distance and the distance to the nearest hundredth of a meter.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q1/5 • NEAREST RELATED

A central angle cuts off an arc of length 6 on a circle of radius 2.

What is its radian measure?

Radians: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q2/5 • NEAREST RELATED

A wheel turns at 3 radians per second.

Through about how many degrees does it turn in 2 seconds?

A. 6°

B. 172°

C. 1,080°

D. 344°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

One radian is the angle that cuts off an arc as long as the radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A rotation of $-\pi \div 2$ radians is a quarter turn counterclockwise.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q5/5 • NEAREST RELATED

Recognize situations in which a quantity grows or decays by a constant percent rate per unit interval

MAIN	Algebra I main question	**HSF-LE.A.1.c**
BACKWARD 1	Read growth and decay rates from exponential functions	**HSF-IF.C.8.b**
BACKWARD 2	Solve percent problems	**7.RP.A.3**
FORWARD 1	Find times in percent-rate models with logarithms	**HSF-LE.A.4**
FORWARD 2	Use natural logarithms with continuous growth	**HSF-BF.B.5**

SET 55 • HSF-LE.A.1.c

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c

A car worth $24,000 loses 15% of its value each year.

Write a function V(t), and find the value after 3 years to the nearest dollar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q1/5

Explain why a quantity that grows by 6% each year grows by equal factors over equal intervals.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q2/5

A town's population increases by 2.5% per year.

By what factor does it grow in 2 years? What is the percent increase over 2 years?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q3/5

Which situation is exponential?

A. A worker earns $15 each hour.

B. A medicine dose decreases by 20% each hour.

C. A tank gains 3 liters each second.

D. A runner covers 200 meters each minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-M-Q4**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q4/5

True or false?

A 10% increase followed by a 10% decrease returns a quantity to its original value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q5/5

SET 55 • HSF-LE.A.1.c

BACKWARD 1

Read growth and decay rates from exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

$V = 24{,}000(0.85)^t$.

What is the yearly percent change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q1/5

$P = 500(1.025)^t$.

Is this growth or decay, and at what rate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q2/5

Which function shows 4% decay per year?

A. $y = 300(1.04)^t$

B. $y = 300(0.04)^t$

C. $y = 300 - 0.04t$

D. $y = 300(0.96)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q3/5

True or false?

$y = 50(1.5)^t$ grows by 50% per unit of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q4/5

True or false?

$y = 80(0.75)^t$ decreases by 75% per unit of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSF-LE.A.1.c

BACKWARD 2

Solve percent problems • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A $24,000 car loses 15% of its value.

What is its new value?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q1/5

What is 8% sales tax on $45?

Tax: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q2/5

What is the price of a $60 item after a 25% discount?

A. $15

B. $45

C. $75

D. $35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q3/5

True or false?

20% of 150 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q4/5

True or false?

A 50% increase of 40 is 60, and a 50% decrease of 60 is 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSF-LE.A.1.c

FORWARD 1

Find times in percent-rate models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$V(t) = 24{,}000(0.85)^t$.

When is V = 12,000? Round to the nearest tenth of a year.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q1/5

Solve $1.025^t = 2$. Round to the nearest tenth.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q2/5

Which expression gives the half-life of $y = (0.9)^t$? (log means base 10.)

A. 0.9 ÷ 2

B. 0.5 ÷ 0.9

C. log 0.9 ÷ log 0.5

D. log 0.5 ÷ log 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q3/5

True or false?

$\log(0.85^t) = t \log 0.85$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q4/5

True or false?

If $2^t = 5$, then $t = 2.5$.

A. True

B. False

SET 55 • HSF-LE.A.1.c

FORWARD 2

Use natural logarithms with continuous growth • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

Write the inverse of $y = e^x$.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q1/5

Solve $e^{0.05t} = 3$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q2/5

What is $\ln(e^4)$? (ln means base e.)

A. 4e

B. 4

C. 1

D. e

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q3/5

True or false?

e raised to the power ln 7 equals 7. (ln means base e.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q4/5

True or false?

ln 1 = 1 (ln means base e.)

A. True

B. False

Construct linear and exponential functions from a graph, a description, two points, or a table

MAIN	Algebra I main question	**HSF-LE.A.2**
BACKWARD 1	Write a linear function from two points or a table	**8.F.B.4**
BACKWARD 2	Write formulas for arithmetic and geometric sequences	**HSF-BF.A.2**
FORWARD 1	Build a sinusoidal model from a description	**HSF-TF.B.5**
FORWARD 2	Find a quadratic through three points with a matrix equation	**HSA-REI.C.8**

SET 56 • HSF-LE.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the linear function whose graph passes through (2, 11) and (6, 23).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q1/5

Write the exponential function $y = ab^x$ $(b > 0)$ whose graph passes through $(1, 6)$ and $(3, 54)$.

$y =$ _____

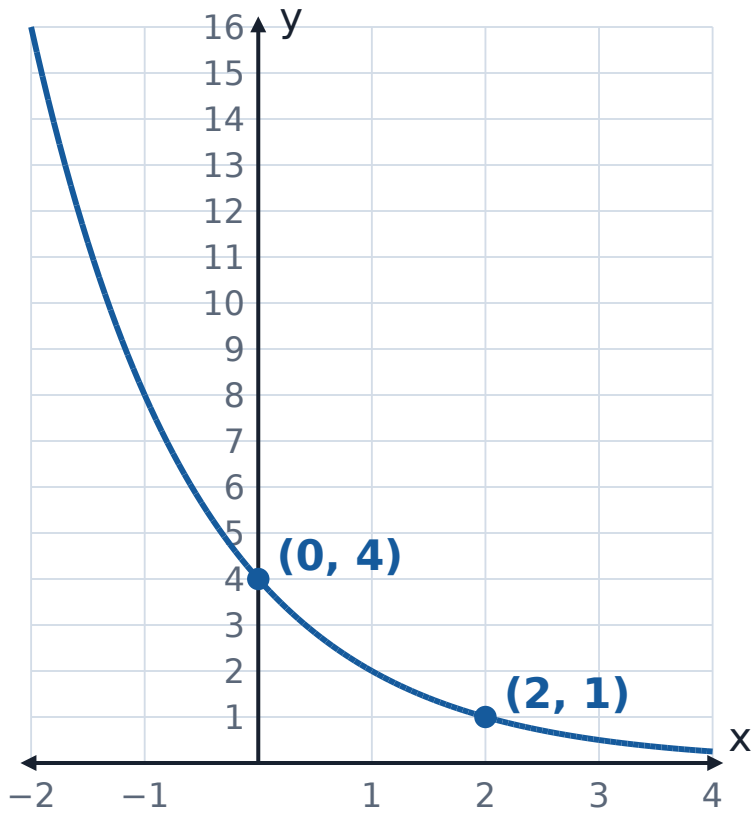
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q2/5

The graph of an exponential function $y = ab^x$ passes through the marked points.

Write the function.



$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q3/5

A table shows x = 0, 1, 2, 3 and y = 1.5, 3, 6, 12.

Which function fits?

A. $y = 2(1.5)^x$

B. $y = 1.5x + 1.5$

C. $y = 3x$

D. $y = 1.5(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q4/5

True or false?

The linear function with $f(0) = 10$ that decreases by 2 for each unit increase in x is $f(x) = 10(0.8)^x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q5/5

SET 56 • HSF-LE.A.2

BACKWARD 1

Write a linear function from two points or a table • Grade 8 • Functions • 8.F.B.4

Write the linear function through (0, −3) and (4, 5).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q1/5

A table shows x = 1, 2, 3 and y = 10, 7, 4.

Write the linear function.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q2/5

Which line passes through $(-1, 2)$ and $(1, 6)$?

A. $y = -2x + 4$

B. $y = 2x + 4$

C. $y = 2x + 2$

D. $y = 4x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q3/5

True or false?

The line through (0, 0) and (3, 6) is $y = 2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q4/5

True or false?

The line through (1, 1) and (2, 3) is $y = 2x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSF-LE.A.2

BACKWARD 2

Write formulas for arithmetic and geometric sequences • High School — Algebra I • Functions: Building Functions • HSF-BF.A.2

A geometric sequence has a(1) = 2 and ratio 3.

Write an explicit formula.

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q1/5

Write an explicit formula for 10, 7, 4, ...

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q2/5

Which is a recursive formula for 6, 3, 1.5, ...?

A. $a(n) = 6 - 3n$

B. $a(1) = 6, a(n) = 0.5a(n - 1)$

C. $a(1) = 6, a(n) = 2a(n - 1)$

D. $a(1) = 6, a(n) = a(n - 1) - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q3/5

True or false?

An arithmetic sequence is a linear function on the positive integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q4/5

True or false?

The geometric sequence 5, 10, 20, ... has ratio 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSF-LE.A.2

FORWARD 1

Build a sinusoidal model from a description • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

A tide has high water 10 ft and low water 2 ft, with period 12 hours and high tide at t = 0.

Write a cosine model h(t).

h(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q1/5

A sinusoid has maximum 10 and minimum 2.

What are its amplitude and midline?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q2/5

A model $y = \cos(Bt)$ has period 12.

What is B?

A. 12

B. $\pi \div 12$

C. $\pi \div 6$

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q3/5

True or false?

A sinusoidal model with maximum 8 and minimum 0 has amplitude 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q4/5

True or false?

$y = 3\cos(2t)$ has period 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSF-LE.A.2

FORWARD 2

Find a quadratic through three points with a matrix equation • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8

$y = ax^2 + bx + c$ passes through (0, 1), (1, 2), and (2, 7).

Write the system for a, b, and c as a matrix equation $AX = B$, with X the column [a, b, c]. Give A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q1/5

A has rows [0 0 1], [1 1 1], and [4 2 1], X is the column [a, b, c], and B is the column [1, 2, 7]. Solve AX = B to find the parabola $y = ax^2 + bx + c$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q2/5

A parabola $y = ax^2 + bx + c$ passes through (0, 1), (1, 2), and (2, 7). It is written as $AX = B$ with X the column $[a, b, c]$.

Which row of A comes from the point (2, 7)?

A. [4 2 1]

B. [2 4 1]

C. [4 2 7]

D. [2 2 1]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q3/5

True or false?

Three points with different x-values that are not on one line determine exactly one parabola $y = ax^2 + bx + c$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q4/5

True or false?

A parabola $y = ax^2 + bx + c$ through (0, 1), (1, 2), and (2, 7) gives a matrix equation $AX = B$ in which A has 2 rows.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q5/5

Observe that exponential growth eventually exceeds linear, quadratic, and polynomial growth

MAIN	Algebra I main question	**HSF-LE.A.3**
BACKWARD 1	Tell equal differences from equal factors	**HSF-LE.A.1.a**
BACKWARD 2	Compare two functions given in different forms	**8.F.A.2**
FORWARD 1	Compare the growth of polynomial functions	**HSF-IF.C.7.c**
FORWARD 2	Compare logarithmic growth with other growth	**HSF-BF.B.5**

SET 57 • HSF-LE.A.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3

$f(x) = 100x$ and $g(x) = 2^x$.

What is the smallest whole number $x \geq 1$ for which $g(x) > f(x)$?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q1/5

$f(x) = x^2$ and $g(x) = 1.5^x$.

Which is larger at $x = 10$? At $x = 20$? What happens for larger x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q2/5

Plan A pays $1,000 per day. Plan B pays 1 cent on day 1, and its daily pay doubles each day.

On which day does Plan B's daily pay first exceed Plan A's?

Day: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q3/5

Which function eventually has the greatest values as x increases?

A. $y = 50x^2$

B. $y = 1.01^x$

C. $y = x^3 + 500$

D. $y = 1{,}000x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q4/5

True or false?

Because $3x^2 > 2^x$ at $x = 5$, $3x^2$ stays greater than 2^x for all $x > 5$.

A. True

B. False

SET 57 • HSF-LE.A.3

BACKWARD 1

Tell equal differences from equal factors • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

A table shows x = 0, 1, 2, 3 and y = 3, 6, 12, 24.

Is it linear or exponential? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q1/5

A table shows x = 0, 1, 2, 3 and y = 3, 6, 9, 12.

Is it linear or exponential? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q2/5

Which function grows by equal factors over equal intervals?

A. $y = 7 + 1.1x$

B. $y = 7(1.1)^x$

C. $y = 7x$

D. $y = 1.1x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q3/5

True or false?

Linear functions grow by equal differences over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q4/5

True or false?

$y = 5x^2$ grows by equal factors over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSF-LE.A.3

BACKWARD 2

Compare two functions given in different forms • Grade 8 • Functions • 8.F.A.2

Function A is $y = 4x + 1$. Function B has $x = 0, 1, 2$ and $y = 3, 5, 7$.

Which has the greater rate of change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q1/5

Function A is $y = -x + 6$. Function B is the line through $(0, 2)$ and $(2, 8)$.

Which has the greater y-intercept?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q2/5

$f(x) = 3x - 2$. Function g has $g(0) = 1$ and $g(1) = 5$, and it is linear.

Which statement is true?

A. f has the greater rate of change.

B. f has the greater initial value.

C. g has the greater rate of change and the greater initial value.

D. f and g have the same rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q3/5

True or false?

A linear function with slope 5 increases faster than one with slope 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q4/5

True or false?

$y = 2x + 10$ has greater values than $y = 3x$ for every x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSF-LE.A.3

FORWARD 1

Compare the growth of polynomial functions • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

For which $x > 0$ is $x^3 > 100x^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q1/5

Which grows faster for large x, x^4 or $1{,}000x^3$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q2/5

As $x \to \infty$, what does $f(x) = x^4 - 50x^3$ do?

A. $f(x) \to -\infty$

B. $f(x) \to -50$

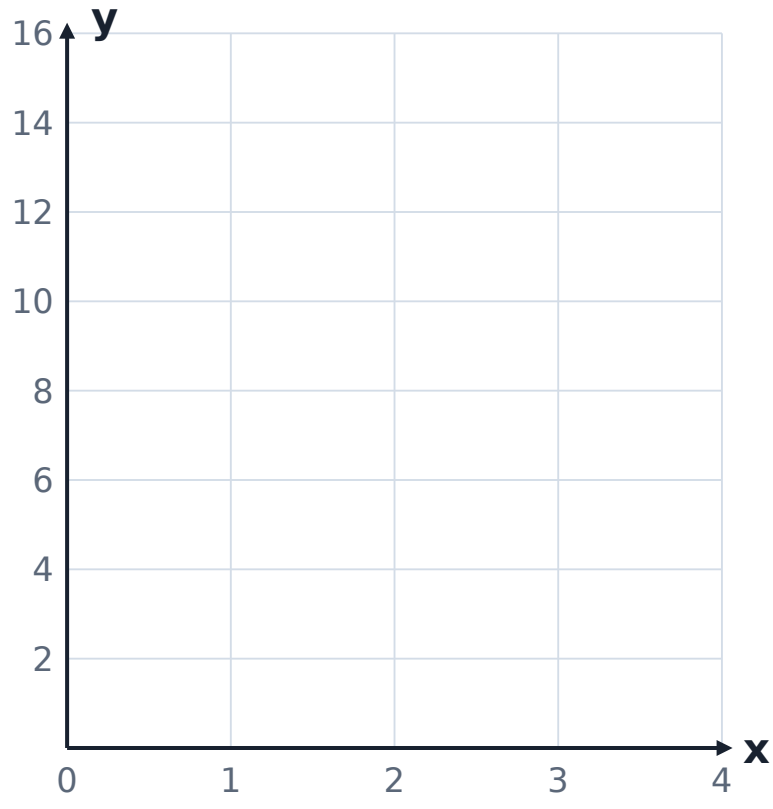
C. $f(x) \to \infty$

D. $f(x) \to 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q3/5

Graph $y = x^2$ and $y = 0.5x^3$ for $0 \le x \le 4$ on the same grid. Mark where they meet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q4/5

True or false?

For every large x, $0.001x^3$ is less than $1{,}000x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSF-LE.A.3

FORWARD 2

Compare logarithmic growth with other growth • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

Find $\log_2 1{,}024$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q1/5

By what factor must x be multiplied for log x to increase by 1? (log means base 10.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q2/5

Which is greatest when x = 1,000,000? (log means base 10; ln means base e.)

A. $\sqrt{x}$

B. $\log_2 x$

C. $\ln x$

D. $\log x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q3/5

True or false?

Eventually, log x grows more slowly than any linear function with positive slope. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q4/5

True or false?

$\log_2 x = 100$ when $x = 200$.

A. True

B. False

Interpret the parameters of a linear or exponential function in context

MAIN	Algebra I main question	**HSF-LE.B.5**
BACKWARD 1	Interpret the parts of an expression in context	**HSA-SSE.A.1.a**
BACKWARD 2	Interpret the rate and initial value of a linear function	**8.F.B.4**
FORWARD 1	Interpret the parameters of a sinusoidal model	**HSF-TF.B.5**
FORWARD 2	Interpret the mean and standard deviation of a normal model — n…	**HSS-ID.A.4**

SET 58 • HSF-LE.B.5

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

$C(m) = 0.35m + 40$ is the cost in dollars of renting a car for one day and driving m miles.

What do 0.35 and 40 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q1/5

$P(t) = 1{,}200(1.04)^t$ is a town's population t years after 2020.

What do 1,200 and 1.04 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q2/5

$A(t) = 80(0.5)^{t/6}$ is the milligrams of a medicine left in the body after t hours.

What do 80, 0.5, and 6 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q3/5

$V(t) = 18{,}000 - 1{,}500t$ is a machine's value in dollars after t years.

What does $-1{,}500$ mean?

A. The machine loses $1,500 of value each year.

B. The machine loses 1,500% of its value each year.

C. The machine was bought for $1,500.

D. The machine is worth $18,000 after 1,500 years.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q4/5

True or false?

In $y = 250(0.92)^t$, the value decreases by 92% each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q5/5

SET 58 • HSF-LE.B.5

BACKWARD 1

Interpret the parts of an expression in context • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A plumber charges 75 + 60h dollars for h hours of work.

What does 75 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q1/5

In $3{,}000(1.05)^t$, what does the factor 1.05 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q2/5

A box with a square base of side s and height h has surface area $2s^2 + 4sh$.

What does 4sh represent?

A. The volume

B. The area of the four side faces

C. The perimeter of the base

D. The area of the top and bottom

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q3/5

True or false?

In 500 − 20w, the term −20w shows a decrease of 20 for each unit of w.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q4/5

True or false?

In 1.5x + 7, the coefficient of x is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSF-LE.B.5

BACKWARD 2

Interpret the rate and initial value of a linear function • Grade 8 • Functions • 8.F.B.4

Cost y = 12x + 30 for x hours of bike rental.

What is the initial value, and what does it mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q1/5

The water in a tank is $y = 500 - 25t$ gallons after t minutes.

What does -25 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q2/5

A plant's height is a linear function of time. It is 8 cm on day 0 and 20 cm on day 4.

What is the rate of change?

A. 5 cm per day

B. 20 cm per day

C. 8 cm per day

D. 3 cm per day

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q3/5

True or false?

In $y = 40x + 100$, 100 is the value of y when $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q4/5

True or false?

In d = 65t, 65 is the starting distance.

A. True

B. False

SET 58 • HSF-LE.B.5

FORWARD 1

Interpret the parameters of a sinusoidal model • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

$T(t) = 15 + 10\sin(\pi t \div 12)$ is the temperature in °C t hours after 8 a.m.

What do 15 and 10 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q1/5

$T(t) = 15 + 10\sin(\pi t \div 12)$ is the temperature in °C t hours after 8 a.m.

What is the period of T, and what does it mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q2/5

$h(t) = 1.5\cos(\pi t \div 6) + 3$ is the depth of water in meters.

What is the greatest depth?

A. 4.5 m

B. 3 m

C. 6 m

D. 1.5 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q3/5

True or false?

In $y = A \sin(Bx) + D$, D gives the midline.

A. True

B. False

True or false?

The amplitude of $y = 1.5\cos(\pi t \div 6) + 3$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSF-LE.B.5

FORWARD 2

Interpret the mean and standard deviation of a normal model • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Interpreting the parameters of a function in context leads to interpreting the parameters of a normal model.

Battery life is normal with mean 400 hours and standard deviation 20 hours.

What does the 20 tell you?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q1/5 • NEAREST RELATED

Battery life is normal with mean 400 hours and standard deviation 20 hours.

About what percent of batteries last between 360 and 440 hours?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 58 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which change makes a normal curve wider?

A. Increasing the standard deviation

B. Increasing the mean

C. Decreasing the standard deviation

D. Decreasing the mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Changing the mean of a normal model shifts the curve left or right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

In a normal model with mean 50 and standard deviation 5, the value 60 is 10 standard deviations above the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 58 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 59 • HSS-ID.A.1 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Represent data with dot plots, histograms, and box plots

MAIN	Algebra I main question	**HSS-ID.A.1**
BACKWARD 1	Display numerical data in plots	**6.SP.B.4**
BACKWARD 2	Compute measures of center and variation	**6.SP.B.5.c**
FORWARD 1	Decide whether a normal model fits a distribution	**HSS-ID.A.4**
FORWARD 2	Decide whether results are consistent with a model by simulation…	**HSS-IC.A.2**

SET 59 • HSS-ID.A.1

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Make a dot plot of the numbers of books read by 10 students:

2, 3, 3, 4, 4, 4, 5, 5, 6, 9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q1/5

Find the five-number summary of 2, 3, 3, 4, 4, 4, 5, 5, 6, 9. Use the median of each half (lower five and upper five values) as Q1 and Q3.

Summary: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q2/5

For the data 2, 3, 3, 4, 4, 4, 5, 5, 6, 9 (Q1 and Q3 are the medians of the lower five and upper five values), use the 1.5 × IQR rule to decide whether 9 is an outlier.

Answer: _____

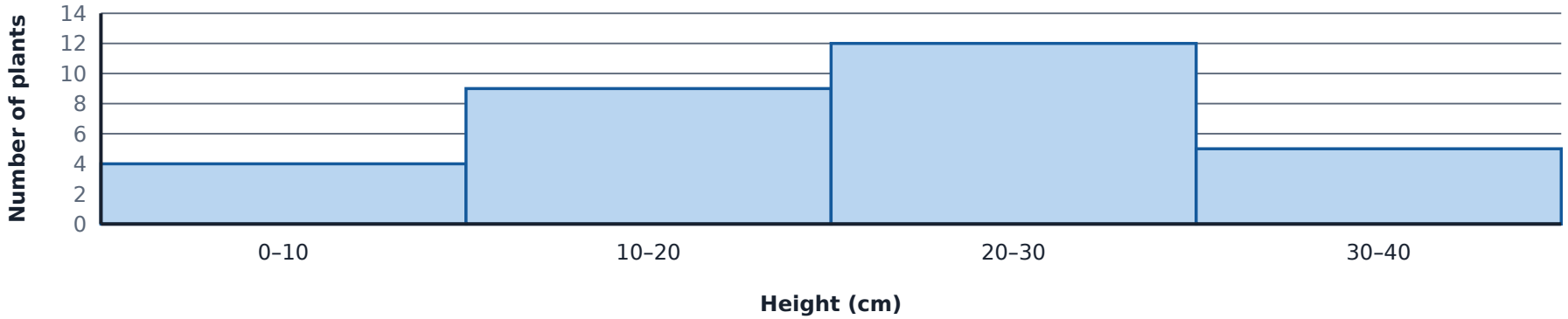
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q3/5

The histogram shows the heights of 30 plants. Each bar includes its lower edge but not its upper edge.

How many plants are at least 20 cm tall?



A. 21 **B.** 17 **C.** 5 **D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q4/5

True or false?

A box plot shows the mean of a data set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q5/5

SET 59 • HSS-ID.A.1

BACKWARD 1

Display numerical data in plots • Grade 6 • Statistics & Probability • 6.SP.B.4

Which display shows every data value, a dot plot or a box plot?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q1/5

Make a dot plot of 1, 2, 2, 3, 3, 3, 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q2/5

Which display groups data values into intervals?

A. Box plot

B. Dot plot

C. Histogram

D. Ordered list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q3/5

True or false?

A box plot is made from the five-number summary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q4/5

True or false?

A histogram shows each individual data value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • HSS-ID.A.1

BACKWARD 2

Compute measures of center and variation • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median of 3, 8, 5, 10, 4.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q1/5

Find the mean of 3, 8, 5, 10, 4.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q2/5

Q1 = 12 and Q3 = 20.

What is the interquartile range?

A. 32

B. 8

C. 4

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q3/5

True or false?

The range of 2, 7, 9, 15 is 13.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q4/5

True or false?

The median of 1, 2, 3, 100 is 26.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSS-ID.A.1

FORWARD 1

Decide whether a normal model fits a distribution • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

A data set is strongly skewed to the right.

Is a normal model appropriate? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q1/5

Scores are normal with mean 75 and standard deviation 8.

What score is 1 standard deviation above the mean?

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q2/5

Values are normal with mean 20 and standard deviation 4.

About what percent are between 12 and 28?

A. 99.7%

B. 50%

C. 68%

D. 95%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q3/5

True or false?

In a normal distribution, the mean and the median are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 59 • FORWARD 1 • Q4/5

True or false?

About 50% of a normal distribution lies within 1 standard deviation of the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSS-ID.A.1

FORWARD 2

Decide whether results are consistent with a model by simulation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Simulation results are summarized with the data displays of the MAIN set and then compared with a model.

In 100 simulated sets of 10 fair coin flips, 8 or more heads happened in 5 sets.

Estimate the probability of 8 or more heads in 10 fair flips.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 59 • FORWARD 2 • Q1/5 • NEAREST RELATED

A spinner model says red comes up 25% of the time. In 200 spins, red came up 52 times.

What proportion is that? Is it consistent with the model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 59 • FORWARD 2 • Q2/5 • NEAREST RELATED

A model says a coin lands heads 50% of the time.

Which result most strongly questions the model?

A. 52 heads in 100 flips

B. 6 heads in 10 flips

C. 95 heads in 100 flips

D. 48 heads in 100 flips

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 59 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A result that rarely happens in simulations of a model is evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 59 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

If a coin lands heads 6 times in 10 flips, the coin must be unfair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 59 • FORWARD 2 • Q5/5 • NEAREST RELATED

Make histograms from data

MAIN	Algebra I main question	**HSS-ID.A.1**
BACKWARD 1	Display data in a histogram	**6.SP.B.4**
BACKWARD 2	Report the number of observations	**6.SP.B.5.a**
FORWARD 1	Use a histogram's shape to decide whether a normal model fits	**HSS-ID.A.4**
FORWARD 2	Use a histogram of simulated results to judge a model	**HSS-IC.A.2**

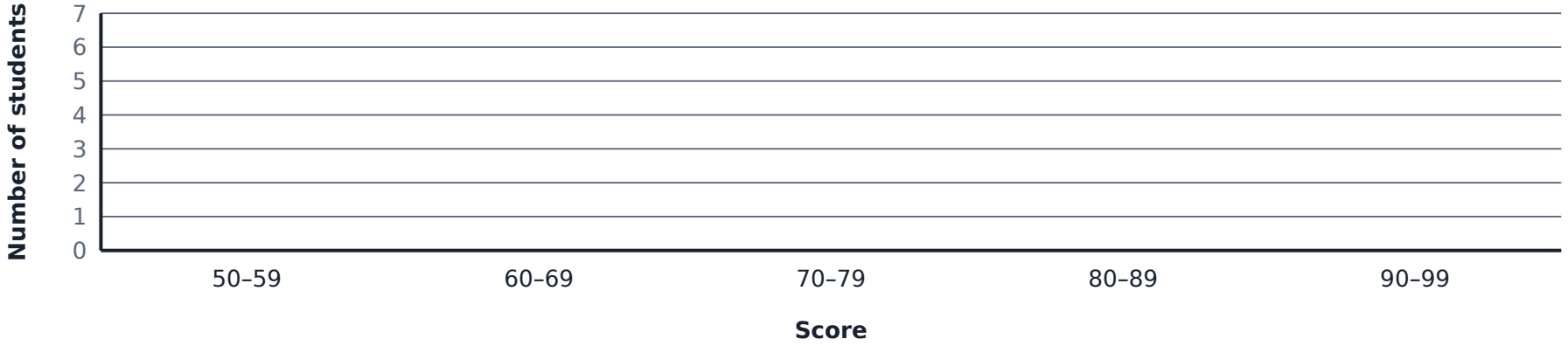
SET 76 • HSS-ID.A.1

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Make a histogram of these 20 test scores. Use the intervals 50–59, 60–69, 70–79, 80–89, and 90–99.

52, 58, 61, 64, 67, 69, 70, 72, 74, 75, 77, 78, 81, 83, 85, 86, 88, 91, 94, 97



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q1/5

Using the intervals 50–59, 60–69, 70–79, 80–89, and 90–99 for the scores 52, 58, 61, 64, 67, 69, 70, 72, 74, 75, 77, 78, 81, 83, 85, 86, 88, 91, 94, 97:

how many scores are in the 70–79 interval, and which interval has the tallest bar?

Answer: _____

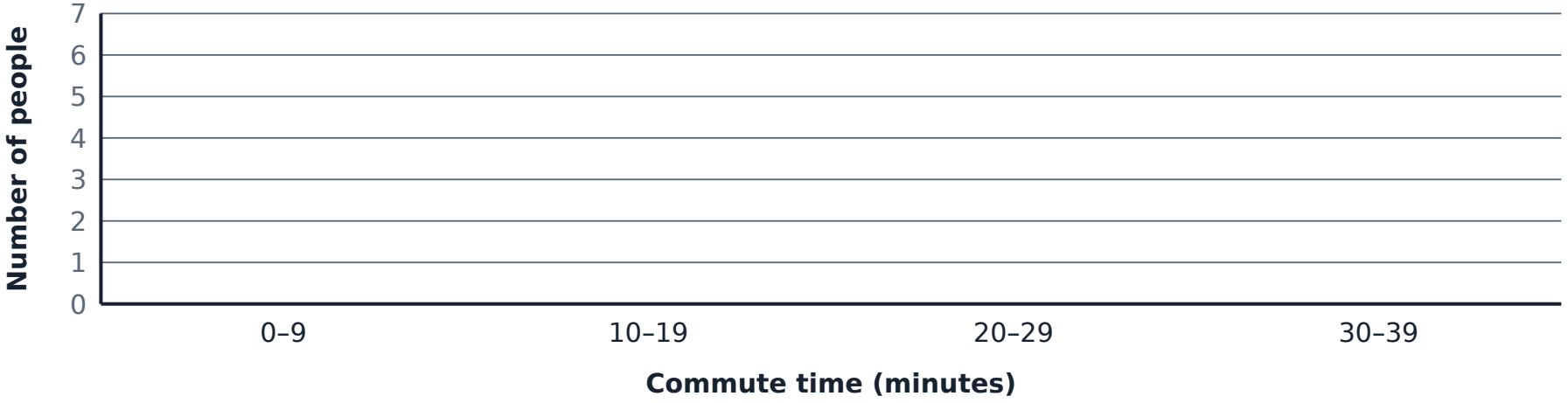
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q2/5

Make a histogram of these 16 commute times, in minutes. Use the intervals 0–9, 10–19, 20–29, and 30–39.

4, 8, 12, 14, 15, 18, 19, 21, 22, 22, 25, 28, 29, 33, 36, 38



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q3/5

A histogram uses the intervals 10–19 and 20–29 for whole-number data.

In which bar is the value 20 counted?

A. 20–29

B. 10–19

C. Both

D. Neither

True or false?

Using wider intervals for the same data gives a histogram with more bars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q5

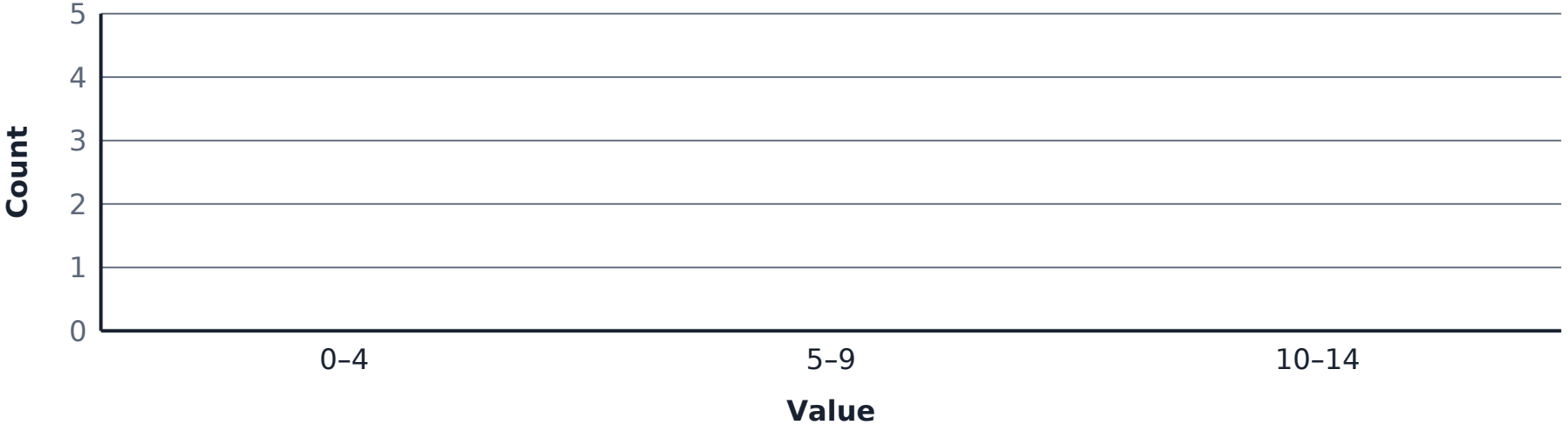
High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q5/5

SET 76 • HSS-ID.A.1

BACKWARD 1

Display data in a histogram • Grade 6 • Statistics & Probability • 6.SP.B.4

Make a histogram of 1, 3, 4, 6, 7, 8, 9, 12 with the intervals 0–4, 5–9, and 10–14.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q1/5

A histogram has bars of heights 3, 5, and 2. How many data values does it show?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q2/5

Which data are best shown in a histogram?

A. The heights of 200 students

B. Each student's favorite color

C. One test score

D. The names of 5 cities

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q3/5

True or false?

In a histogram, the height of a bar shows how many values are in that interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q4/5

True or false?

A histogram is the best display for categories such as favorite colors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q5/5

SET 76 • HSS-ID.A.1

BACKWARD 2

Report the number of observations • Grade 6 • Statistics & Probability • 6.SP.B.5.a

How many observations are in the data set 4, 7, 7, 9, 10, 12?

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q1/5

A histogram has bars of heights 2, 4, 6, 5, and 3. How many observations does it show?

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q2/5

A survey recorded the number of pets owned by each of 25 students. How many observations are there?

A. 1

B. 25

C. 100

D. The total number of pets

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q3/5

True or false?

The number of observations is the number of data values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q4/5

True or false?

If 8 students each report 2 siblings, there are 16 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSS-ID.A.1

FORWARD 1

Use a histogram's shape to decide whether a normal model fits • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

A histogram of 1,000 adult heights is single-peaked and symmetric, with mean 170 cm and standard deviation 8 cm. About how many heights are between 162 cm and 178 cm?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 76 • FORWARD 1 • Q1/5

A histogram of household incomes is strongly skewed right. Would a normal model give good estimates of percents? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q2/5

Which histogram shape suggests that a normal model may fit?

A. Two separate peaks

B. Flat (uniform)

C. Single-peaked and symmetric

D. Strongly skewed to the left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q3/5

True or false?

In a normal model, about 95% of values are within 2 standard deviations of the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 76 • FORWARD 1 • Q4/5

True or false?

Any histogram with one peak is normal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSS-ID.A.1

FORWARD 2

Use a histogram of simulated results to judge a model • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

100 simulated trials of 20 fair coin flips give a histogram of the number of heads centered near 10. What does the histogram show?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 76 • FORWARD 2 • Q1/5

In that simulation, 2 of the 100 trials had 15 or more heads. Estimate the probability of 15 or more heads in 20 fair flips.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 76 • FORWARD 2 • Q2/5

A spinner is claimed to land on blue 50% of the time. In 50 spins it landed on blue 12 times. In 200 simulated sets of 50 spins with a 50% chance, 12 or fewer blue never happened.

What do you conclude?

A. Nothing can be concluded.

B. The result supports the 50% claim.

C. The spinner lands on blue exactly 24% of the time.

D. The result is not consistent with the 50% claim.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 76 • FORWARD 2 • Q3/5

True or false?

A simulation repeats a random process many times to show which results are typical under a model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 76 • FORWARD 2 • Q4/5

True or false?

If a result never happens in 200 simulated trials, it is impossible under the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 76 • FORWARD 2 • Q5/5

Make box plots from data

MAIN	Algebra I main question	**HSS-ID.A.1**
BACKWARD 1	Draw and read box plots	**6.SP.B.4**
BACKWARD 2	Find medians, quartiles, and ranges	**6.SP.B.5.c**
FORWARD 1	Estimate the quartiles of a normal distribution	**HSS-ID.A.4**
FORWARD 2	Compare two treatments with box plots and a randomization test	**HSS-IC.B.5**

SET 77 • HSS-ID.A.1

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Make a box plot of 3, 5, 6, 7, 8, 9, 10, 12, 15. (Use the median of each half, not counting the middle value, as Q1 and Q3.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q1/5

Find the five-number summary of 12, 15, 15, 18, 20, 22, 25, 30. (Q1 and Q3 are the medians of the lower four and upper four values.)

Summary: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q2/5

Make a box plot of 12, 15, 15, 18, 20, 22, 25, 30. (Q1 and Q3 are the medians of the lower four and upper four values.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q3/5

In a box plot, what does the length of the box show?

A. The interquartile range

B. The mean

C. The range

D. The median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q4/5

True or false?

In a box plot whose whiskers reach the minimum and maximum, each whisker covers about a quarter of the data values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q5/5

SET 77 • HSS-ID.A.1

BACKWARD 1

Draw and read box plots • Grade 6 • Statistics & Probability • 6.SP.B.4

Draw a box plot with minimum 2, Q1 4, median 5, Q3 8, and maximum 10.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q1/5

A box plot has Q1 = 20 and Q3 = 35. What is the interquartile range?

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q2/5

Which part of a box plot marks the median?

A. The line inside the box

B. The left end of the box

C. The right whisker

D. The left whisker

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q3/5

True or false?

A box plot shows the minimum and the maximum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q4/5

True or false?

A box plot shows how many data values there are.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSS-ID.A.1

BACKWARD 2

Find medians, quartiles, and ranges • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median of 4, 9, 2, 7, 5, 8.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q1/5

The ordered data are 2, 4, 5, 7, 8, 9. Find Q1, the median of the lower half 2, 4, 5.

Q1 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q2/5

What is the range of 14, 3, 9, 21?

A. 21

B. 18

C. 7

D. 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q3/5

True or false?

The interquartile range is Q3 − Q1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q4/5

True or false?

The median of 1, 3, 100 is 34.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSS-ID.A.1

FORWARD 1

Estimate the quartiles of a normal distribution • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

For a normal distribution, the quartiles are about $\mu - 0.67\sigma$ and $\mu + 0.67\sigma$. Estimate Q1 and Q3 for mean 100 and standard deviation 15.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 77 • FORWARD 1 • Q1/5

Using those quartiles, estimate the interquartile range of that normal distribution.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 77 • FORWARD 1 • Q2/5

In a normal distribution, about what percent of values are between Q1 and Q3?

A. 25%

B. 95%

C. 50%

D. 68%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q3/5

True or false?

A box plot of data from a normal distribution is roughly symmetric about the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 77 • FORWARD 1 • Q4/5

True or false?

For normal data, the median is far from the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSS-ID.A.1

FORWARD 2

Compare two treatments with box plots and a randomization test • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

Students were randomly assigned to a new study method (group A) or the usual method (group B). Box plots show a median of 78 for A and 70 for B, with similar spreads. In 1,000 random regroupings, a difference in means at least as large as observed happened 30 times.

Using a 5% threshold, is the difference significant?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q1/5

Why is it useful to draw side-by-side box plots of the two groups before running the test?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q2/5

In another randomized comparison, random regroupings matched or beat the observed difference 22% of the time.

Using a 5% threshold, what is the conclusion?

A. The difference is significant.

B. The methods are proved to have the same effect.

C. The new method is proved to work.

D. The difference is not significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q3/5

True or false?

In a randomized comparison, a difference that rarely appears in random regroupings is evidence of a treatment effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q4/5

True or false?

Overlapping box plots prove that two treatments have the same effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q5/5

SET 60 • HSS-ID.A.2 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Compare the center and spread of two or more data sets

MAIN	Algebra I main question	**HSS-ID.A.2**
BACKWARD 1	Tell measures of center from measures of variation	**6.SP.A.3**
BACKWARD 2	Compare two distributions using center and variability	**7.SP.B.3**
FORWARD 1	Use simulation to decide whether a difference between treatment…	**HSS-IC.B.5**
FORWARD 2	Evaluate reports based on data	**HSS-IC.B.6**

SET 60 • HSS-ID.A.2

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Class A scores: 70, 75, 80, 85, 90. Class B scores: 60, 70, 80, 90, 100.

Compare the centers and the spreads.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q1/5

Find the population standard deviation (divide by n) of 2, 4, 4, 4, 5, 5, 7, 9.

Standard deviation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q2/5

Store X's daily sales have median 50 and IQR 10. Store Y's have median 45 and IQR 25.

Which store typically sells more? Which is more consistent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q3/5

Which measure of spread is best for skewed data with outliers?

A. Range

B. Standard deviation

C. Mean

D. Interquartile range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q4/5

True or false?

Two data sets with the same mean must have the same standard deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q5/5

SET 60 • HSS-ID.A.2

BACKWARD 1

Tell measures of center from measures of variation • Grade 6 • Statistics & Probability • 6.SP.A.3

Which describes a typical value of a data set, the median or the IQR?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q1/5

What is the range of the data set 4, 4, 4, 4?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q2/5

Which is a measure of variation?

A. Median

B. Mode

C. Mean

D. Interquartile range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q3/5

True or false?

A measure of center summarizes all of the values with a single number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q4/5

True or false?

The range is a measure of center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • HSS-ID.A.2

BACKWARD 2

Compare two distributions using center and variability • Grade 7 • Statistics & Probability • 7.SP.B.3

Two classes have mean scores 78 and 84, and each has MAD 3.

How many MADs apart are the means?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q1/5

Team A's mean height is 170 cm and Team B's is 180 cm; each has MAD 5 cm.

Express the difference of the means as a multiple of the MAD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q2/5

Two means differ by 12, and each data set has MAD 4.

How many MADs is the difference?

A. 48

B. 12

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q3/5

True or false?

When two distributions have little variability, a small difference in their centers is easier to notice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q4/5

True or false?

If two data sets overlap, their means must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSS-ID.A.2

FORWARD 1

Use simulation to decide whether a difference between treatments is significant • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

Plants were randomly assigned to a fertilizer group or a control group with no fertilizer. The fertilizer group averaged 24 cm and the control group 20 cm. To test chance alone, the same heights were randomly regrouped 1,000 times, and a difference of 4 cm or more (first group minus second) happened 12 times.

Using a 5% threshold, is the difference significant? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.5 • Set 60 • FORWARD 1 • Q1/5

Why are subjects randomly assigned to treatments in an experiment?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.5 • Set 60 • FORWARD 1 • Q2/5

In a regrouping simulation, a difference as large as the observed one happened 400 times out of 1,000.

What does that suggest?

A. The treatment definitely works.

B. The difference could easily be due to chance.

C. The treatment definitely does not work.

D. The experiment had no control group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q3/5

True or false?

In a randomized experiment, a small simulated probability for the observed difference is evidence that the treatment had an effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q4/5

True or false?

In an experiment, the control group receives the new treatment being tested.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.5 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSS-ID.A.2

FORWARD 2

Evaluate reports based on data • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

An ad says "9 out of 10 dentists recommend Brand X," based on 10 dentists chosen by Brand X.

What is the flaw?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q1/5

An online poll of a website's visitors finds 80% support a policy.

Can the result be applied to all adults? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 60 • FORWARD 2 • Q2/5

Which sample is most likely to represent a school's students?

A. The first 50 students to arrive

B. The 50 students in the chess club

C. A random sample of 50 students from the school roster

D. 50 volunteers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions

HSS-IC.B.6 • Set 60 • FORWARD 2 • Q3/5

True or false?

A larger random sample generally gives a smaller margin of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 60 • FORWARD 2 • Q4/5

True or false?

A survey with many volunteers is free of bias.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSS-ID.A.3 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Interpret differences in shape, center, and spread, accounting for outliers

MAIN	Algebra I main question	**HSS-ID.A.3**
BACKWARD 1	Describe a distribution by its center, spread, and shape	**6.SP.A.2**
BACKWARD 2	Choose measures of center and spread that fit the data	**6.SP.B.5.d**
FORWARD 1	Use z-scores to judge how unusual a value is	**HSS-ID.A.4**
FORWARD 2	Distinguish a population parameter from a sample statistic — nea…	**HSS-IC.A.1**

SET 61 • HSS-ID.A.3

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Salaries at a small company, in thousands of dollars: 40, 42, 45, 45, 48, 50, 250.

Find the mean and the median. Which better describes a typical salary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q1/5

Salaries at a small company, in thousands of dollars: 40, 42, 45, 45, 48, 50, 250.

If the salary of 250 is removed, what happens to the mean and to the median?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q2/5

A histogram of wait times has a long tail on the right.

Describe its shape and the usual relationship between its mean and median.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q3/5

Which data set has the largest standard deviation?

A. 28, 29, 30, 31, 32

B. 25, 30, 30, 30, 35

C. 10, 20, 30, 40, 50

D. 30, 30, 30, 30, 30

True or false?

An outlier affects the median more than the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q5/5

SET 61 • HSS-ID.A.3

BACKWARD 1

Describe a distribution by its center, spread, and shape • Grade 6 • Statistics & Probability • 6.SP.A.2

Data: 1, 2, 2, 3, 3, 3, 4, 4, 5.

Describe the shape of the distribution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q1/5

Which value in 2, 3, 3, 3, 4, 15 is unusual?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q2/5

A dot plot has most values on the left and a few values far to the right.

What is its shape?

A. Skewed right

B. Symmetric

C. Uniform

D. Skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q3/5

True or false?

A distribution can be described by its center, its spread, and its shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q4/5

True or false?

A symmetric distribution must have an outlier.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSS-ID.A.3

BACKWARD 2

Choose measures of center and spread that fit the data • Grade 6 • Statistics & Probability • 6.SP.B.5.d

For data with an extreme outlier, should you report the mean or the median as the center? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q1/5

For symmetric data with no outliers, which measures of center and spread are usually reported?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q2/5

Home prices in a town include one mansion.

Which measure best describes a typical price?

A. The mean

B. The median

C. The maximum

D. The range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q3/5

True or false?

The median is not changed much by a single extreme value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q4/5

True or false?

The IQR changes greatly when the largest value is made even larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSS-ID.A.3

FORWARD 1

Use z-scores to judge how unusual a value is • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

A distribution has mean 50 and standard deviation 4.

What is the z-score of 58?

$z =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

A score of 92 on a test with mean 80 and standard deviation 6, or a score of 85 on a test with mean 70 and standard deviation 10: which is better relative to its test?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

In a normal distribution, which z-score is the most unusual?

A. -3.1

B. -0.5

C. 1.2

D. 2.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

True or false?

A z-score of 0 means the value equals the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

True or false?

A z-score of -2 is above the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

FORWARD 2

Distinguish a population parameter from a sample statistic • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Centers and spreads computed from data are sample statistics that estimate population parameters.

What is the difference between a population and a sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.1 • Set 61 • FORWARD 2 • Q1/5 • NEAREST RELATED

A random sample of 200 students has a mean of 7.1 hours of sleep.

What does 7.1 estimate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.1 • Set 61 • FORWARD 2 • Q2/5 • NEAREST RELATED

A parameter describes

A. a graph

B. one individual

C. a population

D. a sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.1 • Set 61 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Different random samples from the same population can give different sample means.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.1 • Set 61 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A sample statistic always equals the population parameter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.1 • Set 61 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 62 • HSS-ID.B.5 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Summarize categorical data in two-way frequency tables and interpret relative frequencies

MAIN	Algebra I main question	**HSS-ID.B.5**
BACKWARD 1	Read two-way tables for association	**8.SP.A.4**
BACKWARD 2	Write a part of a whole as a percent	**7.RP.A.3**
FORWARD 1	Use a two-way table to find conditional probabilities and test inde…	**HSS-CP.A.4**
FORWARD 2	Explain conditional probability and independence in everyday lang..	**HSS-CP.A.5**

SET 62 • HSS-ID.B.5

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

The table shows a survey of 200 students.

What percent of all the students play a sport?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q1/5

What percent of Grade 10 students play a sport? What percent of Grade 9 students do?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q2/5

Is there an association between grade and playing a sport? Explain.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q3/5

What is the joint relative frequency of "Grade 9 and does not play a sport"?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. 50%

B. 57.9%

C. 55%

D. 27.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q4/5

True or false?

Of the students who play a sport, more than half are in Grade 10.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q5/5

SET 62 • HSS-ID.B.5

BACKWARD 1

Read two-way tables for association • Grade 8 • Statistics & Probability • 8.SP.A.4

12 of 20 boys and 9 of 15 girls in a class like math.

How many students in all like math?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q1/5

12 of 20 boys and 9 of 15 girls in a class like math.

Is there an association between gender and liking math? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q2/5

What are the row and column totals of a two-way table called?

A. Joint frequencies

B. Conditional frequencies

C. Marginal frequencies

D. Outliers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q3/5

True or false?

A two-way table shows data on two categorical variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q4/5

True or false?

In a two-way table, every row must have the same total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSS-ID.B.5

BACKWARD 2

Write a part of a whole as a percent • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Write 45 out of 100 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q1/5

Write 60 out of 105 as a percent, rounded to the nearest tenth.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q2/5

27 is what percent of 60?

A. 45%

B. 60%

C. 33%

D. 27%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q3/5

True or false?

55 out of 200 is 27.5%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q4/5

True or false?

40 out of 95 is 40%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSS-ID.B.5

FORWARD 1

Use a two-way table to find conditional probabilities and test independence • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

A student from the table is chosen at random.

Find P(plays a sport | Grade 10).

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.4 • Set 62 • FORWARD 1 • Q1/5

Are "Grade 10" and "plays a sport" independent? Explain with probabilities.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.4 • Set 62 • FORWARD 1 • Q2/5

What is P(Grade 9 | plays a sport)?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. About 0.43

B. 0.225

C. 0.525

D. 0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q3/5

True or false?

If P(B) > 0, events A and B are independent exactly when P(A | B) = P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.4 • Set 62 • FORWARD 1 • Q4/5

True or false?

P(A | B) always equals P(B | A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.4 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSS-ID.B.5

FORWARD 2

Explain conditional probability and independence in everyday language • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

Explain in everyday language what P(late | rain) > P(late) means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.5 • Set 62 • FORWARD 2 • Q1/5

A and B are independent, P(A) = 0.3, and P(B) = 0.5.

Find P(A and B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.5 • Set 62 • FORWARD 2 • Q2/5

Which pair of events is most likely independent?

A. Height and shoe size

B. Studying and passing a test

C. Flipping heads and rolling a 6

D. Rain and carrying an umbrella

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q3/5

True or false?

If A and B are independent, then P(A and B) = P(A) · P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.5 • Set 62 • FORWARD 2 • Q4/5

True or false?

If P(A | B) = 0.8 and P(A) = 0.4, then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.5 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSS-ID.B.6.a • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Fit a linear, quadratic, or exponential function to data and use it to solve problems

MAIN	Algebra I main question	**HSS-ID.B.6.a**
BACKWARD 1	Describe patterns in scatter plots	**8.SP.A.1**
BACKWARD 2	Write linear and exponential functions through given points	**HSF-LE.A.2**
FORWARD 1	Use logarithms with an exponential model fitted to data	**HSF-LE.A.4**
FORWARD 2	Fit a sinusoidal model to periodic data	**HSF-TF.B.5**

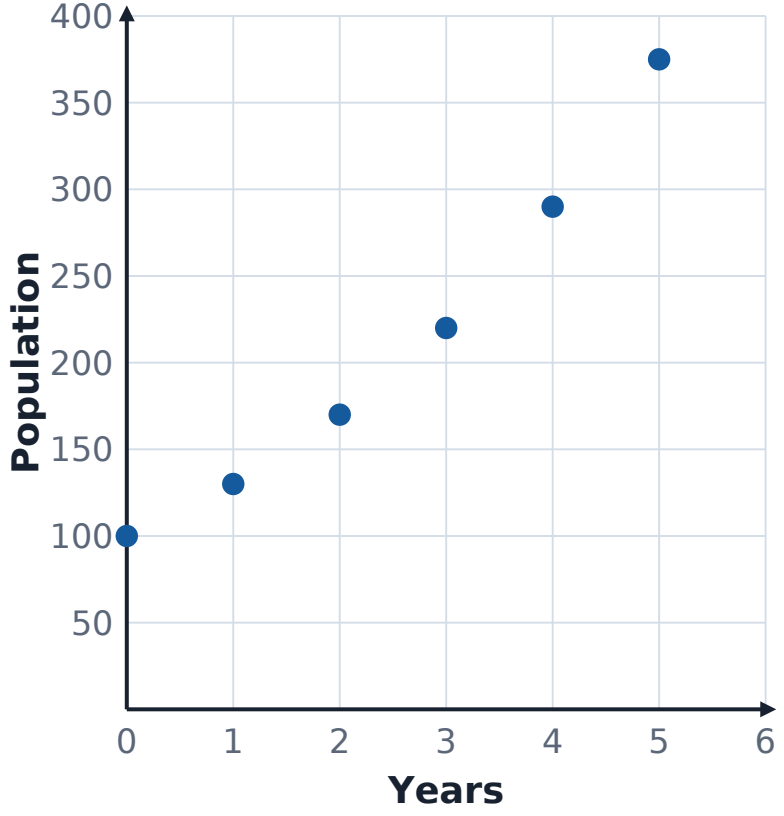
SET 63 • HSS-ID.B.6.a

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

The scatter plot shows a population over 5 years.

Which type of function fits best? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q1/5

An exponential model for the data is $P(t) = 100(1.3)^t$.

Use it to predict the population at $t = 8$, to the nearest whole number.

$P(8) \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q2/5

Data: x = 0, 1, 2, 3, 4 and y = 2, 5, 10, 17, 26.

Which quadratic function fits these data exactly? Write it.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q3/5

A scatter plot rises and then falls in a symmetric curve.

Which model fits best?

A. Quadratic

B. Exponential growth

C. Exponential decay

D. Linear

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q4/5

True or false?

Using a model to predict far outside the range of the data is always reliable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q5/5

SET 63 • HSS-ID.B.6.a

BACKWARD 1

Describe patterns in scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

Points rise from left to right, close to a line.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

What is an outlier in a scatter plot?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

Points fall from left to right in a curved pattern.

Which describes the association?

A. Positive and nonlinear

B. Positive and linear

C. Negative and nonlinear

D. No association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

True or false?

A scatter plot shows the relationship between two quantitative variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

True or false?

If the points show no pattern, there is a strong association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSS-ID.B.6.a

BACKWARD 2

Write linear and exponential functions through given points • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the exponential function through (0, 100) and (1, 130).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q1/5

Write the linear function through (0, 2) and (4, 26).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q2/5

Which exponential function $y = ab^x$ $(b > 0)$ passes through $(0, 50)$ and $(2, 12.5)$?

A. $y = 50(0.25)^x$

B. $y = 12.5(2)^x$

C. $y = 50(0.5)^x$

D. $y = 50 - 18.75x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q3/5

True or false?

$y = 3(2)^x$ passes through (2, 12).

A. True

B. False

True or false?

$y = 5(1.2)^x$ passes through (0, 6).

A. True

B. False

SET 63 • HSS-ID.B.6.a

FORWARD 1

Use logarithms with an exponential model fitted to data • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$P(t) = 100(1.3)^t$.

When does P reach 1,000? Round to the nearest tenth.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q1/5

Solve $2(1.5)^x = 30$. Round to the nearest hundredth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q2/5

Which is the exact solution of $100(1.3)^t = 500$? (log means base 10.)

A. t = log 1.3 ÷ log 5

B. t = 500 ÷ 130

C. t = log 5 ÷ log 1.3

D. t = 5 ÷ 1.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q3/5

True or false?

If $1.3^t = 1.69$, then $t = 2$.

A. True

B. False

True or false?

If $10^t = 50$, then $t = 5$.

A. True

B. False

SET 63 • HSS-ID.B.6.a

FORWARD 2

Fit a sinusoidal model to periodic data • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

Monthly average temperatures peak at 30 °C and bottom out at 10 °C.

What amplitude and midline fit a sinusoidal model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q1/5

What period fits monthly temperatures that repeat every year?

Period: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q2/5

Daylight hours vary from 9 to 15 over a year.

Which model fits?

A. A linear model

B. A sinusoidal model with amplitude 3 and period 12 months

C. A quadratic model with vertex at month 12

D. An exponential model

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q3/5

True or false?

A repeating pattern in data suggests a sinusoidal model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q4/5

True or false?

Data that vary between 9 and 15 have amplitude 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q5/5

SET 83 • HSS-ID.B.6.a • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Construct scatter plots from paired data and describe how the variables are related

MAIN	Algebra I main question	**HSS-ID.B.6.a**
BACKWARD 1	Make and describe scatter plots	**8.SP.A.1**
BACKWARD 2	Plot points and find distances on the coordinate plane	**6.NS.C.8**
FORWARD 1	Choose an exponential model from a scatter plot and solve it with…	**HSF-LE.A.4**
FORWARD 2	Evaluate claims based on scatter plots	**HSS-IC.B.6**

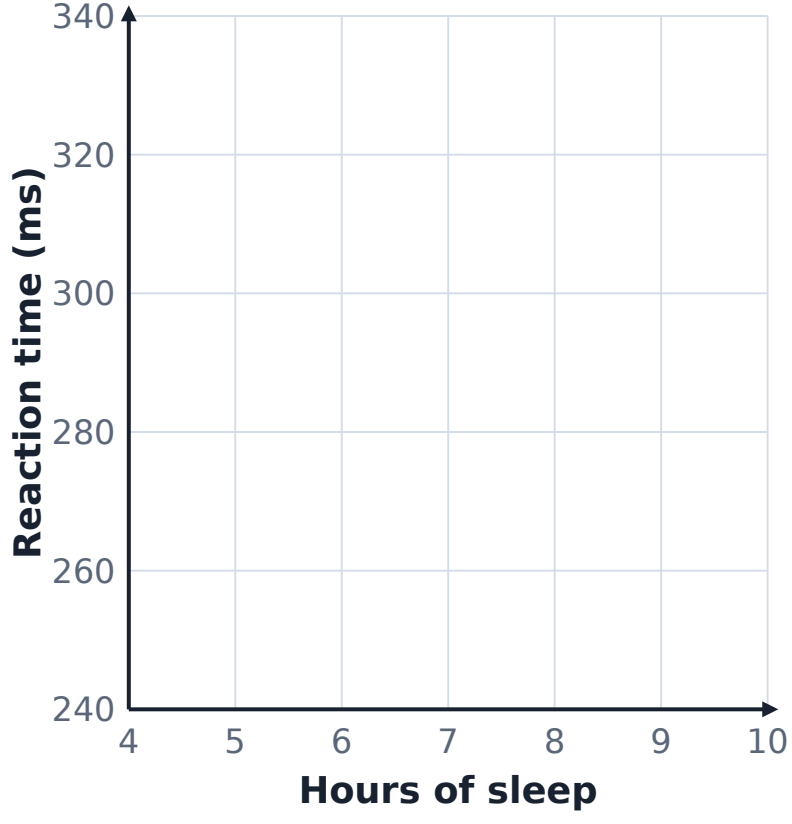
SET 83 • HSS-ID.B.6.a

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

Make a scatter plot of hours of sleep (x) and reaction time in milliseconds (y):

(5, 320), (6, 300), (6.5, 290), (7, 280), (8, 260), (9, 255)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q1/5

Describe the association between hours of sleep and reaction time in the data (5, 320), (6, 300), (6.5, 290), (7, 280), (8, 260), (9, 255): its direction, form, and strength.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q2

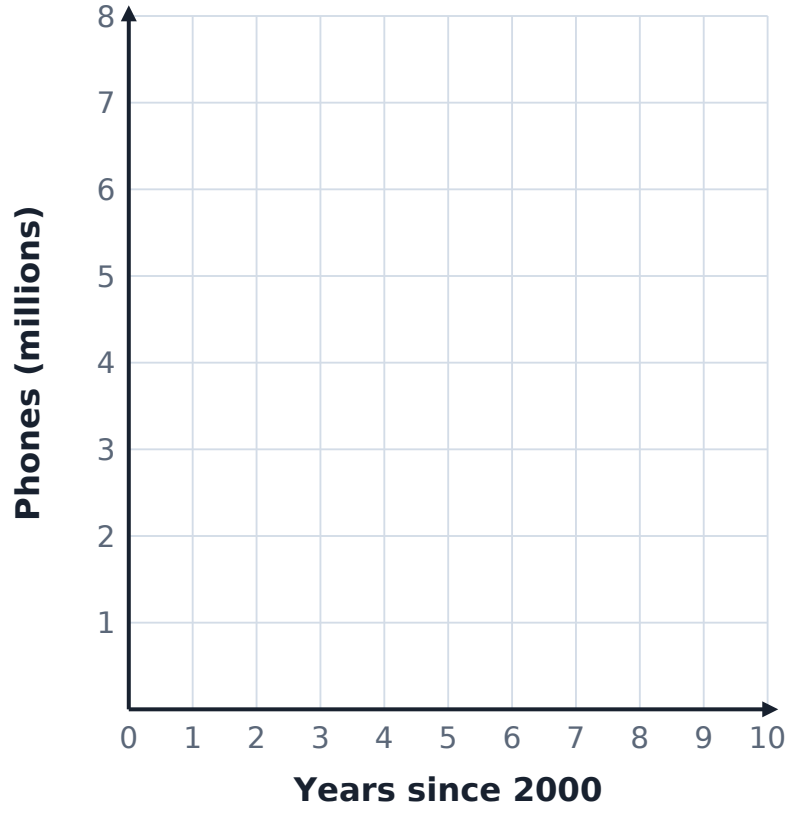
High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q2/5

Make a scatter plot of years since 2000 (x) and phones in millions (y):

(0, 1), (2, 1.5), (4, 2.3), (6, 3.4), (8, 5.1), (10, 7.6)

Which type of model does the plot suggest?

Model: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q3/5

When you make a scatter plot to see whether one variable helps explain another, which variable usually goes on the horizontal axis?

A. Whichever variable has larger numbers

B. Whichever variable has fewer values

C. The explanatory variable (the one that may explain the other)

D. The response variable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q4/5

True or false?

Both axes of a scatter plot must start at 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q5

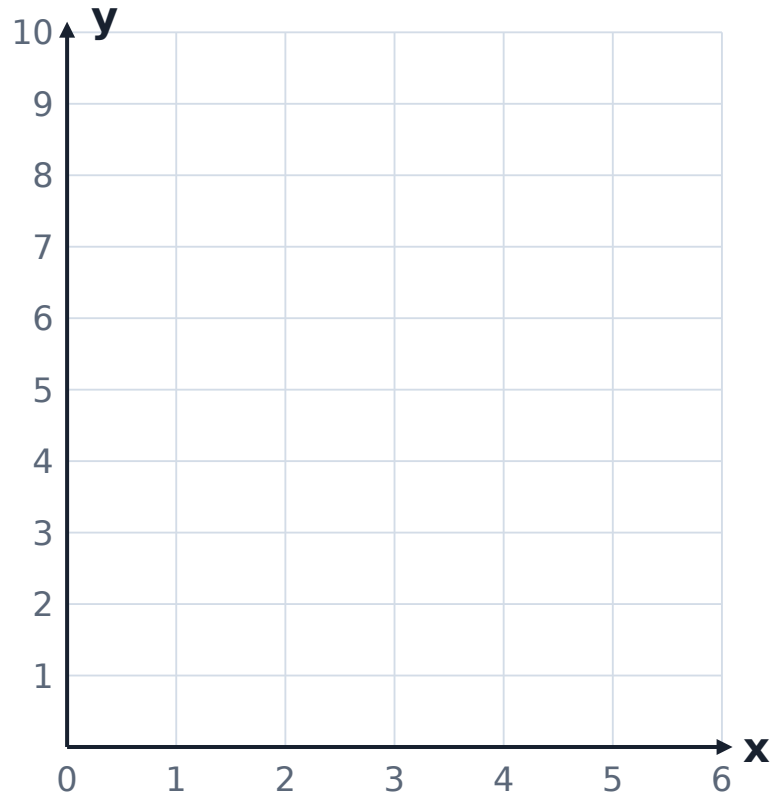
High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q5/5

SET 83 • HSS-ID.B.6.a

BACKWARD 1

Make and describe scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

Plot (1, 2), (2, 3), (3, 5), (4, 6), and (5, 8) as a scatter plot.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q1/5

Points rise from left to right but curve upward. Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q2/5

Which describes a scatter plot whose points show no trend?

A. Linear association

B. Strong positive association

C. Strong negative association

D. No association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q3/5

True or false?

Each point in a scatter plot represents one pair of values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q4/5

True or false?

A cluster in a scatter plot is the same thing as an outlier.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q5

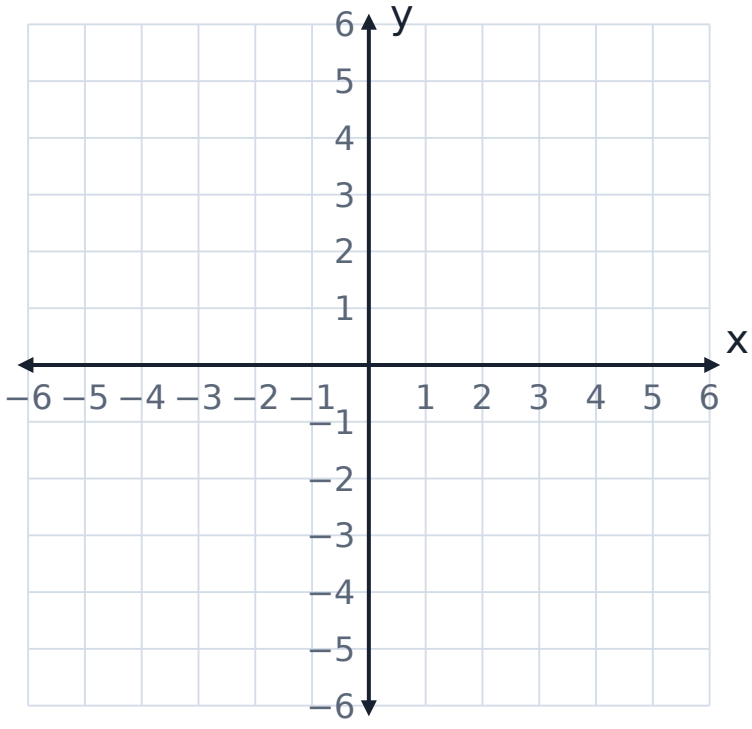
Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q5/5

SET 83 • HSS-ID.B.6.a

BACKWARD 2

Plot points and find distances on the coordinate plane • Grade 6 • The Number System • 6.NS.C.8

Plot the points (2, 5), (−3, 1), and (0, −4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q1/5

Find the distance between $(-3, 2)$ and $(5, 2)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q2/5

Which point lies on the y-axis?

A. $(1, -1)$

B. $(0, -4)$

C. $(-4, 0)$

D. $(4, 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q3/5

True or false?

The point $(-2, 6)$ is in Quadrant II.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q4/5

True or false?

The distance between $(1, -3)$ and $(1, 4)$ is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q5/5

SET 83 • HSS-ID.B.6.a

FORWARD 1

Choose an exponential model from a scatter plot and solve it with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

A model for the phone data is $y = 1(1.22)^x$, where x is years since 2000 and y is millions of phones. When does it predict 20 million? Round to the nearest tenth. (log means base 10.)

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q1/5

Using $y = 1(1.22)^x$, how many years does it take the number of phones to double? Round to the nearest tenth. (log means base 10.)

Years: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q2/5

Solve $4(1.25)^x = 20$. (log means base 10.)

A. $x = \log 5 \div \log 1.25 \approx 7.21$

B. $x = 5 \div 1.25$

C. $x = 4$

D. $x = \log 20 \div \log 1.25 \approx 13.43$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q3/5

True or false?

If $y = a \cdot b^x$ with $a > 0$ and $b > 1$, then y doubles every $\log 2 \div \log b$ units of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q4/5

True or false?

The model $y = 1(1.22)^x$ predicts 0 phones in 2000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSS-ID.B.6.a

FORWARD 2

Evaluate claims based on scatter plots • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A website plots shoe size against reading score for students in Grades 1 to 8, finds a strong positive association, and claims that bigger feet improve reading. What is wrong?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q1/5

A scatter plot of 6 students from one class shows a strong trend. Why should you be careful applying it to all students?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q2/5

Which detail would make a claim based on a scatter plot more trustworthy?

A. Both axes start at 0.

B. The plot uses bright colors.

C. There are only 5 points.

D. The data come from a large random sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q3/5

True or false?

Extrapolating a trend far beyond the plotted data can give misleading predictions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q4/5

True or false?

If a scatter plot shows a strong trend, one variable causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q5/5

SET 64 • HSS-ID.B.6.b • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Assess the fit of a function by plotting and analyzing residuals

MAIN	Algebra I main question	**HSS-ID.B.6.b**
BACKWARD 1	Use a fitted function to make predictions	**HSS-ID.B.6.a**
BACKWARD 2	Subtract rational numbers	**7.NS.A.1.c**
FORWARD 1	Use logarithms to solve exponential models fitted to data	**HSF-LE.A.4**
FORWARD 2	Evaluate reports that use fitted models	**HSS-IC.B.6**

SET 64 • HSS-ID.B.6.b

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

The model $\hat{y} = 2.2x + 0.8$ is fitted to the data (1, 3), (2, 5), (3, 8), (4, 9), (5, 12).

Find the residual for each point.

Residuals: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q1/5

For that model, what does the residual -0.6 at $x = 4$ mean?

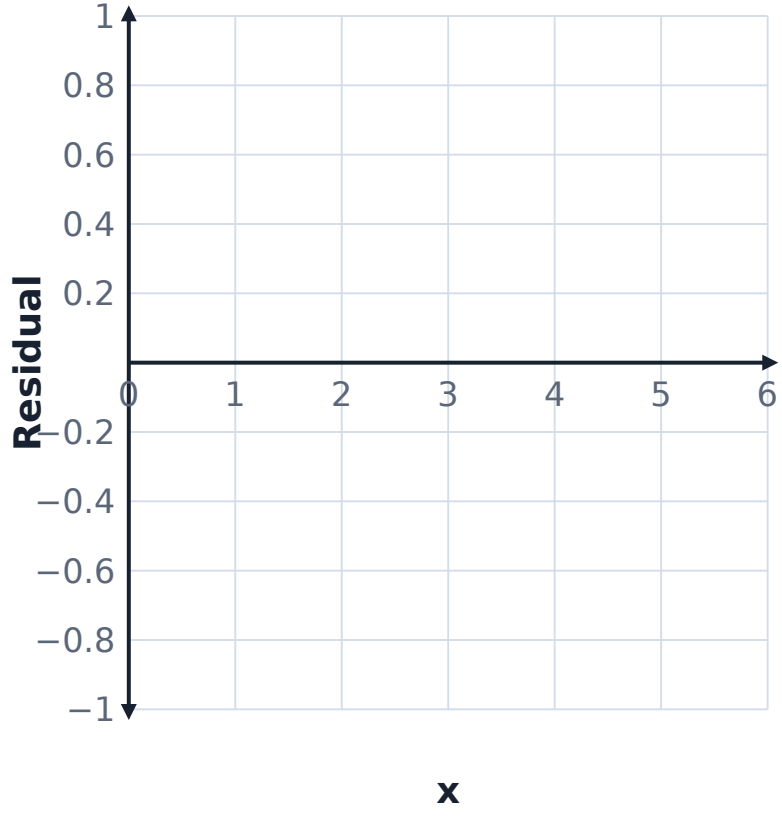
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q2/5

Make a residual plot for the residuals 0, −0.2, 0.6, −0.6, 0.2 at x = 1, 2, 3, 4, 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q3/5

A residual plot shows a clear U-shaped pattern.

What does that suggest?

A. The correlation is exactly 1.

B. A linear model is not a good fit; a curved model may fit better.

C. The linear model fits well.

D. There are no residuals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q4/5

True or false?

A good linear fit has residuals scattered randomly around 0, with no pattern.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q5/5

SET 64 • HSS-ID.B.6.b

BACKWARD 1

Use a fitted function to make predictions • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

The model is $\hat{y} = 2.2x + 0.8$.

Predict y when $x = 6$.

$\hat{y} =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q1/5

The model is $y = 3x^2 - 2$.

Predict y when x = 4.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q2/5

The model is $y = 50(0.9)^x$.

Predict y when x = 2.

A. 50

B. 40.5

C. 45

D. 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q3/5

True or false?

A predicted value comes from the model, not from the data table.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q4/5

True or false?

If a model predicts 14 and the actual value is 14, the residual is 14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q5/5

SET 64 • HSS-ID.B.6.b

BACKWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.c

Find $8 - 7.4$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q1/5

Find $9 - 9.6$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q2/5

Find $5 - 5.2$.

A. -10.2

B. 10.2

C. 0.2

D. -0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q3/5

True or false?

$3 - (-2) = 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q4/5

True or false?

$-4 - 6 = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q5/5

SET 64 • HSS-ID.B.6.b

FORWARD 1

Use logarithms to solve exponential models fitted to data • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

A model fitted to bacteria counts is $N = 150(1.4)^t$, with t in hours. When does N reach 1,000? Give the exact expression and the value to the nearest tenth of an hour. (log means base 10.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q1/5

A line fitted to the logarithms of data is log y = 0.3x + 1 (log means base 10).

Write y as an exponential function of x.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q2/5

A fitted model is $y = 20e^{0.08x}$. For which x is y = 60? (ln means base e.)

A. $x = 3 \div 0.08 = 37.5$

B. $x = \ln 60 \div 0.08 \approx 51.2$

C. $x = 0.08 \ln 3 \approx 0.09$

D. $x = \ln 3 \div 0.08 \approx 13.7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q3/5

True or false?

For the model $y = 5(1.5)^x$, the solution of $5(1.5)^x = 50$ is $x = \log 10 \div \log 1.5 \approx 5.68$. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q4/5

True or false?

The doubling time of $N = 150(1.4)^t$ is $\log 2 \div \log 1.4 \approx 0.49$ hour. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q5/5

SET 64 • HSS-ID.B.6.b

FORWARD 2

Evaluate reports that use fitted models • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A report fits a line to the heights of children aged 5 to 15 and uses it to predict the height of a 40-year-old.

What is the problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 64 • FORWARD 2 • Q1/5

A report finds a strong correlation between ice cream sales and drownings and concludes that ice cream causes drowning.

Do these observational data establish that conclusion? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 64 • FORWARD 2 • Q2/5

A residual plot shows a strong curve, but a report says the linear model is excellent.

What should you conclude?

A. The report is correct.

B. The data must be wrong.

C. The residuals are all 0.

D. The linear model may not be appropriate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 64 • FORWARD 2 • Q3/5

True or false?

A report should state how its data were collected.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q4/5

True or false?

A model with a high correlation coefficient is always the right type of model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 64 • FORWARD 2 • Q5/5

SET 65 • HSS-ID.B.6.c • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Fit a linear function to a scatter plot that suggests a linear association

MAIN	Algebra I main question	**HSS-ID.B.6.c**
BACKWARD 1	Judge a line of fit informally	**8.SP.A.2**
BACKWARD 2	Write a line through two points	**HSF-LE.A.2**
FORWARD 1	Evaluate conclusions drawn from a line of fit	**HSS-IC.B.6**
FORWARD 2	Use a randomized experiment and simulation to test whether mo…	**HSS-IC.B.5**

SET 65 • HSS-ID.B.6.c

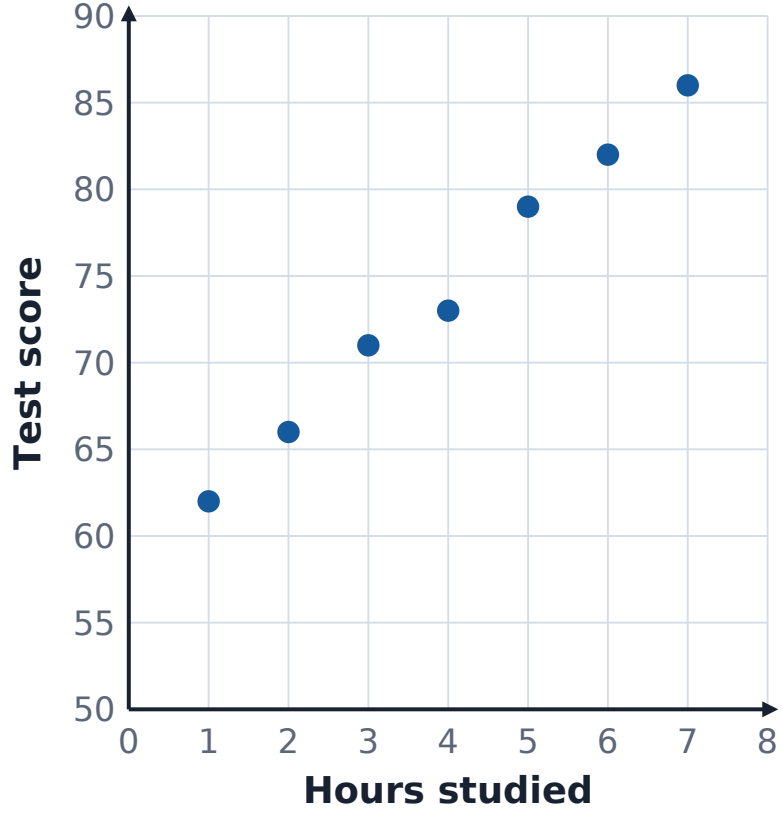
MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

The scatter plot shows hours studied and test scores.

Draw a line of fit and write its equation.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q1/5

The least-squares line for the data is $\hat{y} = 4x + 58.14$.

Predict the score for 4.5 hours, to the nearest whole number.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q2/5

Write the equation of the line through the data points (1, 62) and (7, 86), to use as a line of fit.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q3/5

Which line fits the points (0, 1), (1, 3), (2, 5), (3, 7) exactly?

A. $y = 3x + 1$

B. $y = 2x + 1$

C. $y = 2x$

D. $y = x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q4/5

True or false?

A line of fit should pass through every data point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q5/5

SET 65 • HSS-ID.B.6.c

BACKWARD 1

Judge a line of fit informally • Grade 8 • Statistics & Probability • 8.SP.A.2

Is a straight line a good model for points that curve strongly upward? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q1/5

How can you judge whether a line of fit is good?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q2/5

Which is the best sign of a good line of fit?

A. All of the points are below it.

B. All of the points are above it.

C. The points are close to the line and follow its trend.

D. Exactly two points are on it.

True or false?

Lines of fit are used for data with a linear association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q4/5

True or false?

A line of fit must pass through the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q5/5

SET 65 • HSS-ID.B.6.c

BACKWARD 2

Write a line through two points • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the equation of the line through (2, 66) and (6, 82).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q1/5

Write the equation of the line through (0, 7) and (3, 13).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q2/5

Which line passes through (0, 10) and (5, 0)?

A. $y = 2x + 10$

B. $y = 10x - 2$

C. $y = -2x + 10$

D. $y = -0.5x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q3/5

True or false?

The line through (0, 3) with slope 4 is $y = 4x + 3$.

A. True

B. False

True or false?

The line through (2, 2) and (4, 8) has slope 2.

A. True

B. False

SET 65 • HSS-ID.B.6.c

FORWARD 1

Evaluate conclusions drawn from a line of fit • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

In a study of 7 volunteer students, each extra hour of study went with about 4 more points.

Can a school claim this holds for all of its students? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions

HSS-IC.B.6 • Set 65 • FORWARD 1 • Q1/5

A study of 7 students who studied from 1 to 7 hours found the line $\hat{y} = 4x + 58$, where x is hours of study and $\hat{y}$ is the test score out of 100.

A report uses this line to predict a score of 138 for 20 hours of study. Why is that unreasonable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 65 • FORWARD 1 • Q2/5

Which detail makes a regression conclusion more trustworthy?

A. The prediction is far outside the data.

B. The line was drawn by hand.

C. The data came from a large random sample.

D. The sample was 5 volunteers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q3/5

True or false?

Observational data alone cannot prove that one variable causes another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q4/5

True or false?

A line of fit from 7 volunteer students proves that extra study raises the score of every student.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q5/5

SET 65 • HSS-ID.B.6.c

FORWARD 2

Use a randomized experiment and simulation to test whether more study raises scores • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

To test whether extra study raises scores, 40 volunteers were randomly assigned: 20 studied 3 extra hours and 20 did not. The extra-study group averaged 5 points higher. In 1,000 random regroupings of the 40 scores, a difference of 5 points or more (first group minus second) happened 18 times.

Using a 5% threshold, is the difference significant? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

In a randomized experiment, a tutoring group averaged 6 points more than a control group. Random regroupings of the scores gave differences of 6 points or more 30% of the time.

Using a 5% threshold, what is the conclusion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

What is the purpose of random assignment in an experiment?

A. To balance other variables between the groups, on average

B. To make the sample smaller

C. To guarantee a significant result

D. To remove the need for a control group

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

True or false?

A result is statistically significant when a difference that large would rarely happen by chance alone, using a threshold chosen in advance such as 5%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

True or false?

Using a 5% threshold, a difference that happens in 40% of random regroupings is statistically significant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

Interpret the slope and the intercept of a linear model in the context of the data

MAIN	Algebra I main question	**HSS-ID.C.7**
BACKWARD 1	Interpret the parameters of linear and exponential functions	**HSF-LE.B.5**
BACKWARD 2	Use a linear model to solve problems	**8.SP.A.3**
FORWARD 1	Interpret the parameters of continuous exponential models	**HSF-LE.A.4**
FORWARD 2	Understand a fitted slope as an estimate of a population value	**HSS-IC.A.1**

SET 66 • HSS-ID.C.7

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

$\hat{y} = 4x + 58$ predicts a test score from x hours of study.

Interpret the slope and the intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q1/5

A model for used cars is $\hat{y} = 22{,}000 - 1{,}800x$, where x is the age in years and $\hat{y}$ is the price in dollars.

Interpret $-1{,}800$ and $22{,}000$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q2/5

$\hat{y} = 0.9x - 20$ predicts an adult's weight in pounds from height in inches.

Does the intercept make sense in context? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q3/5

$\hat{y} = 2.5x + 10$ predicts the cost in dollars of a pizza with x toppings.

What does 2.5 mean?

A. Each additional topping adds about \$2.50 to the predicted cost.

B. A pizza has 2.5 toppings.

C. A pizza costs \$2.50.

D. A plain pizza costs \$2.50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q4/5

True or false?

The slope of a fitted line gives the exact change in y for every individual.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q5/5

SET 66 • HSS-ID.C.7

BACKWARD 1

Interpret the parameters of linear and exponential functions • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

A technician charges $C = 50 + 20h$ dollars for h hours.

What does 20 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q1/5

$P = 3{,}000(0.9)^t$ is a population after t years.

What does 0.9 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q2/5

$d = 300 - 60t$ is a driver's distance from home, in miles, after t hours.

What does 300 mean?

A. The starting distance from home

B. The speed

C. The distance driven each hour

D. The time to arrive home

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q3/5

True or false?

In y = 1.5x + 4, increasing x by 2 increases y by 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q4/5

True or false?

In $y = 8(1.25)^x$, y increases by 1.25 for each unit of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • HSS-ID.C.7

BACKWARD 2

Use a linear model to solve problems • Grade 8 • Statistics & Probability • 8.SP.A.3

$y = 2x + 5$ models a plant's height in cm after x weeks.

Predict the height at 6 weeks.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q1/5

Using $y = 2x + 5$, after how many weeks is the plant 21 cm tall?

Weeks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q2/5

$y = 2x + 5$ models a plant's height in cm after x weeks.

In this model, what does 5 mean?

A. The height at week 0

B. The weeks needed to grow 1 cm

C. The final height

D. The growth per week

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q3/5

True or false?

The slope of a linear model is a rate of change.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q4/5

True or false?

In y = 2x + 5, the plant grows 5 cm each week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSS-ID.C.7

FORWARD 1

Interpret the parameters of continuous exponential models • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$A(t) = 500e^{0.04t}$.

What does 0.04 mean? When does A double, to the nearest tenth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q1/5

$N(t) = 1{,}000e^{-0.1t}$.

What does -0.1 mean? When is $N = 500$, to the nearest tenth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q2/5

Solve $e^{2x} = 9$ exactly. (ln means base e.)

A. $x = \ln 3$

B. $x = 4.5$

C. $x = e^{4.5}$

D. $x = \ln 9 \div 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q3/5

True or false?

$\ln(e^{0.04t}) = 0.04t$ (ln means base e.)

A. True

B. False

True or false?

If $e^k = 1$, then $k = 1$.

A. True

B. False

SET 66 • HSS-ID.C.7

FORWARD 2

Understand a fitted slope as an estimate of a population value • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

A random sample of 50 used cars of one model gives $\hat{y} = 22{,}000 - 1{,}800x$.

What population value does $-1{,}800$ estimate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q1/5

Would another random sample of 50 cars give exactly the same slope? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.1 • Set 66 • FORWARD 2 • Q2/5

Which of these is a parameter?

A. The mean price of the 50 sampled cars

B. One car's price

C. The slope from the sample

D. The mean price of all used cars of this model

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions

HSS-IC.A.1 • Set 66 • FORWARD 2 • Q3/5

True or false?

Larger random samples tend to give statistics closer to the parameter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q4/5

True or false?

A statistic is a number that describes a whole population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.1 • Set 66 • FORWARD 2 • Q5/5

Compute (using technology) and interpret the correlation coefficient of a linear fit

MAIN	Algebra I main question	**HSS-ID.C.8**
BACKWARD 1	Describe the direction of an association	**8.SP.A.1**
BACKWARD 2	Fit and use a line for linear data	**HSS-ID.B.6.c**
FORWARD 1	Use a randomized experiment and simulation to test whether an a…	**HSS-IC.B.5**
FORWARD 2	Evaluate claims based on a correlation	**HSS-IC.B.6**

SET 67 • HSS-ID.C.8

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Use technology to find the correlation coefficient r for the data

(1, 9), (2, 8), (3, 8), (4, 6), (5, 5), (6, 5), (7, 3), (8, 2),

to the nearest hundredth. Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q1/5

$r = -0.85$ for hours of TV and test score.

Interpret this value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q2/5

Which value of r shows the strongest linear relationship?

A. 0.10

B. −0.92

C. −0.40

D. 0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q3/5

Can a correlation coefficient equal 1.3? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q4/5

True or false?

r = 0 means there is no relationship of any kind between the variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q5/5

SET 67 • HSS-ID.C.8

BACKWARD 1

Describe the direction of an association • Grade 8 • Statistics & Probability • 8.SP.A.1

Points rise from left to right, fairly close to a line.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q1/5

Points for hours of sleep and number of errors fall from left to right.

Is the association positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q2/5

Which pair of variables likely has a positive association?

A. Height and arm span

B. Outdoor temperature and heating cost

C. A car's age and its value

D. Speed and travel time for a fixed trip

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q3/5

True or false?

In a negative association, as one variable increases, the other tends to decrease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q4/5

True or false?

Clusters in a scatter plot mean the data are linear.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • HSS-ID.C.8

BACKWARD 2

Fit and use a line for linear data • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

Write a line of fit through (2, 5) and (8, 14).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q1/5

The least-squares line is $\hat{y} = 1.5x + 2$.

Predict y when x = 6.

$\hat{y} =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q2/5

What does the least-squares line minimize?

A. The y-intercept

B. The sum of the squared residuals

C. The number of points

D. The slope

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q3/5

True or false?

The least-squares line passes through the point $(\bar{x}, \bar{y})$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q4/5

True or false?

Two people drawing lines of fit by eye always get the same line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSS-ID.C.8

FORWARD 1

Use a randomized experiment and simulation to test whether an association is causal • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

Observational data show r = −0.6 between evening screen time and hours of sleep. To test cause, 50 volunteers were randomly assigned to no screens after 8 p.m. or to their usual habits. The no-screen group slept 0.5 hour more on average. In 1,000 random regroupings, a difference of at least 0.5 hour (no-screen minus usual) happened 2 times.

Using a 5% threshold, is the difference significant? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.5 • Set 67 • FORWARD 1 • Q1/5

Why can this randomized experiment, but not the observational correlation alone, support the claim that evening screens reduce sleep?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.5 • Set 67 • FORWARD 1 • Q2/5

In a randomization test, the observed difference was matched or exceeded in 350 of 1,000 regroupings.

Using a 5% threshold, what is the conclusion?

A. The treatment definitely has no effect.

B. The difference is not significant; chance could easily explain it.

C. The difference is significant.

D. The experiment proves causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions

HSS-IC.B.5 • Set 67 • FORWARD 1 • Q3/5

True or false?

In a randomization test, the regroupings show how large the difference between groups tends to be when the treatment has no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q4/5

True or false?

If an experiment's difference is not significant, that proves the treatment has no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSS-ID.C.8

FORWARD 2

Evaluate claims based on a correlation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A news story says, "r = 0.6, so 60% of the change in sales is caused by advertising."

Name two errors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

A study of 8 cities finds r = 0.7 between the number of parks and happiness scores.

Why should you be cautious?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

Which statement about $r = -0.9$ is correct?

A. There is a strong negative linear association.

B. The slope of the line is -0.9.

C. 90% of the points are on the line.

D. An increase in x causes y to fall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

True or false?

A correlation can be strong even when the relationship is not causal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

True or false?

If r = 0.8, the slope of the least-squares line is 0.8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

SET 68 • HSS-ID.C.9 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Distinguish between correlation and causation

MAIN	Algebra I main question	**HSS-ID.C.9**
BACKWARD 1	Interpret the correlation coefficient	**HSS-ID.C.8**
BACKWARD 2	Use random samples to make valid generalizations	**7.SP.A.1**
FORWARD 1	Distinguish surveys, experiments, and observational studies	**HSS-IC.B.3**
FORWARD 2	Use random methods to make fair decisions — nearest related	**HSS-MD.B.6**

SET 68 • HSS-ID.C.9

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Data show that cities with more firefighters have more fire damage.

Do these data establish that hiring firefighters causes damage? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q1/5

Ice cream sales and sunburn rates are strongly correlated.

Name a lurking variable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q2/5

How could you test whether a new study app causes higher test scores?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q3/5

Which study design can show that one variable causes changes in another?

A. A survey of volunteers

B. A scatter plot of existing data

C. An observational study

D. A randomized experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q4/5

True or false?

A strong correlation between two variables proves that one causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q5/5

SET 68 • HSS-ID.C.9

BACKWARD 1

Interpret the correlation coefficient • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

$r = 0.88$.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q1/5

Which shows the stronger linear association, $r = -0.7$ or $r = 0.5$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q2/5

What is r for points that lie exactly on a line with negative slope?

A. 1

B. 0

C. −0.5

D. −1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q3/5

True or false?

The correlation coefficient has no units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q4/5

True or false?

Switching the roles of x and y changes the value of r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSS-ID.C.9

BACKWARD 2

Use random samples to make valid generalizations • Grade 7 • Statistics & Probability • 7.SP.A.1

Why is a random sample more likely to represent a population than a convenience sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q1/5

A survey at a gym asks about the exercise habits of all town residents.

What is the problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q2/5

Which is a random sample of a school's students?

A. Drawing 40 names from a list of all students

B. Asking the first 40 students in line

C. Asking your friends

D. Asking one math class

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q3/5

True or false?

Generalizations about a population are valid only if the sample represents the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q4/5

True or false?

A sample of volunteers is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • HSS-ID.C.9

FORWARD 1

Distinguish surveys, experiments, and observational studies • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Researchers record students' screen time and grades without changing anything.

What kind of study is this?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.3 • Set 68 • FORWARD 1 • Q1/5

Researchers randomly assign plants to receive fertilizer or no fertilizer.

What kind of study is this?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.3 • Set 68 • FORWARD 1 • Q2/5

What is the main purpose of a survey?

A. To test whether a treatment causes an effect

B. To assign treatments to subjects

C. To control lurking variables

D. To estimate a characteristic of a population from a sample

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.3 • Set 68 • FORWARD 1 • Q3/5

True or false?

Random selection lets you generalize to a population; random assignment lets you conclude cause and effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions

HSS-IC.B.3 • Set 68 • FORWARD 1 • Q4/5

True or false?

In an observational study, the researchers decide which treatment each subject gets.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.3 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSS-ID.C.9

FORWARD 2

Use random methods to make fair decisions • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Random assignment, which is needed to show causation, is a fair random method.

Describe a fair way to split 20 volunteers into two groups of 10 using random numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q1/5 • NEAREST RELATED

A coach picks a captain by drawing one name from a hat with 12 names.

What is the probability that a given player is chosen?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which method is fair for deciding who goes first among 4 players?

A. The tallest goes first

B. Each rolls a number cube; ties roll again; the highest goes first

C. The coach's favorite goes first

D. Whoever asks first goes first

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A fair lottery gives every participant the same probability of being chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

Flipping one fair coin once is a fair way to choose one person from 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q5/5 • NEAREST RELATED